
Repair Manual

911 Carrera **(993)**

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Preface

Structure

The "Technical Literature" for the "911 Carrera (993)" model is basically structured as before, i.e. the structure follows the familiar repair groups.

A new feature is that the structure includes the main groups 0 to 9 and the main group D.

Main groups:	0	Complete vehicle – General
	1	Engine
	2	Fuel, exhaust, engine electrical system
	3	Transmission
	4	Chassis
	5	Body
	6	Body equipment, outside
	7	Body equipment, interior
	8	Air conditioning
	9	Electrical system
	D	Diagnosis

Layout

The layout in the below items remains unchanged throughout the repair manual

1. Table of tightening torques
2. Special tools required
3. Exploded views
4. Legends for the exploded views
5. Assembly notes / use of special tools

As a new feature, however, the former item 6 (Repair group diagnosis) is no longer filed in the volume corresponding to the respective repair group. The **Diagnosis test plans / diagnosis procedures** have been combined in a **separate Diagnosis volume** broken down according to the main groups 0 to 9.

Another new feature is that the contents of the "Service Information Technik" are indicated in the Repair Manual. This brochure concentrates on a description of the design and function of components and of the new features introduced for a particular model year.

Service Number

All major repair procedures and repair descriptions are identified by a two- or four-digit **Service Number** completed by two additional digits to identify the work that corresponds to the first six digits of the working position number in the Working Times and Damage Catalog.

Example: 30 37 37 Dismantling and assembling clutch control shaft

Explanation:	30	37	37	50 (full working position number)
Repair group here: Clutch, control	_____	_____	_____	_____
Component designation here: Clutch control shaft	_____	_____	_____	_____
Activity here: Dismantling and assembling	_____	_____	_____	_____
Index here: Removed	_____	_____	_____	_____

Presentation in the various documents

30 37 37 50	Working position no. from Working Times and Damage Catalog , consisting of repair group, component designation, activity and index
30 37 37	Six-digit number in Repair Manual , consisting of repair group, component designation and activity
30 37	Service number in Service Information , consisting of repair group and component designation

Goal

The introduction of a service number in the "technical literature" is intended to facilitate standardization and positive identification to allow direct cross-referencing among the various documents. This is of particular importance with regard to the use of electronic media.

II Manual transmission

The Repair Manual of the 911 Carrera (993) also includes the 911 Carrera 4 manual (993 four-wheel drive). The 911 Carrera (993) is the basic model covered by the repair operations described in this Manual. "911 Carrera (993)" is also indicated in the header of each page.

Descriptions of repair operations that deviate for the 911 Carrera 4 will be included after the respective 911 Carrera section. The repair descriptions of both models are separated by a cover page. All pages included after the cover page (separation sheet) have the "911 Carrera 4" heading. To facilitate distinction, the page numbering will start with 200.

3 Transmission

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Survey of contents of Service Information Technik '95

The Service Information gives a detailed description of the technical features of the new 911 Carrera.

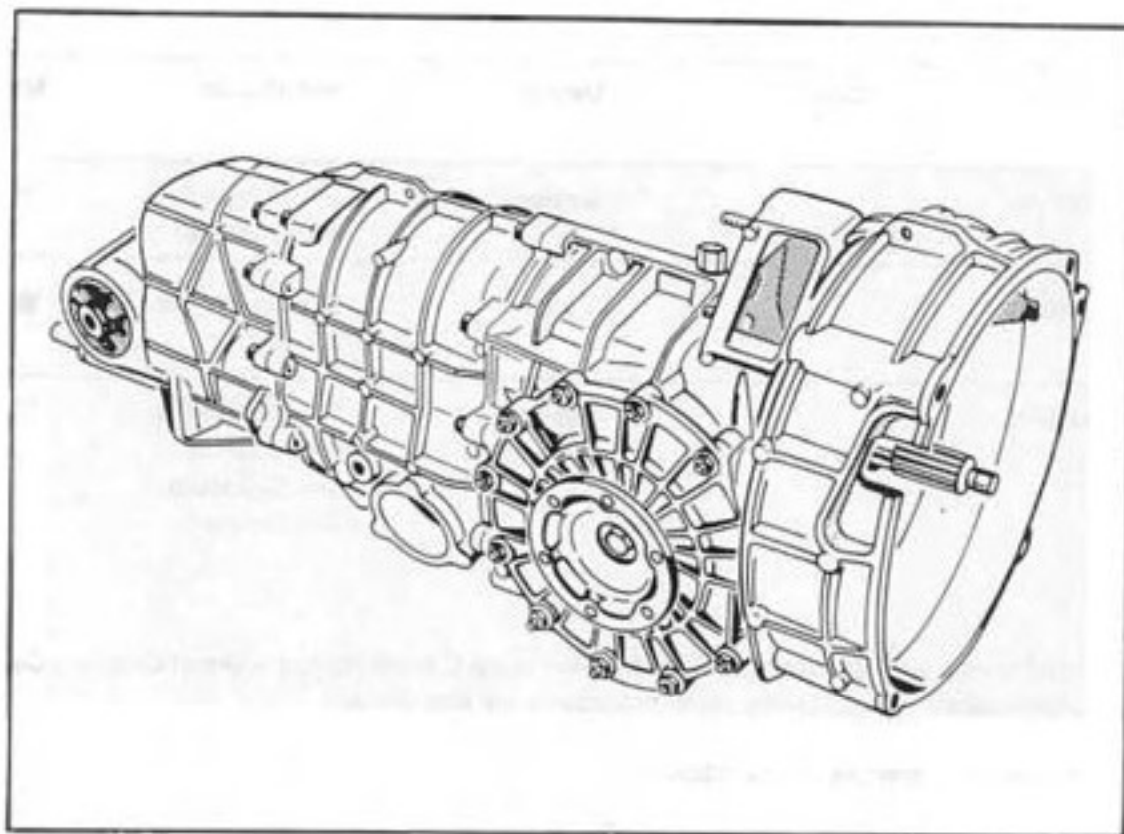
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3 Technical data

6 speed manual transmission G 50 / 20 / 21 for 911 Carrera (993)



Type	Equipment	installed in:	Model year
G 50/21	6 speed	911 Carrera worldwide except USA, CH, A	'94...'96
G 50/20	6 speed	911 Carrera USA, CH, A	'94...'96
		911 Carrera worldwide	'97

3 Technical data

6-speed manual transmission, type G 50/31/32/33 for 911 Carrera RS

Type	Code	Design	installed in:	Model year
G50/31		6-speed	911 Carrera RS (basic vehicle)	'95/'96
G50/32		6-speed	911 Carrera RS (Clubsport)	'95/'96
G50/33		6-speed	911 Carrera RS (basic vehicle and Clubsport) Switzerland	'95/'96

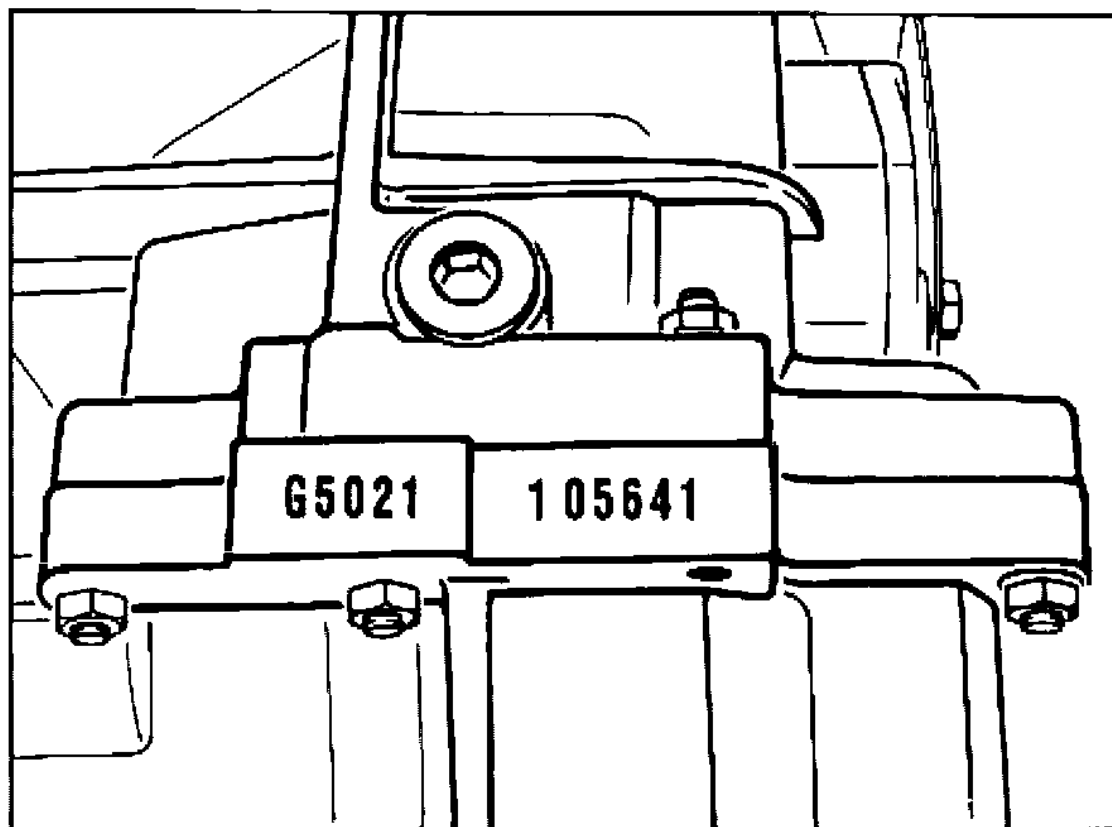
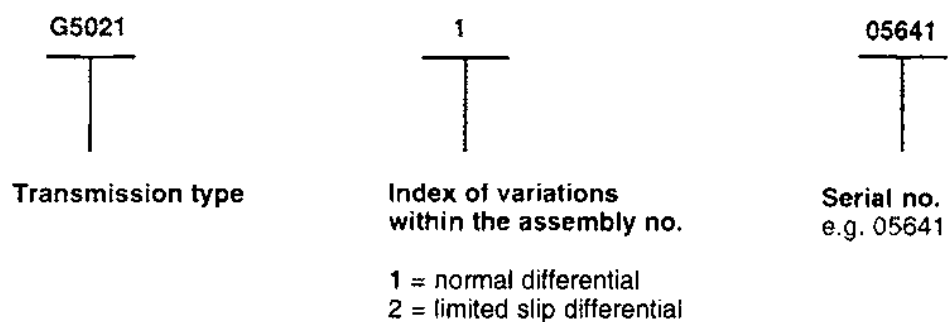
Apart from a few differences, the transmission of the Carrera RS has a similar design to Carrera transmissions G50/20/21; the repair procedures are also similar.

The major differences are as follows:

- different gear ratios
- 2nd gear wheel splined to drive shaft
- steel/molybdenum synchronizer ring (type G50/32 only)
- limited-slip differential as standard equipment (locking values 40 % under acceleration, 65% under deceleration)
- harder gearbox bearing

3 Technical data

Structure of transmission numbers



3 Technical data

Manual transmission for 911 Carrera (1993)

General data	Manual transmission G 50 / 21 / 20	
Transmission ratios	G50/21 $Z_1 : Z_2 = Z_2 : Z_1$	G50/20 $Z_1 : Z_2 = Z_2 : Z_1$
1st gear	11 : 42 = 3.818	11 : 42 = 3.818
2nd gear	20 : 43 = 2.150	21 : 43 = 2.048
3rd gear	25 : 39 = 1.560	27 : 38 = 1.407
4th gear	33 : 41 = 1.242	34 : 38 = 1.118
5th gear	37 : 38 = 1.027 41 : 42 = 1.024*	38 : 35 = 0.921 42 : 39 = 0.928**
6th gear	39 : 32 = 0.820	40 : 31 = 0.775
Rev. gear	14 : 40 = 2.857	14 : 40 = 2.857
Final drive	Hypoid bevel-gear drive with 16 mm offset	
Transmission Final drive	9 : 31 = 3.444	9 : 31 = 3.444
Capacity	3.6 liter multigrade transmission oil SAE 75 W 90 of API classification GL 5 (or MIL-L 2105 B)	

* introduction 15 December 1994

** introduction 7 December 1994

3 Technical data

Manual transmission for Carrera RS

General data		Manual transmission G 50 / 31 / 32 / 33	
Ratios	G50/31 $Z_1 : Z_2 = Z_2 : Z_1$	G50/32 $Z_1 : Z_2 = Z_2 : Z_1$	G50/33 $Z_1 : Z_2 = Z_2 : Z_1$
1st gear	13 : 41 = 3.154	13 : 41 = 3.154	13 : 41 = 3.154
2nd gear	20 : 40 = 2.000	20 : 40 = 2.000	20 : 40 = 2.000
3rd gear	23 : 35 = 1.522	23 : 35 = 1.522	27 : 38 = 1.407
4th gear	33 : 41 = 1.242	29 : 36 = 1.241	34 : 38 = 1.118
5th gear	41 : 42 = 1.024	32 : 33 = 1.031	37 : 36 = 0.973
6th gear	39 : 32 = 0.821	35 : 29 = 0.829	39 : 32 = 0.821
Reverse	14 : 40 = 2.857	14 : 40 = 2.857	14 : 40 = 2.857
Final drive	Hypoid bevel final drive with 16 mm offset		
Final drive ratio	9 : 31 = 3.444	9 : 31 = 3.444	9 : 31 = 3.444
Filling capacities*	3.6 l multi-grade transmission oil SAE 75 W 90, API specification GL 5 (or MIL-L 2105 B)		

*933 GT 2 (transmission type G 50/53) only Veedol SAF 44.

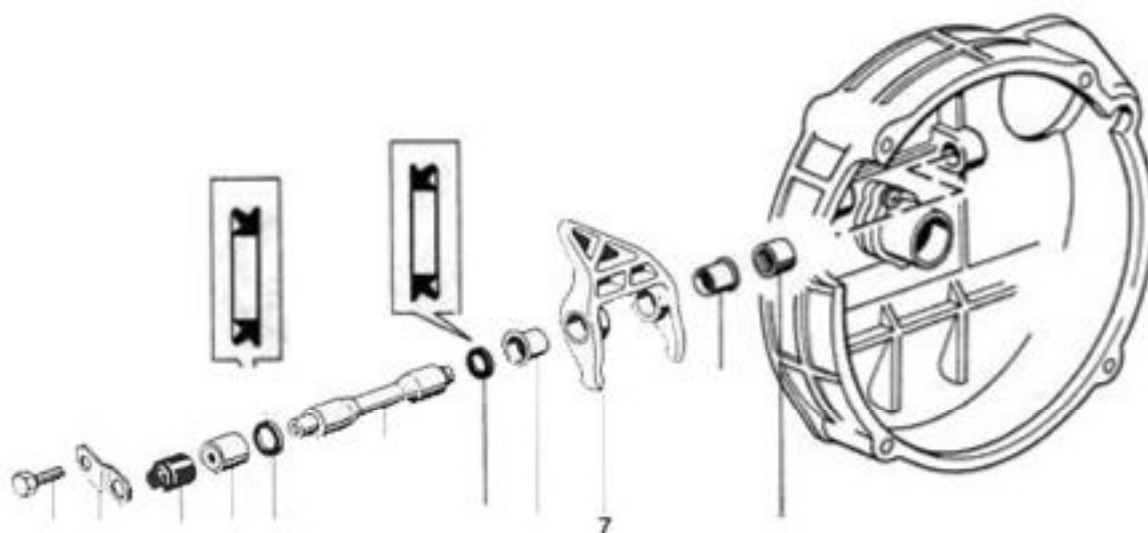
3 Technical data

Tightening torques for manual transmission and transmission mounts

Location	Thread	Tightening torques Nm (ftlb.)
Oil drain and oil filler plugs	M 22 x 1.5	30 (22)
Hexagon head nuts at: Front and side transmission cover, gear and transmission housings, Tensioning plate	M 8	23 (17)
Clamping plate to front transmission cover	M 6	10 (7)
Hexagon head nut / input shaft	M 22 x 1.5	200 (147)
Hexagon head nut / input shaft	M 30 x 1.5	250 (184)
Hexagon head nut / output shaft	M 30 x 1.25	300 (220)
Hexagon head nut / shift fork	M 8	23 (17)
Reverse light switch to gear housing	M 18 x 1.5	35 (26)
Guide tube to transmission housing	M 6	10 (7)
Selector gate to tensioning plate	M 6	10 (7)
Collar nut / return gear II	M 8	35 (26)
Vent to transmission housing	M 14 x 1.5	35 (26)

Location	Thread	Tightening torques Nm (ftlb.)
Hexagon head bolt / halfshaft flange	M 10	44 (32)
Oil cup to tensioning plate	M 5 (micro-sealed)	6 (4)
Ring gear to differential housing (Hexagon head bolt with serrated mating surface)	M 12 x 1.25	200 (148)
Transmission mount to body	M 10 x 70	46 (34)
Transmission mount to transmission (lock nut)	M 12 x 1.5	85 (63)
Longitudinal mount to transmission	M 12 x 1.5 x 65	85 (63)
Longitudinal mount to transmission (lock nut)	M 10	30 (22)

30 37 37 D smantling assem lin clutch trol haft



No.	Designation	Qty.	Note:	
			Removal	Installation
1	Hexagon head bolt M 6 x 16	1		Pack with grease (use Olista Longtime 3 EP)
2	Bracket	1		
3	Bearing cover	1		
4	Needle-roller bearing with assembly hole	1		
5	Release lever shaft	1	Pull out with hexagon head screw M 6 x 40	Install in correct posi- tion, assembly hole points towards outside
6	Sealing ring	2		Do not grease plastic bushing bearing surface
7	Release lever	1		Check, replace if re- quired, install in correct position
8	Plastic bushing	2		Do not grease
9	Needle-roller bearing with- out assembly hole	1		

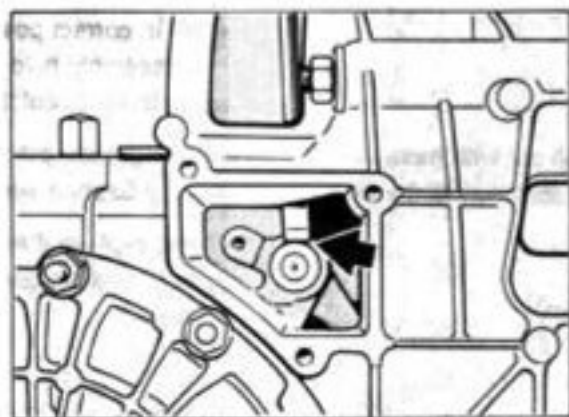
Note

Lubricate all sliding surfaces of the clutch release mechanism - **except** for the items indicated in the list - with Olista Longtime 3 EP, Part No. 000.043.024.00.

Assembly notes

Removing bearing cover

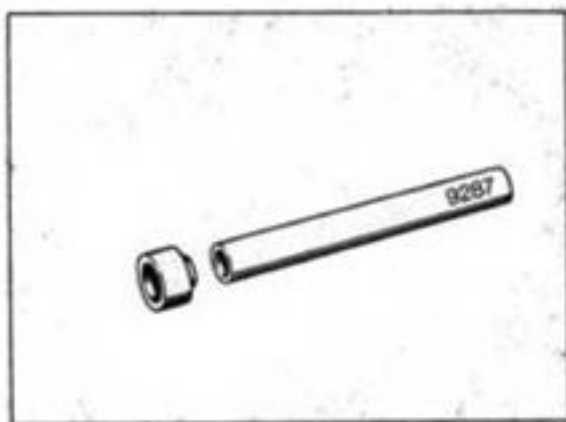
Insert a suitable screwdriver across rectangular cutout (of clutch bell housing) and push bearing cover out of bearing hole.



1865-30

Installing bearing cover

Locate bearing cover with assembly mandrel, e.g. Special Tool 9287 (drift for headlamp beam range adjusters) and carefully press in to stop.



1866-30

30 56 01 Checking clutch drive plate for wear

(clutch drive plate removed)

Checking:

Using a depth gage or sliding calipers, measure distance X from surface of lining to rivet head.

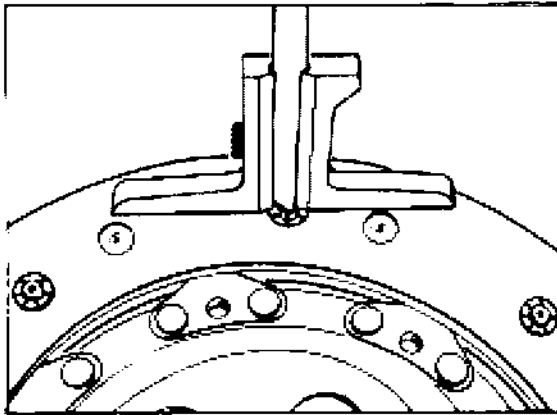
Note

Measure only at the beaded-over side of the rivet head.

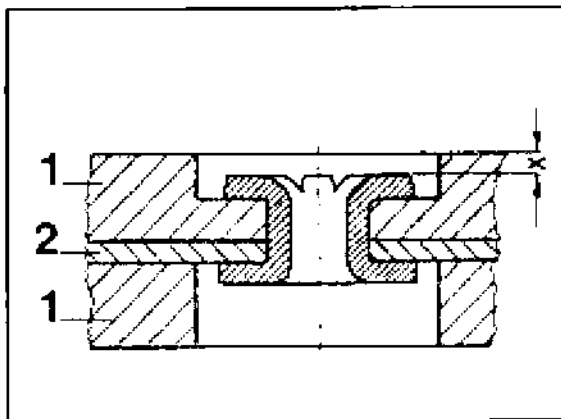
Wear limit X is 0.3 mm (see drawing).

When the drive plate is new, this dimension is approx. 1.0 mm.

Note that lining wear is **not linear**, i.e. not at a steady rate throughout the clutch's operating life. The way in which the lining material beds down and is compressed initially suggests that greater lining wear is taking place, but as the car is driven further, the rate of wear declines considerably.



645-30

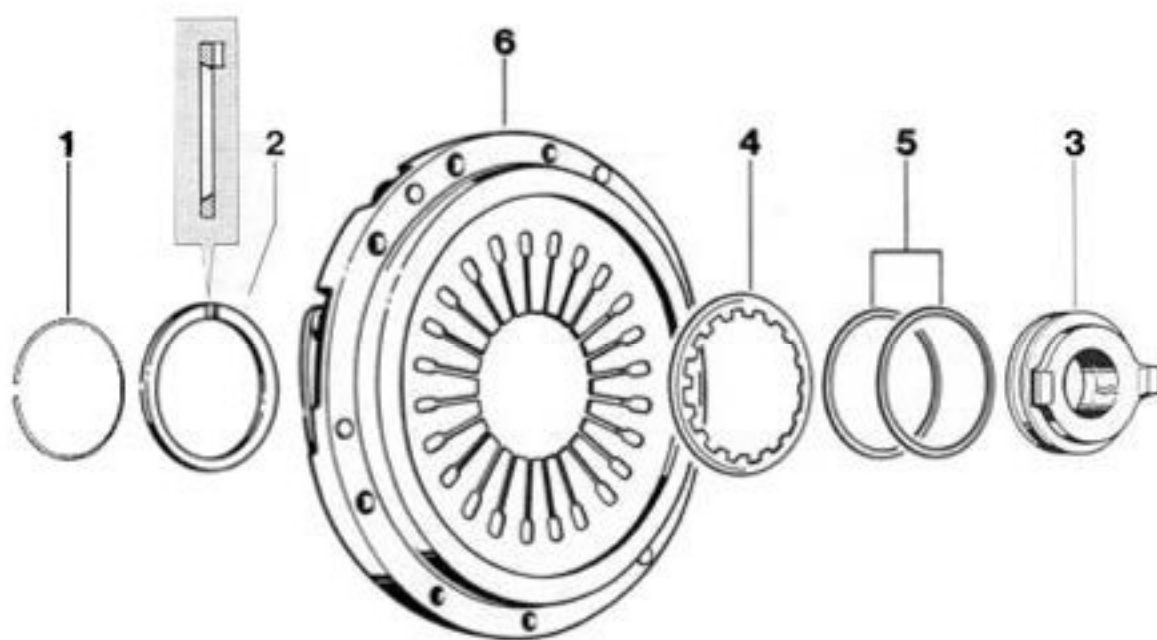


646-30

1 - Lining

2 - Drive plate/lining spring

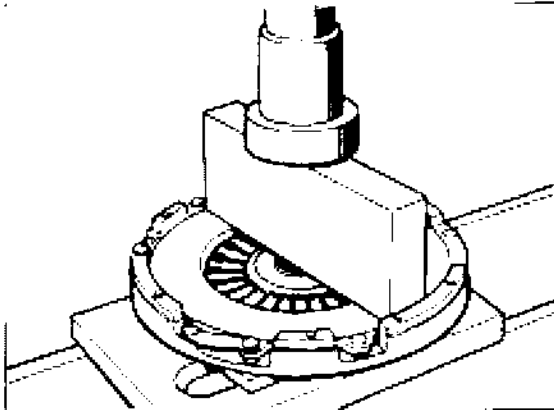
30 52 19 Removing and installing release bearing



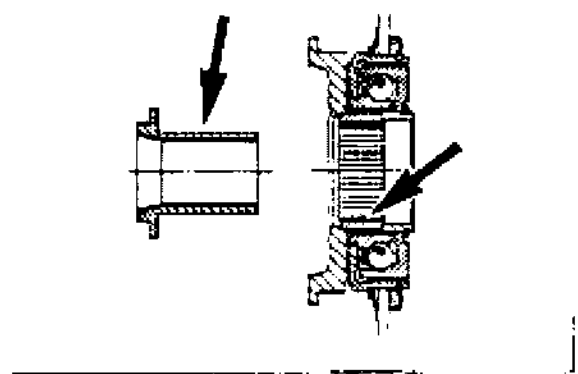
No.	Designation	Qty.	Note	
			Removal	Installation
	Snap ring	1	use pointed pliers to remove, pressing down thrust plate in a press	observe correct seating of snap ring
2	Thrust washer	1		groove must face snap ring
3	Release bearing	1	check, do not wash out, just wipe with a dry cloth	
4	Spring washer	1	install in correct position	
5	Spacer	2		always insert 2 spacers
6	Thrust plate	1		check for wear, replace if required

Assembly notes

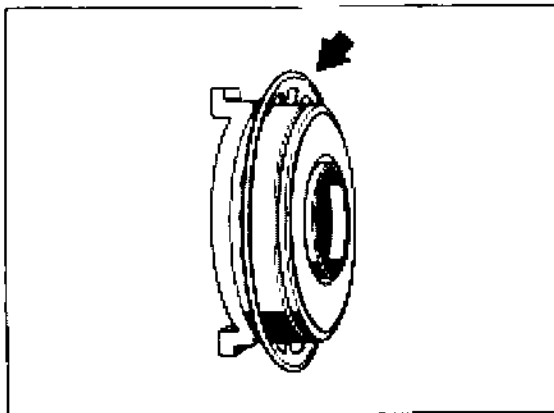
1. The snap ring can only be fitted with a shop press. Use a wooden block as a thrust piece.
2. Install spring washer in correct position.
3. Do not grease sliding surfaces of release bearing and guide tube.
4. The guide sleeve is slotted (uneven outline - see arrow).



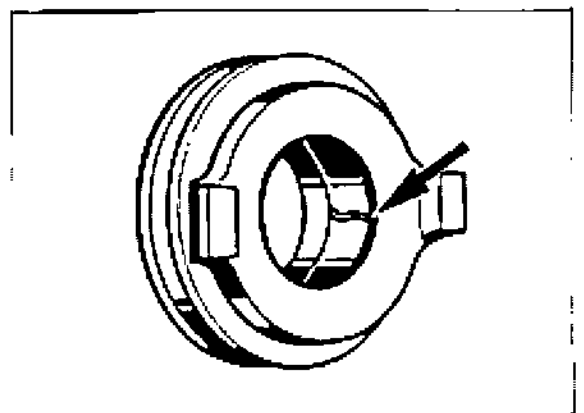
1000-30



1867-30



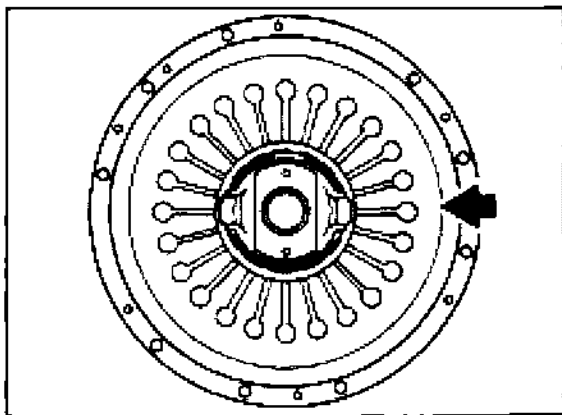
1001-30



1868-30

30 52 Identification of clutch pressure plates

To avoid confusion, the pressure plates are color-marked for positive identification.



836-30

Color marking:

968	Green
911 Carrera 2/4	Blue
911 Carrera (993)	No mark
911 Turbo 3.6	Red

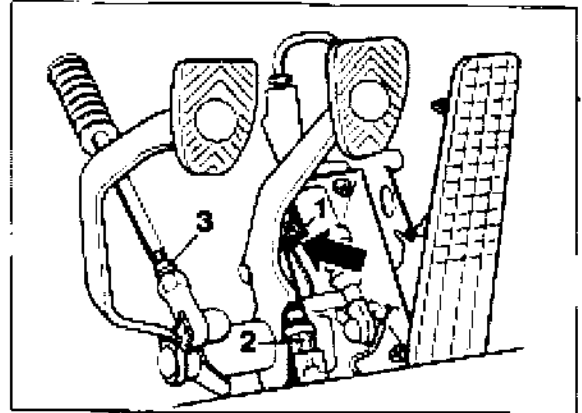
30 Pedals - adjustment overview

Adjusting clutch pedal

Note

The following points are **prerequisites** for proper operation of the clutch and must be observed when adjusting the pedals:

- Proper bleeding of the hydraulic clutch system
- Tightness of the hydraulic system
- Proper setting of the servo spring
- Returning of pedal to initial position
- Pedal travel limited by properly adjusted floor board stopper
- Installation position of pedals in accordance with standard situation.



Checking and adjusting work

1. Checking the pedal return (returning force)

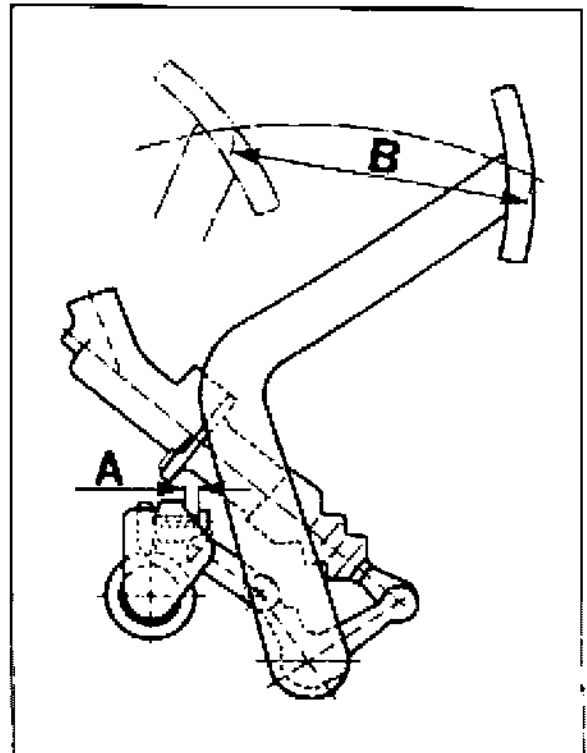
Try to return the pedal to its initial position by applying at least 30 N (spring scale). The pedal must not yield. If it does yield (= insufficient returning force), the servo spring pre-load must be reduced or set to the dimension A = 6 mm (see 2.).

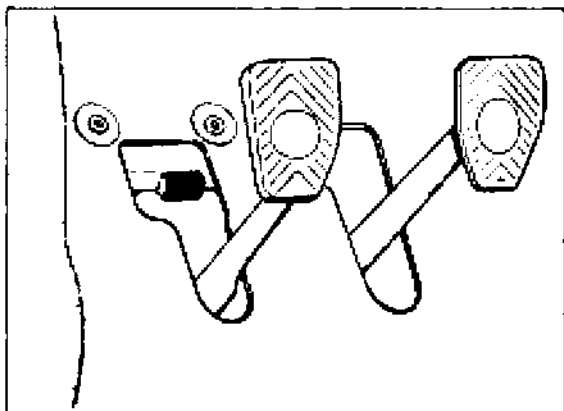
2. Setting servo spring (dim. A = 6 mm).

The servo spring is set by turning the hexagon socket head bolt **arrow / no. 1** (chart 1097-30). The distance (dimension A/ figure 1113-30) between the inner face of the bolt head and the rocker must be 6 mm.

3. Checking pedal travel (dim. B = 150 mm).

Measured from the center of the pedal plate, the pedal travel must be 150 mm + 5 mm. If necessary, adjust or limit the pedal travel on the floor board stopper (fig. 1098-30 on p. 30-14).





1098-30

4. Clutch play.

The clutch master cylinder is fitted with an internal stopper. The push rod and the clutch pedal are always pressed against this stopper by the servo spring. Due to automatic hydraulic readjustment of the clutch, the clutch play cannot be assessed by checking the clutch pedal.

Adjusting microswitch

The microswitch (on vehicles with cruise control) is adjusted by a sliding movement after the fastening screws have been slackened. The microswitch must be triggered (clicking sound) on the first few millimeters of pedal travel.

Adjusting brake light switch

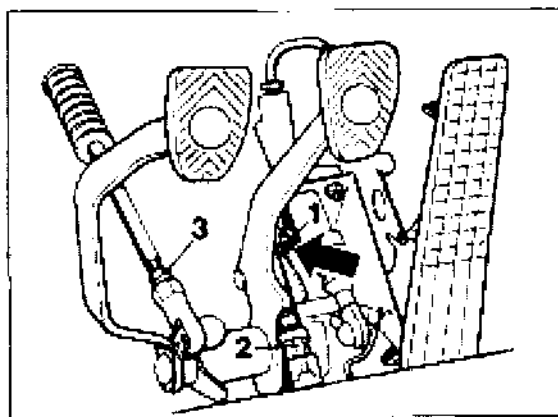
The brake light must light up after a brake pedal travel of 6-16 mm (measured from the center of the pedal plate).

If the brake light lights up after a brake travel of less than 6 mm, the brake light switch (2) must be turned to the right until it responds within the range of tolerance. Do not damage the electric wire and the plug. If the adjustment range of the brake light switch is not sufficient, the brake pedal must be adjusted at the ball joint of brake push rod 3 (by shortening push rod).

If the brake light lights up after a pedal travel of more than 16 mm, the brake pedal must be adjusted at the ball joint of the brake push rod (by extending push rod) until the brake pedal switch responds within the range of tolerance.

Note

Usually, the brake pedal plate is roughly at the same height as the clutch pedal plate (plus/minus 3 mm).



1097-30

Throttle operation

Check for smooth operation and at wide open throttle.

30 Disassembling and assembling pedals

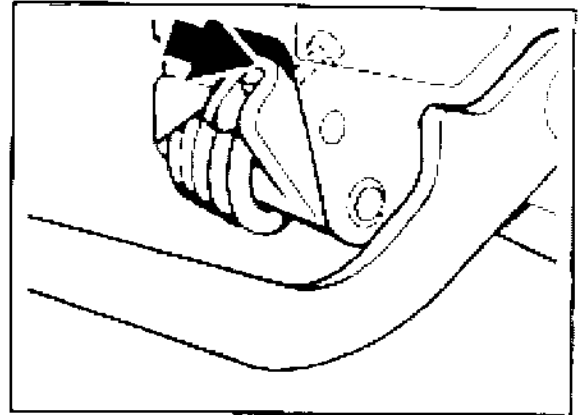
Includes: removing and installing pedals

Important note

Remove the pedals together with the clutch master cylinder.

To do so, slacken the hydraulic lines on the clutch master cylinder. **Release servo spring partially** (move clutch pedal into de-clutched position).

Plug the hydraulic lines or clamp off the supply line using a commercially available hose clamp. Also plug the connecting bores of the clutch master cylinder.



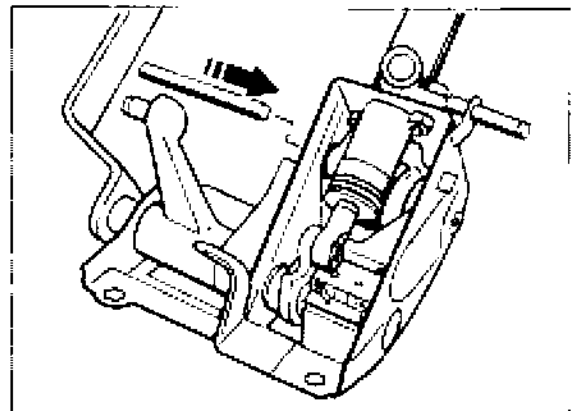
2358-30

3. Remove clutch master cylinder. Then lock rocker / servo spring. To do so, move clutch pedal into **de-clutched position**.

In this position, operate clutch pedal (by pulling it backwards), until an auxiliary tool (steel pin with 8 mm Ø) can be inserted from the side into the bore of the rocker.

Disassembling pedals

1. Mount brake pedal of pedal system onto a vise – **using protective jaws** –.
2. Turn adjustment screw for the servo spring to the left (counter-clockwise), until the leg of the servo spring fits closely to the rocker (arrow).



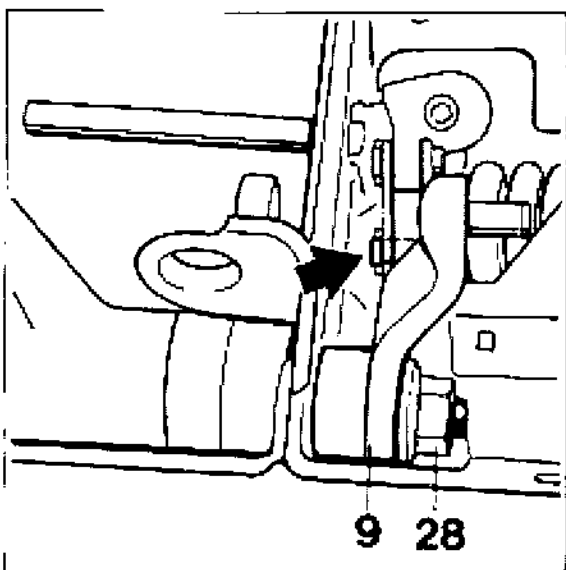
2357-30

4. Release servo spring in **two steps**.
Use an appropriate spring tensioning tool, e.g. brake spring pincers by Hazet (Hazet no. 797) to load and release the servo spring. To protect the mounting block, use a liner when applying the pincers.

First step:

Remove the support pin (arrow) on lever no. 9 using a screw driver.

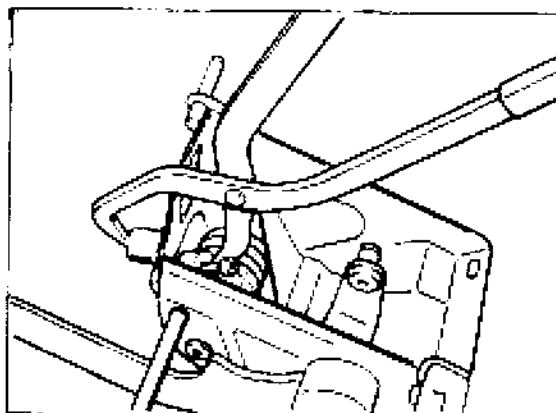
Caution: Do not get into the operating range of the rocker, the servo spring or the levers with your fingers or hands.



2370-30

Second step:

First load the servo spring through the rocker until the auxiliary tool (steel pin with 8 mm Ø) can be removed easily from the bore of the rocker. Then release the servo spring by opening the pincers.

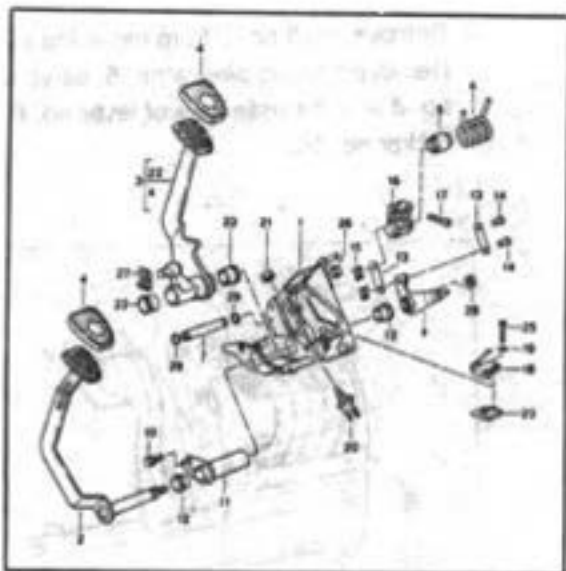


2366-30

5. On vehicles with cruise control:
Mark position of microswitch.
Remove microswitch.
6. Slacken fastening nut no. 28 on operating lever no. 9.
Remove lever from clutch pedal shaft – **through the lateral bore on the mounting block** – using a copper or brass mandrel.
If lever can be removed, continue with item 8.

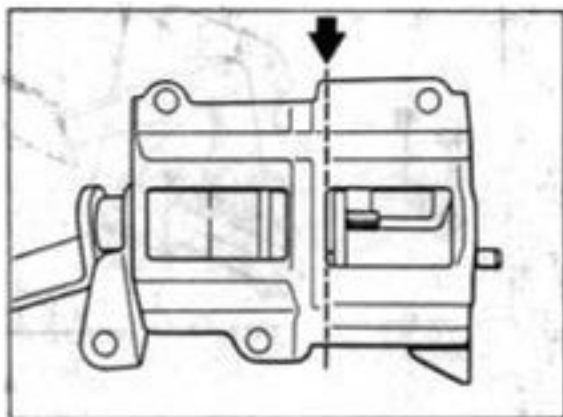
Important note

If lever no. 9 cannot be removed from the clutch pedal shaft without damaging the clutch pedal shaft (which may occur due to tolerances), continue with item 7.

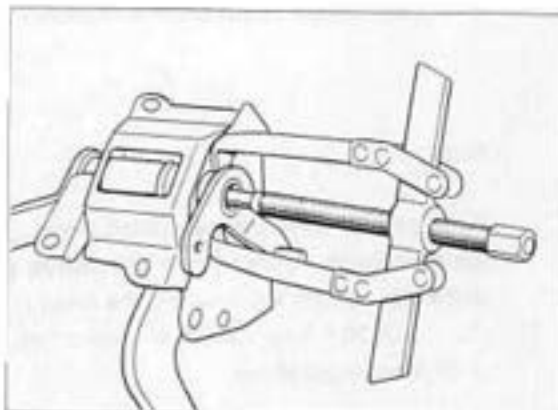


2348-30

7. If **lever no. 9** cannot be removed from the clutch pedal shaft without damaging the clutch pedal shaft (which may occur due to tolerances), part of the mounting block must be sawed off (arrow / dotted line). Then remove the lever using a commercially available two-claw puller.



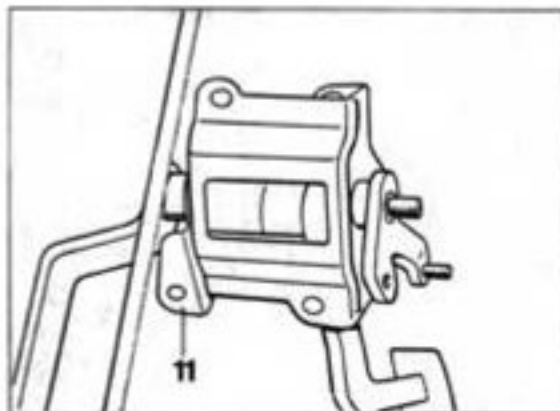
2359-30



2360-30

If the lever cannot be removed with a puller, the clutch pedal shaft must be destroyed. To do so, separate clutch pedal from pedal shaft (by sawing) with bearing tube no. 11 slackened (screw no. 10 removed).

If lever no. 9 can be reused (see – Note – on page 30-18), remove the lever from the pedal. To do so, use plate VW 401 and an appropriate bushing, e.g. Matra 40-503.



2361-30

8. Lift out clutch pedal shaft with clutch pedal.

Note

If the pedals were disassembled because of the complaint – **sometimes the clutch pedal does not return fully** –, replace lever no. 9 (fig. 2348-30 / new, modified version with optimized kinematics).

> Distinguishing old and new levers

Part number on part.

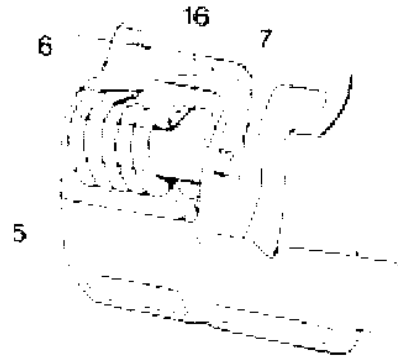
> Previous version:

Part no. 993 423 519 00 or
993 423 519 01 (gold-colored).

> Modified / new version:

Part no. 993 423 519 02 (olive).
In the beginning, the modified, new levers were golden as well.

9. Remove shaft no. 7 from mounting block.
Remove bearing sleeve no. 5, servo spring no. 6 and the assembly of lever no. 9 and rocker no. 16.



2365-30

10. Remove bearing pipe no. 11 and brake pedal.

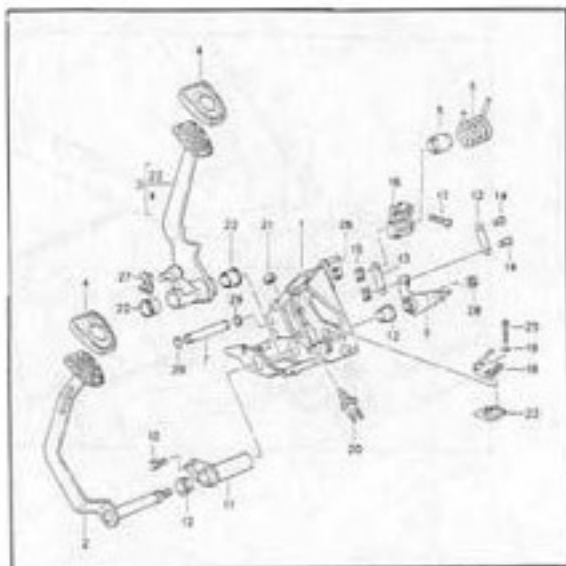


2363-30

Assembling pedals

1. Check all parts, replace any parts, if required.

If the bushings must be replaced, use an appropriate mandrel for removal or installation. Coat all bearings and slide faces with a Teflon-compatible low temperature grease – e.g. Glissando by DEA (Texaco).



2348-30

Note

Check the mounting block for deformations (might stem from removal of lever no. 9). If necessary, remove the mounting block.

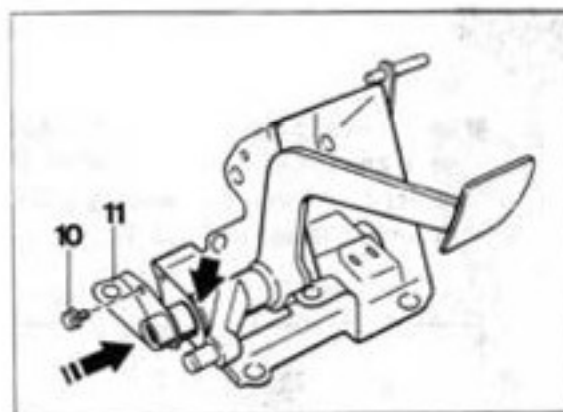
2. Insert brake pedal into mounting block. If necessary, fit a shim (arrow) between the brake pedal and the mounting block (to compensate for the axial play). The shim is not always present.

When replacing the mounting block or the brake pedal, determine the thickness of the shim (if present and/or required): With the brake pedal installed, determine by trial and error which shim can be inserted. There are shims with two different thicknesses available.

Insert bearing tube no. 11.

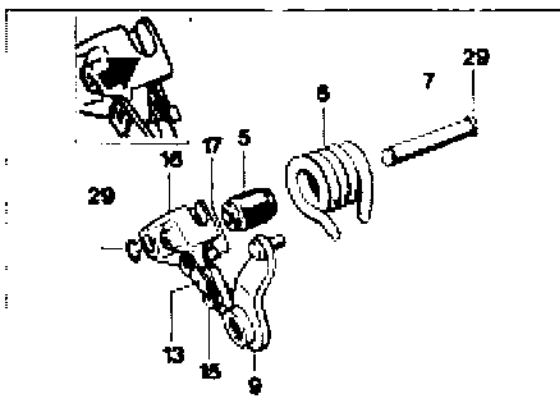
Insert screw no. 10, but do not tighten it yet.

Important: Tighten screw no. 10 only after installing the pedals.



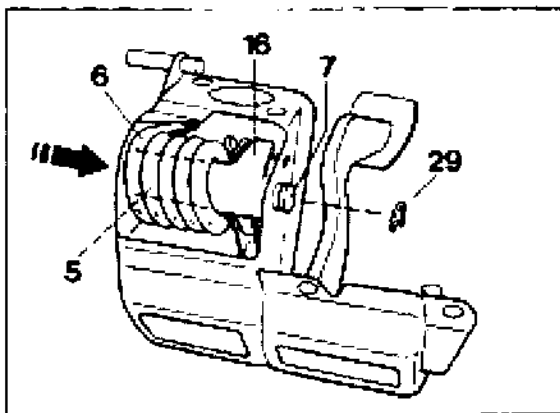
2363/1-30

3. Preassemble lever no. 9, rocker no. 16 and connecting pieces no. 13. If necessary (after installation of a new rocker), turn adjustment screw no. 17 until the leg of the servo spring fits closely to the rocker (arrow). **Install only levers no. 9 with optimized kinematics. See note on p. 30-18.**



2364-30

4. Insert bearing sleeve no. 5, servo spring no. 6 and the assembly of lever no. 9 and rocker no. 16 into the mounting block. Insert and fasten shaft no. 7.

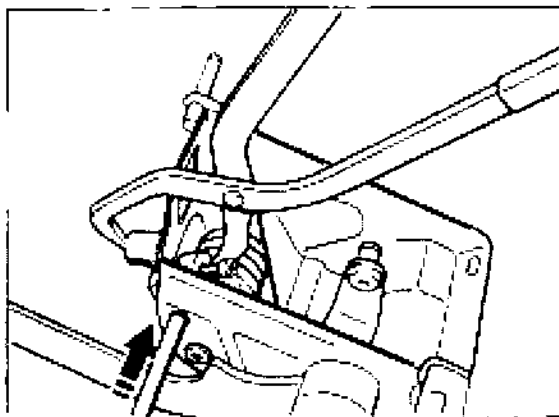


2365/1-30

5. To preload the servo spring, mount the brake pedal of the completed pedal system onto a vise – **using protective jaws.**

6. Preload servo spring as follows:

Use an appropriate spring tensioning tool, e.g. brake spring pincers by Hazet (Hazet no. 797), to preload the servo spring until an auxiliary tool (steel pin with 8 mm Ø) can be inserted into the bore of the rocker from the side. To protect the mounting block, use a liner when applying the pincers.



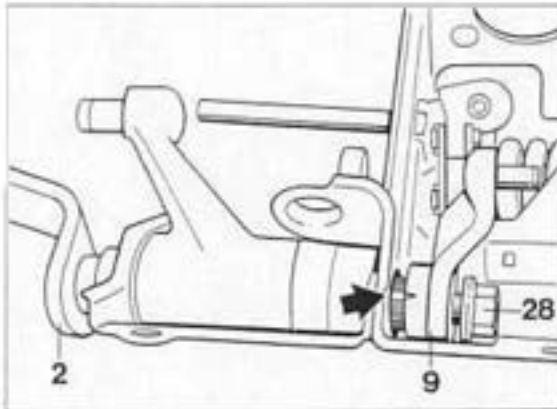
2366/1-30

7. Insert clutch pedal shaft with clutch pedal no. 2.

Fit lever no. 9 in **its correct position** to the clutch pedal shaft (notch on lever must face tooth gap of shaft).

Fit fastening nut no. 28.

Do not tighten fastening nut yet, (see next item).



2367-30

8. Remove **auxiliary tool** (steel pin to preload servo spring) from rocker. To do so, move clutch pedal backwards (towards clutch position) **carefully and only slightly**. Remove the steel pin as soon as it is no longer under load.

Then tighten fastening nut no. 28, tightening torque **34 Nm**.

Note

If the auxiliary tool (steel pin) is not removed with utmost care or if the fastening nut is tightened with the auxiliary tool still inserted, the rocker may get damaged in the area of the bore.

9. On vehicles with cruise control, install the microswitch.
10. Install the clutch master cylinder.
On vehicles with cruise control, adjust the microswitch by sliding it into the proper position.
The microswitch must be triggered (clicking sound) on the first few millimeters of pedal travel.

If the clutch master cylinder does not fit closely to the mounting block, rework the mounting block's contact surface using a plastic hammer.

11. When replacing the mounting block, re-install those parts of the old mounting block that are required.
12. Set adjustment screw of servo spring (no. 17) to the dimension of 6 mm (see page 30-13).
13. Install pedals **with the servo spring released partially** (clutch pedal in de-clutched position).

Important note

Then tighten screw of bearing tube (see item 2 on page 30-19).

14. Connect lines of hydraulic clutch system to clutch master cylinder. Bleed clutch system.

Note

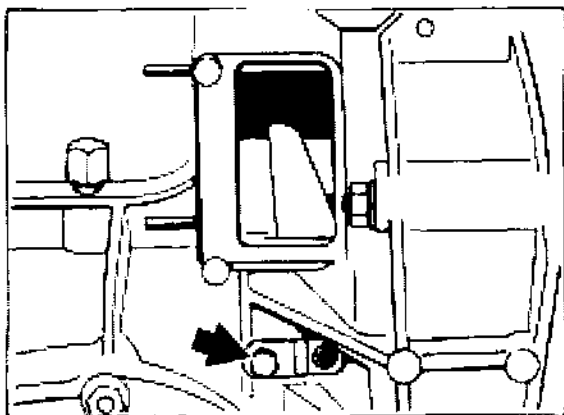
Carry out adjustments, e.g. adjusting brake light switch, and inspections, e.g. limiting / adjusting pedal travel.

The adjustment jobs are described on pages 30-13 and 30-14.

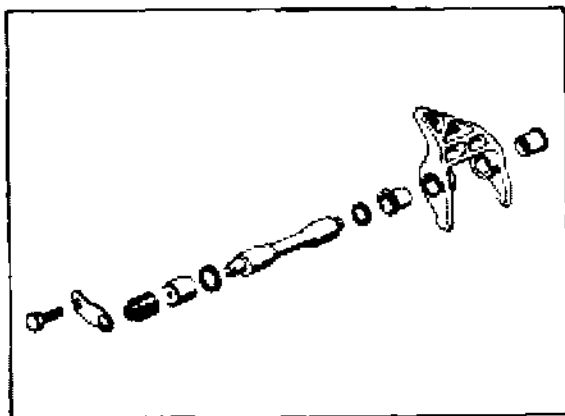
34 35 27 Removing and refitting the transmission

Removing the transmission

- Remove wire from starter and wire from reverse light switch.
2. Remove M 6 x 16 hexagon head bolt, bracket and mount cover. Pull out release lever shaft using an M 6 x 40 hexagon head bolt.
 3. Undo four fastening nuts and separate transmission from engine. Undo upper fastening nut from starter using Special Tool P 119.



1391-10



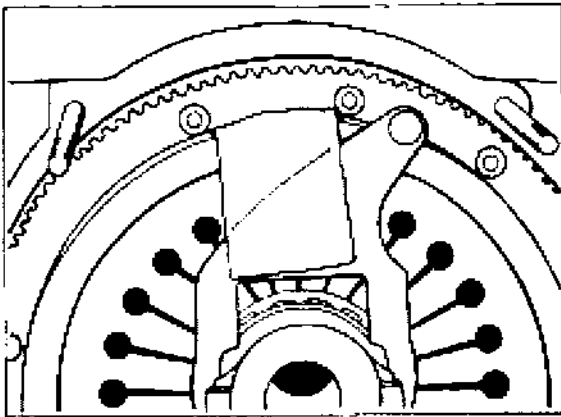
1388-10

Refitting the transmission

Note:

Apply a very thin coat of Olistra Longtime 3 EP to input shaft splines (for corrosion protection only).

Engage release fork into release bearing and use a suitable length of adhesive tape to locate it provisionally in installation position.

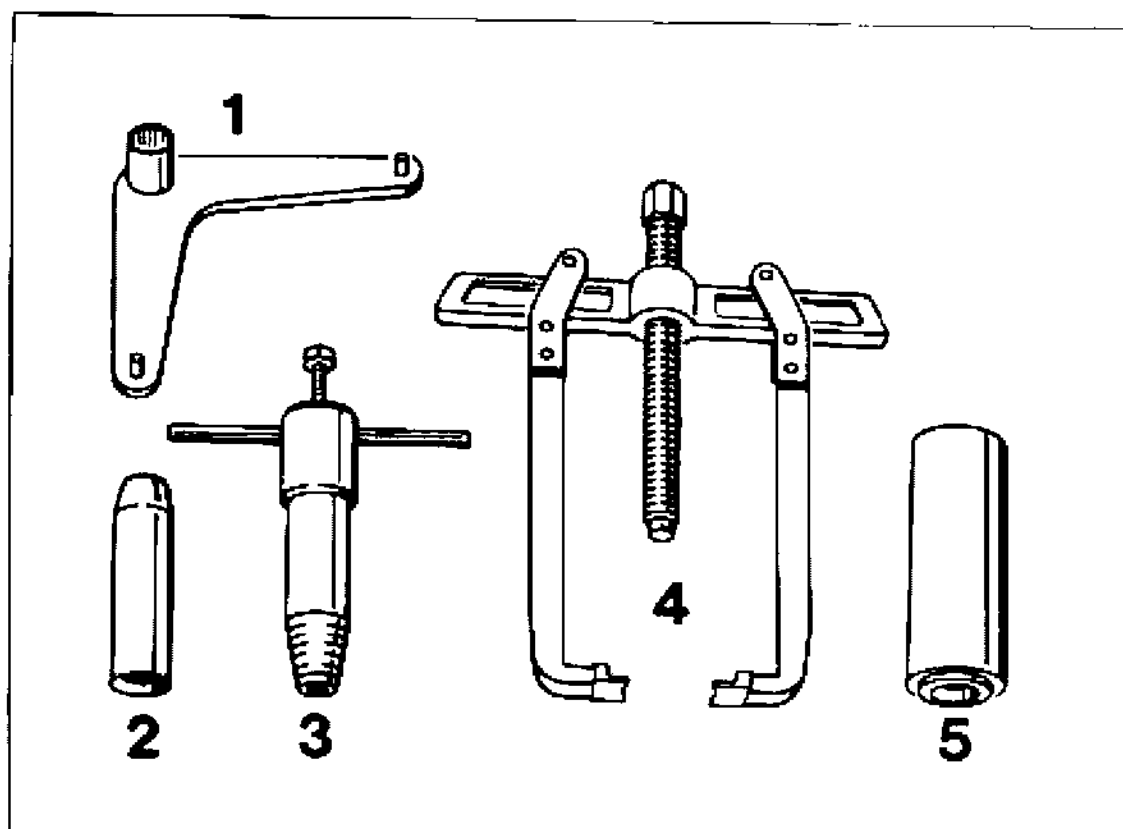


1390-10

2. Refit transmission to engine.
3. Engage release lever shaft with seals into release fork.
4. Remove locating tape from release fork across assembly hole.
5. Install needle-roller bearing, mount cover and bracket and tighten down with M 6 x 16 hexagon head bolt (also refer to assembly notes, Group 30, page 30 - 1).

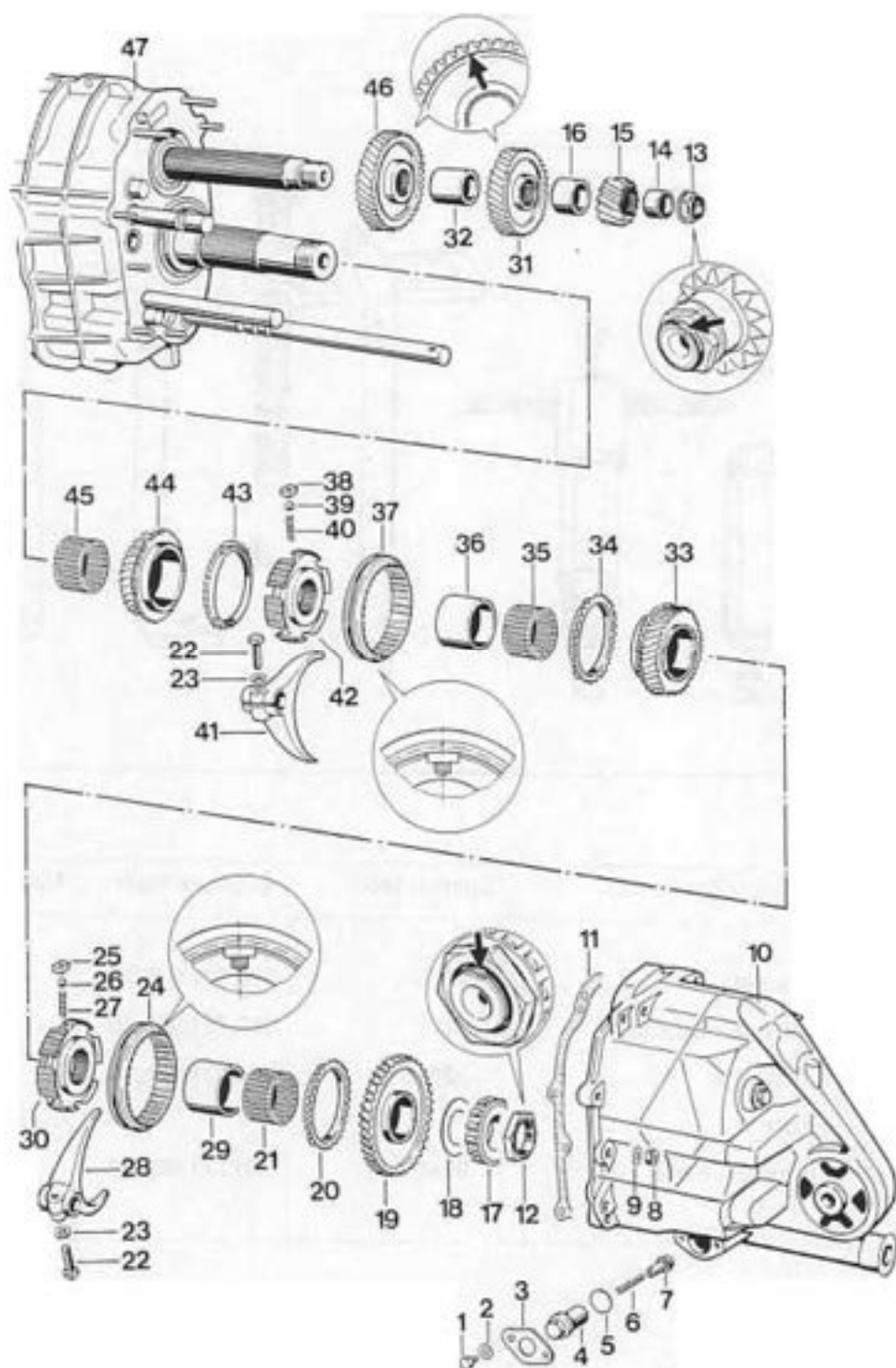
34 35 37 Dismantling and assembling transmission

Tools



No.	Designation	Special tool	Order number	Explanation
1	Bracket	9253	000.721.925.30	use arms of putler 92B4
2	Assembling sleeve	9255	000.721.925.50	
3	Puller set	9251	000.721.925.10	
4	Puller	-	-	
5	Pressure piece	9256	000.721.925.60	

34 35 37 Dismantling and assembling transmission



No.	Designation	Qty.	Note:	
			Removal	Installation
1	Hexagon head bolt	2		Tighten to 10 Nm (7 ftlb.)
2	Washer	2		
3	Clamping plate	1		
4	Eccentric bushing	1		Adjust
5	Seal	1		Replace
6	Thrust spring	1		
7	Locking bushing	1		
8	Hexagon head nut	10		Tighten to 23 Nm (17 ftlb.)
9	Washer	10		
10	Front transmission cover	1		
11	Seal	1		Replace
12	Hexagon nut*	1	Lock input shaft with Special Tool 9253 and engage 6th gear	Tighten to 250 Nm (184 ftlb.). Upset flange to lock
13	Hexagon nut*	1	Lock input shaft with Special Tool 9253 and engage 6th gear	Tighten to 140 Nm (103 ftlb.). Upset flange to lock
14	Inner bearing race	1	Pull off across fixed gear-wheel No. 15	Heat to approx. 120 °C
15	Fixed gear (reverse)	1		
16	Spacer sleeve	1		
17	Cylindrical roller bearing	1	Pull off across loose gear-wheel No. 19	Heat to approx. 120 °C
18	Thrust washer	1		
19	Loose gearwheel (reverse)	1		
20	Synchronizer ring	1	Mark for reinstallation	Check for wear. Fit with the same gear-wheel (cogs must face driver dogs)

No.	Designation	Qty.	Note:	
			Removal	Installation
21	Needle bearing cage	1	Mark for reinstallation	Fit with the same gear-wheel
22	Hexagon head bolt	2		Tighten to 23 Nm (17 ftlb.)
23	Washer	2		
24	Shift sleeve	1	Lift off along with shift fork No. 28. Take care not to lose synchromesh parts.	Center the centerpunch marks relative to the driver dogs. Offset side faces loose gearwheel No. 19
25	Driver dog	3		Domed end faces shift sleeve
26	Ball	3		
27	Spring	3		
28	Shift fork (reverse)	1		Adjust so that play is barely felt with reverse engaged. When neutral is engaged, the synchronizing ring must be able to rotate freely.
29	Inner race**	1	Pull off across guide sleeve No. 30	Heat to approx. 120 °C
30	Guide sleeve (with snap ring)	1		Open end of snap ring must not be located in the driver dog area. Install in correct position complete with shift sleeve and shift fork (snap ring faces loose gearwheel No. 33)
31	Fixed gearwheel (5th gear)	1		Identification groove faces fixed gearwheel No. 15. Do not confuse with fixed gearwheel No. 46

No.	Designation	Qty.	Note:	
			Removal	Installation
32	Spacer sleeve	1		
33	Loose gearwheel (5th gear)	1		Do not confuse with loose gearwheel No. 44
34	Synchronizing ring	1	Mark for reinstallation	Check for wear. Fit with the same gearwheel (cogs face driver dogs)
35	Needle bearing cage	1	Mark for reinstallation	Fit with the same gearwheel
36	Inner race**	1	Mark for reinstallation. Pull off with arms of Special Tool 9284 across loose gearwheel No. 44	Fit with the same gearwheel. Heat to approx. 120 °C
37	Shift sleeve	1		Center the centerpunch marks relative to the driver dogs
38	Driver dog	3		Domed side faces shift sleeve
39	Ball	3		
40	Spring	3		
41	Shift fork (5th and 6th gears)	1		Adjust. In the neutral position, the shift sleeve must be set exactly in the middle between the loose gearwheels
42	Guide sleeve	1		Install complete with shift sleeve and shift fork

No.	Designation	Qty.	Note:	
			Removal	Installation
43	Synchronizing ring	1	Mark for reinstallation	Check for wear. Fit with the same gear-wheel (cogs face driver dogs)
44	Loose gearwheel (6th gear)	1		Do not confuse with loose gearwheel No. 33
45	Needle roller bearing cage	1	Mark for reinstallation	Fit with the same gear-wheel
46	Fixed gearwheel (6th gear)	1		Identification groove faces fixed gearwheel No. 31. Do not confuse with fixed gearwheel No. 31
47	Transmission	1		

Transmission shafts without recesses for securing the hexagon nuts have been installed since September 1995.

Self-locking hexagon nuts are used on these shafts, and these nuts must always be replaced in every transmission repair.

** As a result of tolerances, the bearing inner rings (no. 29 and 36) may either form a force fit on the shaft or have a certain amount of play.

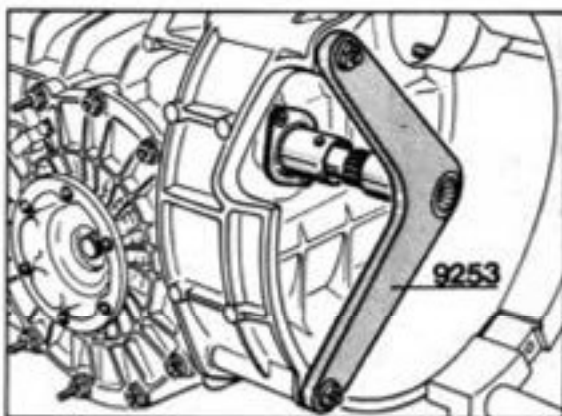
Dismantling and assembly notes

Note

If the shift rods are moved across the neutral or gear latching positions when the transmission is dismantled or assembled, the small intermediate locks (see page 35 - 13) may drop out inadvertently.

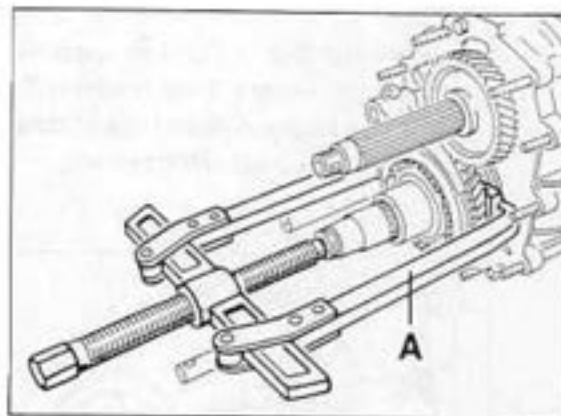
Dismantling

1. Engage 6th gear, lock input shaft with Special Tool 9253 and release hexagon head nuts for output and input shafts.



503-34

2. Engage fourth gear and pull off inner race of needle roller bearing (5th gear).



1641-34

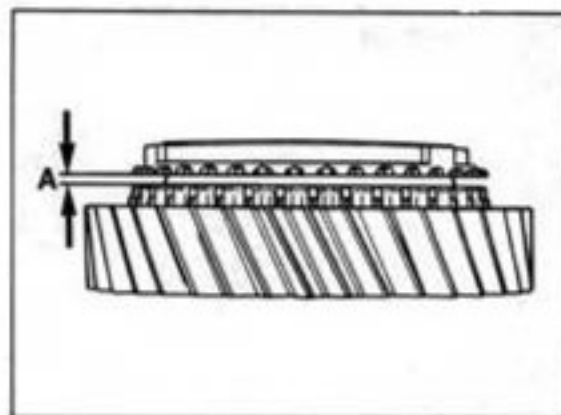
A = Arms of pulier 9284

Assembly

1. To check synchronizing rings, push rings onto the tapers of the gearwheels and measure gap "A" with a feeler gauge.

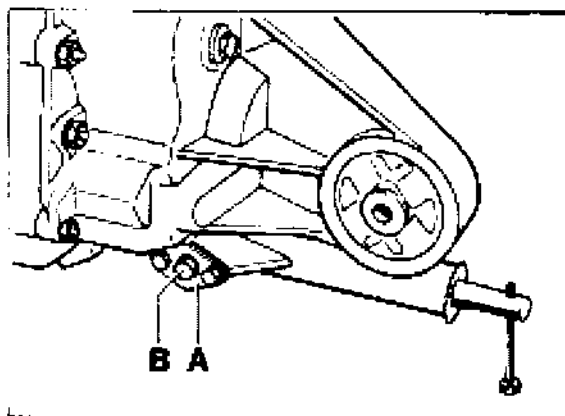
Installation dimension (new) = min. 0.9 mm

Wear limit = 0.6...0.7 mm



518-35

2. Engage fourth gear and fit all gearwheels.
3. Adjust inner shift rod until all gears may be preselected freely without notching. To adjust, undo hexagon head bolts of tensioning plate and rotate eccentric bushing.

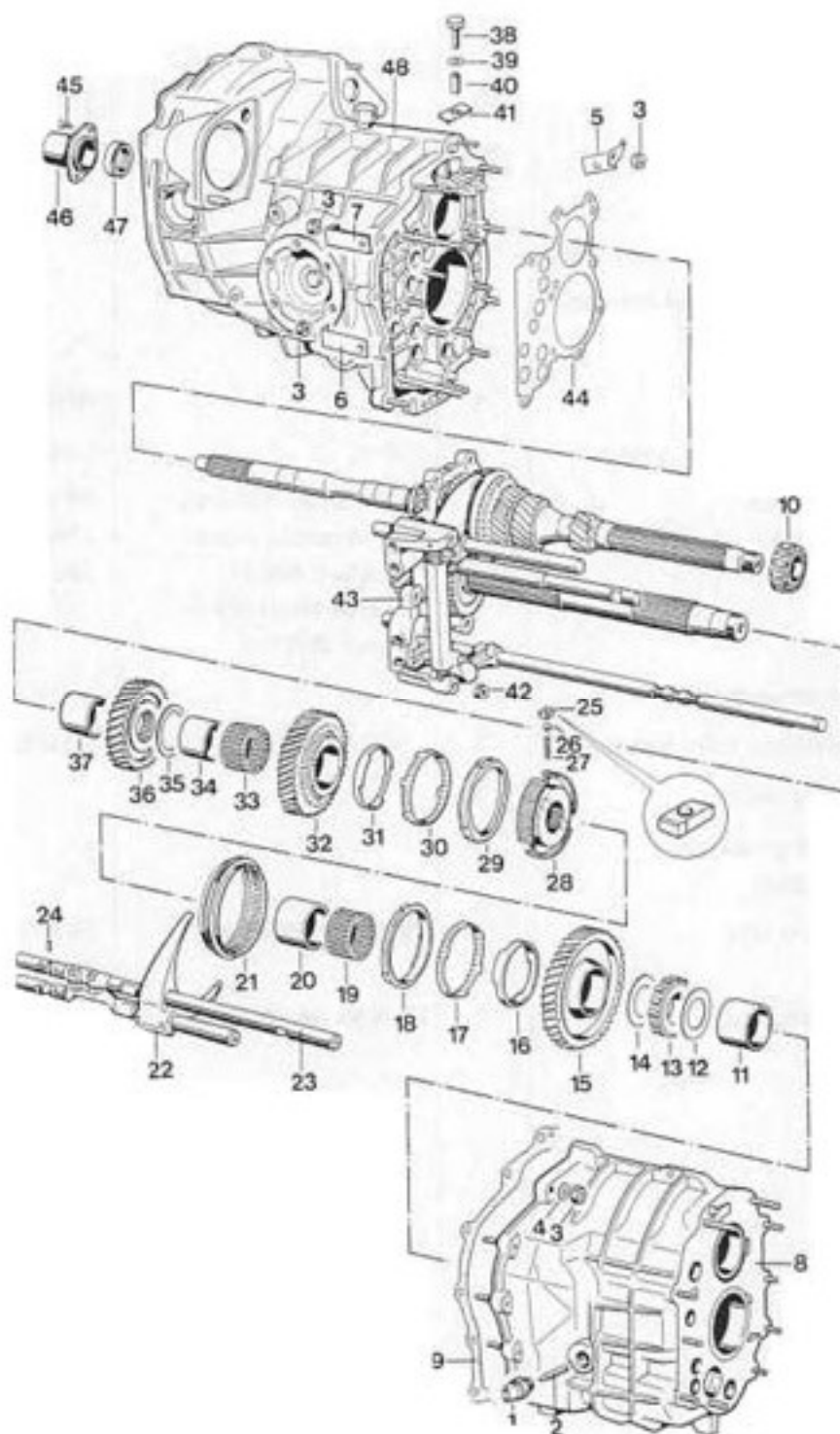


527-34

A = Tensioning plate
B = Eccentric bushing

4. To check, shift through all gears, using a long M 8 bolt that is screwed into the inner shift rod.

34 35 37 Dismantling and assembling transmission



No.	Designation	Qty.	Note:	
			Removal	Installation
1	Reverse light switch	1		Tighten to 35 Nm (26 ftlb.)
2	Plunger	1		Stepped end faces switch
3	Hexagon head nut	10		Tighten to 23 Nm (17 ftlb.)
4	Washer	10		
5	Retaining bracket	1		
6	Retaining bracket (straight)	1		
7	Retaining bracket (domed)	1		
8	Gear housing	1		
9	Gasket	1		Replace
10	Cylindrical roller bearing	1	Pull off	Heat to approx. 120 °C
11	Inner race	1	Mark for reinstallation. Pull off across loose gearwheel No. 15. Observe clearance at reverse shift rod	Fit with the same gear-wheel. Heat to approx. 120 °C
12	Thrust washer	1		
13	Cylindrical roller bearing	1		Heat to approx. 120 °C
14	Thrust washer	1		
15	Loose gearwheel (1st gear)	1		
16	Friction ring	1	Mark for reinstallation	Fit with the same gear-wheel
17	Tapered ring	1	Mark for reinstallation	Fit with the same gear-wheel. Tabs must engage in the cutouts in the loose gearwheel

No.	Designation	Qty.	Note:	
			Removal	Installation
18	Synchronizing ring	1	Mark for reinstallation	Check for wear. Fit with the same gearwheel. Drivers must engage in the cutouts in the tapered ring. Three lugs face driver dogs
19	Needle roller bearing		Mark for reinstallation	Fit with the same gear-wheel
20	Inner race	1	Shift rods in neutral position. Pull out shift rod for reverse gear. Pull off across loose gearwheel No. 32. Mark for reinstallation.	Fit with the same gear-wheel. Heat to approx. 120 °C
21	Shift sleeve	1		Insert complete with guide sleeve and shift rods. The missing tooth of the internal teeth of the guide sleeve (No. 28) must be aligned exactly above the oil bore of the output shaft.
22	Shift rod with pinned shift fork	1		
23	Shift rod (reverse)	1		
24	Spacer	1		Apply stiff grease to insert into shift rod
25	Driver dog	3		Place into correct position
26	Ball	3		
27	Spring	3		

No.	Designation	Qty.	Note:	
			Removal	Installation
28	Guide sleeve	1		Missing tooth of internal teeth must be placed exactly above oil bore of output shaft
29	Synchronizing ring	1	Mark for reinstallation	Check for wear. Fit with the same gearwheel. Drivers must engage into the cutouts in the tapered ring. The lugs face the driver dogs.
30	Tapered ring	1	Mark for reinstallation	Fit with the same gearwheel. Tabs must engage into the cutouts in the loose gearwheel
31	Friction ring	1	Mark for reinstallation	Fit with the same gearwheel
32	Loose gearwheel (2nd gear)	1		
33	Needle roller bearing	1	Mark for reinstallation	Fit with the same gearwheel
34	Inner race	1	Mark for reinstallation. Pull off across fixed gearwheel No. 36	Fit with the same gearwheel. Heat to approx. 120 °C
35	Thrust washer	1		
36	Fixed gearwheel (3rd gear)	1		Large flange faces thrust washer No. 35
37	Spacer sleeve	1		
38	Hexagon head bolt (micro-sealed)	1		Must always be replaced. Tightening torque 6 Nm (4 ftlb.)
39	Washer	1		
40	Spacer sleeve	1		

No.	Designation	Qty.	Note:	
			Removal	Installation
41	Retaining plate	1		
42	Hexagon head nut	10		Tighten to 23 Nm (17 ftlb.)
43	Tensioning plate with gear set	1		
44	Adjusting washer "S3"	X	Note number and thickness for reinstallation	Determine thickness if required
45	Oval-head screw	2		Tighten to 10 Nm (7 ftlb.)
46	Guide tube	1		
47	Shaft seal	1	Refer to page 35 - 1	Do not fit until gear set has been fitted (refer to page 35 - 1)
48	Transmission housing	1		

Note

Due to the tolerances (transition fit), the inner bearing rings (nos. 11, 20 and 34) may have clearance or press-fit on the shaft.

Modifications to the synchromesh and the shift sleeve for 1st and 2nd gear are described on page 35 - 14.

Dismantling and assembly notes

Dismantling

Note

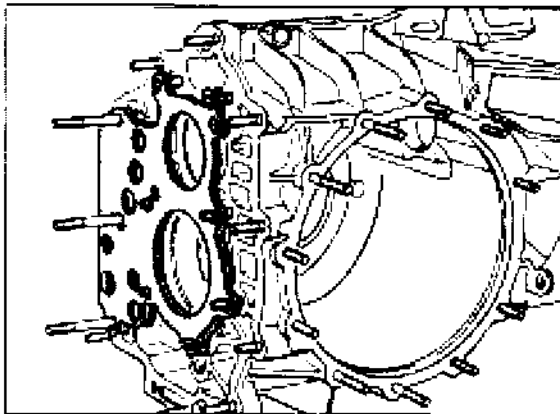
To facilitate assembly, do not remove the complete gear set assembly but partially dismantle the installed drive pinion.

Note

Do not move shift rods across the neutral or gear stop position as this may cause the small spacers to drop out inadvertently (refer to page 35 - 13).

Assembly

1. The number of adjusting shims "S3" noted during dismantling or the number of adjusting shims noted when the drive pinion was adjusted should be placed onto the studs in the housing until the equivalent thickness is obtained.

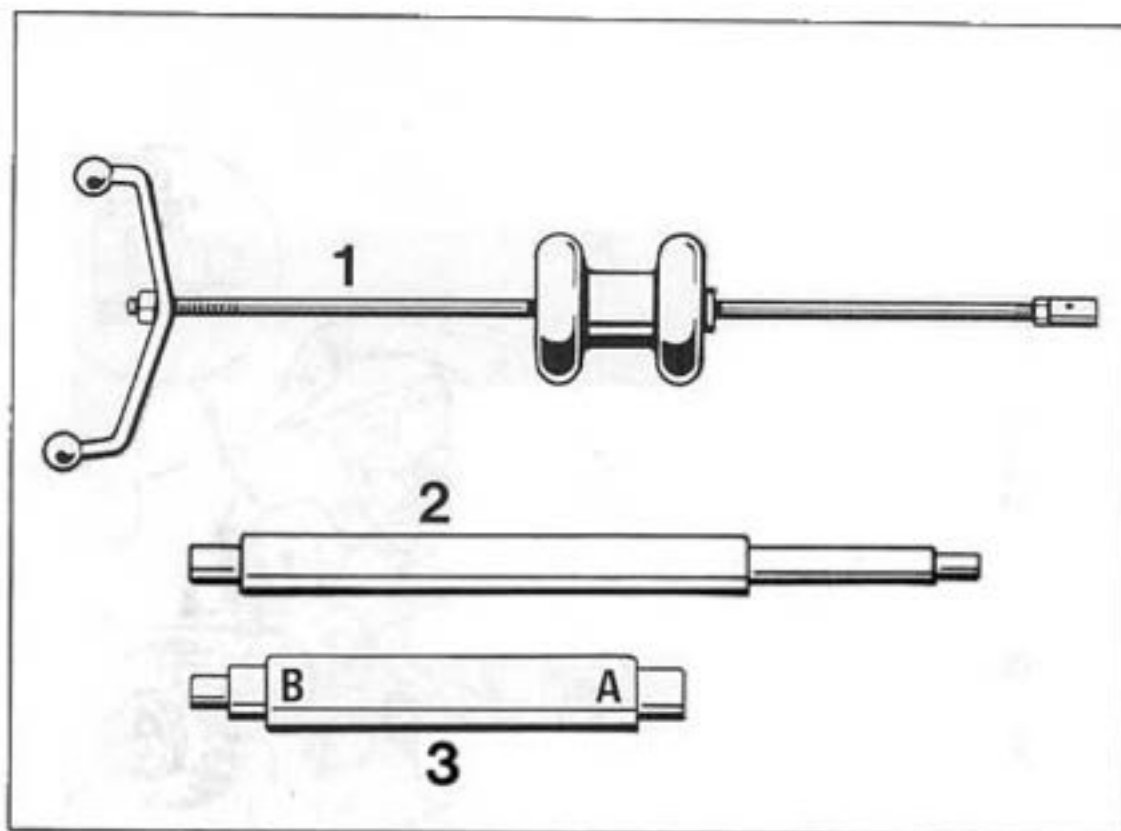


506-34

2. Fit fully assembled gear set complete with inner shift rod and tighten tensioning plate fastening nuts to **23 Nm** (17 ftlb.) (refer to page 35 - 7).

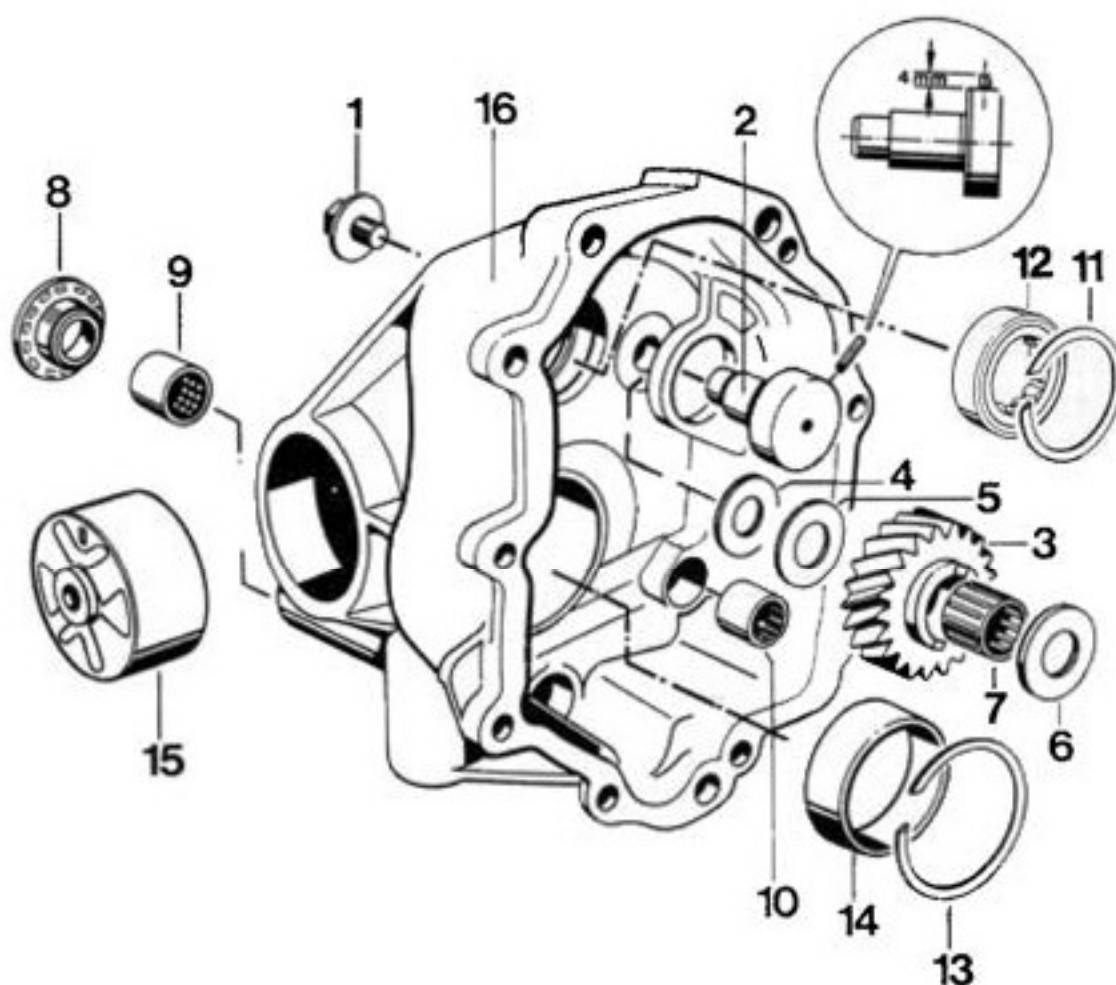
34 55 37 Dismantling and assembling front transmission cover

Tools



No.	Designation	Special tool	Order number	Explanation
1	Punch	VW 771		
2	Mandrel	9515	000.721.951.50	
3	Mandrel	9254	000.721.925.40	

34 55 19 Dismantling and assembling front transmission cover

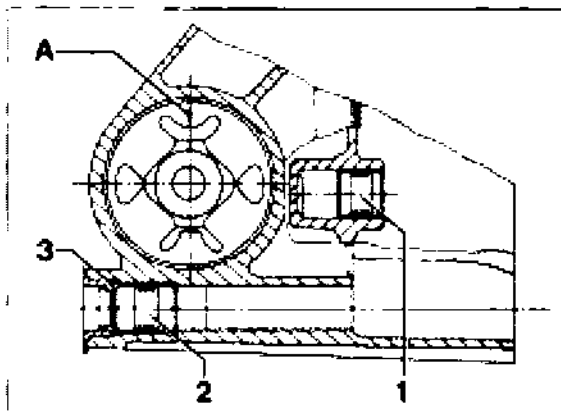


No.	Designation	Qty.	Removal	Note:	
				Installation	
1	Hexagon-head bolt	1		Torque: 35 Nm (26 ftlb.)	
2	Bolt with pin	1		Insert in correct position	
3	Reverse idler gear	1		Small collar points towards hexagon-head bolt (no. 1)	
4	Thrust washer (2.0 mm)	1			
5	Thrust washer (1.5 mm)	1		Do not confuse with washer no. 4	
6	Thrust washer (1.5 mm)	1		Do not confuse with washer no. 4	
7	Needle-roller assembly	1			
8	Shaft seal	1		Pack the area between dust lip and sealing lip with grease (e.g. Optimol HT2 + 2 EP); press in to stop with mandrel 9254 (p. A)	
9	Ball sleeve (long)	1	Press out from the inside, using a suitable mandrel (e.g. 9515)	Press in to stop with Special tool 9254 (p. B)	
10	Ball sleeve (short)	1	Pull out with a suitable internal puller (e.g. Schrem 14 - 20) and punch VW 771	Press in to stop with Special tool 9254 (p. A)	
11	Retaining ring	1			
12	Cylindrical roller bearing	1	Pull out with suitable internal puller (e.g. Schrem 30 - 40)	Heat cover to approx. 120 °C and press in to stop	
13	Retaining ring	1			

No.	Designation	Qty.	Note:	
			Removal	Installation
14	Outer bearing race	1	Pull out with suitable internal puller (e.g. Schrem 50 - 60)	Heat cover to approx. 120 °C and press in to stop
15	Transmission bearing	1	Press out with suitable pipe section	Press in to correct position
16	Front transmission cover	1		

Notes on assembly

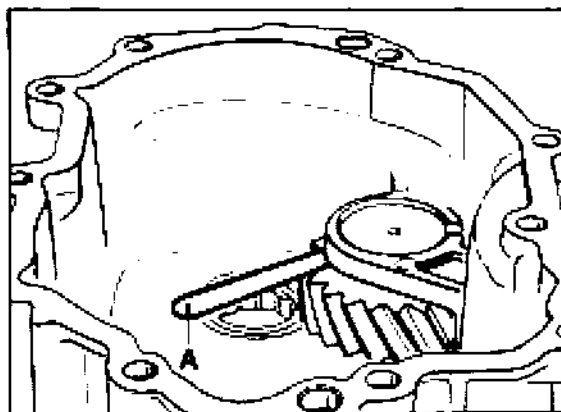
1. Press in ball sleeves, shaft seal and transmission bearing to correct position.



1642-34

- 1 - Short ball sleeve (press in flush)
- 2 - Long ball sleeve
- 3 - Shaft seal
- A - Assembly mark (must point upwards)

2. Reassemble reverse idler gear with thrust washers and bolt in correct position (refer to exploded view). With thrust washers and bolt located correctly, the end clearance of the reverse idler gear is 0.15...0.35 mm.



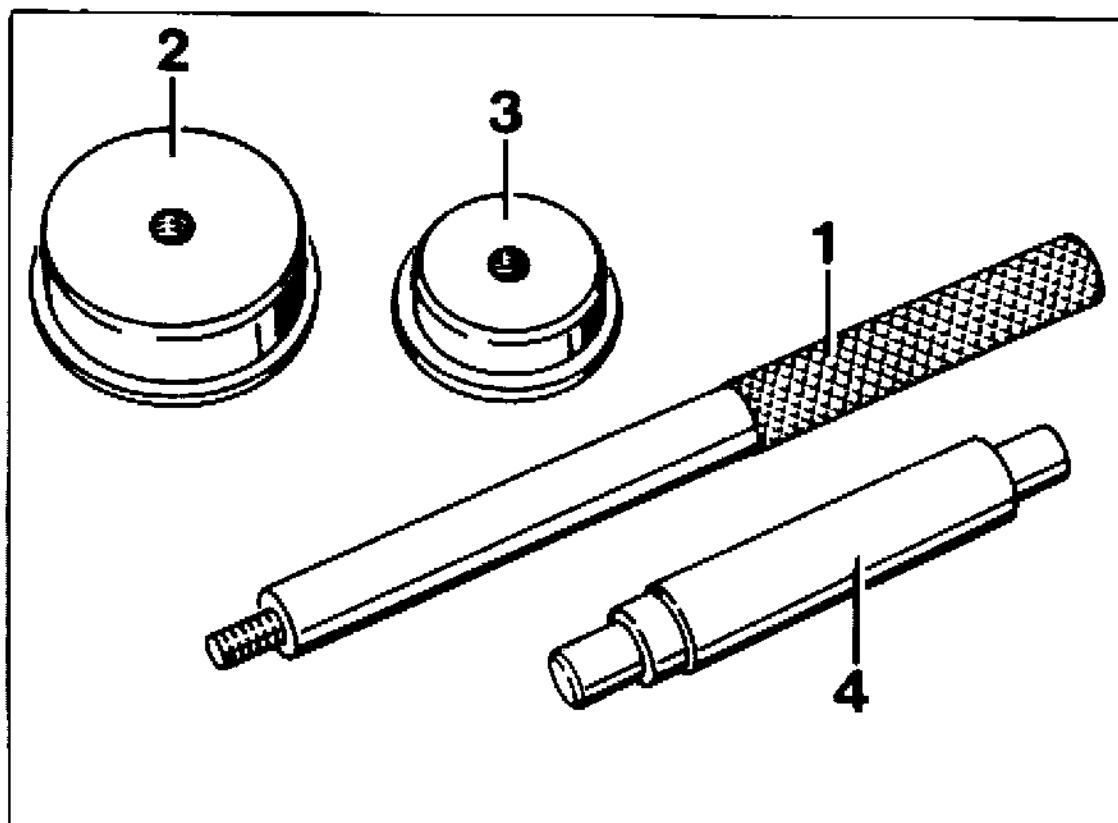
520-34

A = Feeler gauge

3. Check end clearance of reverse idler gear with feeler gauge.

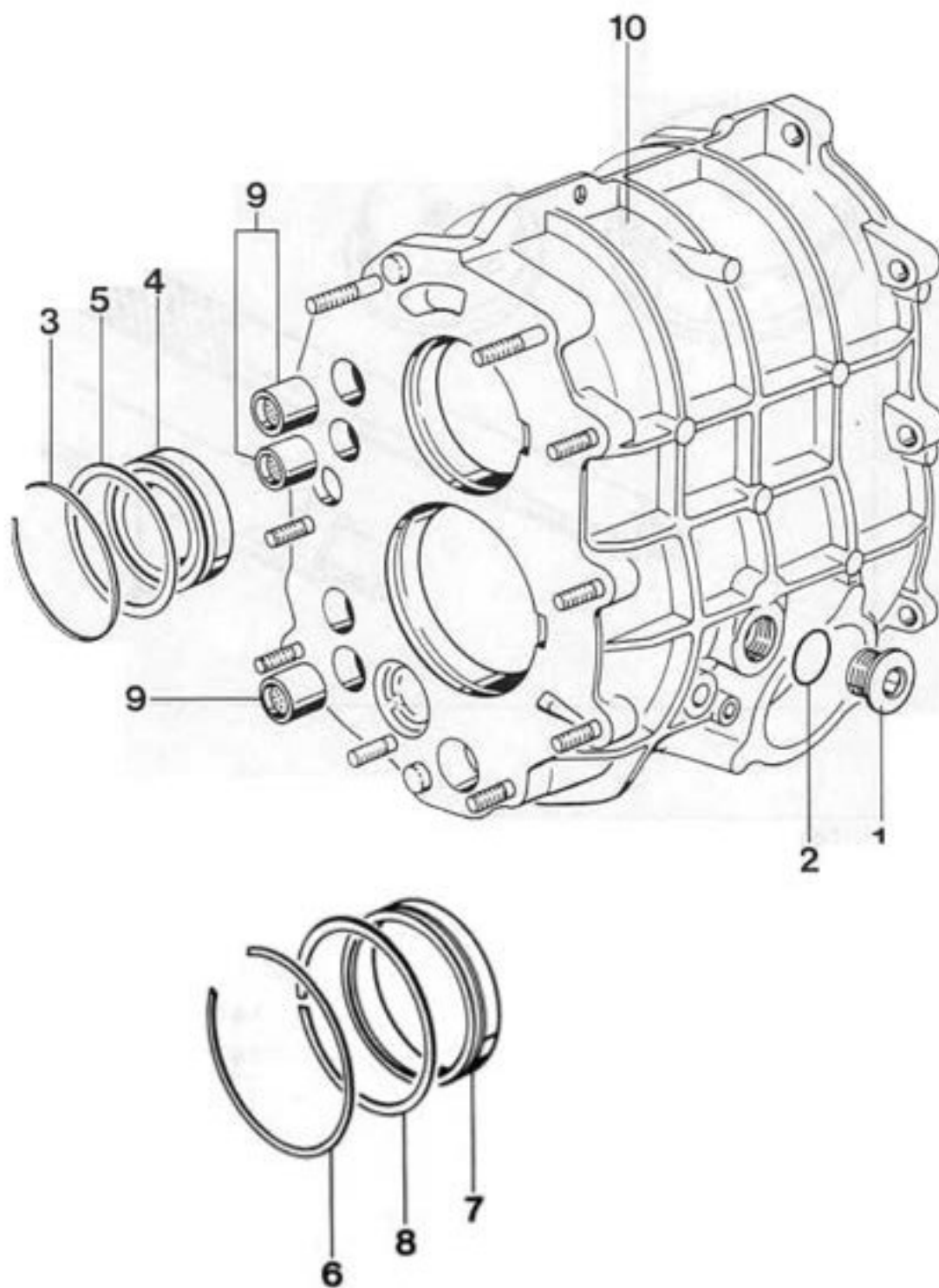
34 52 37 Dismantling and assembling gear housing

Tools



No.	Designation	Special tool	Order number	Explanation
1	Mandrel	P 254	000.721.254.00	
2	Thrust piece	P 254 a	000.721.254.10	
3	Thrust piece	P 254 b	000.721.254.20	
4	Mandrel	9254	000.721.925.40	

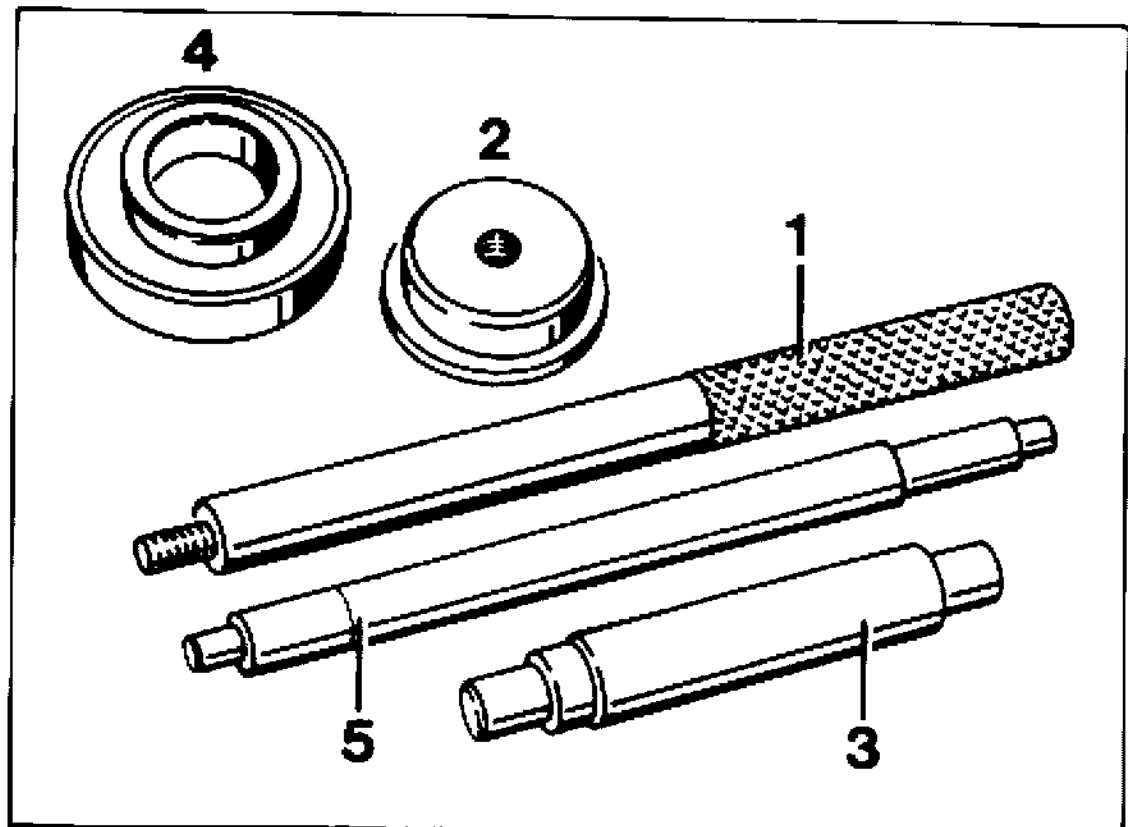
34 52 37 Dismantling and assembling gear housing



No.	Designation	Qty.	Note:	
			Removal	Installation
1	Plug	1		Torque: 30 Nm (22 ftlb.)
2	Seal	1		Replace
3	Retaining ring	1		
4	Bearing outer race	1	Working from inside, press out with Special Tools P 254 and P 254 b	Press in with Special Tools P 254 and P 254 b
5	Snap ring	1		
6	Retaining ring	1		
7	Bearing outer race	1	Working from inside, press out with Special Tools P 254 and P 254 b	Press in with Special Tools P 254 and P 254 b
8	Snap ring	1		
9	Ball sleeve	3	Press out with suitable mandrel	Press in flush with Special Tool 9254
10	Gear housing	1		

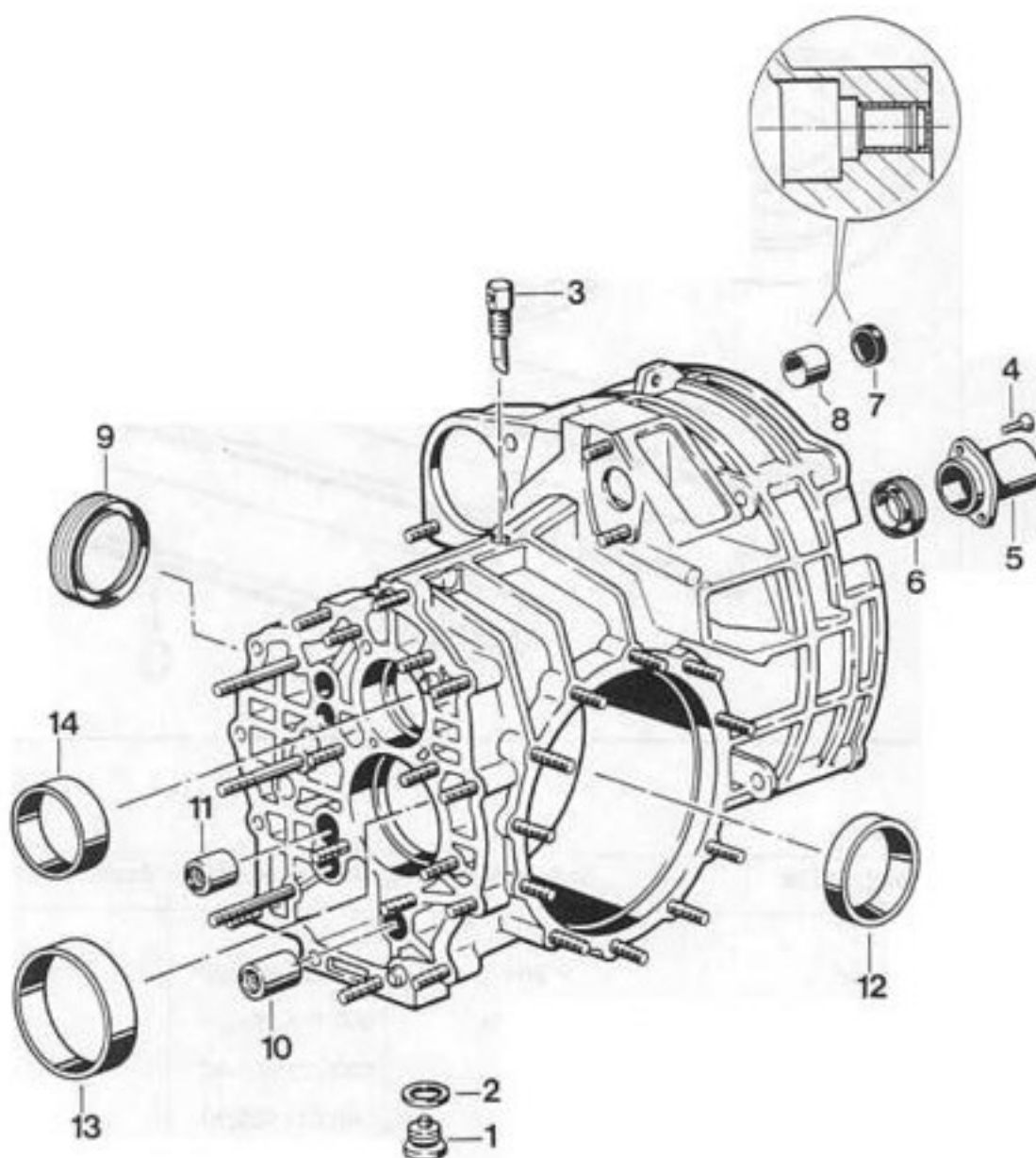
34 37 37 Dismantling and assembling transmission housing

Tools



No.	Designation	Special tool	Order number	Explanation
1	Mandrel	P 254	000.721.254.00	
2	Thrust piece	P 254 b	000.721.254.20	
3	Mandrel	9254	000.721.925.40	
4	Thrust piece	9252	000.721.925.20	
5	Mandrel	9515	000.721.951.50	

34 37 37 Dismantling and assembling transmission housing

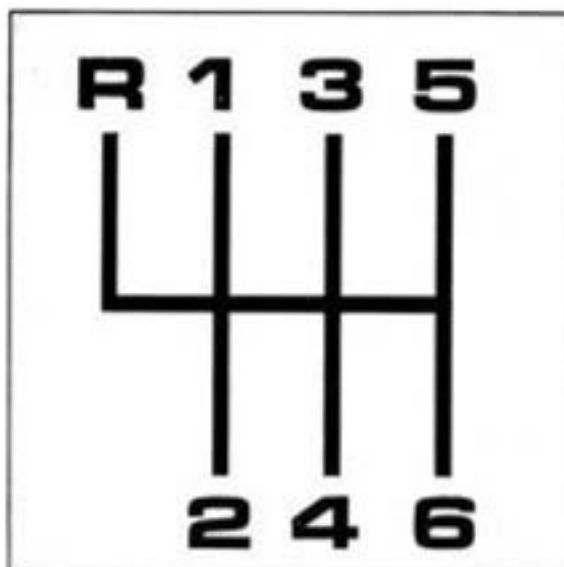


No.	Designation	Qty.	Removal	Note:
				Installation
	Plug	1		Clean, tighten to 30 Nm (22 ftlb.)
2	Seal	1		Replace
3	Vent	1		Tighten to 35 Nm (26 ftlb.). The hole in the hexagon head must face the front transmission cover
4	Oval-head screw	2		Tighten to 10 Nm (7 ftlb.)
5	Guide tube			
6	Shaft seal	1		Do not fit until the gear set has been fitted (also refer to page 35 - 1)
7	Cover	1		
8	Bushing	1	Push out with drift 9515, working from inside towards outside	Do not grease or oil and do not clean with solvents (e.g. benzene). Push in from inside with drift 9515 until it is seated against the stop.
9	Shaft seal			Push in with Special Tool 9252 until it is seated against the stop. Pace space between dust lip and sealing lip with grease (e.g. Optimol HT2 + 2EP)
10	Ball sleeve (long)	1	Pull out with suitable internal puller (e.g. Schrem 14 - 20)	Press in flush with Special Tool 9254
11	Ball sleeve (short)	4	Pull out with suitable internal puller (e.g. Schrem 14 - 20)	Press in flush with Special Tool 9254

No.	Designation	Qty.	Note:	
			Removal	Installation
12	Bearing outer race	1	Heat transmission housing to approx. 120 °C and remove race with suitable drift	Heat transmission housing to approx. 120 °C and press in with suitable thrust piece
13	Bearing outer race	1	Heat transmission housing to approx. 120 °C and remove race with suitable thrust piece (e.g. VW 513)	Heat transmission housing to approx. 120 °C and press in with suitable thrust piece (e.g. 9247/4) until it is seated against the stop
14	Bearing outer race	1	Heat transmission housing to approx. 120 °C and pull out with suitable internal puller (e.g. Schrem 50 - 60)	Heat transmission housing to approx. 120 °C and press in with Special Tools P 254 and P 254 b until it is seated against the stop
15	Transmission housing	1		

34 Adjusting gear shift

1. Set the shift lever to the neutral position.
The lever will automatically move to the 3rd/4th gear plane (no adjustment possible).



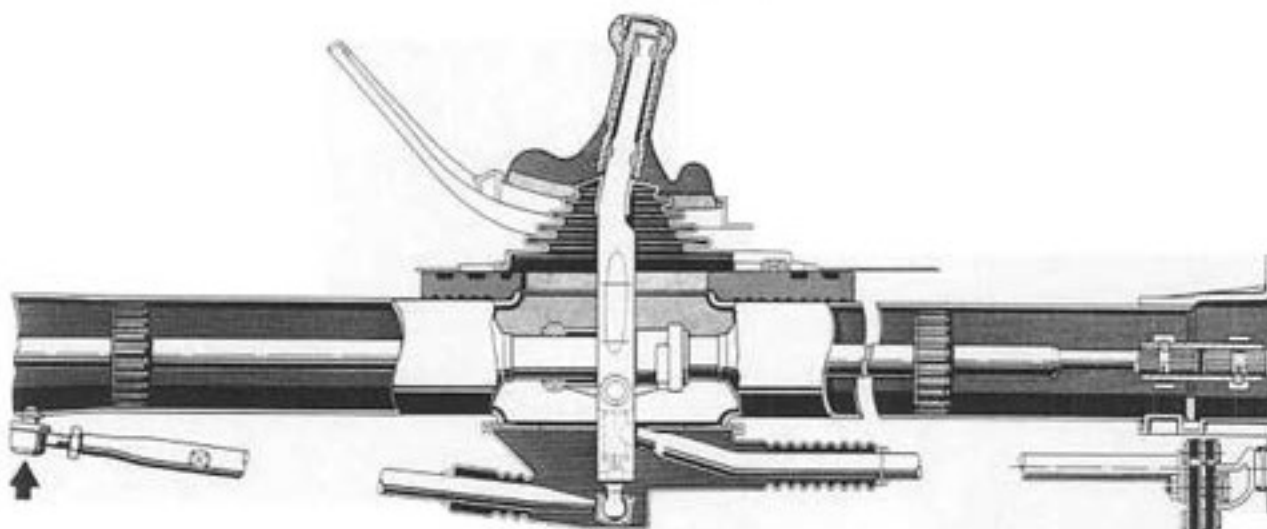
2049A-34

2. If the gearshift is correctly adjusted, there should be no longitudinal inclination on the lower part of the lever, which is curved at the top. The lower 90° part should be at right angles to the central tube or transmission tunnel.

3. Checking the adjustment: shift through all the gears. whichever gear is selected, there must be about the same space between the front and rear consoles.

4. If necessary, correct the adjustment on the ball of the adjustable guide tube (arrow). One turn of the ball corresponds to a change of about 3 to 5 mm in the angle of the shift lever.

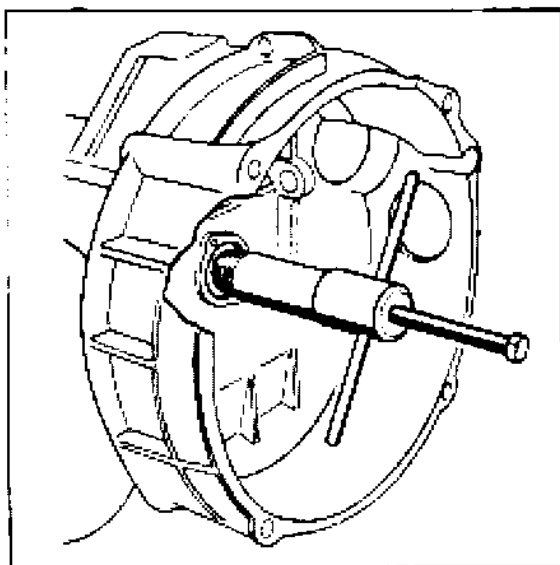
Caution: When tightening the lock nut, make sure that you do not twist the balls in relation to each other.



Drawing 644A shows the four-wheel drive version. On the rear-wheel drive version, the ball of the guide tube (arrow) is installed on a transverse strut attached to the front axle cross member.

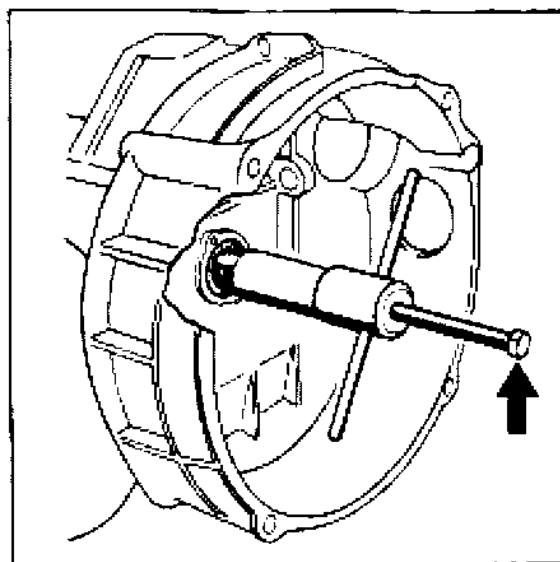
35 50 19 Removing and installing input shaft oil seal**Removing**

1. Remove engine/transmission assembly and separate transmission from engine.
2. Remove release bearing guide tube.
3. Screw Special Tool 9251 firmly into the oil seal.



1652-34

4. Screw in the hexagon-head bolt and pull out oil seal.



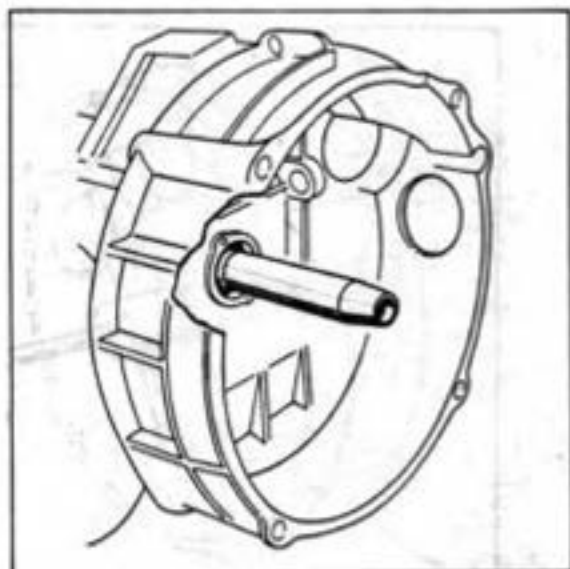
1653-34

Note

if the helical tension spring jumps out when removing the seal, use a wire hook to pull it off the input shaft.

Installing

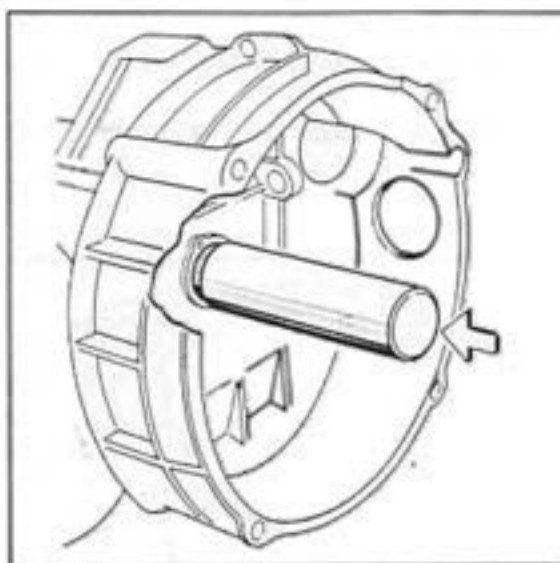
1. Push assembling sleeve **9255** onto the input shaft splines.



1654-34

2. Pack oil seal area between dust lip and sealing lip with grease (e.g. Optimol HT 2 + 2 EP).

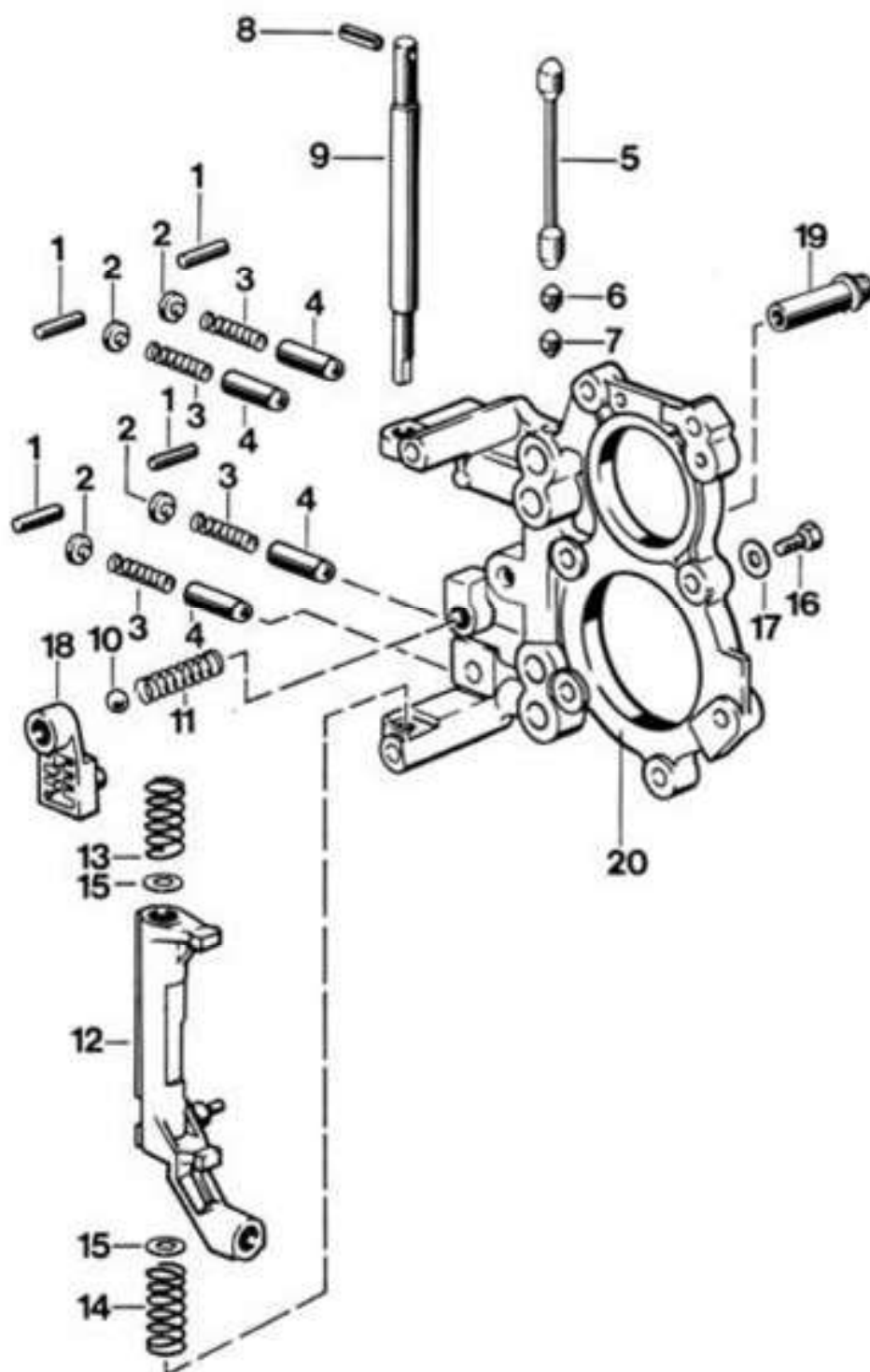
3. Use Special Tool **9256** to drive the oil seal into place.



1655-34

4. Install guide tube and tighten mounting screws to **10 Nm** (7 ftlb.).

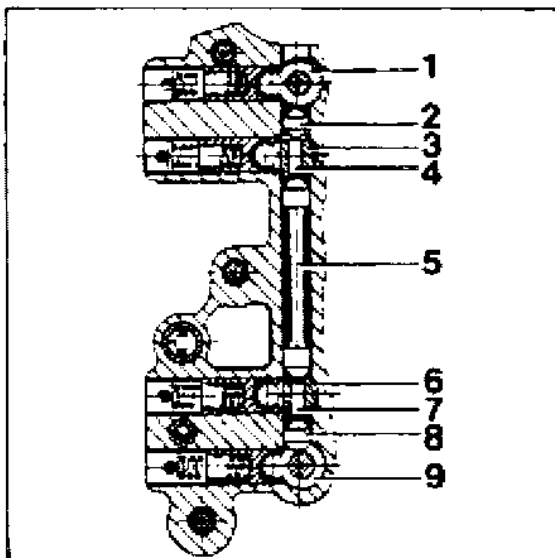
35 19 37 Dismantling and assembling tensioning plate



No.	Designation	Qty.	Note:	
			Removal	Installation
1	Straight pin	4	With the shift rods fitted, the springs (No. 3) are under load	
2	Washer	4		
3	Thrust spring	4		
4	Locking bush	4		
5	Lock (long)	1		
6	Lock (short)	1		
7	Lock (short)	1		
8	Roll pin	1		
9	Bearing shaft	1		Install in correct position, use 8 mm dia. locating mandrel to locate relative to tensioning plate hole
10	Ball	1		
11	Thrust spring	1		
12	Deflection lever	1		
13	Thrust spring	1	Mark for reinstallation	Free length = 51.3 + 0.5 mm wire thickness = 1.5 mm. Do not confuse with spring No. 14
14	Thrust spring	1	Mark for reinstallation	Free length = 72.6 + 0.5 mm wire thickness = 1.1 mm. Do not confuse with spring No. 13
15	Shim	2		

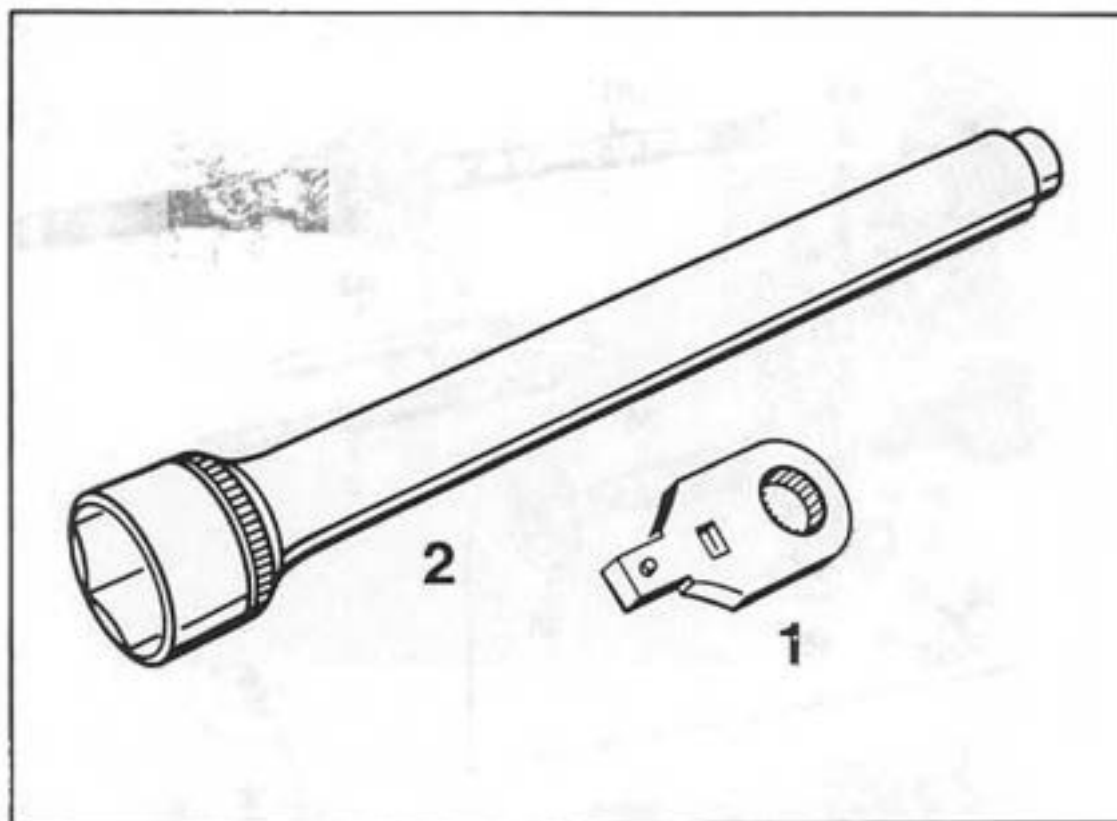
No.	Designation	Qty.	Note:	
			Removal	Installation
16	Hexagon head bolt	1		Tighten to 10 Nm (7 ftlb.)
17	Washer	1		
18	Selector gate	1		
19	Adapter sleeve with snap ring	1	Press out with suitable drift	Press in with suitable drift until it is sealed against the stop
20	Tensioning plate	1		

Installation position of the locks



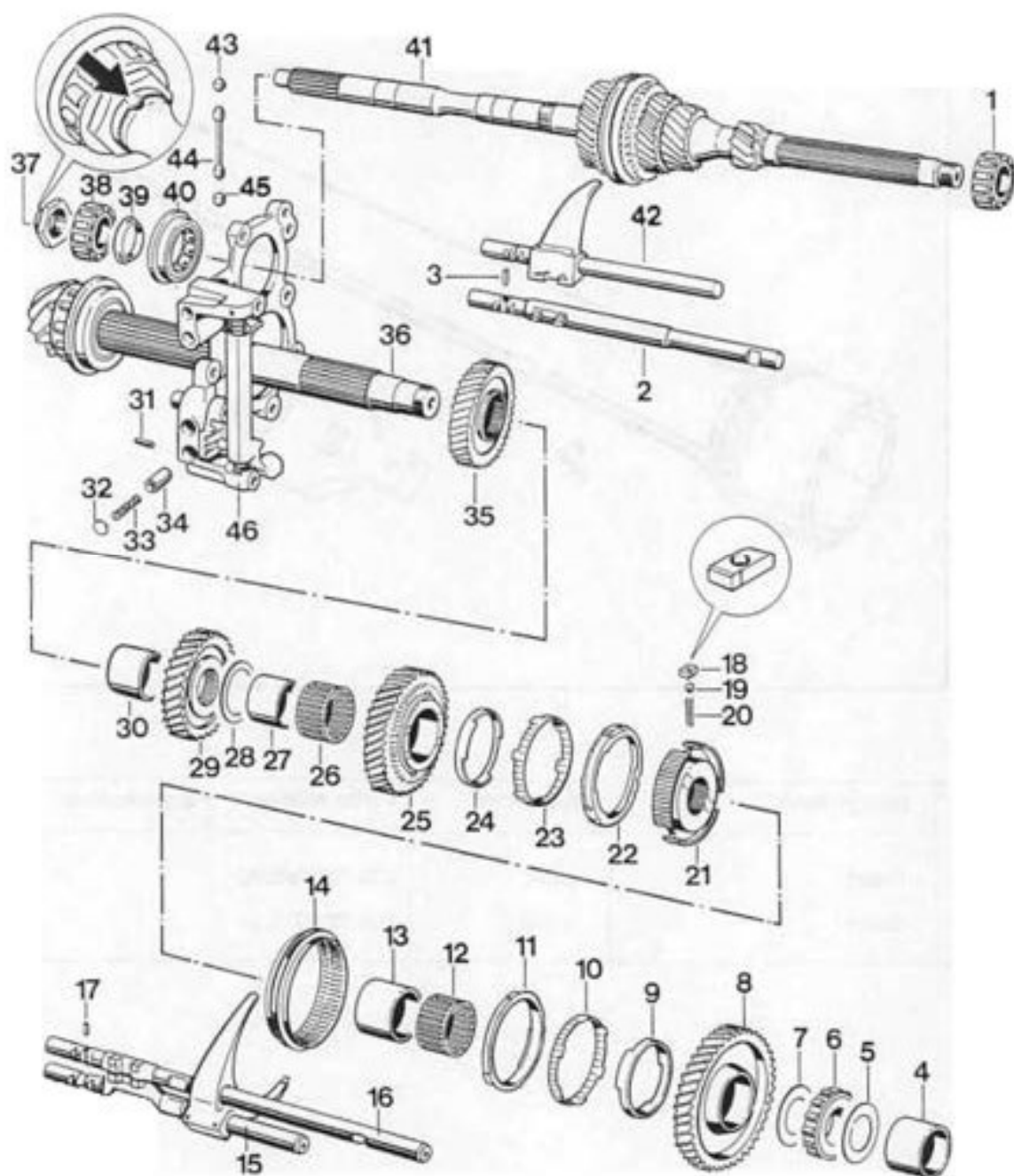
35 40 19 Removing and installing input shaft

Tools



No.	Designation	Special tool	Order number	Explanation
1	Insert	9282	000.721.928.20	
2	Socket	9105	000.721.910.50	

35 40 19 Removing and installing input shaft



No.	Designation	Qty.	Note:	
			Removal	Installation
1	Cylindrical roller bearing	1	Shift rods in neutral position	Heat to approx. 120 °C
2	Shift rod (5th and 6th gear)	1		
3	Spacer	1		Coat with stiff grease to insert
4	Inner race	1		Fit with the same gear-wheel. Heat to approx. 120 °C
5	Thrust washer	1		
6	Cylindrical roller bearing	1		Heat to approx. 120 °C
7	Thrust washer	1		
8	Loose gearwheel (1st gear)	1		
9	Friction ring*	1		Fit with the same gear-wheel
10	Tapered ring	1		Fit with the same gear-wheel. Tabs must engage into the cutouts in the loose gearwheel
11	Synchronizing ring*	1		Check for wear. Fit with the same gearwheel. Drivers must engage in the cutouts in the tape-red ring. Three lugs face the driver dogs.
12	Needle roller bearing	1		Fit with the same gear-wheel
13	Inner race	1		Fit with the same gear-wheel. Heat to approx. 120 °C

* see page 35 - 14

No.	Designation	Qty.	Note:	
			Removal	Installation
14	Shift sleeve*	1		Insert complete with guide sleeve and shift rods. Make sure the missing tooth of the internal teeth of the guide sleeve (No. 21) is aligned exactly above the oil bore of the output shaft.
15	Shift rod with pinned shift fork	1		
16	Shift rod (reverse)	1		
17	Intermediate lock	1		Coat with stiff grease to insert
18	Driver dog	3		Install in correct position
19	Ball	3		
20	Spring	3		
21	Guide sleeve	1		Missing tooth of the internal splines must be exactly above the oil bore of the output shaft
22	Synchronizing ring*	1		Check for wear. Fit with the same gearwheel. Drivers must engage into the cutouts in the tapered sleeve. Three lugs face the driver dogs.

* see page 35 - 14

No.	Designation	Qty.	Note:	
			Removal	Installation
23	Tapered ring	1		Fit with the same gear-wheel. Tabs must engage into the cutouts in the loose gearwheel
24	Friction ring	1		Fit with the same gear-wheel
25	Loose gearwheel (2nd gear)	1		
26	Needle roller bearing	1		Fit with the same gear-wheel
27	Inner race	1		Fit with the same gear-wheel. Heat to approx. 120 °C
28	Thrust washer	1		
29	Fixed gearwheel (3rd gear)	1		Large collar faces thrust washer No. 35
30	Spacer sleeve	1		
31	Straight pin	4		
32	Washer	4		Guide lug faces spring
33	Thrust spring	4		
34	Latch	4		
35	Fixed gearwheel (4th gear)	1		Large collar faces four-point bearing
36	Output shaft	1		
37	Flange nut**	1	Undo with Special Tools 9282 and 9105	Tighten to 250 Nm (184 ftlb.). Upset the flange to lock
38	Cylindrical roller bearing	1	Press off across 2nd gear fixed gearwheel	Heat to approx. 120 °C
39	Bearing inner race	1		Heat to approx. 120 °C

No.	Designation	Qty.	Note:	
			Removal	Installation
40	Four-point bearing	1		
41	Output shaft	1		
42	Shift rod with pinned shift fork	1		Insert complete with pinned shift rod / shift fork (No. 42)
43	Lock (short)	1		
44	Lock (long)	1		
45	Lock (short)	1		
46	Tensioning plate	1		

- ** Transmission shafts without recesses for securing the hexagon nuts have been installed since September 1995.
Self-locking hexagon nuts are used on these shafts, and these nuts must **always** be replaced in every transmission repair.

Removal and installing notes

Removal

Note

Parts No. 1 to 30 are removed with the tensioning plate remaining fitted (refer to page 34 - 11).

Installation

Using a suitable flat iron bar, clamp tensioning plate in a vise in such a manner that the hole for the shift rod locks is horizontal.

2. Check synchromesh of 1st and 2nd gears.
To do so, place friction ring, tapered ring and synchronizing ring in correct position onto gearwheel. Check gap "A" with a feeler gauge.

Old design:

installation dimension (new) =

1.3 to 1.95 mm

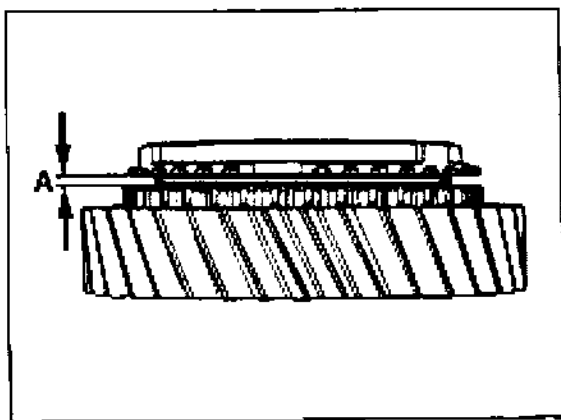
wear dimension = 1.0 mm

New design:

installation dimension (new) = 1.5 to 2.0 mm

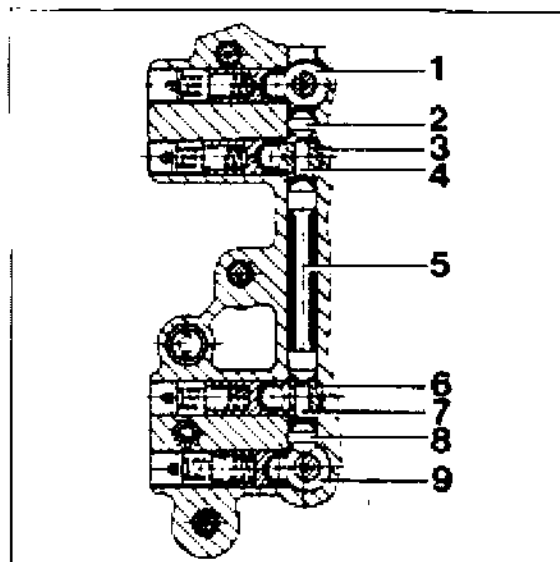
wear dimension = 1.2 mm

(see page 35 - 14)



1701-36

3. Observe installation position of locks.



1709-35

- 1 = Shift rod 3rd and 4th gear
- 2 = Lock (short)
- 3 = Shift rod 5th and 6th gear
- 4 = Intermediate lock
- 5 = Lock (long)
- 6 = Reverse shift rod
- 7 = Intermediate lock
- 8 = Lock (short)
- 9 = Shift rod 1st and 2nd gear

Note

After fitting the shift rods, do not move them across the neutral or gear latch positions as this may cause the small intermediate locks to drop out inadvertently.

To avoid inadvertent movement of the shift rods, lock shift rods by engaging 4th gear.

Modified synchromesh on 1st and 2nd gear

The following parts have been modified:

shift sleeve (with three plunge-milled grooves on the inside teeth).

- friction ring (changed dimensions and marking 94)

From transmission no.:

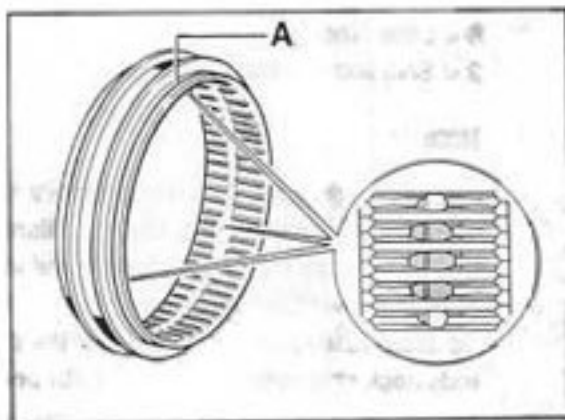
G 5020 1 000 884
G 5020 2 000 883
G 5021 1 002 186
G 5021 2 001 230

compression spring
(unloaded length 16.22 mm)

From transmission no.:

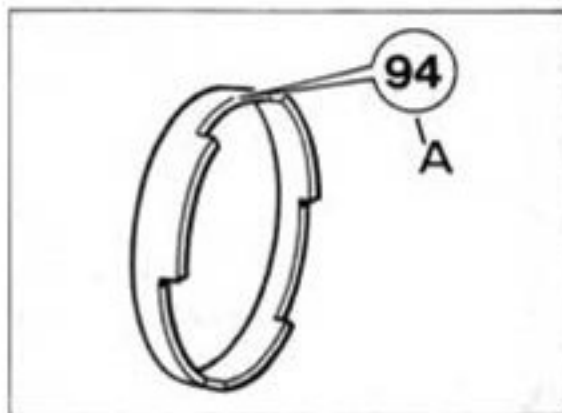
G 5020 1 000960
G 5020 2 000927
G 5021 1 002238
G 5021 2 001294

Shift sleeve



A = distinctive groove
(must point towards 2nd gear)

Friction ring



A = marking

1904-35

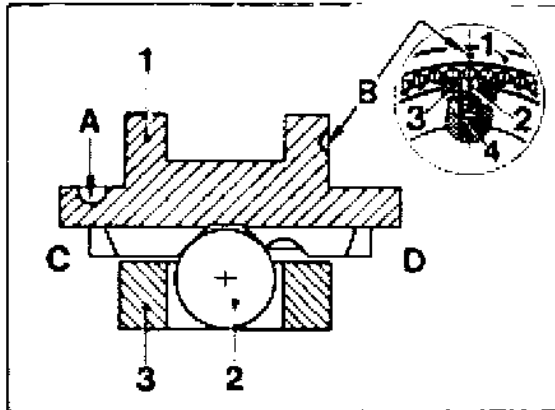
Note

Old and new parts must **not** be installed together in the same transmission. As only the modified parts are now available, all the modified parts must be installed in the event of repair work. The following points must be noted:

Install the shift sleeve on the guide sleeve with the punch marks "B" centered between the balls.

- New installation and wear dimensions
installation (new) = 1.5...2.0 mm
wear dimension = 1.2 mm
(see also page 35 - 13)
- Continuous groove "A" must point towards 2nd gear.

1905-35

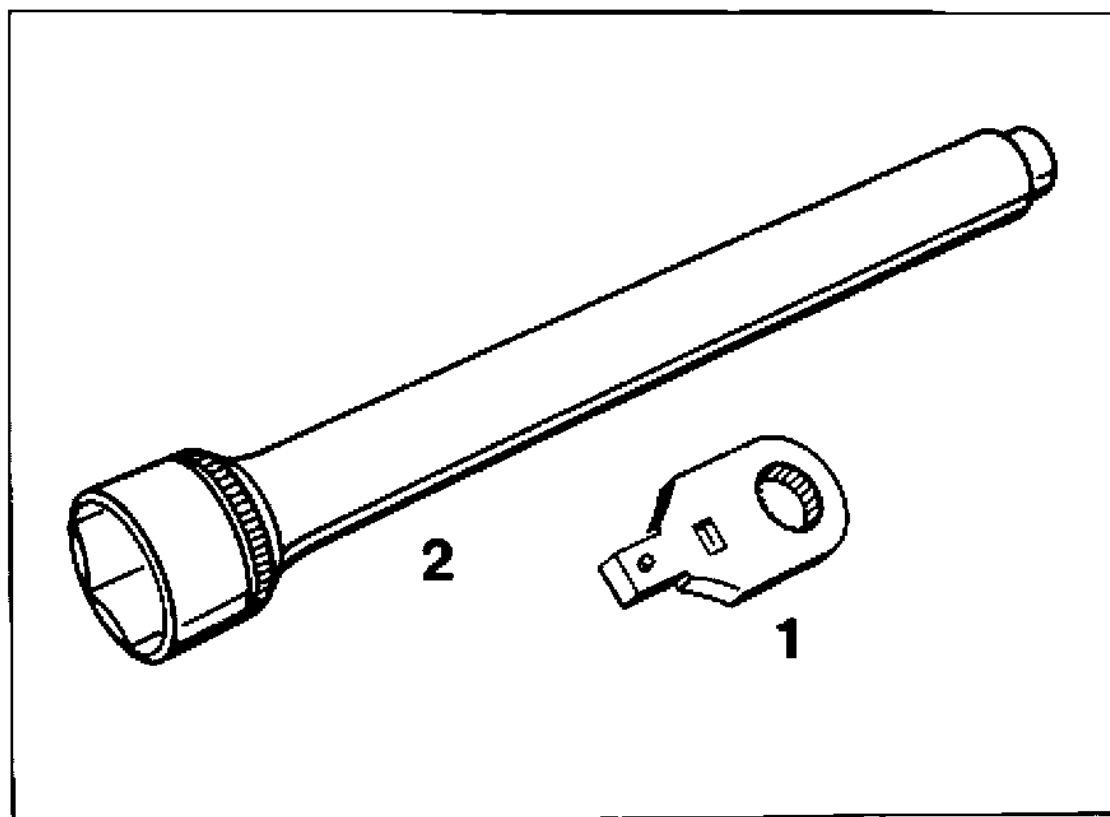


1906-35

- 1 = shift sleeve
- 2 = ball
- 3 = driver dog
- 4 = spring
- A = groove
- B = punch mark
- C = 2nd gear side
- D = 1st gear side

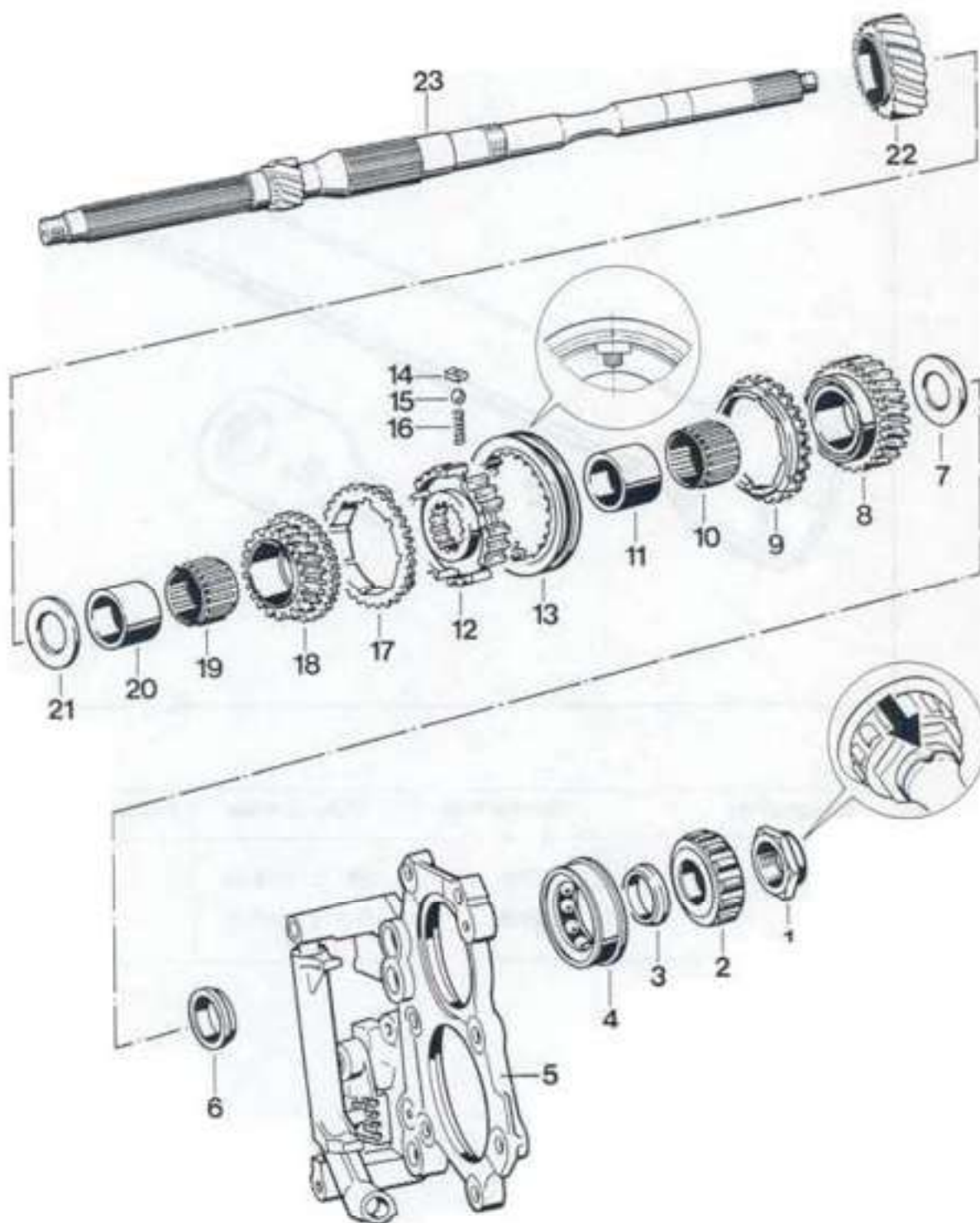
35 40 37 Dismantling and assembling input shaft

Tools



No.	Designation	Special tool	Order number	Explanation
1	Insert	9282	000.721.928.20	
2	Socket	9105	000.721.910.50	

35 40 37 Dismantling and assembling input shaft



No.	Designation	Qty.	Note:	
			Removal	Installation
1	Hexagon head nut (see note on Page 35 - 19)	1	Undo with Special Tools 9282 and 9105	Use Special Tools 9282 and 9105 and tighten to 250 Nm (184 ftlb.). Upset flange to lock
2	Cylindrical roller bearing	1	Press off with suitable separating device (e.g. Kukko 17-1) across 2nd gear fixed gearwheel	Heat to approx. 120 °C
3	Bearing inner race	1	Press off with suitable separating device (e.g. Kukko 17-1) across 2nd gear fixed gearwheel	Heat to approx. 120 °C
4	Four-point bearing	1		
5	Tensioning plate	1		Clamp in a vise (use protective jaws), insert lock (long), engage 5th gear and install input shaft with pinned shift rod/shift fork
6	Bearing inner race	1	Press off with suitable separating device (e.g. Kukko 17-1)	Heat to approx. 120 °C
7	Thrust washer	1		Large, face-ground side faces needle cage
8	Loose gearwheel (4th gear)	1		Replace only in pairs
9	Synchronizing ring	1	Mark for reinstallation	Check for wear, install in correct position using the same gearwheel (lugs face the driver dogs)

No.	Designation	Qty.	Note:	
			Removal	Installation
10	Needle-roller assembly	1	Mark cage	Install with the same gear-wheel
11	Inner race	1		Heat to approx. 120 °C
12	Guide sleeve	1	Remove complete with shift sleeve	Install complete with shaft sleeve and synchro-mesh components
13	Shift sleeve (3rd and 4th gear)	1	Make sure synchromesh components do not pop out.	Observe installation position, install complete with guide sleeve and synchro-mesh components. Center the centerpunch marks relative to the driver dogs
14	Driver dogs	3		Install in correct position, domed side faces shift sleeve
15	Ball	3		
16	Spring	3		
17	Synchronizing ring	1	Mark for reinstallation	Check for wear, install in correct position with the same gearwheel (lugs face the driver dogs)
18	Loose gearwheel (3rd gear)	1		Replace only in pairs
19	Needle-roller assembly*	1	Mark for reinstallation	Fit with the same gear-wheel
20	Inner race*	1		Heat to approx. 120 °C and press into place
21	Thrust washer (1.85 mm thick)*	1		
22	Fixed gearwheel (2nd gear)*	1	Press off with suitable separating device (e.g. Kukko 17-0)	Replace only in pairs. Flange faces gearwheel of 1st gear
23	Input shaft	1		

* See note on page 35 - 19

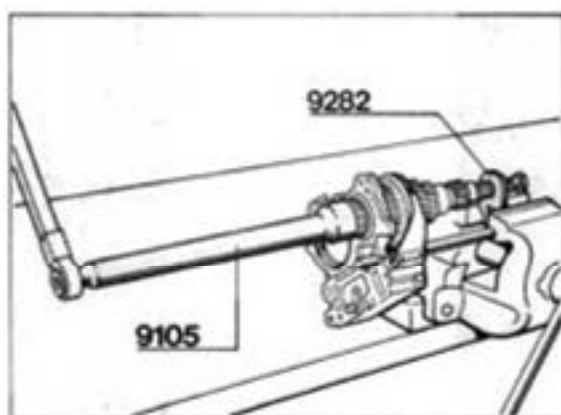
Dismantling and assembling notes

Note

Since 1 October 1994, the fixed wheel of second gear (no. 22) has been firmly fixed to the drive pinion; the needle cage (no. 19) runs directly on the polished shaft. The thrust washer (no. 21) and the inner ring (no. 20) are no longer fitted.

Dismantling

1. Mount retaining plate **9282** in vise, fit input shaft and undo hexagon-head nut with Special Tool **9105**.



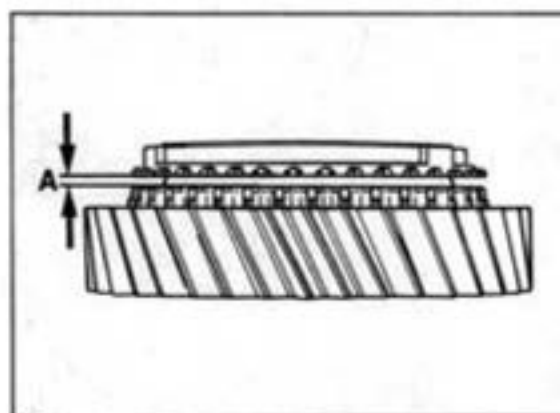
535-35

2. Use a suitable tool (e.g. Kukko 17 - 1) to press all parts off the input shaft over the 2nd gear wheel.

Assembling

1. To inspect synchronizing rings, press rings onto the gear wheel tapers and use a feeler gauge to measure gap "A".

Assembly dimension (new)	= 0.9 mm min.
Wear limit	= 0.6 to 0.7 mm



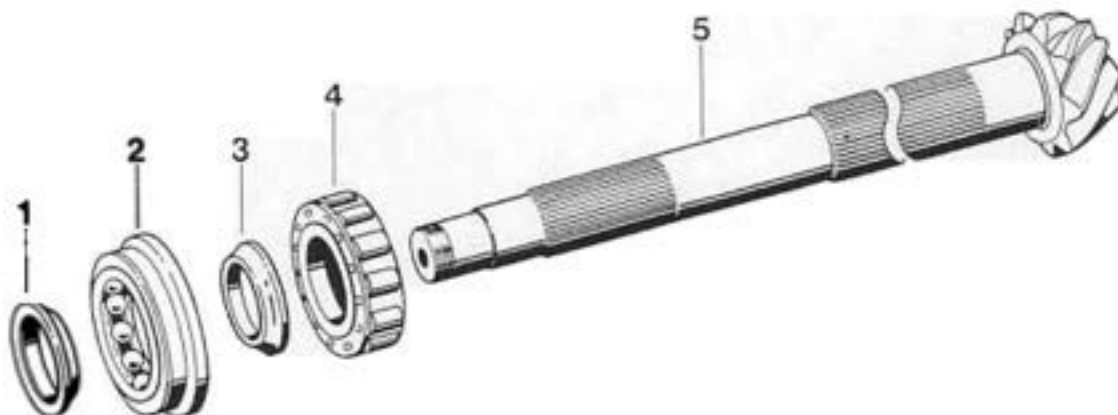
518-35

Note

Transmission shafts without recesses for securing the hexagon nuts have been installed since September 1995.

Self-locking hexagon nuts are used on these shafts, and these nuts must **always** be replaced in every transmission repair.

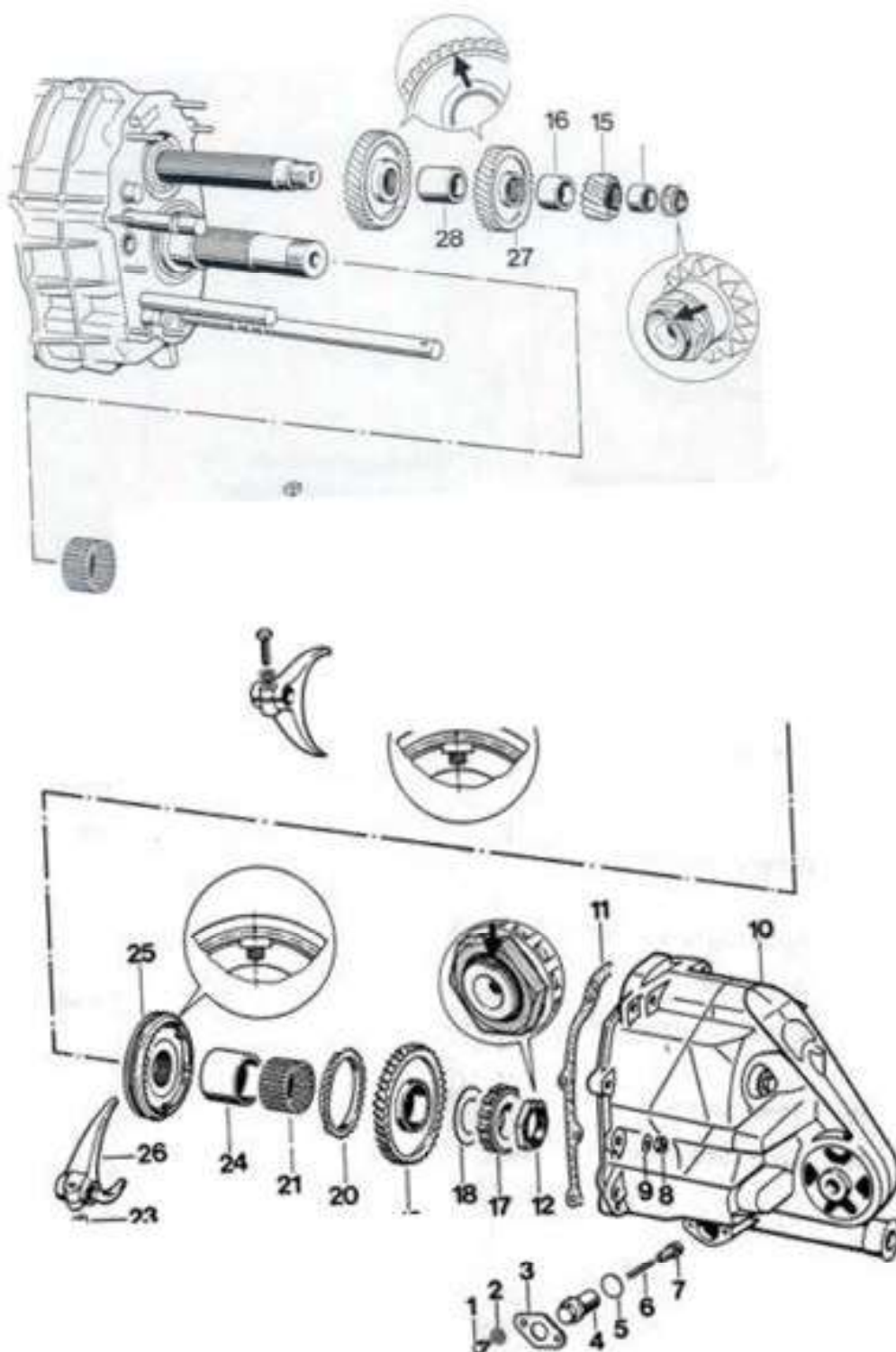
35 59 37 Dismantling and assembling output shaft



1696-35

No.	Designation	Qty.	Note:	
			Removal	Installation
1	Bearing inner race	1	Mark for reinstallation, press off with suitable separating device (e.g. Kukko 15 - 17)	Do not confuse with inner race No. 3, heat to approx. 120 °C and press into place
2	Four-point bearing	1		
3	Bearing inner race	1	Mark for reinstallation, press off with suitable separating device (e.g. Kukko 15 - 17)	Do not confuse with inner race No. 1, heat to approx. 120 °C and press into place
4	Cylindrical roller bearing	1	Press off with suitable separating device (e.g. Kukko 15 - 17)	Heat to approx. 120 °C and press into correct position using a suitable pipe section (e.g. VW 519)
5	Output shaft	1		Observe matching number. Readjust if required

35 75 19 Removing and installing 5th gear (transmission is installed)



No.	Designation	Qty.	Note:	
			Removal	Installation
1	Hexagon-head bolt	2		Tighten to 10 Nm (7 ftlb.)
2	Washer	2		
3	Clamping plate	1		
4	Eccentric bushing	1		Adjust
5	Seal	1		Replace
6	Thrust spring	1		
7	Locking bush	1		
8	Hexagon head nut	10		Tighten to 23 Nm (17 ftlb.)
9	Washer	10		
10	Front transmission cover	1	Remove complete with transmission support	
11	Gasket	1		Replace
12	Hexagon nut*	1	Engage 4th gear and apply handbrake	Replace. Tighten to 300 Nm (221 ftlb.). Upset flange to lock
13	Hexagon nut*	1	Engage 4th gear and apply handbrake	Replace. Tighten to 200 Nm (147 ftlb.). Upset flange to lock
14	Bearing inner race	1	Pull off across fixed gear-wheel No. 15	Heat to approx. 120 °C
15	Fixed gearwheel (reverse)	1		
16	Spacer sleeve	1		
17	Cylindrical roller bearing	1	Pull off across loose gear-wheel No. 19	Heat to approx. 120 °C
18	Thrust washer	1		
19	Loose gearwheel (reverse)	1		

No.	Designation	Qty.	Note:	
			Removal	Installation
20	Synchronizing ring	1	Mark for reinstallation	Check for wear. Fit with the same gearwheel (Cogs face driver dogs)
21	Needle cage	1	Mark for reinstallation	Fit with the same gearwheel
22	Hexagon-head bolt	1	Determine installation position of shift fork, measuring from shift rod face to shift fork stop and record measurement.	Tighten to 23 Nm (17 ftlb.)
23	Washer	1		
24	Inner race	1	Remove, press off across guide sleeve if required and pull off with a suitable puller.	Heat to approx. 120 °C.
25	Shift and guide sleeve	1	Engage 4th gear and take off carefully complete with shift fork. Take care not to lose synchromesh components.	Install in correct position. The stepped side of the shift sleeve faces loose gearwheel No. 19
26	Shift fork (reverse)	1		Adjust so that play at shift sleeve is barely noticeable with reverse engaged. The synchronizing ring must rotate freely in neutral position.
27	Fixed gearwheel (5th gear)	1		Identification groove for fixed gearwheel No. 15
28	Spacer sleeve	1		
29	Loose gearwheel (5th gear)	1		

No.	Designation	Qty.	Note:	
			Removal	Installation
30	Synchronizing ring	1	Mark for reinstallation	Check for wear. Fit with the same gearwheel. Cogs face driver dogs.
31	Needle-roller cage	1	Mark for reinstallation	Fit with the same gear-wheel.

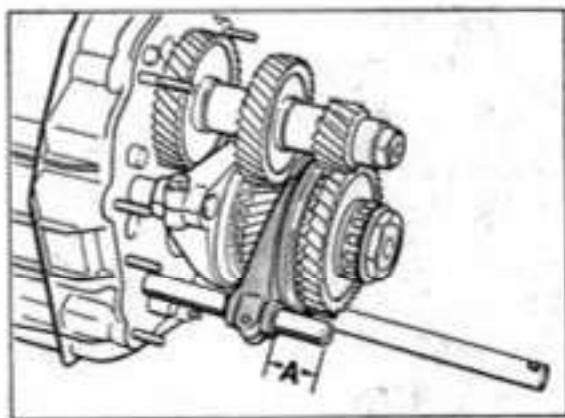
* Transmission shafts without recesses for securing the hexagon nuts have been installed since September 1995.

Self-locking hexagon nuts are used on these shafts, and these nuts must **always** be replaced in every transmission repair.

Removal and assembly notes

Removal

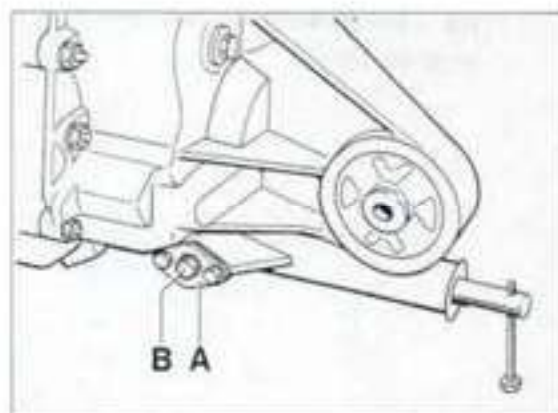
1. Remove engine guard and rear underside panel.
2. Drain transmission oil.
3. Remove crossbrace.
4. Disengage shift rod coupling from inner shift rod.
5. Support transmission on general-purpose transmission jack and unbolt transmission support from body.
6. Lower engine-transmission assembly only so far that the assembly will **not** rest on the stabilizer bar.
7. Determine installation position of reverse shift fork for reinstallation, measuring dimension "A" with depth gauge.



1791-35

Installation

1. Tighten hexagon head nuts of drive pinion and input shaft to specified torque and lock the nuts.
2. Adjust reverse shift fork to the dimension "A" determined during removal and check adjustment (refer to explosion drawing No. 26).
3. Adjust inner shift rod so that all gears can be selected freely without binding. To achieve this, undo hexagon-head bolts of clamping plate and turn eccentric bushing.



527-34

A = Clamping plate
B = Eccentric bushing

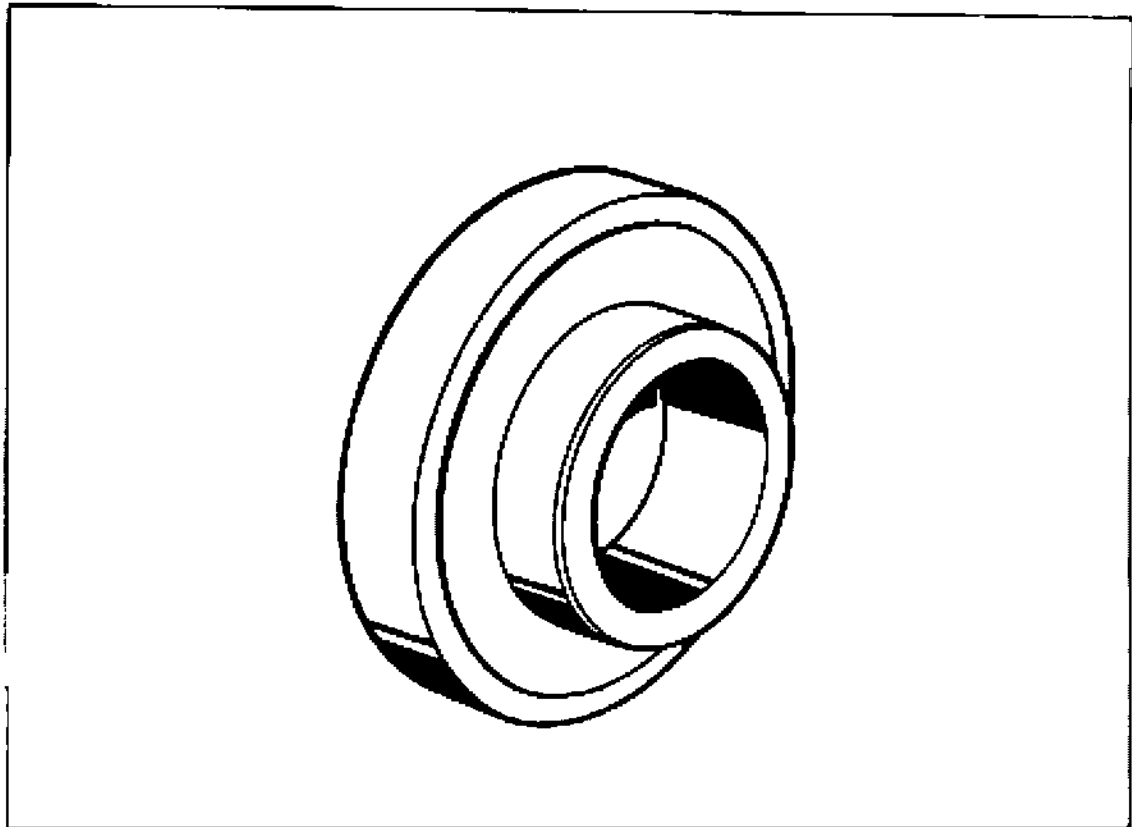
4. Screw a long M 8 bolt into the inner shift rod and shift through all gears to check operation.

5. Place engine-transmission assembly onto general-purpose transmission jack and raise into installation position. Attach transmission support to body ($M_A = 46 \text{ Nm} / 34 \text{ ftlb.}$).

Note

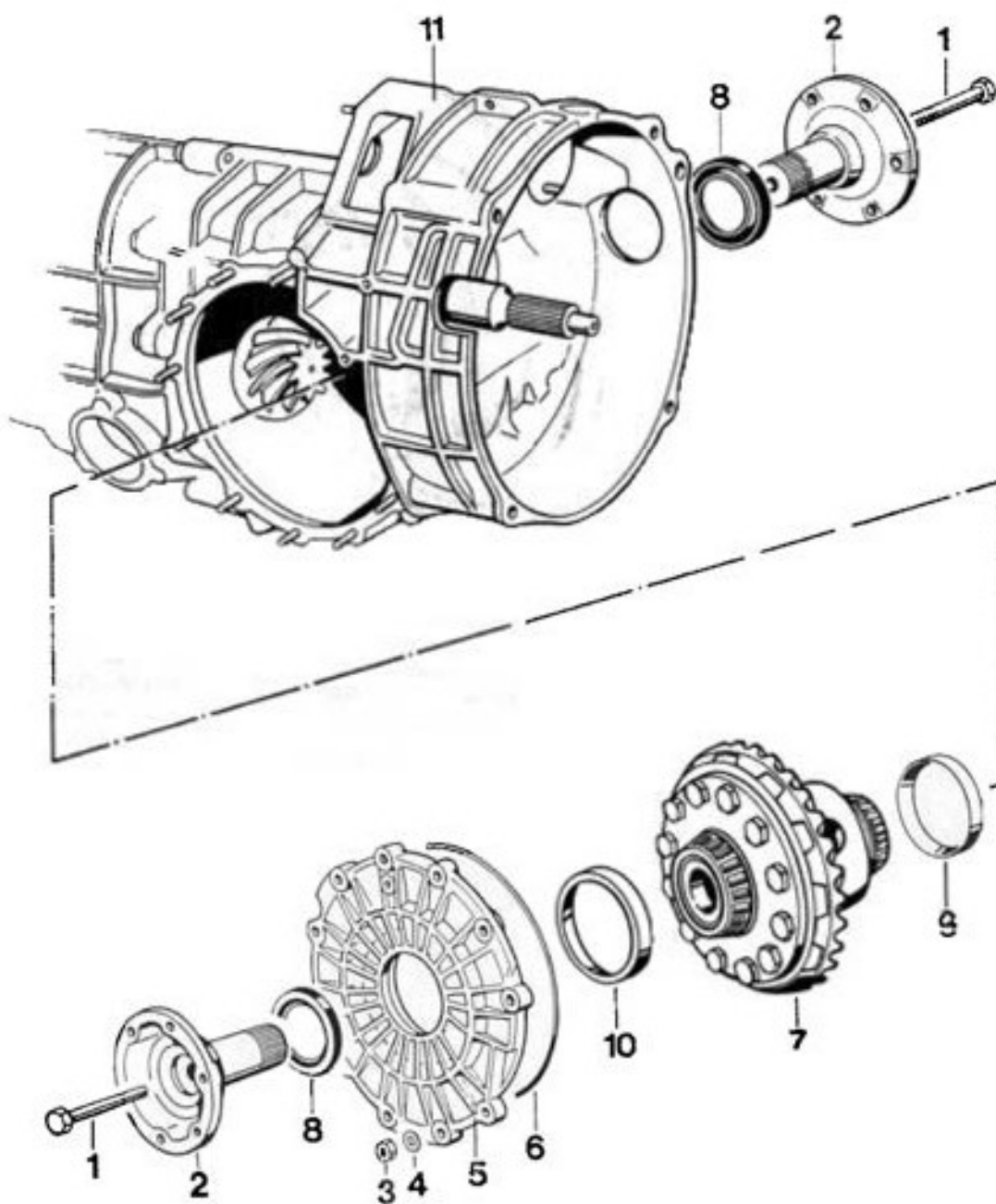
Check engine compartment rubber seal between body and engine paneling for correct seating. If required, unbolt transmission support from body once more, correct seal position and refit transmission support.

6. Fit shift rod coupling to inner shift rod.
 $M_A = 18 \text{ Nm} (13 \text{ ftlb.})$, apply Loctite 270 to lock.
7. Fill in transmission oil (refer to Vol. I, page 03-28).

39 09 19 Removing and installing differential**Tools**

No.	Designation	Special tool	Order number	Explanation
	Thrust piece	9252	000.721.925.20	

39 09 19 Removing and installing differential

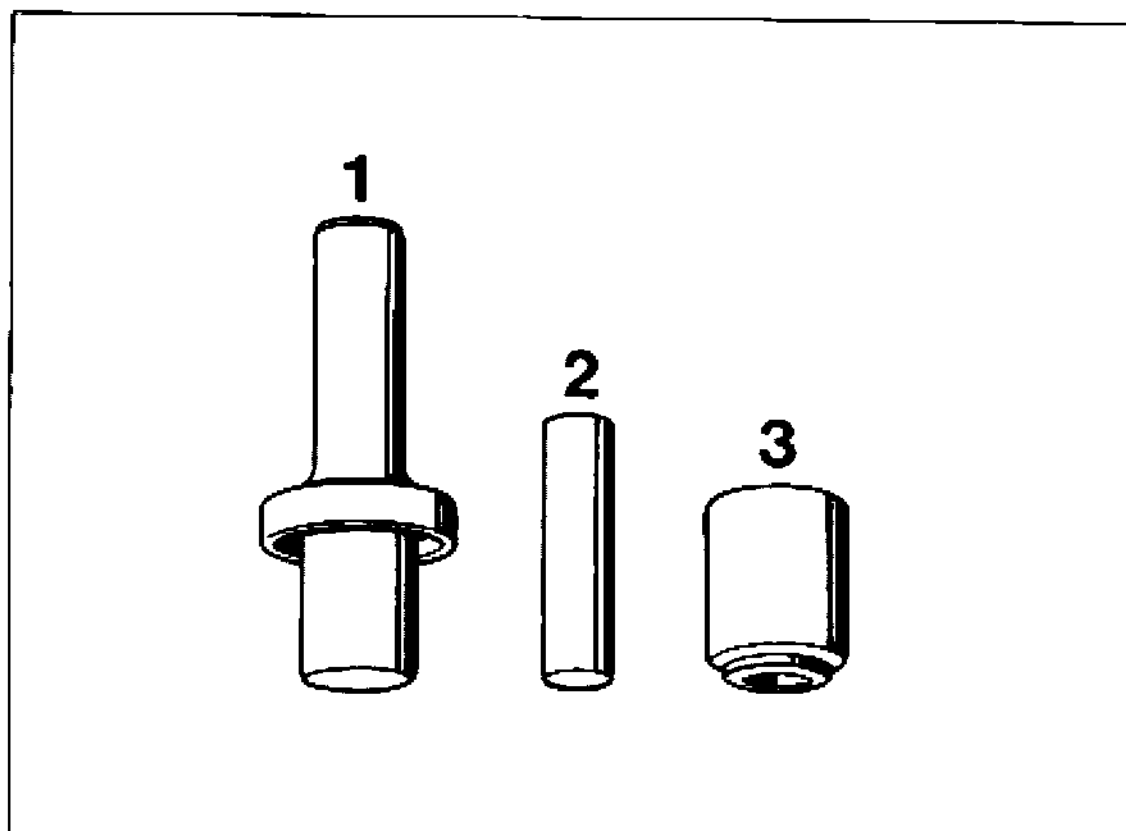


No.	Designation	Qty.	Note:	
			Removal	Installation
1	Hexagon head bolt	2		Tighten to 44 Nm (32 ftlb.)
2	Joint flange*	2		
3	Hexagon head nut	11		Tighten to 23 Nm (17 ftlb.)
4	Washer	11		
5	Cover	1		
6	O-ring	1		Replace oil lightly, do not twist
7	Differential or limited-slip differential	1		Readjust if required
8	Seal	2		Pack space between dust and sealing lips with grease (e.g. Liqui Moly Pu 53). Use thrust piece 9252 to press in until it is seated against the stop
9	Bearing outer race	1	Pull out with suitable internal puller (e.g. Kukko 21 - 8)	Press in with suitable thrust piece
10	Bearing outer race	1	Pull out with suitable internal puller (e.g. Kukko 21 - 9)	Press in with suitable thrust piece
11	Transmission housing	1		

* The halfshaft flange and seal ring can also be removed and installed with the transmission installed.

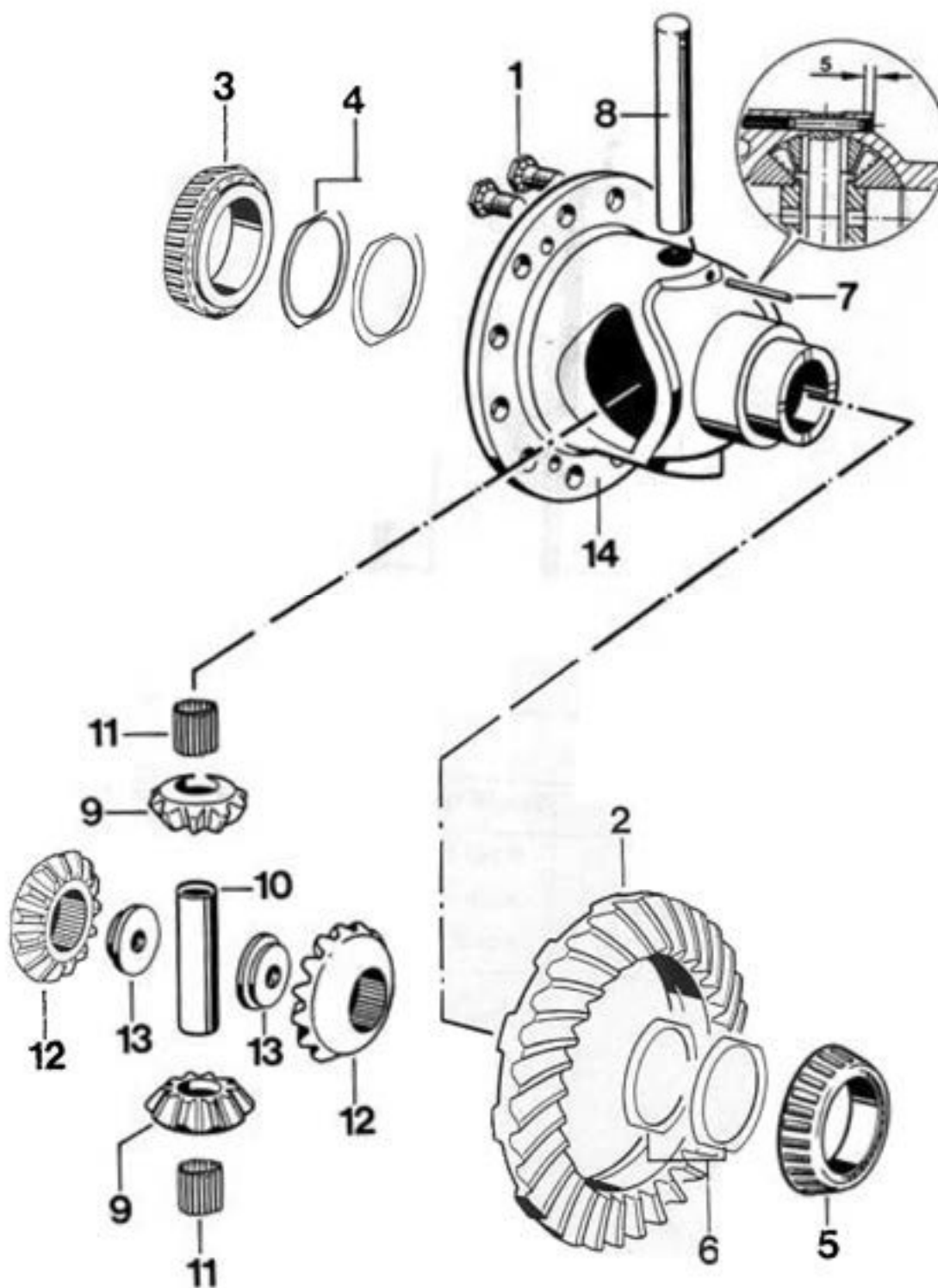
39 09 37 Dismantling and assembling differential

Tools



No.	Designation	Special tool	Order number	Explanation
1	Thrust piece	P 264 b	000.721.264.20	
2	Centering mandrel	9289	000.721.928.90	
3	Thrust piece	P 263	000.721.263.00	

39 09 37 Dismantling and assembling differential



No.	Designation	Qty.	Note:	
			Removal	Installation
1	Hexagon head bolt (with ribbed seating surface)	12		Must always be replaced. Threads must be dry and free from grease. Tighten to 200 Nm (148 ftlb.).
2	Ring gear	1		Threaded holes for ring gear bolts must be dry and free from grease. Observe matching number. Readjust if required.
3	Tapered roller bearing inner race	1	Pull off with suitable puller and P 263	Press on with P 264 b
4	Adjusting shim	X	Mark for reinstallation	Redetermine thickness if required
5	Tapered roller bearing inner race	1	Pull off with suitable puller and P 263	Press on with P 264 b
6	Adjusting shim	X	Mark for reinstallation	Redetermine thickness if required
7	Spiral pin	1		Press into correct position
8	Pin	1		
9	Bevel pinion	2		Always replace as a set (with shaft bevel gears)
10	Spacer sleeve	1		
11	Needle roller sleeve (31 individual needle rollers each)	2	Take care not lose any needle rollers	Apply stiff grease to install
12	Shaft bevel gear	2		Always replace as a set (with bevel pinion)
13	Threaded piece	2		
14	Housing	1		

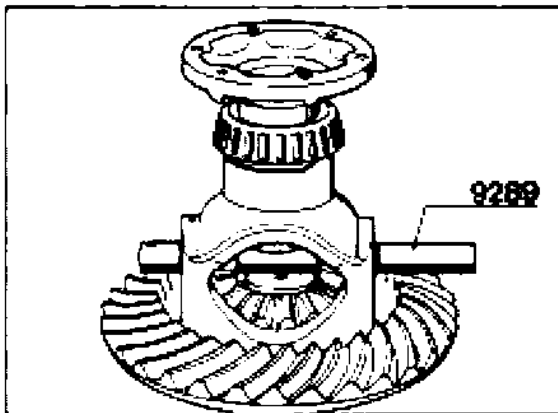
Dismantling and assembly notes

Note

The small bevel pinions are supported in two needle roller sleeves consisting of 31 individual needle rollers each.

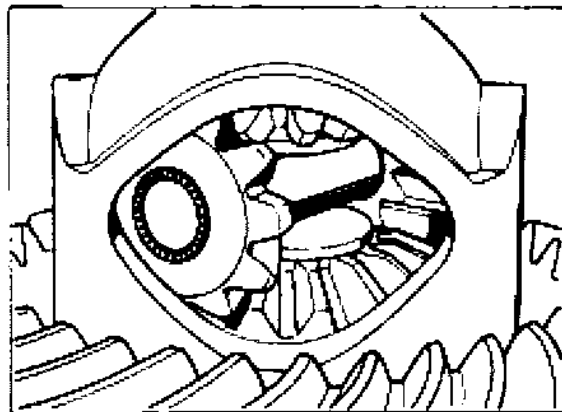
Dismantling

1. Press out pin with centering mandrel 9289.



539-39

2. Using the centering mandrel, rotate bevel pinion relative to housing openings and lift out carefully, taking care not to lose any needle rollers.

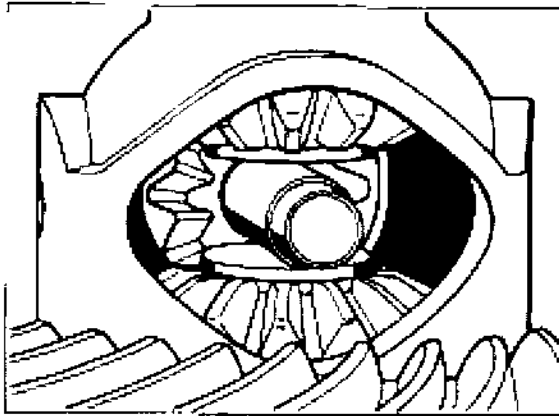


540-39

Assembly

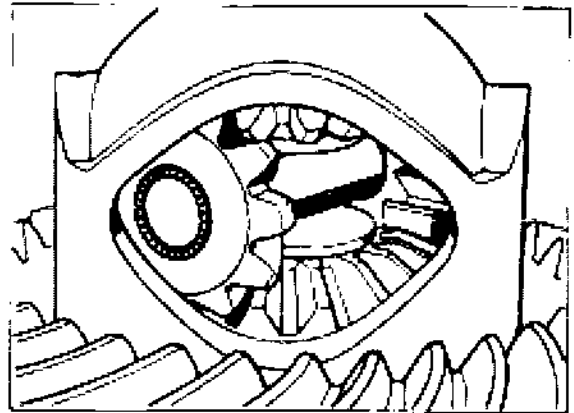
1. Insert differential gears complete with pressed-in threaded washers across the large opening in the differential housing and locate with halfshaft flanges.

2. Preassemble one bevel pinion with 31 needle rollers, centering mandrel 9289 as well as the spacer sleeve and insert across opening in the housing.



541-39

4. Rotate bevel pinion with centering mandrel until the bores in the housing line up with each other.



540-39

3. Complete second bevel pinion with 31 needle rollers and push carefully onto centering mandrel, taking care not to lose any needle rollers.

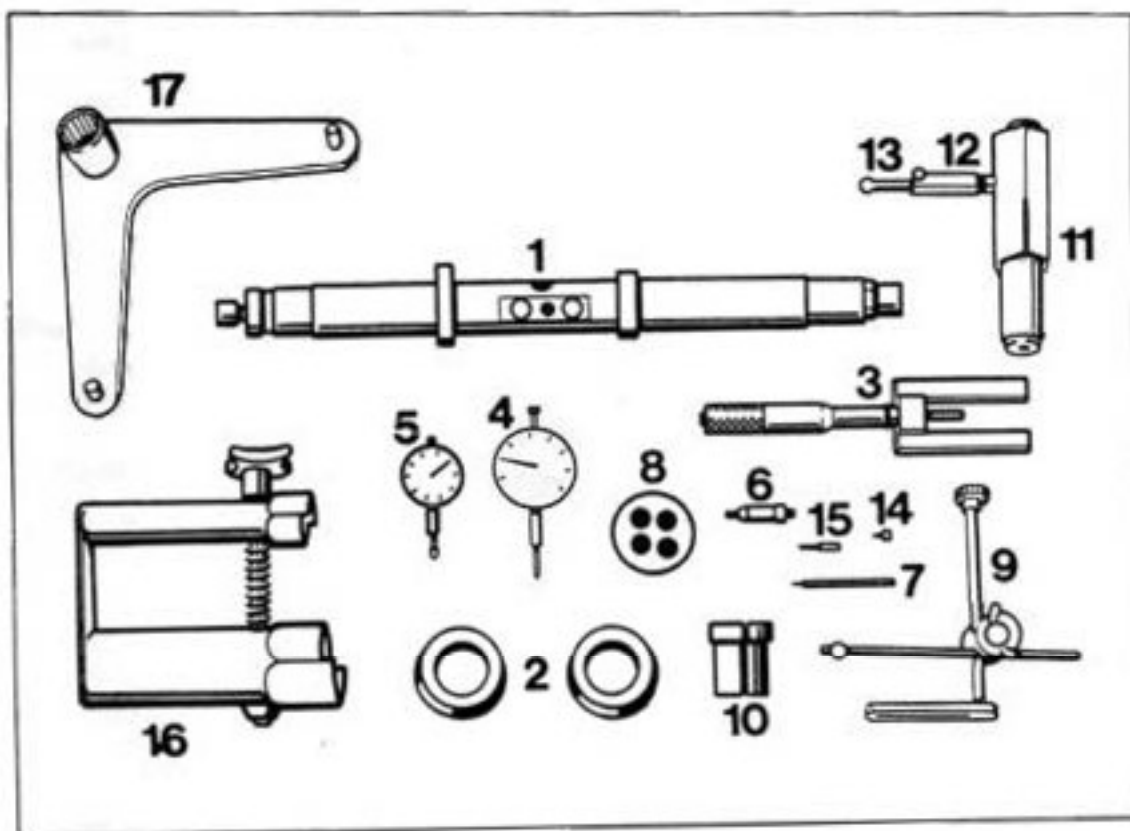
Note

To facilitate assembly, apply stiff grease to place needle rollers into position.

5. Press differential pin into correct position, take off centering mandrel and lock pin according to specifications.

39 08 15 Adjusting drive set

Tools



39 08 15 Adjusting drive set

Tools

No.	Designation	Special tool	Order number	Explanation
1	Measuring mandrel	VW 385/1		
2	Centering disks	9109	000.721.910.90	
3	Master gauge	VW 385/30		
4	Dial gauge	-		Standard
5	Dial gauge	-		Standard, measuring range 3 mm
6	Gauge plunger	VW 385/14		
7	Dial gauge extension	VW 385/56		30 mm long
8	Gauge block plate	9281	000.721.928.10	
9	Dial gauge bracket	VW 387		
10	Clamping sleeve	9145	000.721.914.50	
11	Adjusting device	VW 521/4		
12	Measuring lever	VW 388		
13	Gauge plunger	VW 388		
14	Dial gauge extension	VW 382/10		6.0 mm long
15	Dial gauge extension	VW 385/15		9.3 mm long
16	Clamping device	9562	000.721.956.20	
17	Holder	9253	000.721.925.30	

Practical procedure when readjusting the drive set

If it is necessary to adjust drive pinion and ring gear, follow the below sequence to ensure an efficient working procedure:

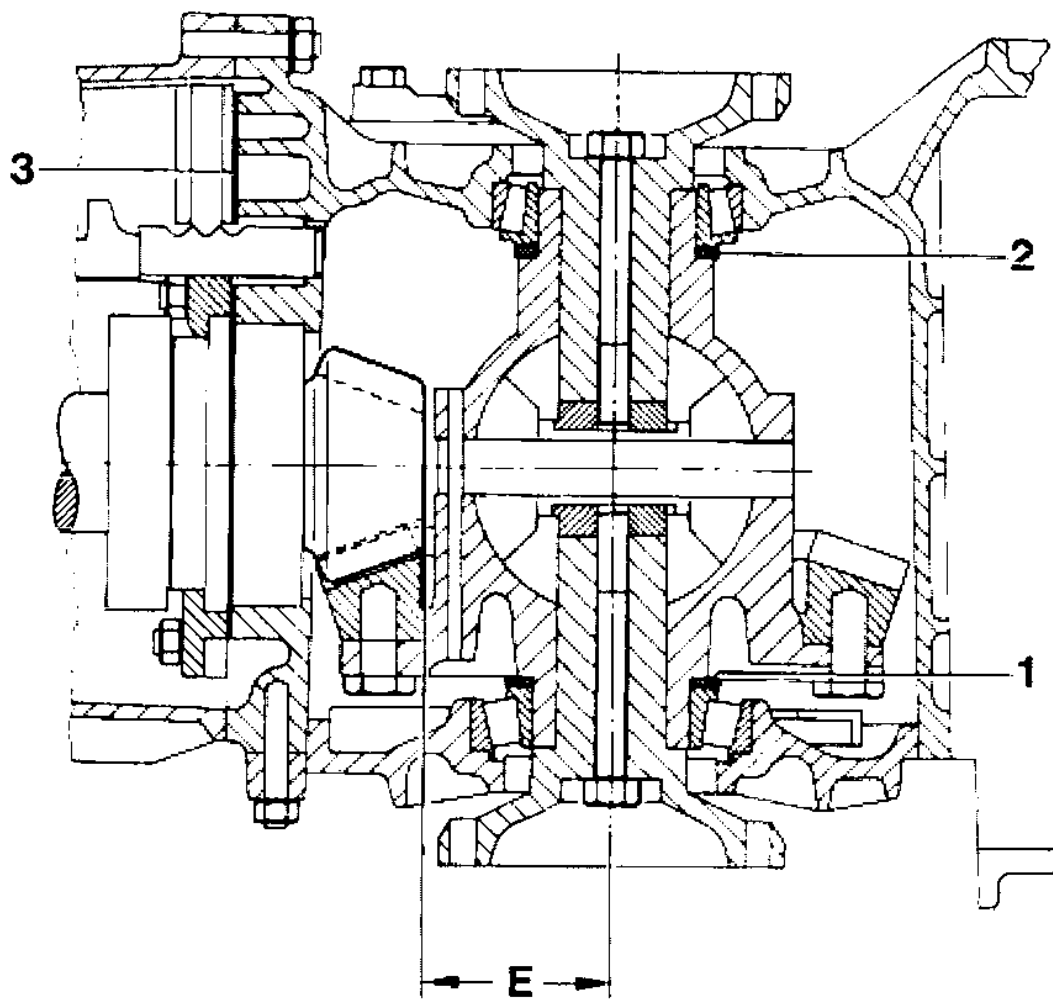
1. Determine the total thickness of shims "Stot" (S_1 plus S_2) for the specified preload on the tapered roller bearings/differential.
2. Determine the thickness of shim " S_3 ".
3. Split the total shim thickness "Stot" into S_1 and S_2 so that the specified circumferential backlash is present between ring gear and drive pinion.

The aim of this adjustment is to restore the smoothest running position which has been achieved on test equipment in the production line.

To achieve correct results, greatest possible cleanliness for all assembly work and measuring procedures is essential.

When assembling the final drive assembly, it is only necessary to readjust drive pinion and ring gear or drive set if components have been replaced which have a direct influence on the adjustment. Refer to the following table to avoid unnecessary adjustment procedures.

Adjust:	Ring gear ($S_1 + S_2$)	Drive pinion (S_3)
Replaced component		
Transmission case	x	x
Lateral transmission cover	x	
Large cylindrical roller bearing and four-point bearing for drive pinion	x	x
Drive set	x	x
Differential housing	x	
Tapered roller bearing for differential	x	



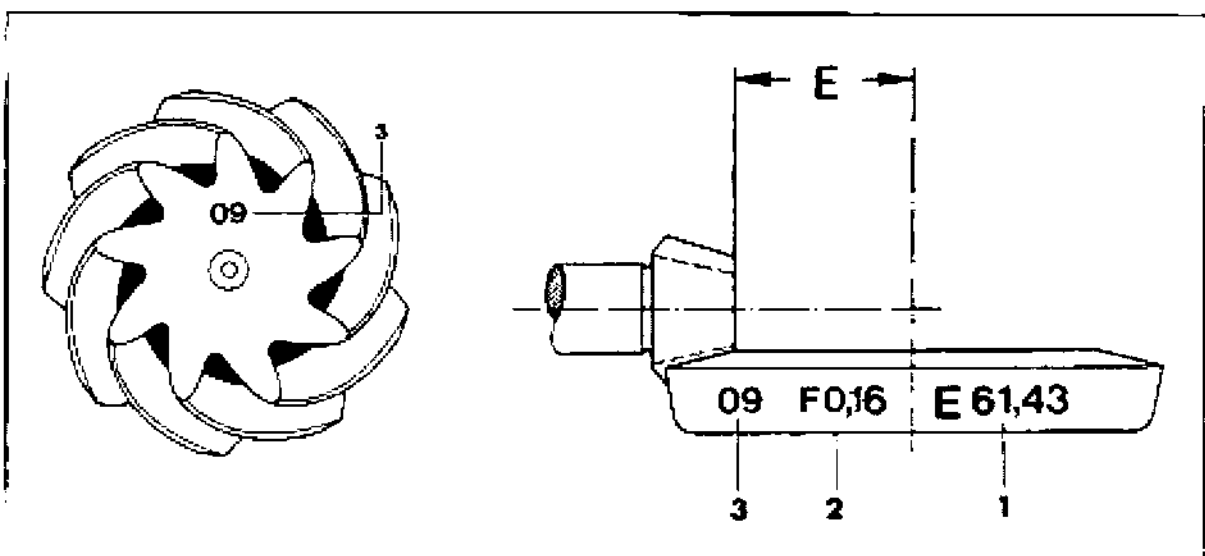
- 1 – Shim S_1
- 2 – Shim S_2
- 3 – Adjusting shim S_3
- E – Setting value

Correct results may only be achieved if assembly work and measuring procedures are carried out carefully and with maximum cleanliness.

Adjusting drive set

General

The setting of drive pinion and ring gear is a determining factor for the service life and smooth running of the rear-axle drive. Drive pinions and ring gears that have been checked for good tooth contact pattern and low noise in both directions of rotation on special test equipment are therefore matched during production. The position at which smoothest running can be achieved is determined by shifting the drive pinion axially, and embossed on the ring gear as setting value "E".

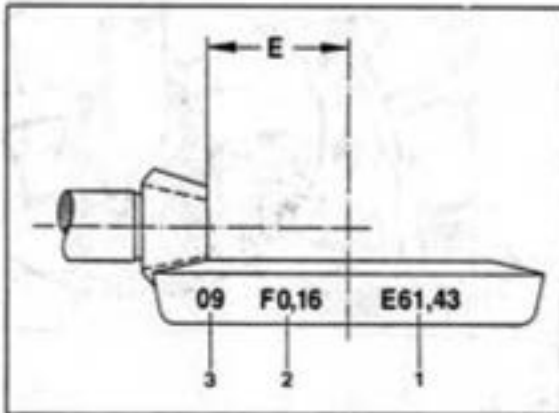


- 1 = setting "E" (e.g. 61.43 mm)
- 2 = circumferential backlash "F"
(e.g. 0.16 mm)
- 3 = matching number

39 08 15 Adjusting drive pinion

Note

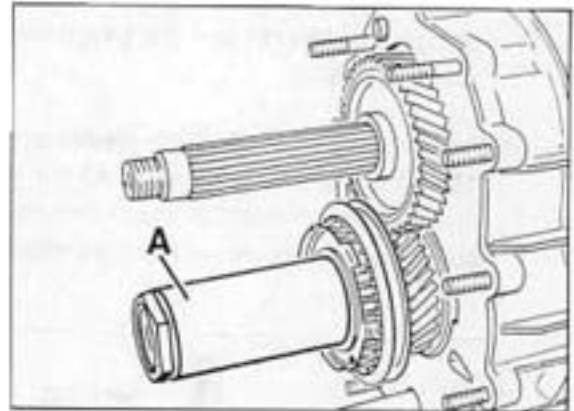
The setting value "E" is indicated on the ring gear.



217-39

1 = Setting value "E"

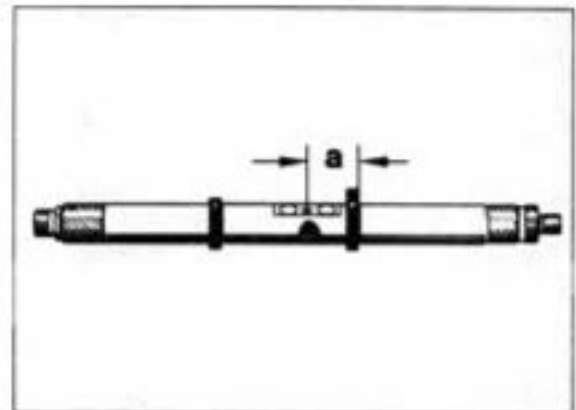
1. Install complete gear set without "S3" shims and tighten all tensioning plate hexagon-head nuts to **23 Nm** (17 ftlb.).
2. Fit gear housing and locate with three nuts.
3. Install and engage sixth gear.
4. Block input shaft with Special Tool 9253 and tighten drive pinion collar nut to **300 Nm** (221 ftlb.).



2018-39

A = Suitable pipe section

5. Set adjusting ring of measuring mandrel VW 385/1 to dimension "a".

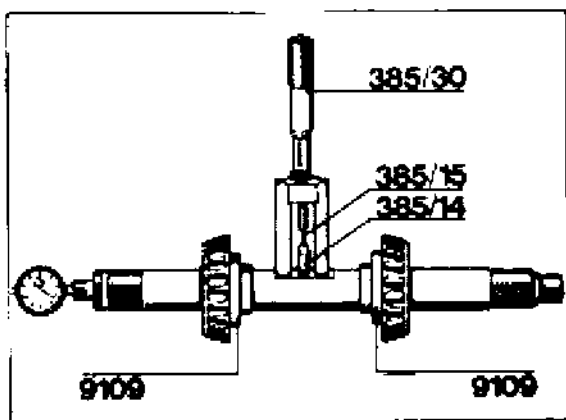


223-39

a = approx. 65 mm

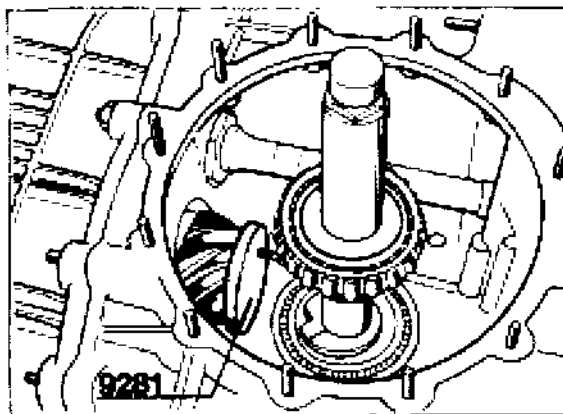
6. Assemble measuring mandrel with centering disks **9109**, tapered roller bearing, plunger **VW 385 /14** and dial gauge extension **VW 385/15**.

7. Set measuring mandrel with master gauge **385/30** to the setting value (61.43 mm in the example). Set dial gauge (3 mm measuring range) to zero with 1 mm preload.



543-39

8. Put gauge block plate **9281** on drive pinion head and insert measuring mandrel into transmission case. Dial gauge extension is located in the area of the gauge block plate.



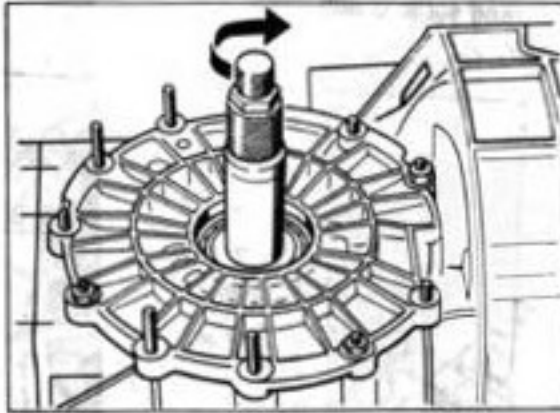
544-39

9. Fit lateral transmission cover without O-ring and tighten crosswise with 4 nuts.

Note

Do not use a hammer when fitting the lateral transmission cover (the gauge block plate held by magnets might fall off). Fit cover in installation position only by tightening the nuts uniformly.

10. Pull second centering disk with spindle towards the outside until the measuring mandrel can just be turned by hand.



545-39

11. Turn measuring mandrel carefully until the dial gauge extension is vertical to the face of the drive pinion head. At this point, the pointer of the dial gauge reaches maximum deflection (reverse point) and the dial gauge must be read.

Note

The measured value always deviates from the set dimension clockwise (the smaller pointer on the dial gauge is between 1 and 2), i.e. if the dial gauge is set with a preload of 1 mm, the value deviating from 1 is taken as shim thickness "S₃".

Example:

If the small pointer on the dial gauge is between 1 and 2 and the large pointer indicates 0.37 mm, then 0.37 mm is the shim thickness (with 1 mm gauge preload) to be inserted. Always round up or down to the nearest 0.05 mm (e.g. 0.37 mm to 0.35 mm).

12. After inserting the necessary shims, check the setting value "E" again. A deviation of ± 0.03 mm is permissible.

Adjusting ring gear

Determine total shim thickness "S tot."
($S_1 + S_2$).

The ring gear must be adjusted, if the:

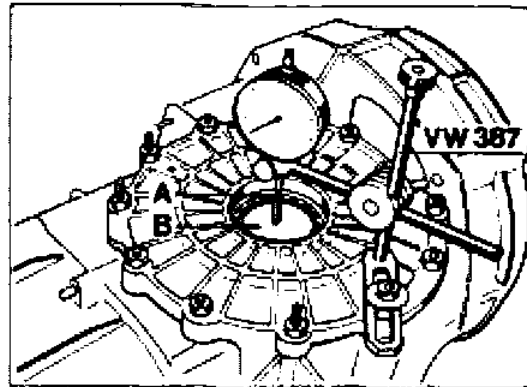
transmission case,
lateral transmission cover,
tapered roller bearing for differential,
differential housing or drive set
have been replaced.

Note

The drive pinion must be removed to determine the preload of the differential tapered roller bearings.

1. Make sure that the bearing outer races of the tapered roller bearings are well seated in the transmission case or lateral transmission cover, respectively.
2. Fit one spacer ring (2.5 mm thick) on the ring gear side and on the opposite side of the differential to be used.
3. Insert differential into transmission case and rotate several times.
4. Fit lateral transmission cover without seal and tighten all hexagon-head nuts to **23 Nm** (17 ftlb.).
5. Put gauge block plate **VW 385/17** on the collar of the differential.

6. Fasten universal dial gauge holder **VW 387** with dial gauge and extension to the case and set to 0 with 2 mm preload.



546-39

A = Dial gauge extension (approx. 30 to 40 mm long)

B = Gauge block plate VW 385/17

7. Move differential up and down. Read off backlash on the dial gauge and note.

Note

Do not turn differential while measuring backlash as this will give an incorrect reading.

8. Calculate "S tot.".

"S tot." = Fitted shim thickness
+ Measured value
+ Pressure fit of tapered roller bearing

Example

Thickness of shims inserted	5.00 mm
Measured value	0.75 mm
Pressing (constant value)	0.40 mm
"S tot."	<u>6.15 mm</u>

9. Remove differential, pull off both tapered roller bearings and split calculated shim thickness "S tot." as follows.
Select spacer S₁ 0.70 mm thinner and S₂ 0.70 mm thicker as a starting point for subsequent adjustment of the backlash.

Example

Total shim thickness of spacers
S₁ + S₂ = 6.15 mm

Thickness of spacer S₁

$$\begin{array}{r} 6.15 \text{ mm} \\ \hline 2 \\ \hline \end{array} = 3.075 \text{ mm}$$

$$\begin{array}{r} - 0.700 \text{ mm} \\ \hline 2.375 \text{ mm} \end{array}$$

Thickness of spacer S₂

$$\begin{array}{r} 6.15 \text{ mm} \\ \hline 2 \\ \hline \end{array} = 3.075 \text{ mm}$$

$$\begin{array}{r} + 0.700 \text{ mm} \\ \hline 3.775 \text{ mm} \end{array}$$

Note

Spacers are available in thicknesses of 1.6 to 3.1 mm in increments of 0.10 mm.

By using a 0.25 mm shim, the shim thicknesses may be graduated in increments of 0.05 mm.

The calculated shim thicknesses must be rounded up or down for plausible dimensions that will not alter the total thickness S₁ and S₂.

Example:

Calculated thicknesses

$$S_1 + S_2 = 2.375 + 3.775 = 6.15 \text{ mm}$$

Rounded thicknesses

$$S_1 + S_2 = 2.35 + 3.80 = 6.15 \text{ mm}$$

Measure shims with a micrometer in several places. Permissible deviation 0.02 mm. Also check shims for burrs and damage.

Adjusting circumferential backlash

Note

The backlash to be set is embossed on the ring gear.

1. Mount gear set using shims "S₃" determined while adjusting the drive pinion.

Note

Make sure that the collar nut of the drive pinion is tightened to **300 Nm** (221 ftlb.) before measuring backlash.

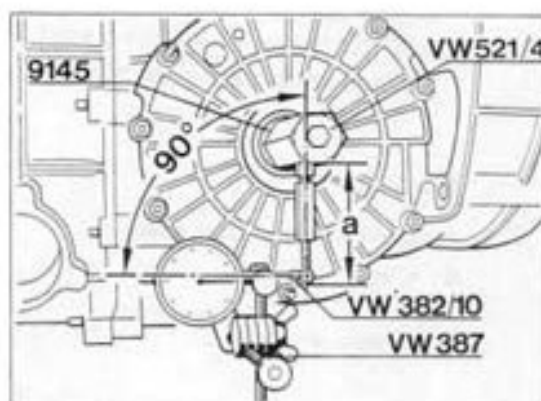
2. Insert differential with tapered roller bearing and shims (S₁ + S₂) into the housing.
3. Fit lateral transmission cover and tighten all hexagon-head nuts to **23 Nm** (17 ftlb.).

Note

Always make sure that there is a certain amount of backlash when tightening the nuts. Never allow the drive pinion to seize.

4. Assemble measuring lever **VW 388** and adjusting device **VW 521/4** and adjust lever length to 80 mm with the plunger. Refer to dimension "a" in the picture.
5. Insert adjusting device with clamping sleeve (Special Tool 9145) into the differential and clamp firmly.
6. Rotate differential in both directions several times to settle the tapered roller bearings.

7. Fit universal dial gauge holder with flat extension in such a way as to produce a right angle between dial gauge axis and lever.



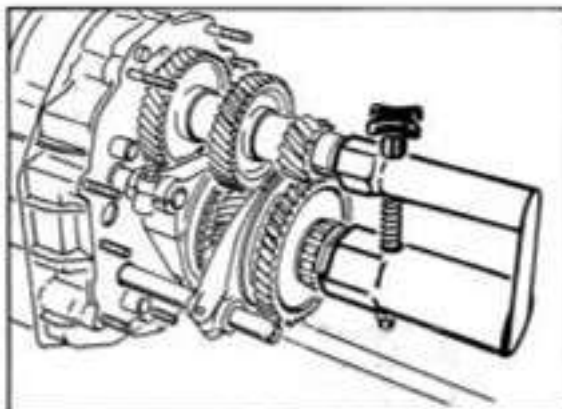
545-39

Dimension "a" = approx. 80 mm

8. Turn ring gear carefully at the clamping screw of the adjusting device as far as the stop and set the dial gauge to zero. Turn back ring gear and read off circumferential backlash. Note down the reading.

Note

When carrying out measurements, the drive pinion must be blocked with Special Tool 9562.



1710-38

9. After turning the ring gear a further 90°, repeat measuring procedures three times. The measured values must not deviate from one another by more than 0.03 mm.

Note

The backlash to be adjusted is embossed on the ring gear. A deviation of ± 0.03 mm is permissible

10. If the required backlash cannot be obtained, replace spacers ($S_1 + S_2$) again. The total shim thickness ("S tot.") must not be altered, however.

Repair Manual

911 Carrera 4 **(993)**

Volume II: Transmission Manual

II Manual Transmission (911 Carrera 4)

3 Transmission

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34 52 37	Dismantling and assembling gear housing	34 - 219
34 37 37	Dismantling and assembling transmission housing	34 - 223

35 Manual Transmission - Gears, Shafts, Internal Controls

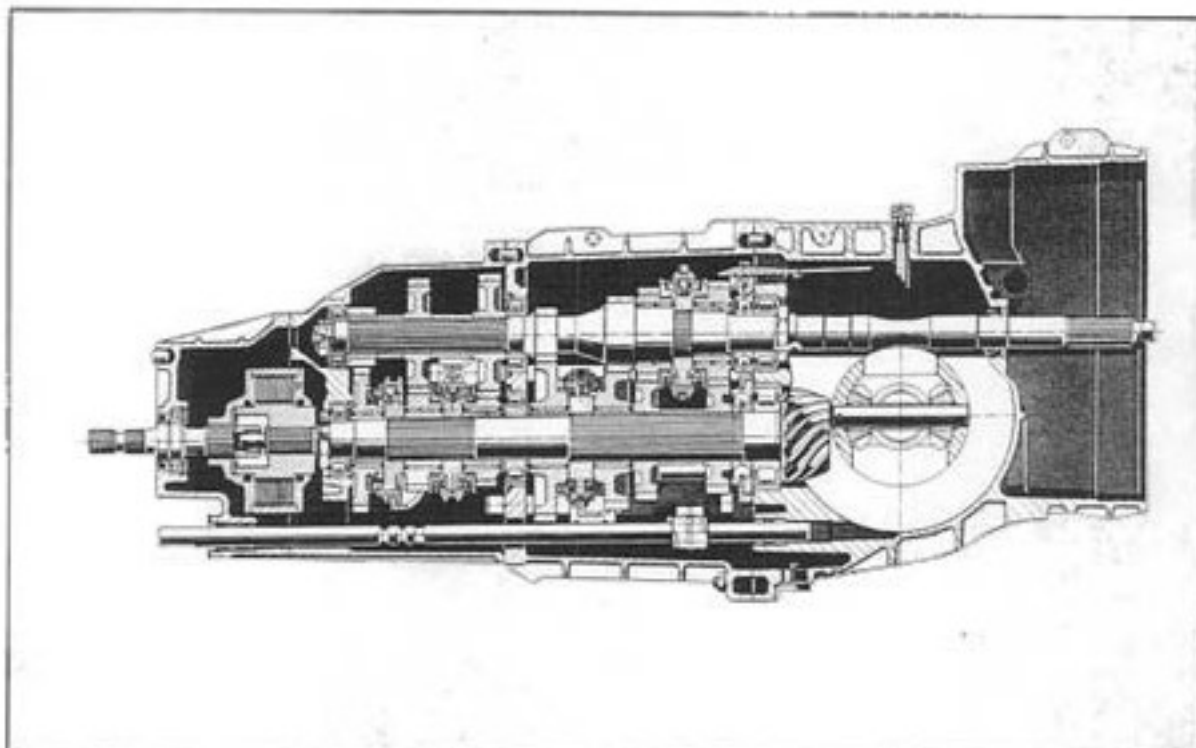
35 50 19	Removing and installing input shaft oil seal	35 - 201
35 19 37	Dismantling and assembling tensioning plate	35 - 203
35 40 19	Removing and installing input shaft	35 - 207
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39 Final Drive, Differential, Differential Lock

39 66 37	Dismantling and assembling transfer casing	39 - 201
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3 Technical data

6 speed manual transmission G 64

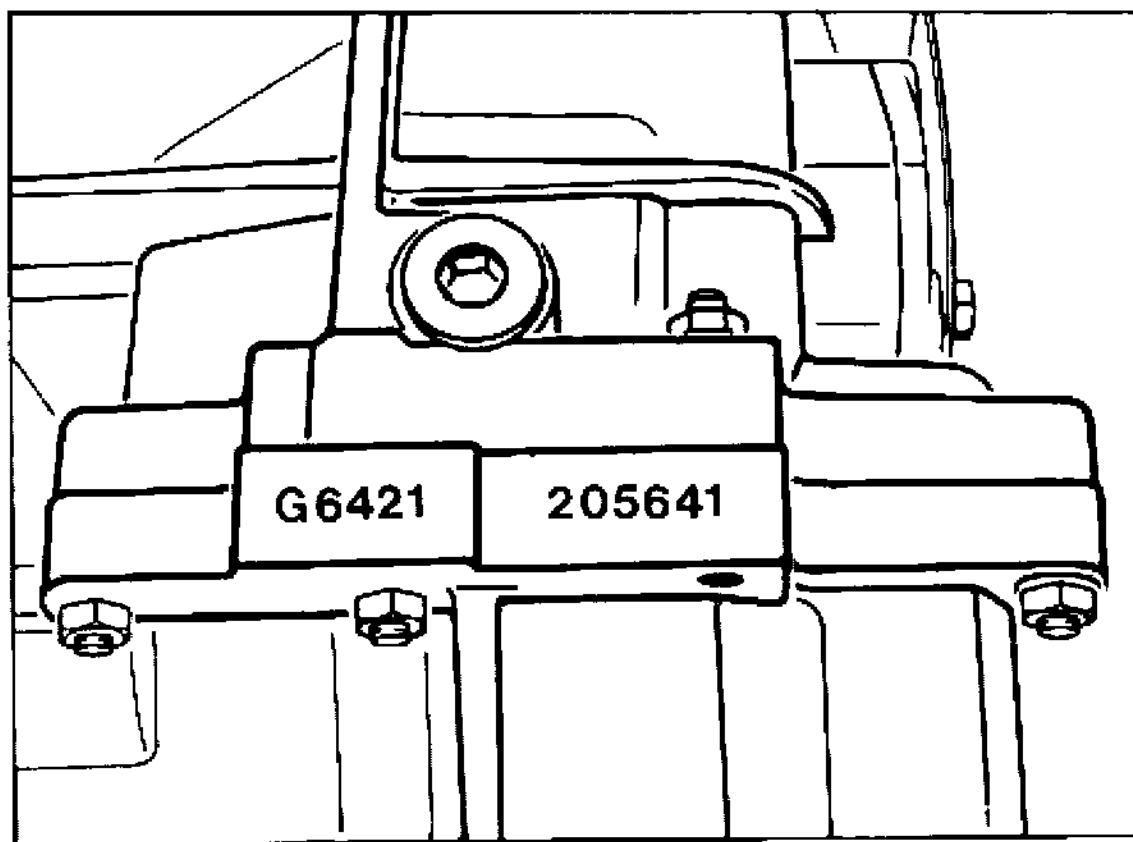
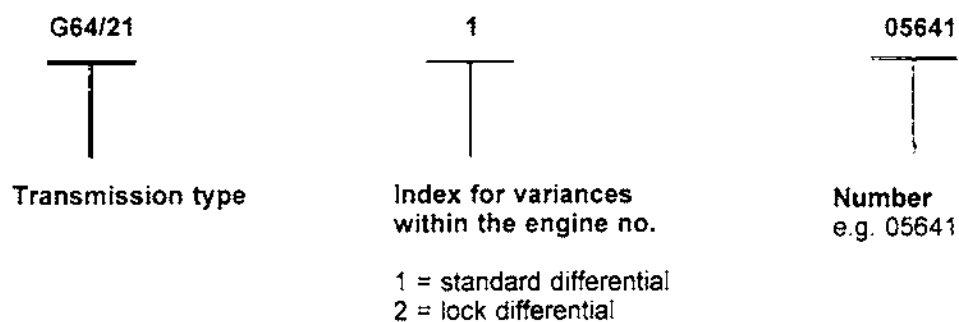


1426-3

Type	Equipment	installed in:	Model year
G 64/21	6 speed	911 Carrera 4 worldwide except USA, CH, A	'95 / '96
G 64/20	6 speed	911 Carrera 4 USA, CH, A	'95 / '96
		911 Carrera 4 worldwide	'97

3 Technical Data

Structure of transmission numbers



3 Technical Data (Manual transmission)

General Data		Manual transmission G 64 / 21 / 20	
Transmission ratios	G64/21 $Z_1 : Z_2 = Z_2 : Z_1$	G64/20 $Z_1 : Z_2 = Z_2 : Z_1$	
1st gear	11 : 42 = 3.818	11 : 42 = 3.818	
2nd gear	20 : 43 = 2.150	21 : 43 = 2.048	
3rd gear	25 : 39 = 1.560	27 : 38 = 1.407	
4th gear	33 : 41 = 1.242	34 : 38 = 1.118	
5th gear	41 : 42 = 1.024	42 : 39 = 0.928	
6th gear	39 : 32 = 0.820	40 : 31 = 0.775	
Reverse	14 : 40 = 2.857	14 : 40 = 2.857	
Final drive	Hypoid bevel-gear drive with 16 mm offset		
Final drive ratio	9 : 31 = 3.444	9 : 31 = 3.444	
Capacity	3.8 liters multigrade transmission oil SAE 75 W 90 of API classification GL 5 (or MIL-L 2105 B)		

3 Technical Data

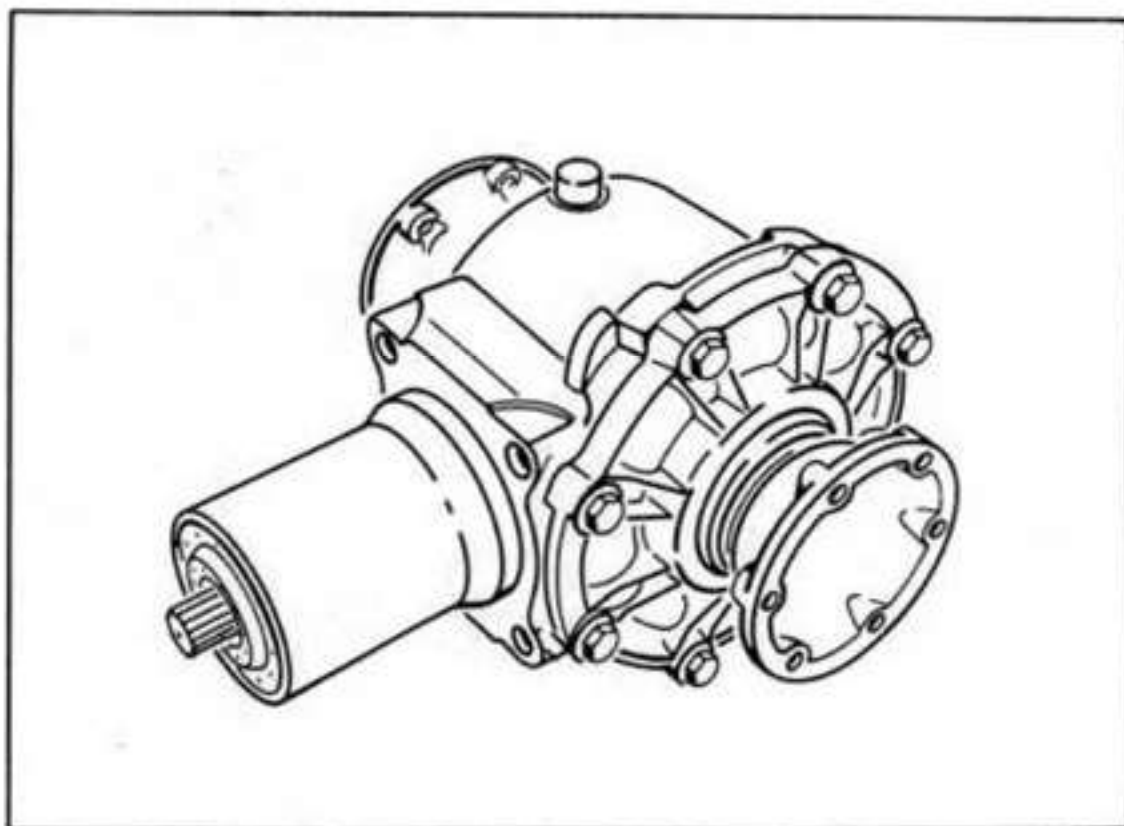
Torque specifications for manual transmission, front axle final drive, transmission suspension and central tube

Location	Thread	Tightening torque Nm (ftlb)
Oil drain and oil filler plugs	M 22 x 1,5	30
Hexagon head nuts at: Front and side transmission cover, gear and transmission housings. Tensioning plate	M 8	23
Clamping plate to intermediate case	M 6	10
Hexagon head nut / input shaft	M 22 x 1,5	200
Hexagon head nut / input shaft	M 30 x 1,5	250
Hexagon head nut / output shaft	M 30 x 1,25	300
Hexagon head nut / shift fork	M 8	23
Reverse light switch to gear housing	M 18 x 1,5	35
Guide tube to transmission housing	M 6	10
Selector gate to tensioning plate	M 6	10
Collar nut / return gear II	M 8	23
Vent to transmission housing	M 14 x 1,5	35

Location	Thread	Tightening torque Nm (ftlb)
Hexagon head bolt / joint flange	M 10	44 (32)
Oil pan to tension plate	M 5 (microencapsulated)	6 (4)
Ring gear to differential housing (hexagon head bolt with ribbed seating surface)	M 12 x 1,25	200 (147)
Front axle final drive		
Oil filler plug	M 22	22 (16)
Oil drain plug	M 22	30 (22)
Final drive mount at rear axle		
Crossmember to body	M 10 x 70	46 (34)
Mount to crossmember (lock nut)	M 10	46 (34)
Mount to transmission (lock nut)	M 10	46 (34)
Final drive mount at front axle		
Front axle final drive mount to front axle crossmember	M 8	23 (17)
Front axle final drive mount to central tube	M 12	85 (63)
Central tube		
Double clamp to central shaft	M 8	35...40 (26...29)
Manual transmission to central tube	M 12	85 (63)
Front axle final drive to central tube	M 10	46 (34)

3 Technical Data

Front axle final drive

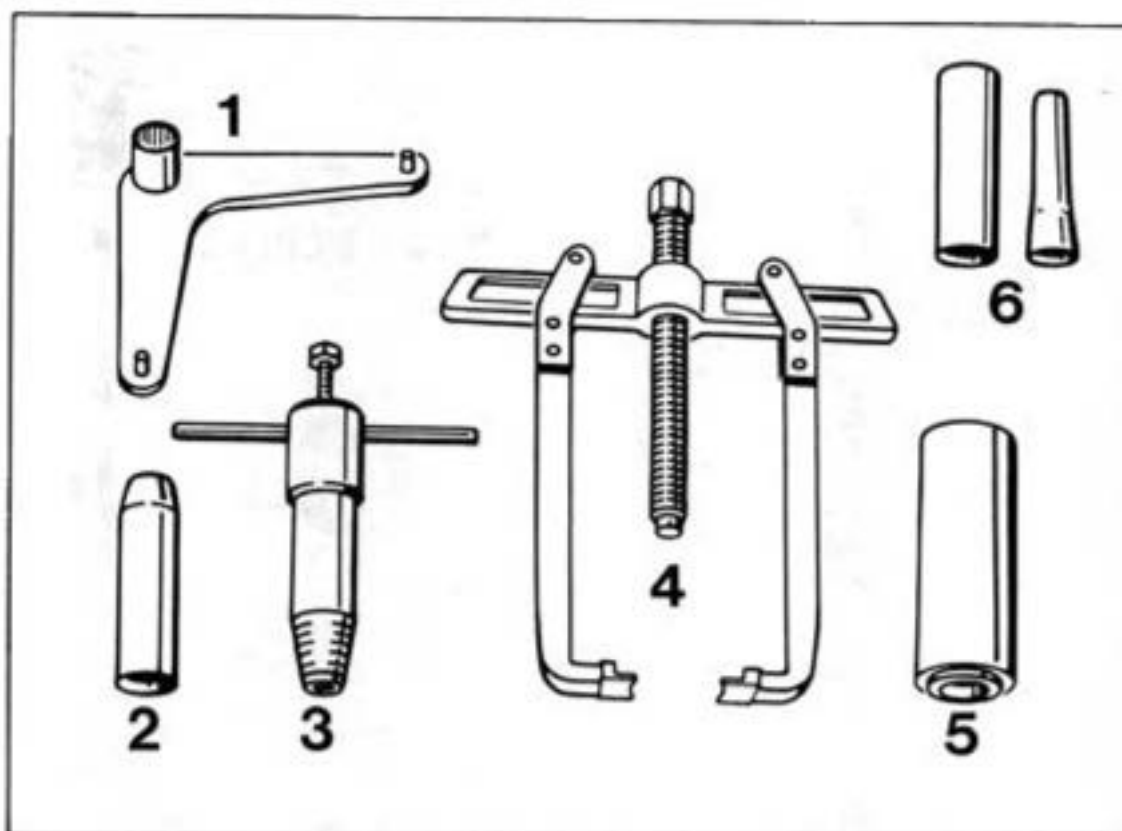


Type	Code letter	Equipment	Installed in:	Model Year
Z64/20			911 Carrera 4 worldwide	'95

General data	Front axle final drive Z64/20
Final drive ratio ($Z_2 : Z_1 = i$)	31 : 9 = 3.444
Final drive	Bevel drive without hypoid offset
Capacity	Approx. 0.6 liters hypoid oil 75 W 90 of API classification GL 5 (or MIL-L 2105 B)

34 35 37 Dismantling and assembling transmission

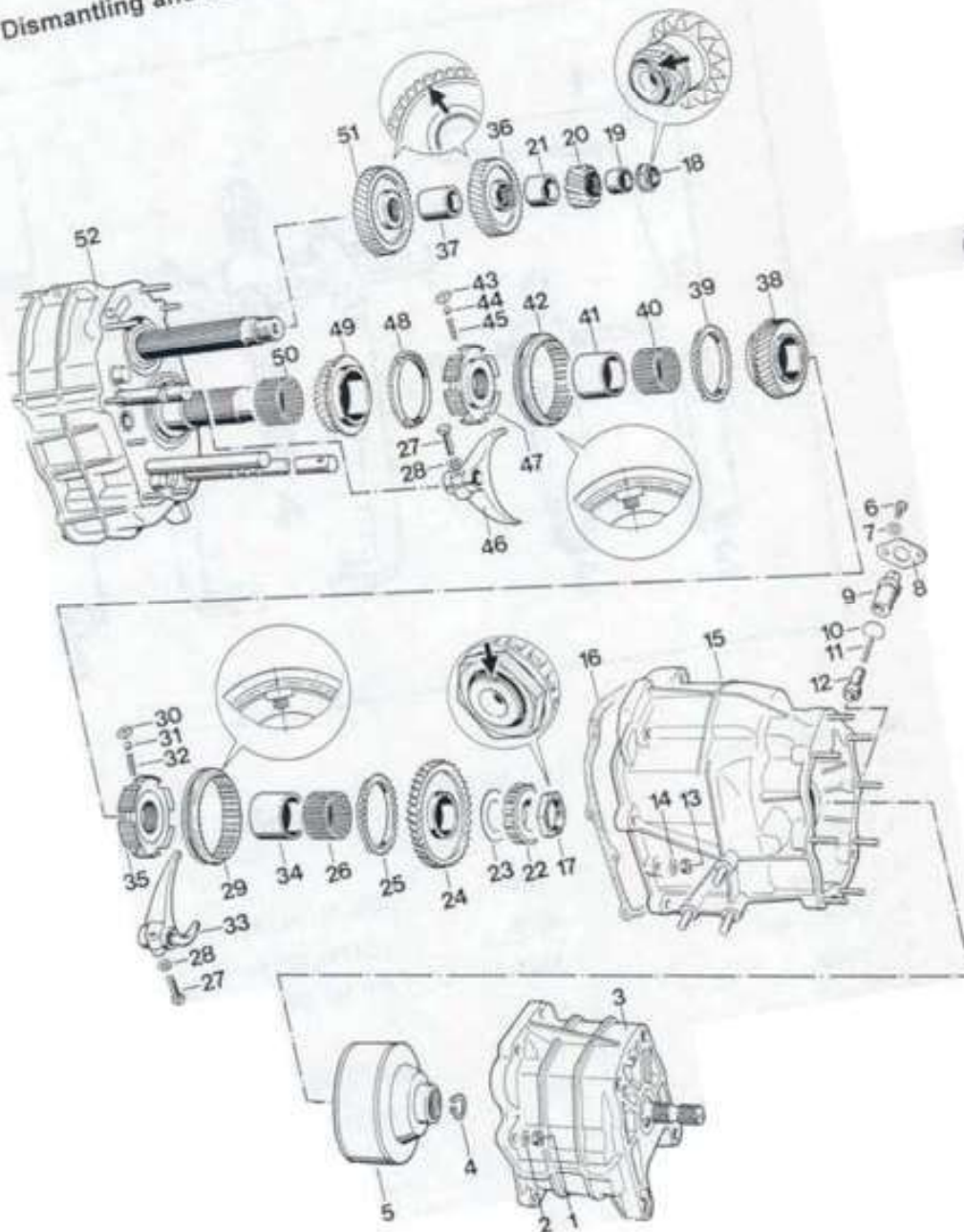
Tools



2009-34

No.	Designation	Special tool	Order number	Explanation
1	Bracket	9253	000.721.925.30	with arms of puller 9284
2	Sleeve	9255	000.721.925.50	
3	Pulling tool	9251	000.721.925.10	
4	Puller	-	-	
5	Thrust piece	9256	000.721.925.60	
6	Assembly tool	9547	000.721.954.70	

34 35 37 Dismantling and assembling transmission



No.	Designation	Qty.	Removal	Note:
				Installation
1	Hexagon head nut	8		MA = 23 Nm (17 ftlb.)
2	Washer	8		
3	Transfer casing	1		Seal between transfer casing and intermediate casing is made with Loctite 573
4	Snap ring	1	Remove with suitable pointed pliers	Install with Special Tool 9547
5	Viscous clutch	1		
6	Hexagon head bolt	2		Tighten to 10 Nm (7 ftlb.)
7	Washer	2		
8	Tensioning plate	1		
9	Eccentric bushing	1		Adjust
10	Seal	1		Replace
11	Thrust spring	1		
12	Locking bushing	1		
13	Hexagon head nut	10		Tighten to 23 Nm (17 ftlb.)
14	Washer	10		
15	Intermediate casing	1		
16	Gasket	1		Replace
17	Hexagon nut*	1	Lock input shaft with Special Tool 9253 and engage 6th gear	Tighten to 300 Nm (221 ftlb.). Upset flange to lock
18	Hexagon nut*	1	Lock input shaft with Special Tool 9253 and engage 6th gear	Tighten to 200 Nm (147 ftlb.). Upset flange to lock
19	Bearing inner race	1	Pull off across fixed gear No. 20	Heat to approx. 120 °C
20	Fixed gear (Reverse)	1		
21	Spacer sleeve	1		
22	Cylindrical roller bearing	1	Pull off across loose gear No. 24	Heat to approx. 120 °C
23	Thrust washer	1		

No.	Designation	Qty.	Note:	
			Removal	Installation
24	Loose gear (Reverse)	1		
25	Synchronizer ring	1	Mark for reinstallation	Check for wear. Fit with the same gear-wheel (Cogs must face driver dogs)
26	Needle bearing cage	1	Mark for reinstallation	Fit with the same gear-wheel
27	Hexagon head bolt	2		Tighten to 23 Nm (17 ftlb.)
28	Washer	2		
29	Selector sleeve	1	Lift off along with shift fork No. 33. Take care not to lose synchromesh parts.	Center the centerpunch marks relative to the driver dogs. Offset side faces loose gear No. 24
30	Driver dog	3		Domed end faces selector sleeve
31	Ball	3		
32	Spring	3		
33	Shift fork (Reverse)	1		Adjust so that play is barely felt at the selector sleeve when reverse is engaged. When neutral is engaged, the synchronizing ring must be able to rotate freely
34	Inner race**	1	Mark for reinstallation. Pull off across guide sleeve No. 35	Heat to approx. 120 °C
35	Guide sleeve (with snap ring)	1		Open end of snap ring must not be located in the driver dog area. Install in correct position complete with selector sleeve and shift fork (Snap ring faces loose gearwheel No. 38)

No.	Designation	Qty.	Note:	
			Removal	Installation
36	Fixed gear (5th gear)	1		Identification groove faces fixed gear No. 20. Do not confuse with fixed gear No. 51
37	Spacer sleeve	1		
38	Loose gear (5th gear)	1		Do not confuse with loose gear No. 49
39	Synchronizing ring	1	Mark for reinstallation	Check for wear. Fit with the same gearwheel. (Cogs face driver dogs)
40	Needle roller bearing cage	1	Mark for reinstallation	Fit with the same gearwheel
41	Inner race**	1	Mark for reinstallation. Pull off with arms of Special Tool 9284 across loose gear No. 49	Fit with the same gearwheel. Heat to approx. 120 °C
42	Selector sleeve	1		Center the centerpunch marks relative to the driver dogs
43	Driver dog	3		Domed side faces selector sleeve
44	Ball	3		
45	Spring	3		
46	Shift fork (5th and 6th gears)	1		Adjust. In neutral position, selector sleeve must be set exactly in the middle between the loose gearwheels
47	Guide sleeve	1		Install complete with selector sleeve and shift fork
48	Synchronizing ring	1	Mark for reinstallation	Check for wear. Fit with the same gear (Cogs face driver dogs)

No.	Designation	Qty.	Note:	
			Removal	Installation
49	Loose gear (6th gear)	1		Do not confuse with loose gear No. 38
50	Needle roller bearing cage	1	Mark for reinstallation	Fit with the same gear-wheel
51	Fixed gear (6th gear)	1		Identification groove faces fixed gear No. 36. Do not confuse with fixed gear No. 36
52	Transmission	1		

* Transmission shafts without recesses for securing the hexagon nuts have been installed since September 1995.

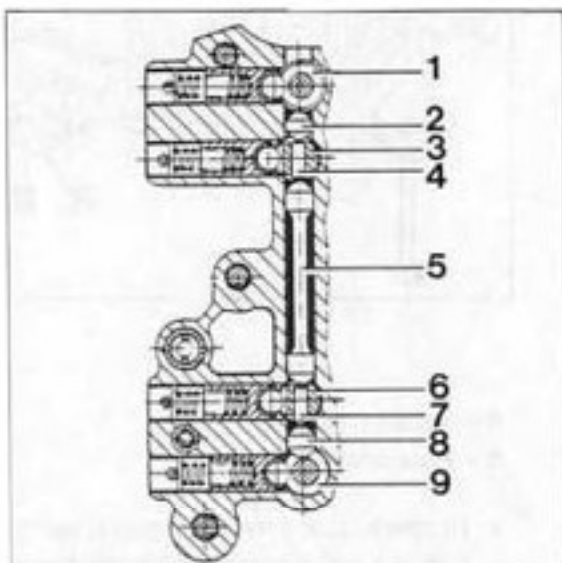
Self-locking hexagon nuts are used on these shafts, and these nuts must **always** be replaced in every transmission repair.

** A result of tolerances, the bearing inner rings (no. 34 and 41) may either from a force fit on the shaft or have a certain amount of play.

Dismantling and assembly notes

Note

If the shift rods are moved across the neutral or gear latching positions when the transmission is dismantled or assembled, the small intermediate locks may drop out inadvertently.

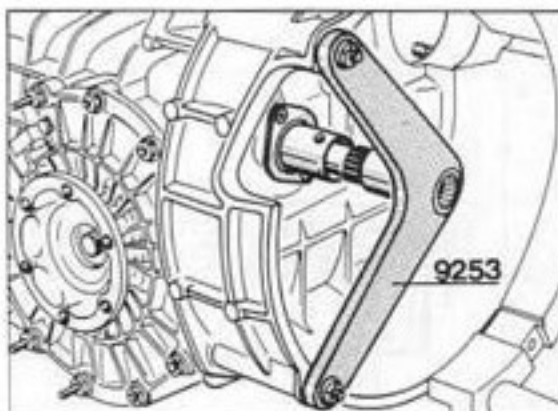


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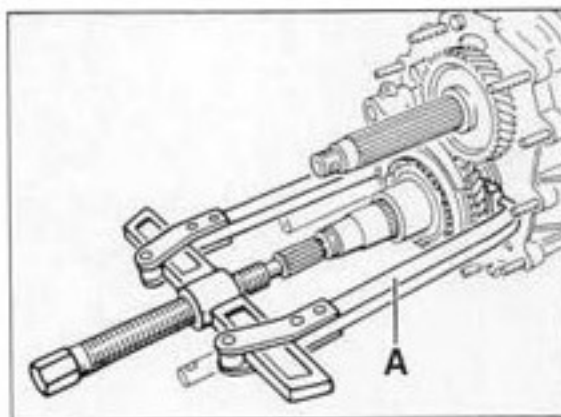
4 and 7 = Intermediate locks

Dismantling

1. Engage 6th gear, lock drive shaft with Special Tool **9253** and unscrew hexagon head nuts for output and input shafts.



2. Engage 4th gear and pull off inner race of needle roller bearing (5th gear).



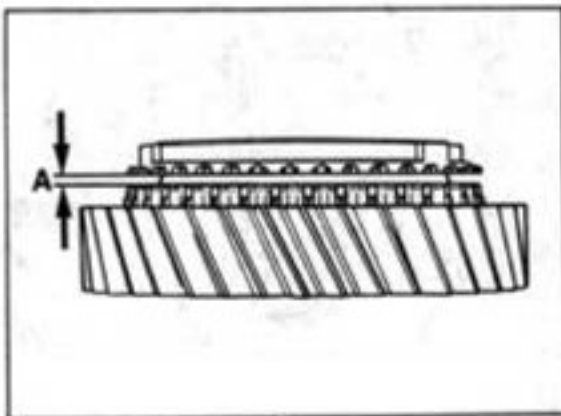
2011-34

A = Arms of puller No. 9284

Assembly

1. Check synchronizing rings by pushing rings onto tapers of gearwheels and measuring gap „A“ with a feeler gauge.

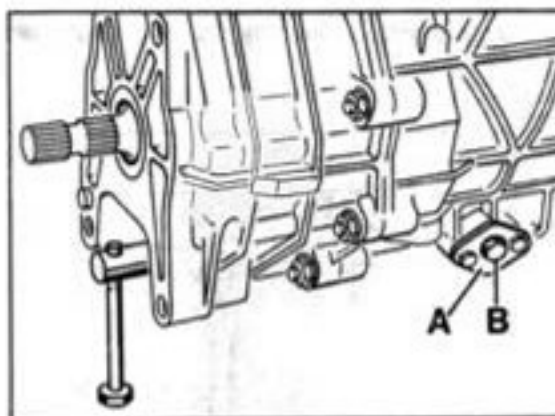
Installation dimension (new) = 0.9 mm min.
Wear limit = 0.6...0.7 mm



518-35

2. Engage 4th gear and fit all gearwheels.

3. Adjust inner shift rod until all gears may be preselected freely without notching. To adjust, undo hexagon head bolts of tensioning plate and rotate eccentric bushing.

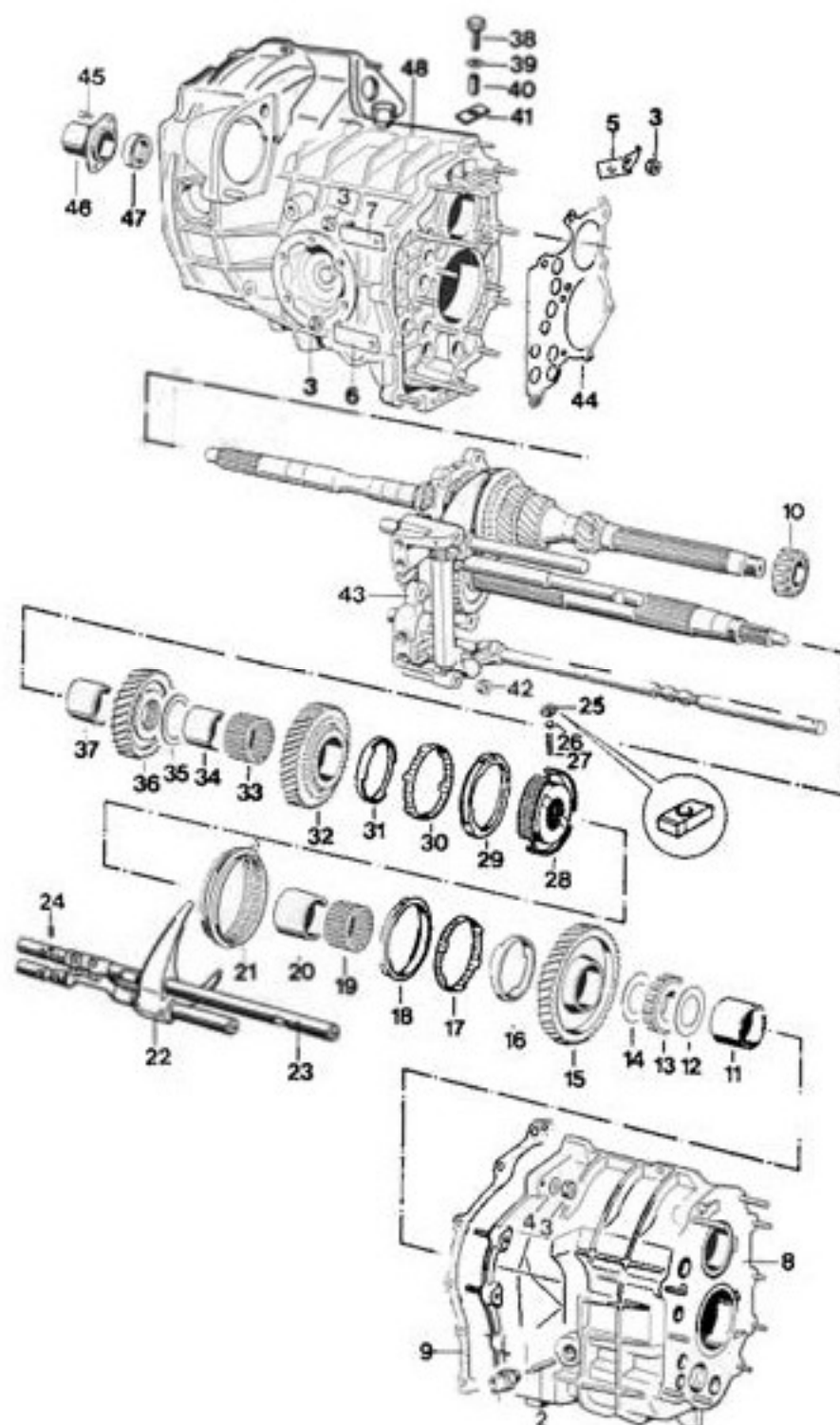


2014-34

A = Tensioning plate
B = Eccentric bushing

4. To check, shift through all gears, using a long M 8 bolt that is screwed into the inner shift rod.

34 35 37 Dismantling and assembling transmission



No.	Designation	Qty.	Removal	Note:	
				Installation	
1	Reverse light switch	1		Tighten to 35 Nm (26 ftlb.)	
2	Plunger	1		Stepped end faces switch	
3	Hexagon head nut	12		Tighten to 23 Nm (17 ftlb.)	
4	Washer	10			
5	Retaining bracket	1	deleted		
6	Retaining bracket (straight)	1			
7	Retaining bracket (domed)	1			
8	Gear housing	1			
9	Gasket	1		Replace	
10	Cylindrical roller bearing	1	Pull off	Heat to approx. 120° C	
11	Inner race	1	Mark for reinstallation. Pull off across loose gear-wheel No. 15. Observe clearance at reverse shift rod	Fit with the same gear-wheel. Heat to approx. 120°C	
12	Thrust washer	1			
13	Cylindrical roller bearing	1		Heat to approx. 120° C	
14	Thrust washer	1			
15	Loose gearwheel (1st gear)	1			
16	Friction ring	1	Mark for reinstallation	Fit with the same gear-wheel	
17	Tapered ring	1	Mark for reinstallation	Fit with the same gear-wheel. Tabs must engage in the cutouts in the loose gearwheel	

No.	Designation	Qty.	Note:	
			Removal	Installation
18	Synchronizing ring	1	Mark for reinstallation	Check for wear. Fit with the same gearwheel. Drivers must engage in the cutouts in the tapered ring. Three lugs face driver dogs
19	Needle roller bearing	1	Mark for reinstallation	Fit with the same gear-wheel
20	Inner race	1	Shift rods in neutral position. Pull out shift rod for reverse gear. Pull off across loose gearwheel No. 32. Mark for reinstallation.	Fit with the same gear-wheel. Heat to approx. 120 °C
21	Shift sleeve	1		Insert complete with guide sleeve and shift rods. The missing tooth of the internal teeth of the guide sleeve (No. 28) must be aligned exactly above the oil bore of the output shaft. For further notes, refer to page 35 - 213
22	Shift rod with pinned shift fork	1		
23	Shift rod (reverse)	1		
24	Spacer	1		Apply stiff grease to insert into shift rod
25	Driver dog	3		Place into correct position
26	Ball	3		
27	Spring	3		

No.	Designation	Qty.	Note:	
			Removal	Installation
28	Guide sleeve	1		Missing tooth of internal teeth must be placed exactly above oil bore of output shaft
29	Synchronizing ring	1	Mark for reinstallation	Check for wear. Fit with the same gearwheel. Drivers must engage into the cutouts in the tapered ring. The lugs face the driver dogs.
30	Tapered ring	1	Mark for reinstallation	Fit with the same gear-wheel. Tabs must engage into the cutouts in the loose gearwheel
31	Friction ring	1	Mark for reinstallation	Fit with the same gear-wheel
32	Loose gearwheel (2nd gear)	1		
33	Needle roller bearing	1	Mark for reinstallation	Fit with the same gear-wheel
34	Inner race	1	Mark for reinstallation. Pull off across fixed gear-wheel No. 36	Fit with the same gear-wheel. Heat to approx. 120 °C
35	Thrust washer	1		
36	Fixed gearwheel (3rd gear)	1		Large flange faces thrust washer No. 35
37	Spacer sleeve	1		
38	Hexagon head bolt (micro-sealed)	1		Must always be replaced. Tightening torque 6 Nm (4 ftlb.)
39	Washer	1		
40	Spacer sleeve	1		

No.	Designation	Qty.	Note:	
			Removal	Installation
41	Retaining plate	1		Tighten to 23 Nm (17 ftlb.)
42	Hexagon head nut	10		
43	Tensioning plate with gear set	1		
44	Adjusting washer "S3"	X		Redetermine thickness if required
45	Oval-head screw	2	Note number and thickness for reinstallation	Tighten to 10 Nm (7 ftlb.)
46	Guide tube	1		
47	Shaft seal	1		Do not fit until gear set has been fitted (refer to page 35 - 1)
48	Transmission housing	1		

Note

Due to the tolerances (transition fit), the inner bearing rings (nos. 11, 20 and 34) may have clearance or press-fit on the shaft.

Dismantling and assembly notes

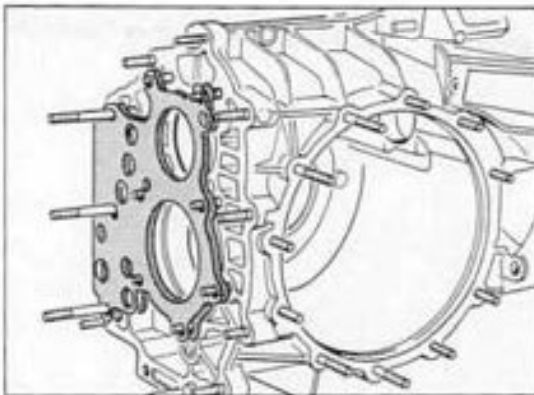
Dismantling

Note

To facilitate assembly, do not remove the complete gear set assembly but partially dismantle the installed drive pinion.

Assembly

1. The number of adjusting shims "S3" noted during dismantling or the number of adjusting shims noted when the drive pinion was adjusted should be placed onto the studs in the housing until the equivalent thickness is obtained.

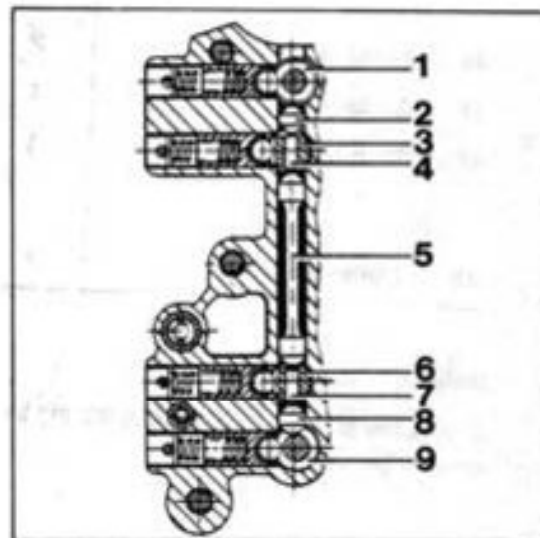


506-34

2. Engage 4th gear and assemble fully preassembled gear set to internal selector rod.
3. Tighten tensioning plate lock nuts to 23 Nm (17 ftlb).

Note

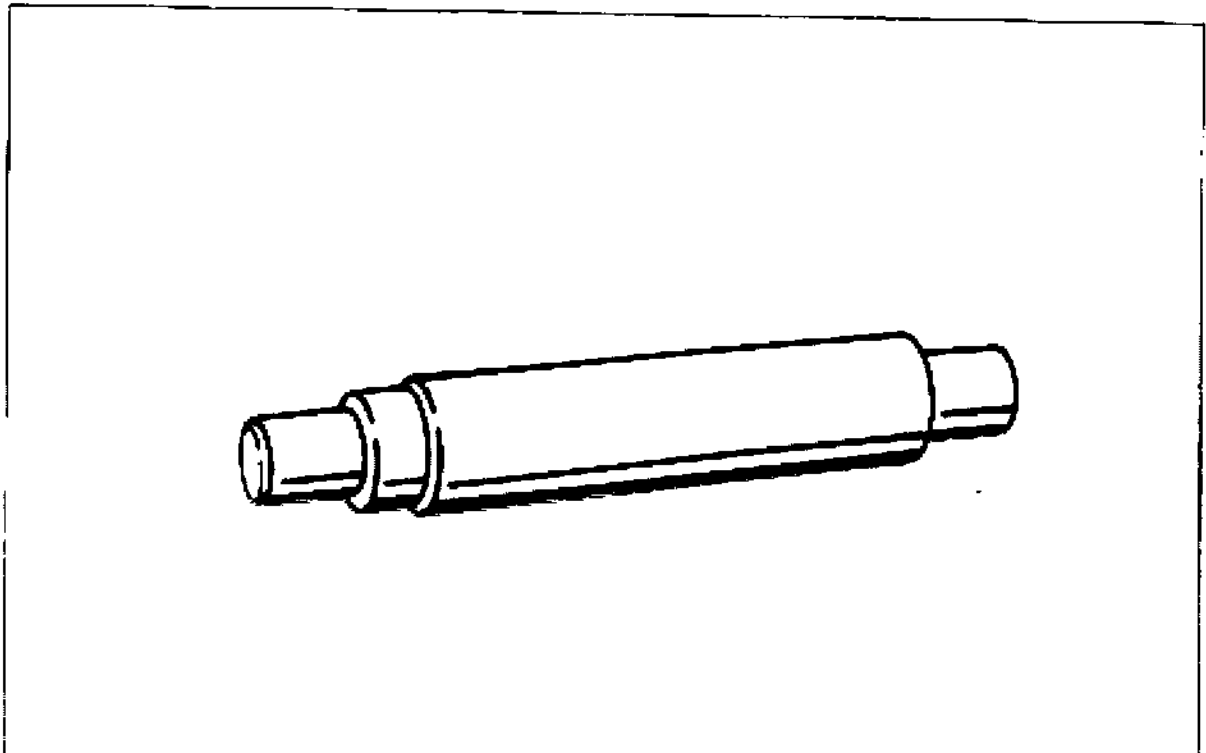
Do not move shift rods across the neutral or gear stop position as this may cause the small intermediate locks to drop out inadvertently.



1709-35

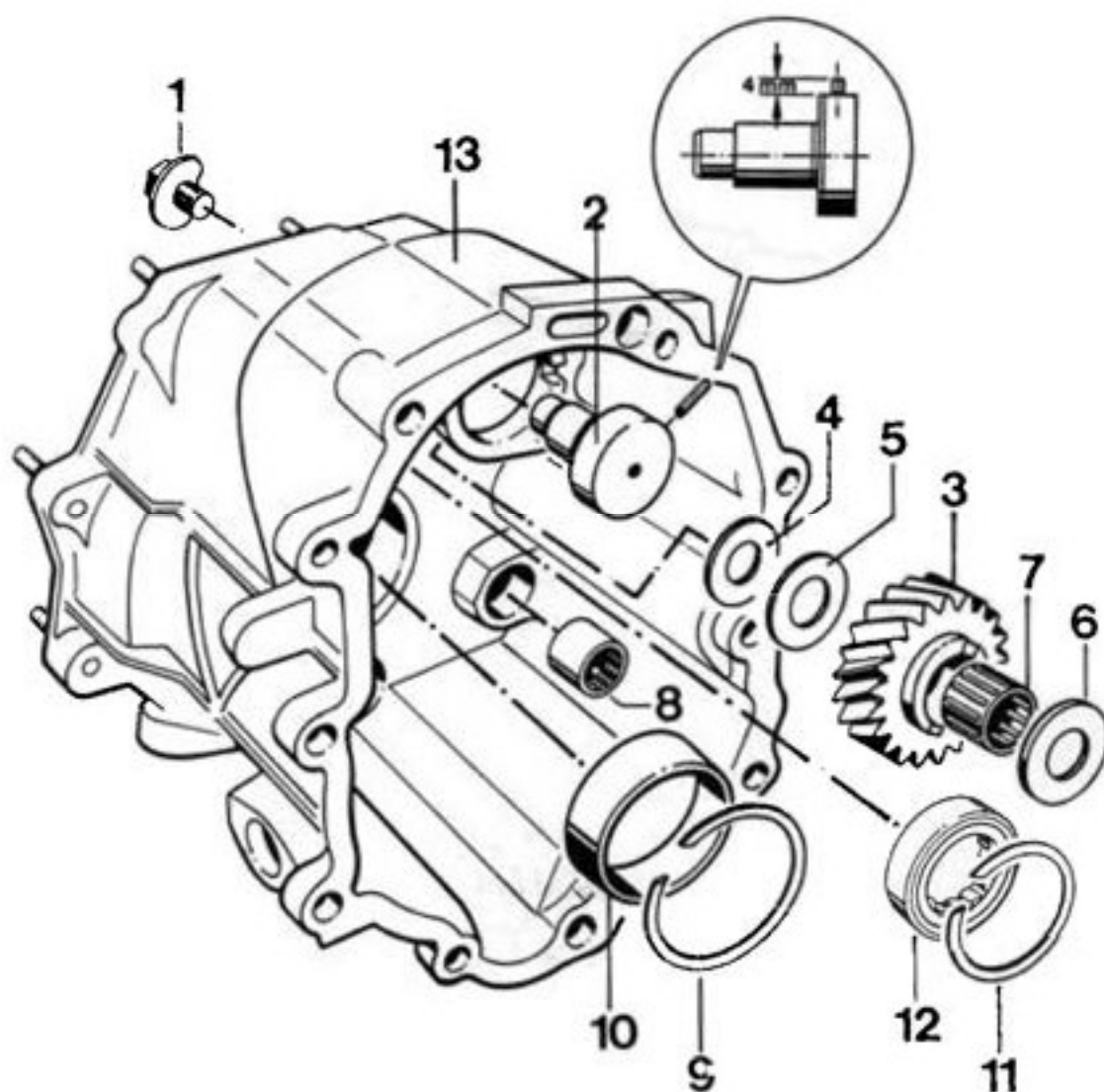
4 and 7 = Intermediate locks

34 50 37 Dismantling and assembling intermediate casing
tools



No.	Designation	Special tool	Order number	Explanation
	Mandrel	9254	000.721.925.40	

Dismantling and assembling intermediate casing



No.	Designation	Qty.	Note:	
			Removal	Installation
1	Hexagon head bolt with washer	1		$M_A = 23 \text{ Nm (17 lbf.)}$
2	Pin with roll pin	1		Install in correct position
3	Reverse idler gear	1		Small flange faces hexagon head bolt (item No. 1)
4	Thrust washer (2.0 mm thick)	1		
5	Thrust washer (1.5 mm thick)	1		Do not confuse with washer No. 4
6	Thrust washer (1.5 mm thick)	1		Do not confuse with washer No. 4
7	Needle roller assembly	1		
8	Ball sleeve	1	Push out with suitable drift or a length of pipe (e.g. VW 423)	Working from inside, press in flush across lettered side using 9254
9	Snap ring	1		
10	Bearing outer race	1	Heat housing to approx 120 °C	Heat housing to approx. 120 °C
11	Snap ring	1		
12	Cylindrical roller bearing	1	Heat housing to approx 120 °C and pull out with suitable internal puller (e.g. Schrem 30 - 40)	As No. 10
13	Housing	1		

Notes on assembly

1. Reassemble reverse idler gear with thrust washers and bolt in correct position (refer to exploded view). With thrust washers and bolt located correctly, the end clearance of the reverse idler gear is 0.15...0.35 mm.

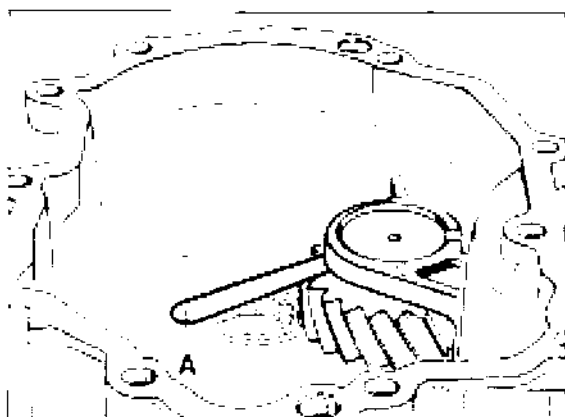


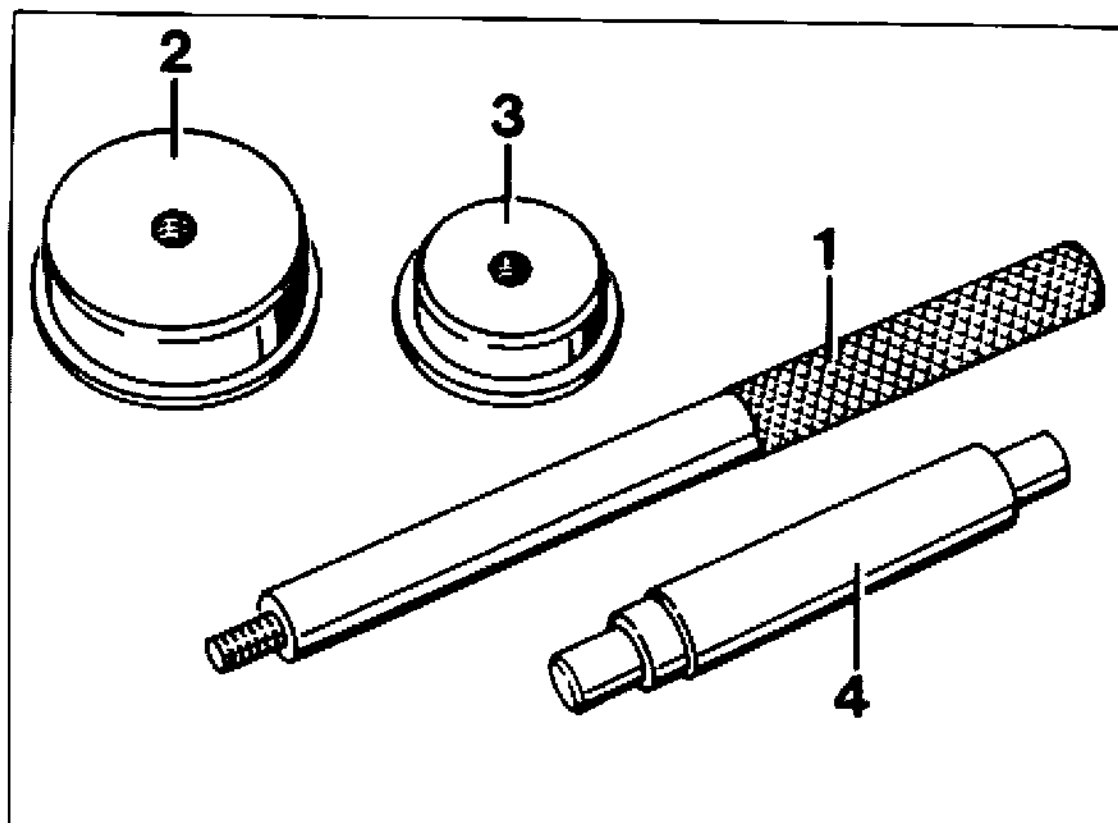
Fig. 34

A = Feeler gauge

2. Check end clearance of reverse idler gear with feeler gauge.

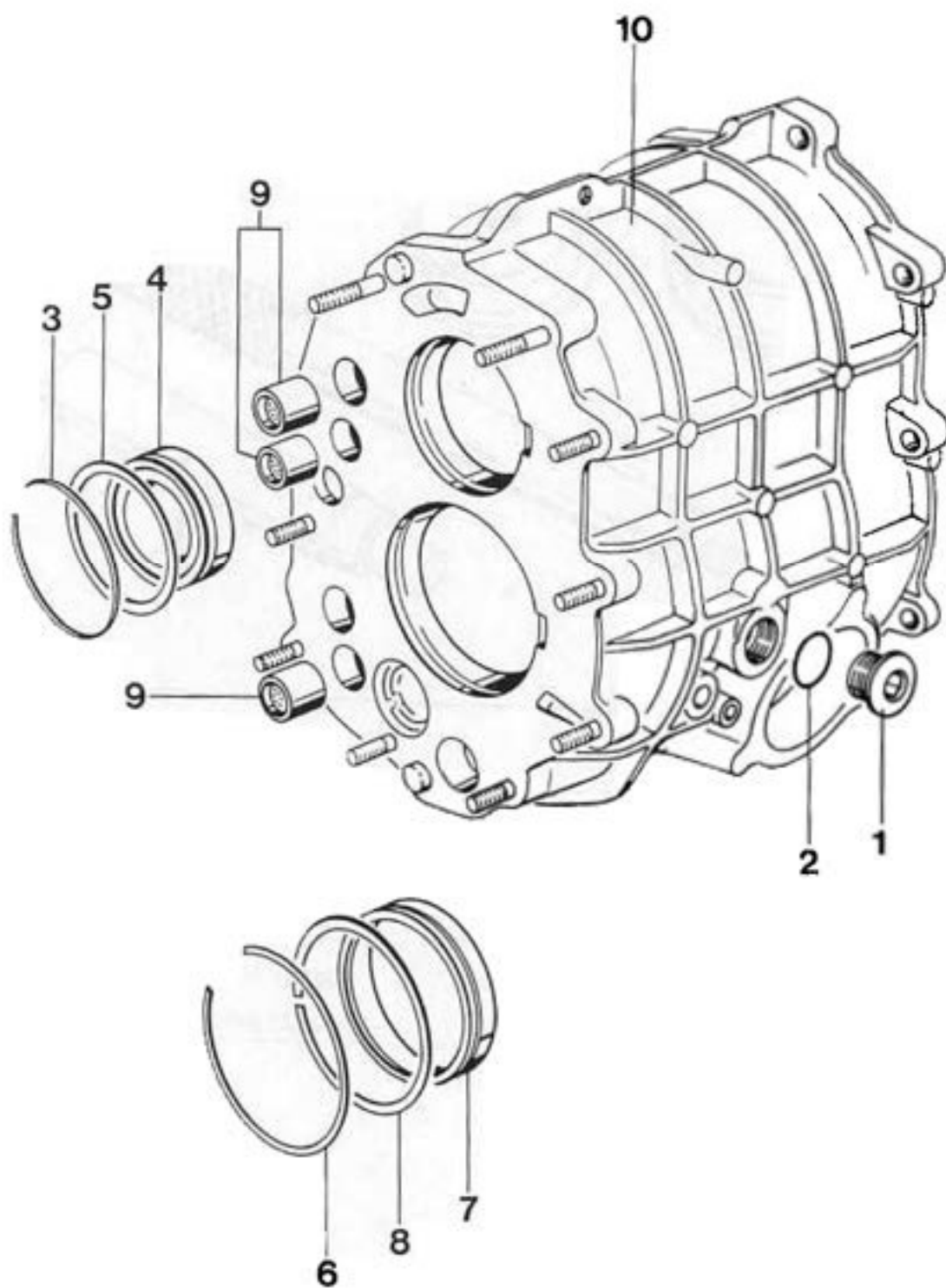
34 52 37 Dismantling and assembling gear housing

Tools



No.	Designation	Special tool	Order number	Explanation
1	Mandrel	P 254	000.721.254.00	
2	Thrust piece	P 254 a	000.721.254.10	
3	Thrust piece	P 254 b	000.721.254.20	
4	Mandrel	9254	000.721.925.40	

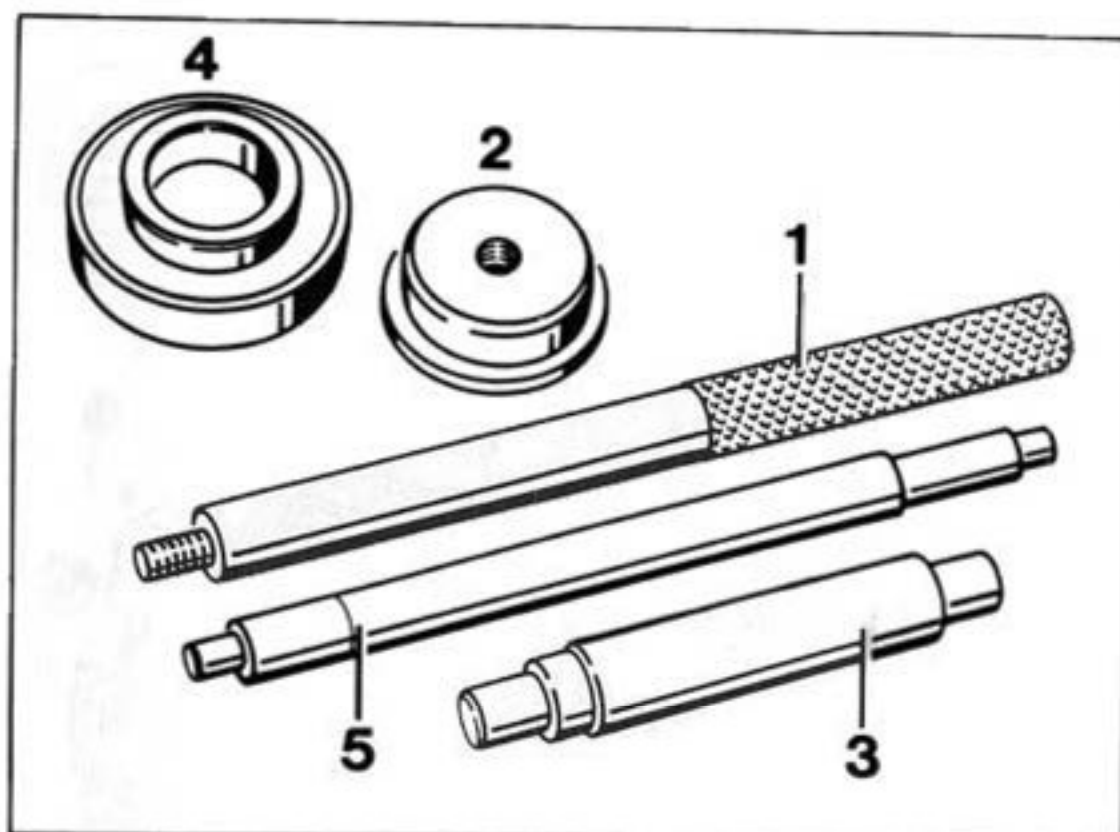
34 52 37 Dismantling and assembling gear housing



No.	Designation	Qty.	Note:	
			Removal	Installation
1	Plug	1		Torque: 30 Nm (22 ftlb.)
2	Seal	1		Replace
3	Retaining ring	1		
4	Bearing outer race	1	Working from inside, press out with Special Tools P 254 and P 254 b	Press in with Special Tools P 254 and P 254 b
5	Snap ring	1		
6	Retaining ring	1		
7	Bearing outer race	1	Working from inside, press out with Special Tools P 254 and P 254 b	Press in with Special Tools P 254 and P 254 b
8	Snap ring	1		
9	Ball sleeve	3	Press out with suitable mandrel	Press in flush with Special Tool 9254
10	Gear housing	1		

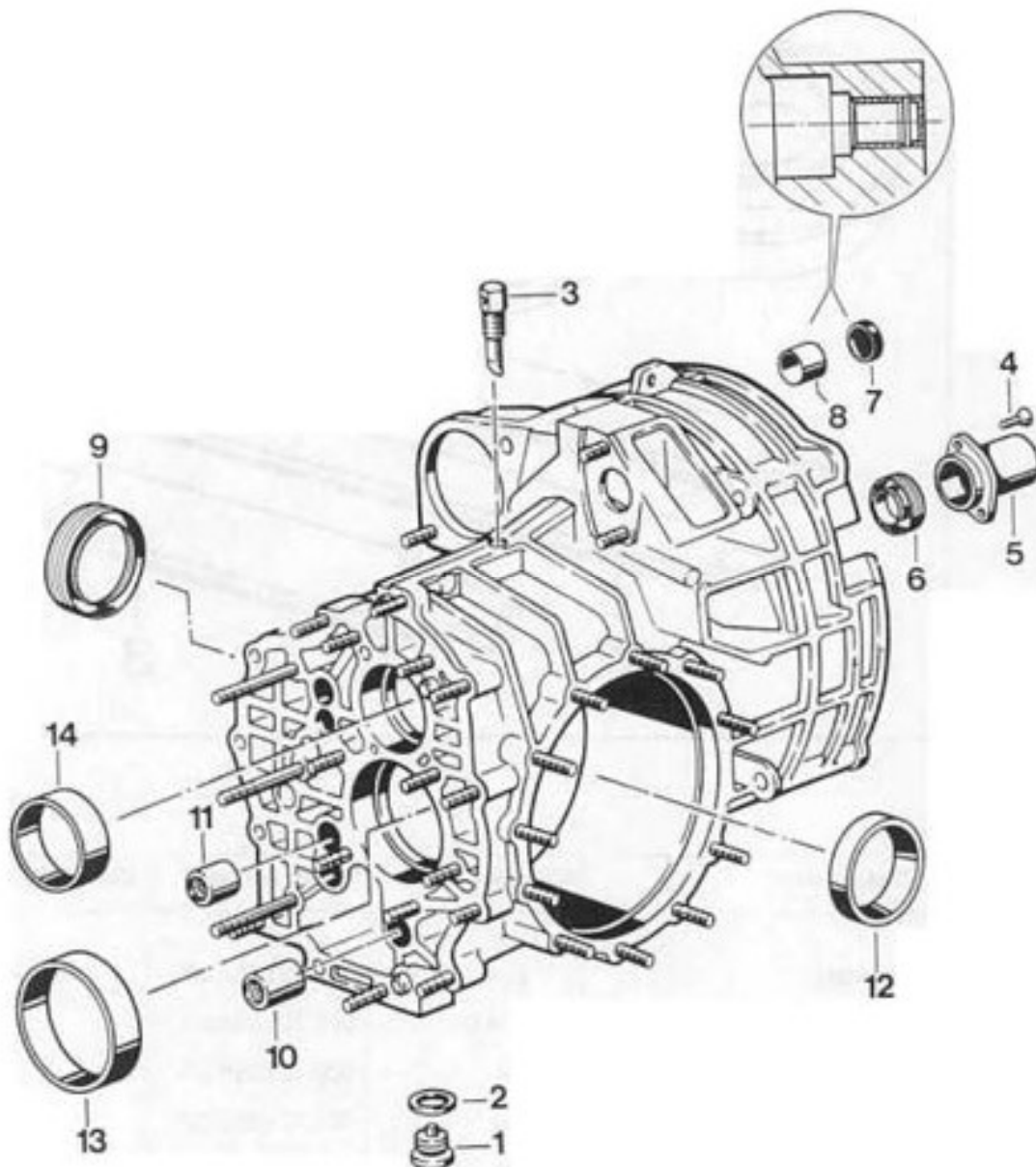
34 37 37 Dismantling and assembling transmission housing

Tools



No.	Designation	Special tool	Order number	Explanation
1	Mandrel	P 254	000.721.254.00	
2	Thrust piece	P 254 b	000.721.254.20	
3	Mandrel	9254	000.721.925.40	
4	Thrust piece	9252	000.721.925.20	
5	Mandrel	9515	000.721.951.50	

34 37 37 Dismantling and assembling transmission housing

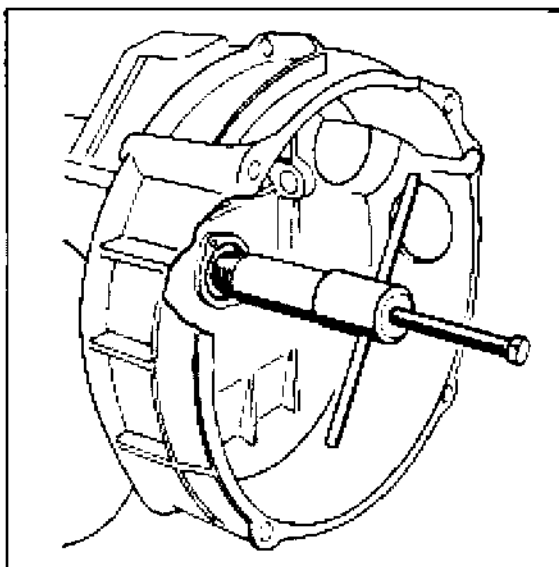


No.	Designation	Qty.	Note:	
			Removal	Installation
1	Plug	1		Clean, tighten to 30 Nm (22 ftlb.)
2	Seal	1		Replace
3	Vent	1		Tighten to 35 Nm (26 ftlb.). The hole in the hexagon head must face the transfer casing
4	Oval-head screw	2		Tighten to 10 Nm (7 ftlb.)
5	Guide tube	1		
6	Shaft seal	1		Do not fit until the gear set has been fitted (also refer to page 35 - 201)
7	Cover	1		
8	Bushing	1	Push out with drift 9515, working from inside towards outside	Do not grease or oil and do not clean with solvents (e.g. benzene). Push in from inside with drift 9515 until it is seated against the stop.
9	Shaft seal	1		Push in with Special Tool 9252 until it is seated against the stop. Pace space between dust lip and sealing lip with grease (e.g. Liqui Moly Pu 53)
10	Ball sleeve (long)	1	Pull out with suitable internal puller (e.g. Schrem 14 - 20)	Press in flush with Special Tool 9254
11	Ball sleeve (short)	4	Pull out with suitable internal puller (e.g. Schrem 14 - 20)	Press in flush with Special Tool 9254

No.	Designation	Qty.	Note:	
			Removal	Installation
12	Bearing outer race	1	Heat transmission housing to approx. 120 °C and remove race with suitable drift	Heat transmission housing to approx. 120 °C and press in with suitable thrust piece
13	Bearing outer race	1	Heat transmission housing to approx. 120 °C and remove race with suitable thrust piece (e.g. VW 513)	Heat transmission housing to approx. 120 °C and press in with suitable thrust piece (e.g. 9247/4) until it is seated against the stop
14	Bearing outer race	1	Heat transmission housing to approx. 120 °C and pull out with suitable internal puller (e.g. Schrem 50 - 60)	Heat transmission housing to approx. 120 °C and press in with Special Tools P 254 and P 254 b until it is seated against the stop
15	Transmission housing	1		

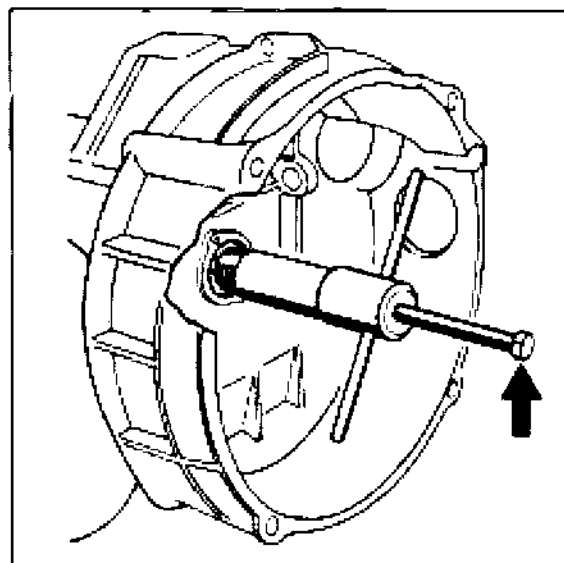
35 50 19 Removing and installing input shaft oil seal**Removing**

1. Remove engine/transmission assembly and separate transmission from engine.
2. Remove release bearing guide tube.
3. Screw Special Tool 9251 firmly into the oil seal.



1652-34

4. Screw in the hexagon-head bolt and pull out oil seal.



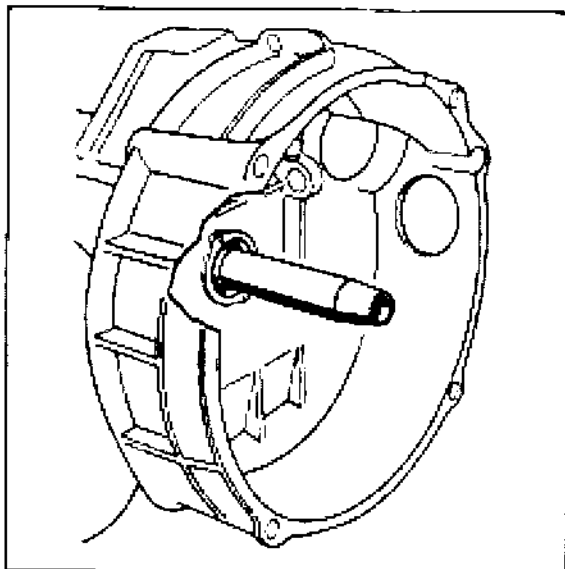
1653-34

Note

If the helical tension spring jumps out when removing the seal, use a wire hook to pull it off the input shaft.

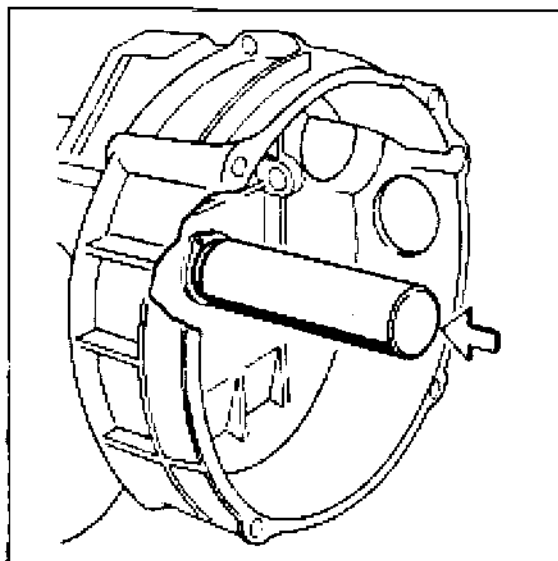
Installing

1. Push assembling sleeve 9255 onto the input shaft splines.



1654-34

3. Use Special Tool 9256 to drive the oil seal into place.

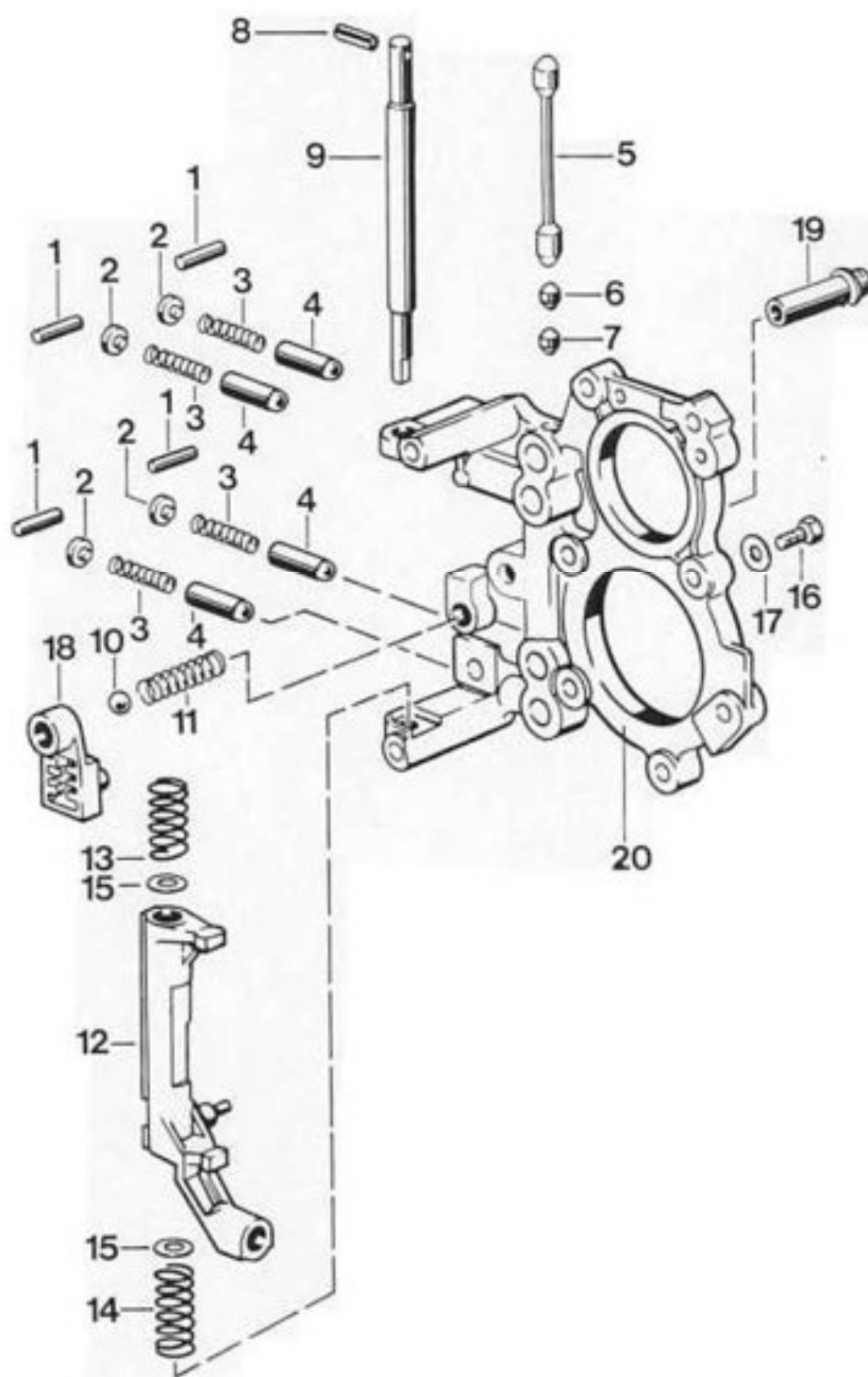


1655-34

2. Pack oil seal area between dust lip and sealing lip with grease (e.g. Liqui Moly Pu 53).

4. Install guide tube and tighten mounting screws to **10 Nm (7 flb.)**.

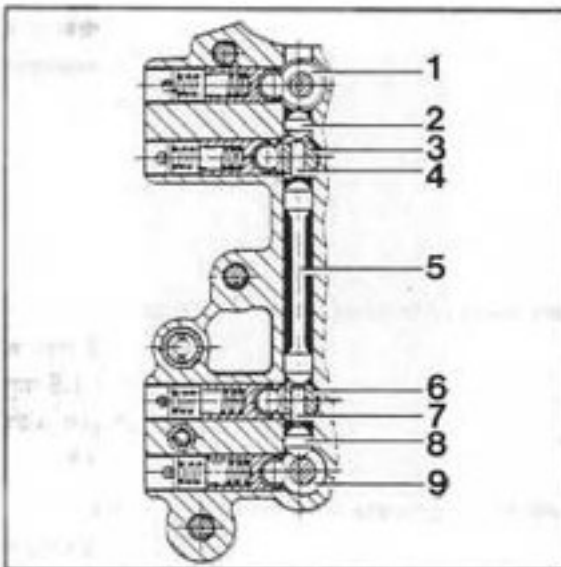
35 19 37 Dismantling and assembling tensioning plate



No.	Designation	Qty.	Removal	Note:	Installation
1	Straight pin	4	With the shift rods fitted, the springs (No. 3) are under load		
2	Washer	4			
3	Thrust spring	4			
4	Locking bush	4			
5	Lock (long)	1			
6	Lock (short)	1			
7	Lock (short)	1			
8	Roll pin	1			
9	Bearing shaft	1			Install in correct position, use 8 mm dia. locating mandrel to locate relative to tensioning plate hole
10	Ball	1			
11	Thrust spring	1			
12	Deflection lever	1			
13	Thrust spring	1	Mark for reinstallation		Free length = 51.3 ± 0.5 mm wire thickness = 1.5 mm. Do not confuse with spring No. 14
14	Thrust spring	1	Mark for reinstallation		Free length = 72.6 ± 0.5 mm wire thickness = 1.1 mm. Do not confuse with spring No. 13
15	Shim	2			

No.	Designation	Qty.	Note:	
			Removal	Installation
16	Hexagon head bolt	1	Press out with suitable drift	Tighten to 10 Nm (7 ftlb.)
17	Washer	1		
18	Selector gate	1		
19	Adapter sleeve with snap ring	1		Press in with suitable drift until it is seated against the stop
20	Tensioning plate	1		

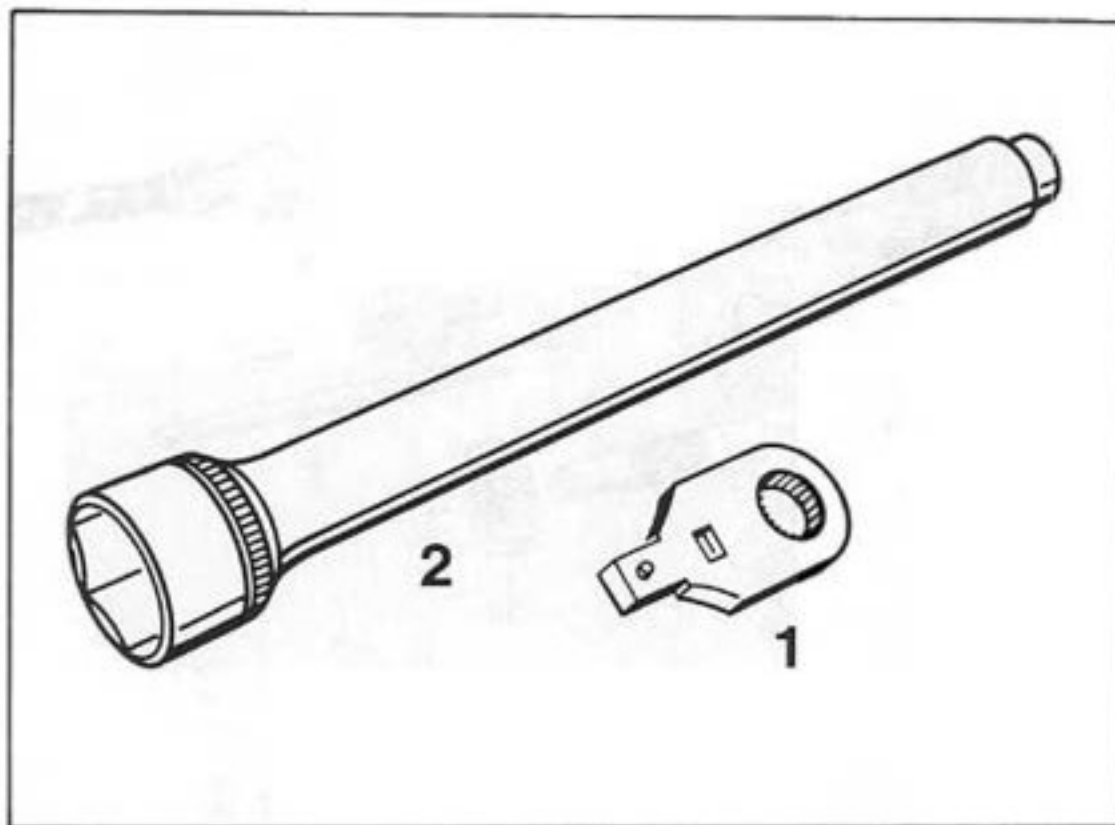
Installation position of the locks



1709-35

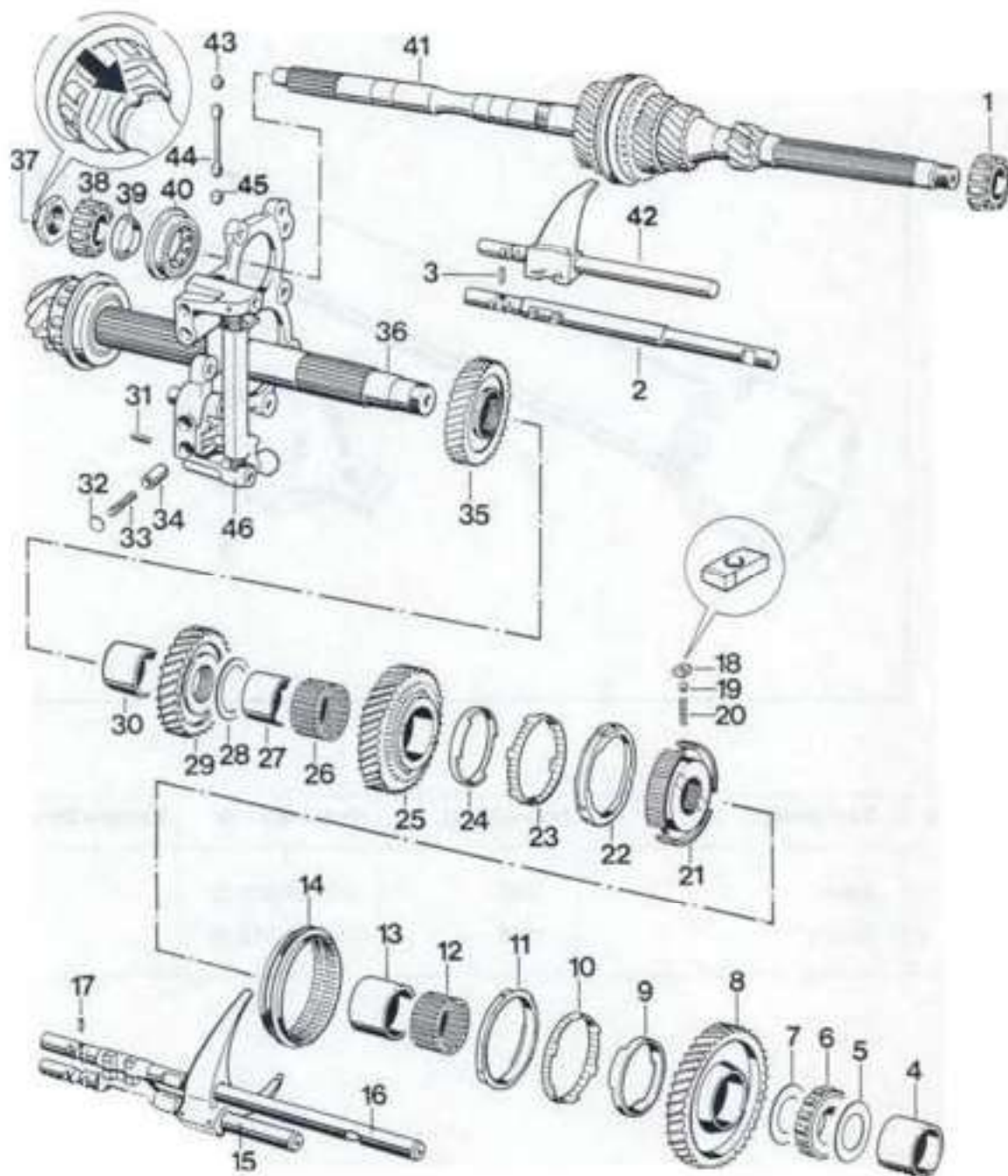
35 40 19 Removing and installing input shaft

Tools



No.	Designation	Special tool	Order number	Explanation
1	Insert	9282	000.721.928.20	
2	Socket	9105	000.721.910.50	

35 40 19 Removing and installing input shaft



No.	Designation	Qty.	Removal	Note:	
				Installation	
1	Cylindrical roller bearing	1		Heat to approx. 120 °C	
2	Shift rod (5th and 6th gear)	1	Shift rods in neutral position		
3	Spacer	1		Coat with stiff grease to insert	
4	Inner race	1		Fit with the same gear-wheel. Heat to approx. 120 °C	
5	Thrust washer	1			
6	Cylindrical roller bearing	1		Heat to approx. 120 °C	
7	Thrust washer	1			
8	Loose gearwheel (1st gear)	1			
9	Friction ring	1		Fit with the same gear-wheel	
10	Tapered ring	1		Fit with the same gear-wheel. Tabs must engage into the cutouts in the loose gearwheel	
11	Synchronizing ring	1		Check for wear. Fit with the same gearwheel. Drivers must engage in the cutouts in the tapered ring. Three lugs face the driver dogs.	
12	Needle roller bearing	1		Fit with the same gear-wheel	
13	Inner race	1		Fit with the same gear-wheel. Heat to approx. 120 °C	

No.	Designation	Qty.	Note:	
			Removal	Installation
14	Shift sleeve	1		Insert complete with guide sleeve and shift rods. Make sure the missing tooth of the internal teeth of the guide sleeve (No. 21) is aligned exactly above the oil bore of the output shaft. The circumferential identification groove must face 2nd gear. The centerpunch marks must be centered relative to the balls (see page 35 - 213)
15	Shift rod with pinned shift fork	1		
16	Shift rod (reverse)	1		
17	Intermediate lock	1		Coat with stiff grease to insert
18	Driver dog	3		Install in correct position
19	Ball	3		
20	Spring	3		
21	Guide sleeve	1		Missing tooth of the internal splines must be exactly above the oil bore of the output shaft
22	Synchronizing ring	1		Check for wear. Fit with the same gearwheel. Drivers must engage into the cutouts in the tapered sleeve. Three lugs face the driver dogs.

No.	Designation	Qty.	Note:	
			Removal	Installation
23	Tapered ring	1		Fit with the same gear-wheel. Tabs must engage into the cutouts in the loose gearwheel
24	Friction ring	1		Fit with the same gear-wheel
25	Loose gearwheel (2nd gear)	1		
26	Needle roller bearing	1		Fit with the same gear-wheel
27	Inner race	1		Fit with the same gear-wheel. Heat to approx. 120 °C
28	Thrust washer	1		
29	Fixed gearwheel (3rd gear)	1		Large collar faces thrust washer No. 28
30	Spacer sleeve	1		
31	Straight pin	4		
32	Washer	4		Guide lug faces spring
33	Thrust spring	4		
34	Latch	4		
35	Fixed gearwheel (4th gear)	1		Large collar faces four-point bearing
36	Output shaft	1		
37	Flange nut*	1	Undo with Special Tools 9282 and 9105	Tighten to 250 Nm (184 ftlb.). Upset the flange to lock
38	Cylindrical roller bearing	1	Press off across 2nd gear fixed gearwheel	Heat to approx. 120 °C
39	Bearing inner race	1		Heat to approx. 120 °C

No.	Designation	Qty.	Note:	
			Removal	Installation
40	Four-point bearing	1		
41	Output shaft	1		Insert complete with pinned shift rod / shift fork (No. 42)
42	Shift rod with pinned shift fork	1		
43	Lock (short)	1		
44	Lock (long)	1		
45	Lock (short)	1		
46	Tensioning plate	1		

Transmission shafts without recesses for securing the hexagon nuts have been installed since September 1995.

Self-locking hexagon nuts are used on these shafts, and these nuts must **always** be replaced in every transmission repair.

Removal and installing notes

Removal

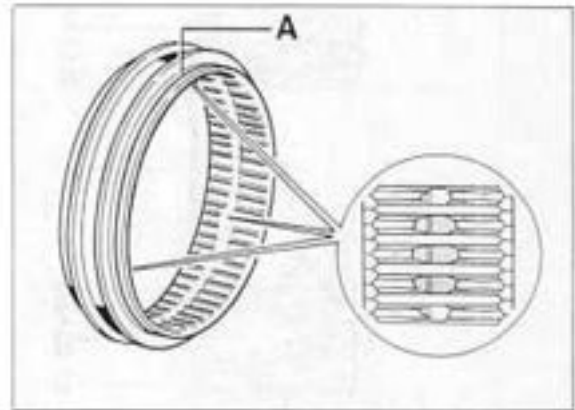
Note

Parts No. 1 to 30 are removed with the tensioning plate remaining fitted (refer to page 34 - 209).

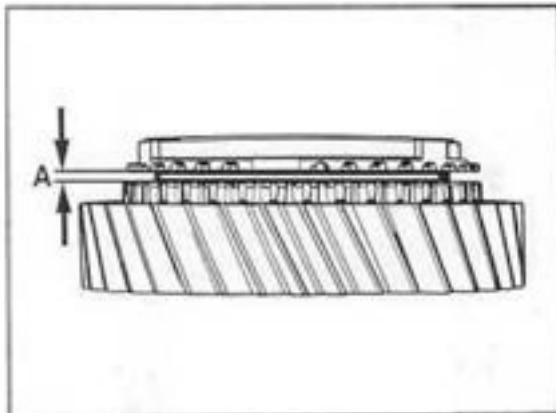
Installation

1. Using a suitable flat iron bar, clamp tensioning plate in a vise in such a manner that the hole for the shift rod locks is horizontal.
2. Check synchromesh of 1st and 2nd gears. To do so, place friction ring, tapered ring and synchronizing ring in correct position onto gearwheel. Check gap "A" with a feeler gauge.

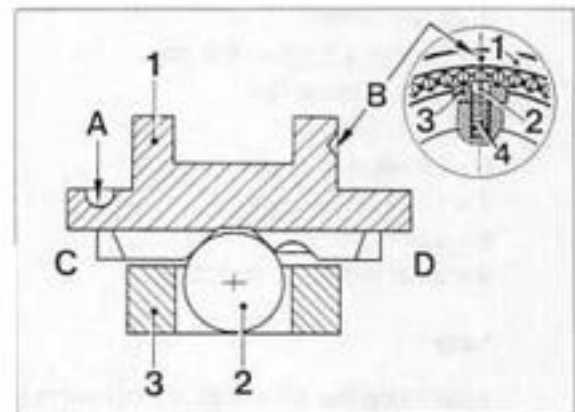
Installation dimension (new) = 1.5 to 2.0 mm
Wear limit = 1.2 mm



A = Identification groove (must face 2nd gear)



1701-35



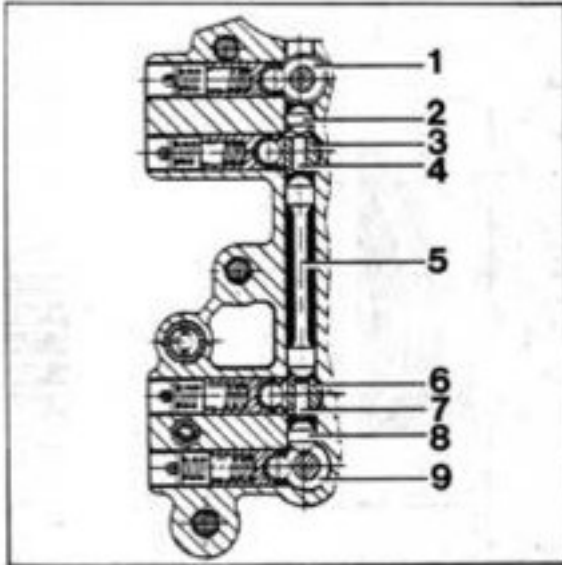
1906-35

3. Place shift sleeve onto guide sleeve so that centerpunch marks "B" are centered relative to the balls.

In addition, the circumferential identification groove "A" must face 2nd gear.

- 1 = Shift sleeve
- 2 = Bail
- 3 = Driver dog
- 4 = Spring
- A = Identification groove
- B = Centerpunch mark
- C = 2nd gear side
- D = 1st gear side

4. Observe installation position of locks.



1799-35

- 1 = Shift rod 3rd and 4th gear
- 2 = Lock (short)
- 3 = Shift rod 5th and 6th gear
- 4 = Intermediate lock
- 5 = Lock (long)
- 6 = Reverse shift rod
- 7 = Intermediate lock
- 8 = Lock (short)
- 9 = Shift rod 1st and 2nd gear

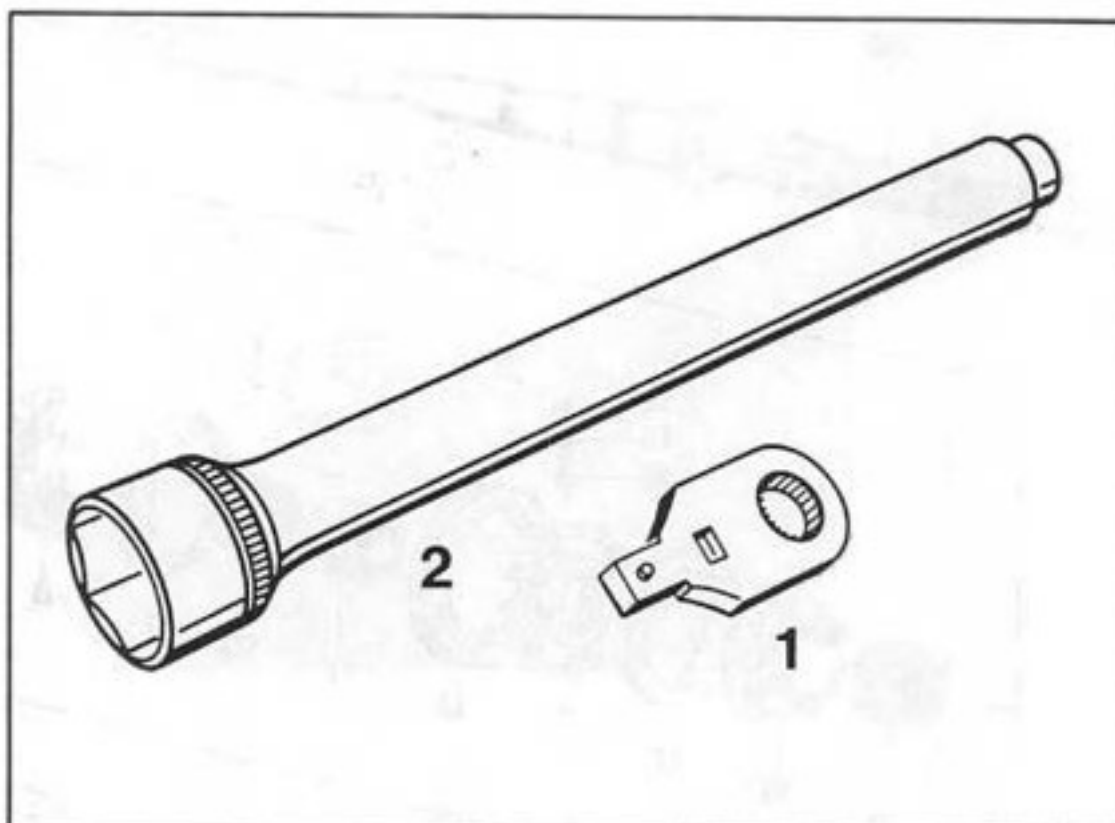
Note

After fitting the shift rods, do not move them across the neutral or gear latch positions as this may cause the small intermediate locks to drop out inadvertently.

To avoid inadvertent movement of the shift rods, lock shift rods by engaging 3th gear.

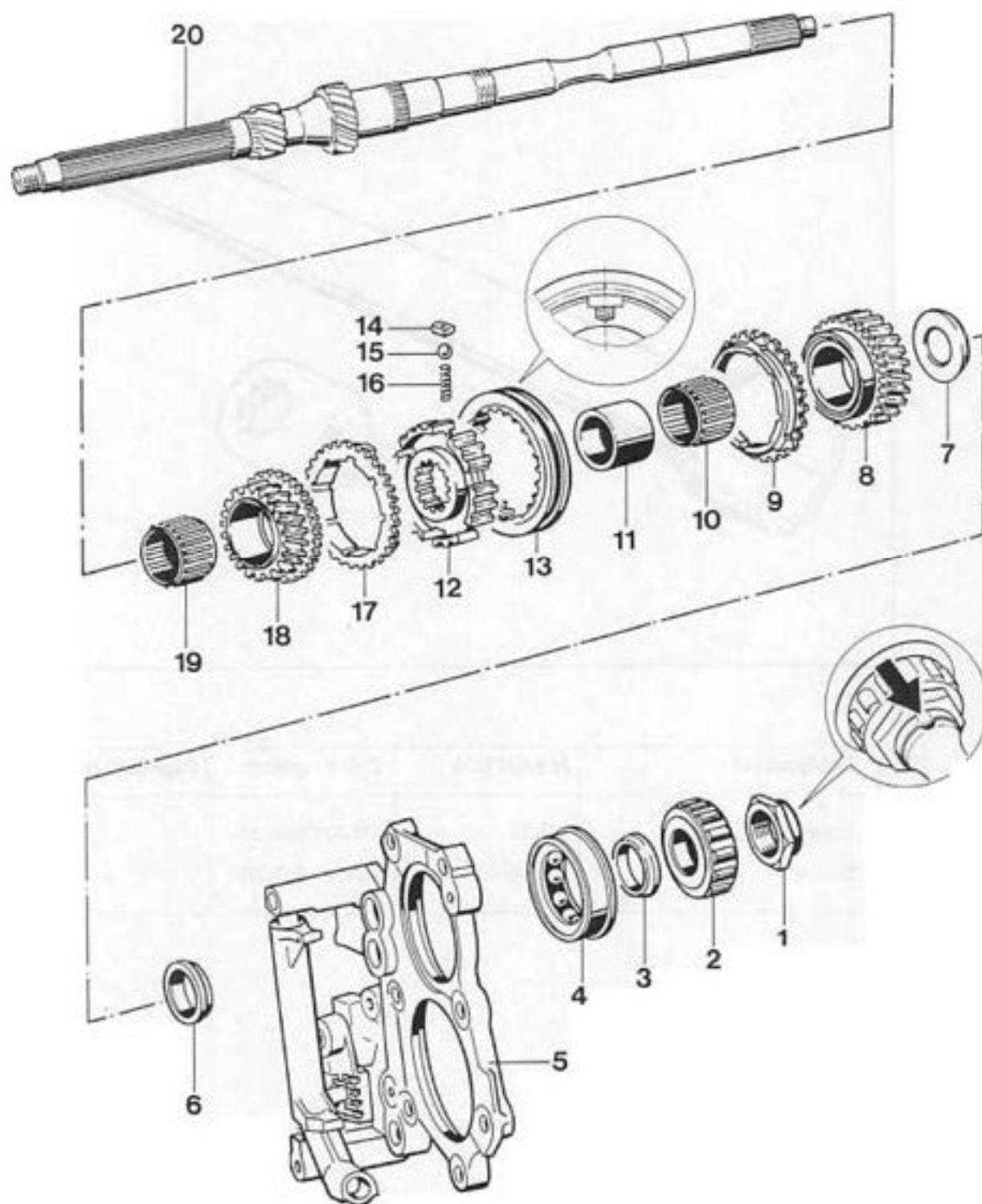
35 40 37 Dismantling and assembling input shaft

Tools



No.	Designation	Special tool	Order number	Explanation
1	Insert	9282	000.721.928.20	
2	Socket	9105	000.721.910.50	

35 40 37 Dismantling and assembling input shaft



No.	Designation	Qty.	Note:	
			Removal	Installation
1	Hexagon nut*	1	Undo with Special Tools 9282 and 9105	Use Special Tools 9282 and 9105 and tighten to 250 Nm (184 ftlb.). Up-set flange to lock
2	Cylindrical roller bearing	1	Press off with suitable separating device (e.g. Kukko 17-1) across 2nd gear fixed gearwheel	Heat to approx. 120 °C
3	Bearing inner race	1	Press off with suitable separating device (e.g. Kukko 17-1) across 2nd gear fixed gearwheel	Heat to approx. 120 °C
4	Four-point bearing	1		
5	Tensioning plate	1		Clamp in a vise (use protective jaws), insert lock (long), engage 5th gear and install input shaft with pinned shift rod/shift fork
6	Bearing inner race	1	Press off with suitable separating device (e.g. Kukko 17-1)	Heat to approx. 120 °C
7	Thrust washer	1		Large, face-ground side faces needle cage
8	Loose gearwheel (4th gear)	1		Replace only in pairs
9	Synchronizing ring	1	Mark for reinstallation	Check for wear, install in correct position using the same gearwheel (lugs face the driver dogs)

No.	Designation	Qty.	Note:	
			Removal	Installation
10	Needle-roller assembly	1	Mark cage	Install with the same gear-wheel
11	Inner race	1		Heat to approx. 120 °C
12	Guide sleeve	1	Remove complete with shift sleeve	Install complete with shaft sleeve and synchromesh components
13	Shift sleeve (3rd and 4th gear)	1	Make sure synchromesh components do not pop out.	Observe installation position, install complete with guide sleeve and synchromesh components. Center the center-punch marks relative to the driver dogs
14	Driver dogs	3		Install in correct position, domed side faces shift sleeve
15	Ball	3		
16	Spring	3		
17	Synchronizing ring	1	Mark for reinstallation	Check for wear, install in correct position with the same gearwheel (lugs face the driver dogs)
18	Loose gearwheel (3rd gear)	1		Replace only in pairs
19	Needle-roller assembly	1	Mark for reinstallation	Fit with the same gear-wheel
20	Drive shaft	1		

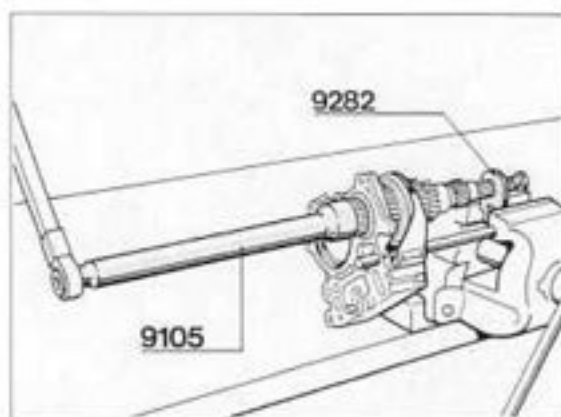
* Transmission shafts without recesses for securing the hexagon nuts have been installed since September 1995.

Self-locking hexagon nuts are used on these shafts, and these nuts must **always** be replaced in every repair.

Dismantling and assembling notes

Dismantling

1. Mount retaining plate 9282 in vise, fit input shaft and undo hexagon-head nut with Special Tool 9105.



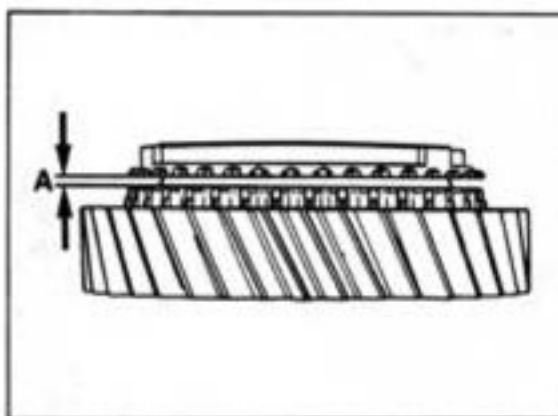
535-35

2. Use a suitable tool (e.g. Kukko 17 - 1) to press all parts off the input shaft over the 2nd gear wheel.

Assembling

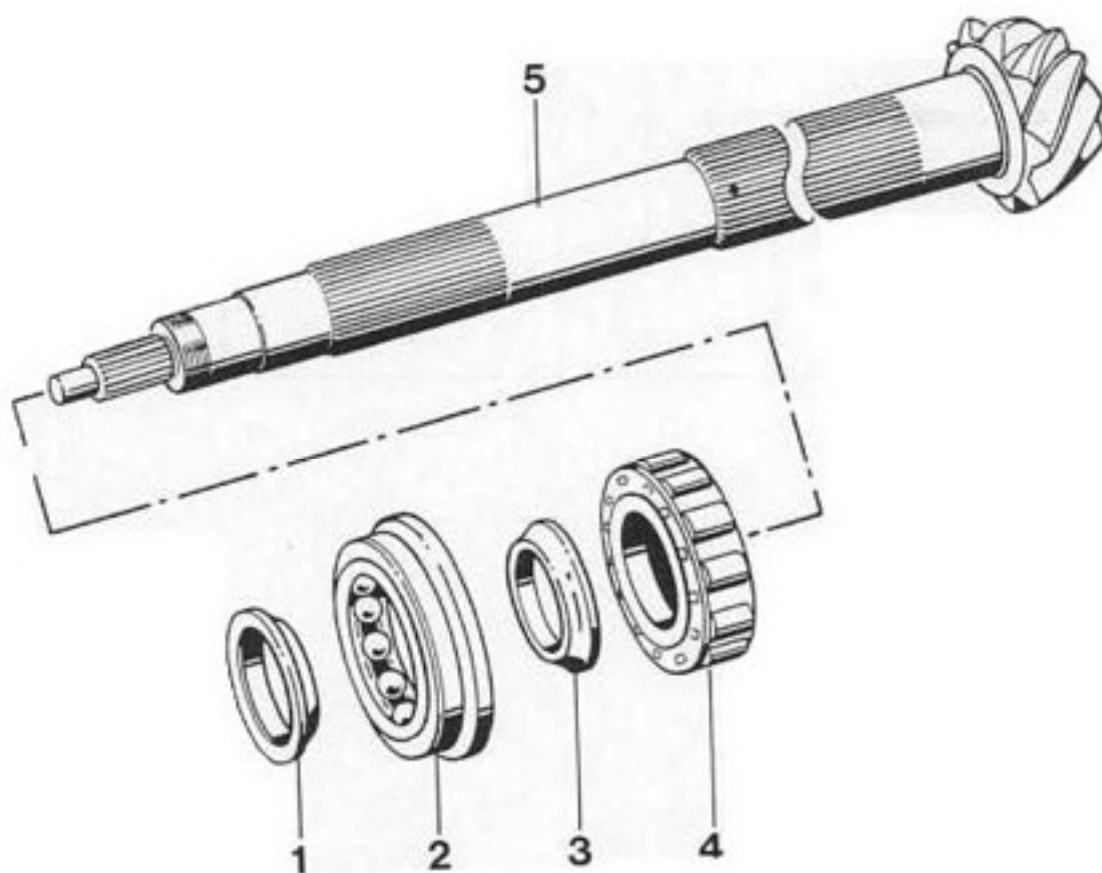
1. To inspect synchronizing rings, press rings onto the gear wheel tapers and use a feeler gauge to measure gap "A".

Assembly dimension (new) = 0.9 mm min.
Wear limit = 0.6 to 0.7 mm



518-35

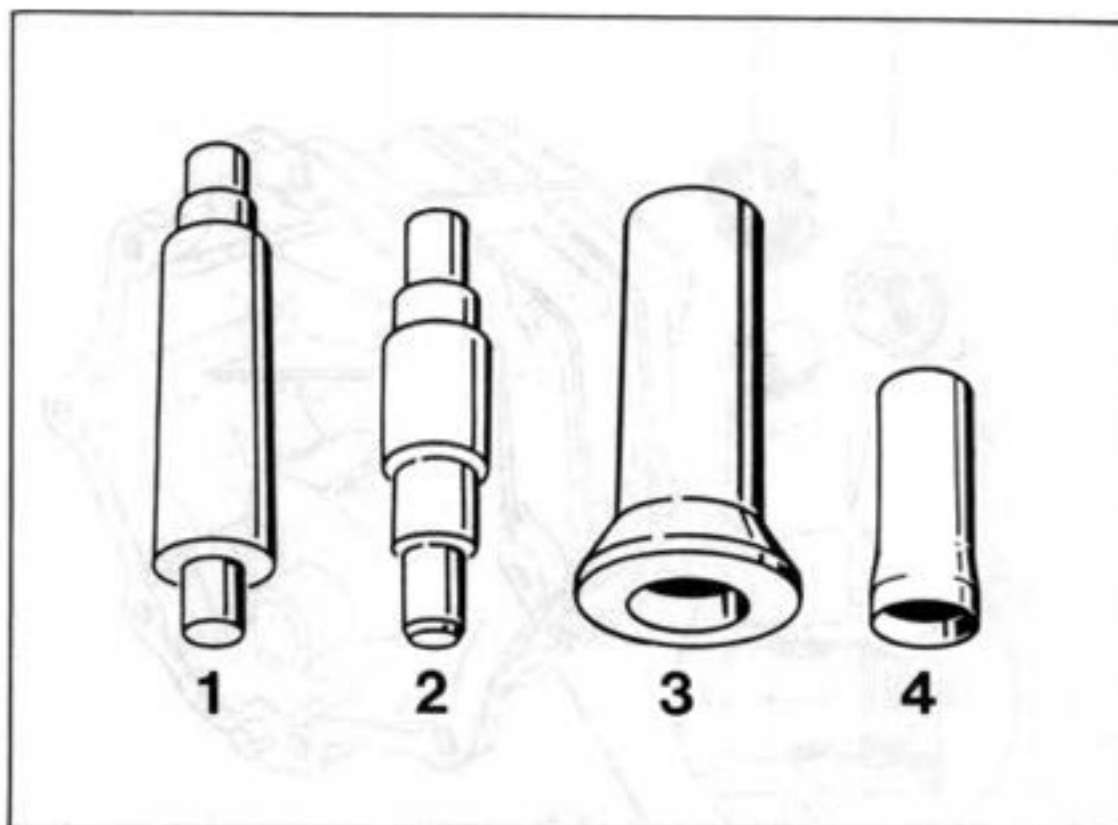
35 59 37 Dismantling and assembling output shaft



No.	Designation	Qty.	Note:	
			Removal	Installation
1	Bearing inner race	1	Mark for reinstallation, press off with suitable separating device (e.g. Kukko 15 - 17)	Do not confuse with inner race No. 3, heat to approx. 120 °C and press into place
2	Four-point bearing	1		
3	Bearing inner race	1	Mark for reinstallation, press off with suitable separating device (e.g. Kukko 15 - 17)	Do not confuse with inner race No. 1, heat to approx. 120 °C and press into place
4	Cylindrical roller bearing	1	Press off with suitable separating device (e.g. Kukko 15 - 17)	Heat to approx. 120 °C and press into correct position using a suitable pipe section (e.g. VW 519)
5	Output shaft	1		Observe matching number. Readjust if required

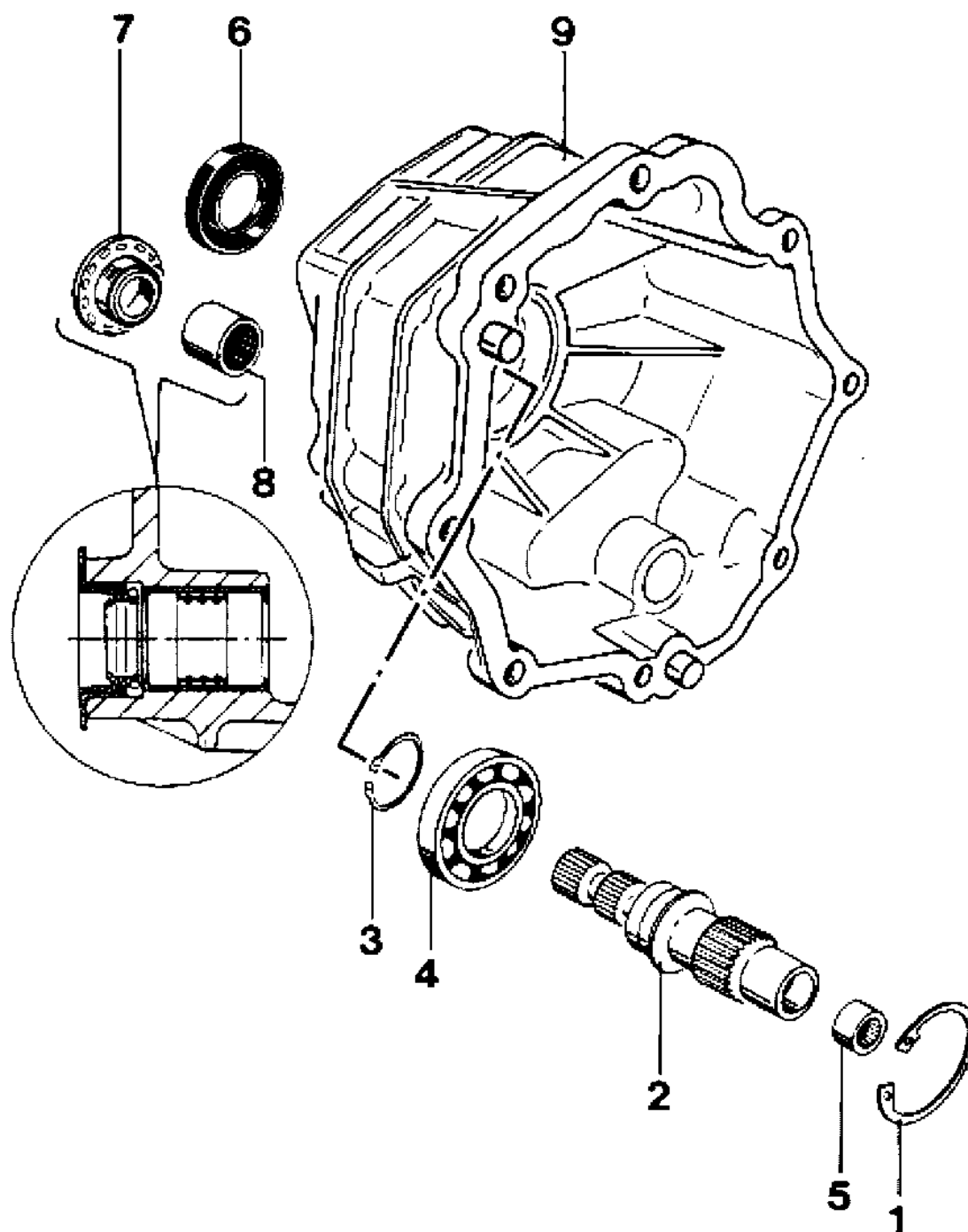
39 66 37 Dismantling and assembling transfer casing

Tools



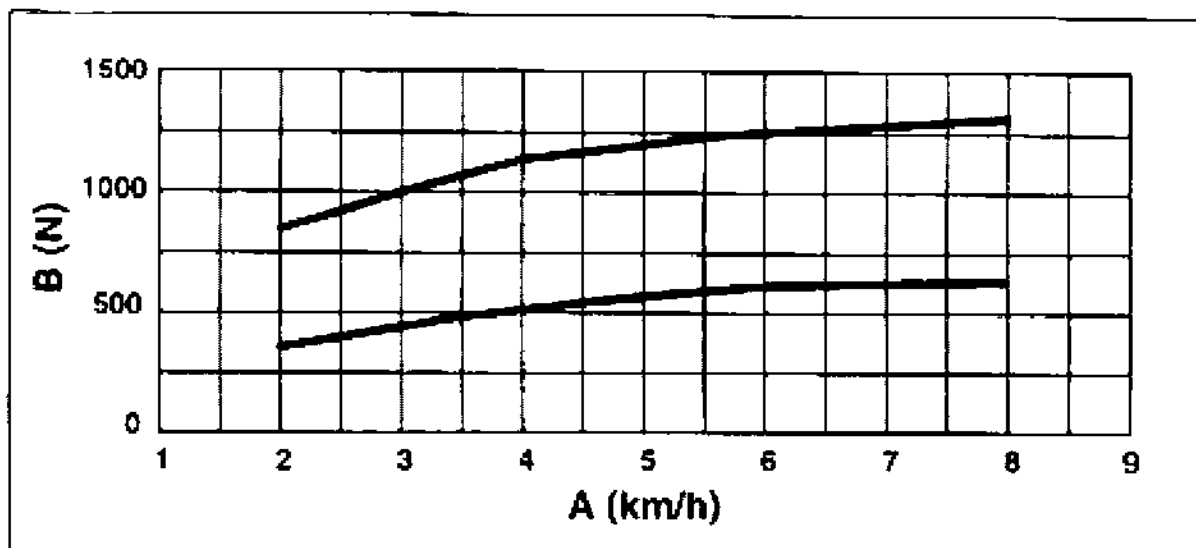
No.	Designation	Special tool	Order number	Explanation
1	Mandrel	9254	000.721.925.40	
2	Mandrel	9223	000.721.922.30	
3	Thrust piece	9234	000.721.923.40	
4	Assembly sleeve	9575	000.721.957.50	

39 66 37 Dismantling and assembling transfer casing



No.	Designation	Qty.	Note:	
			Removal	Installation
1	Snap ring	1		
2	Output shaft	1		
3	Snap ring	1		
4	Deep-groove ball bearing	1	Press off	Heat to approx. 120 °C
5	Needle roller bearing sleeve	1	Pull out with internal puller	Replace, drive in flush with suitable drift (e.g. P 361)
6	Oil seal	1		Replace, drive in with Special Tool 9575 and 9234 after fitting the output shaft
7	Oil seal	1		Replace, pack space between dust lip and sealing lip with grease (e.g. Liqui Moly Pu 53), push in to stop with Special Tool 9254
8	Ball sleeve	1	Press out with suitable drift or pipe section (e.g. VW 423)	Replace, drive home into correct position using Special Tool 9223
9	Housing	1		

39 60 01 Checking operation of installed viscous clutch



2019-39

A = Driving speed of front axle in kilometers per hour

B = Pulling power of front wheels in Newton (N)

Test conditions:

To check, viscous clutch must be at room temperature

Test duration approx. 45 seconds (Determine measurement after 20 seconds)

Repeat measurement after viscous clutch has cooled off for 1 hour

1. Connect system tester and select „Actual values“ menu (ABS speed) (Refer to Vol. VIII, page 45 - 37).
2. Place front wheels of vehicle onto brake dynamometer.
3. Switch off engine, engage parking brake and shift transmission into neutral position.

4. When the rollers have started, switch on ignition and press key > of system tester. The tester will then return to the point where diagnosis was interrupted.

5. Set front wheels into motion with dynamometer rollers (**max. 5 mph**) and determine tractive power of front wheels after 20 seconds.

Note

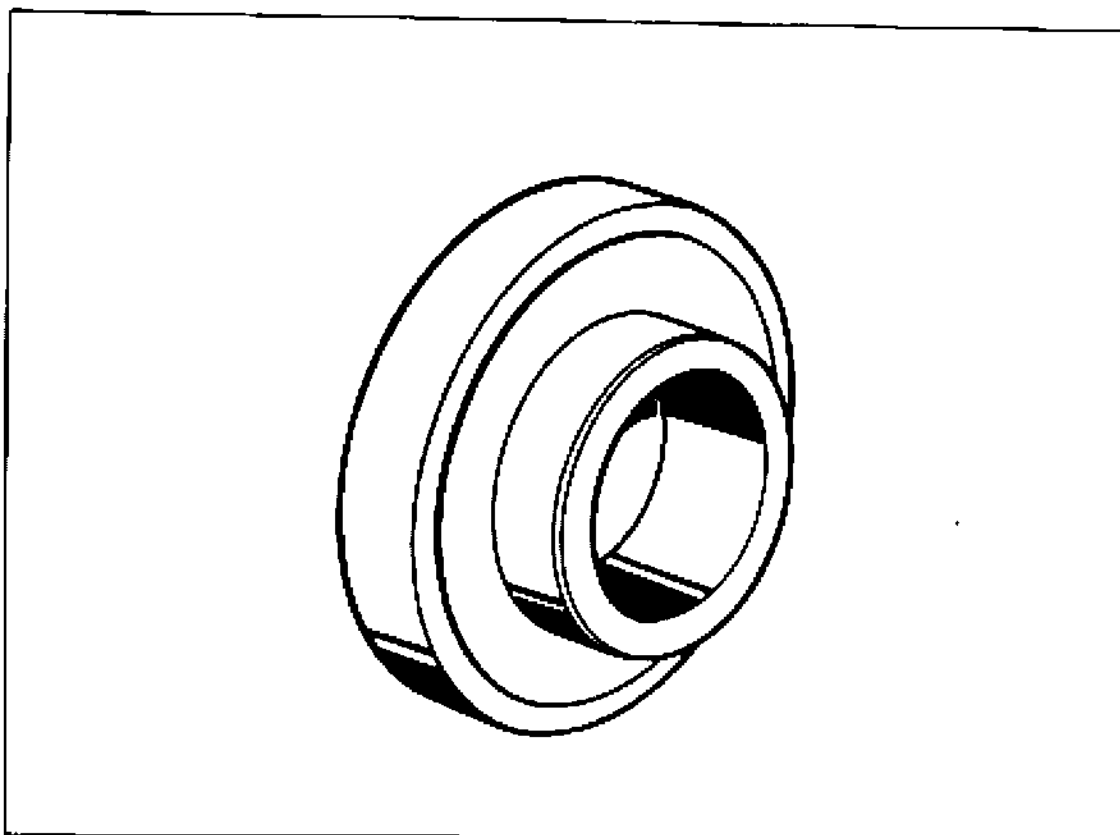
When measuring the wheels individually, add measurements obtained for both front wheels.

5. The viscous clutch is operative if the tractive power of the front wheels is within the boundary curves on the diagram.

The driving speed of the front axle must be taken into account during these measurements (speed can be read off on System Tester 9288).

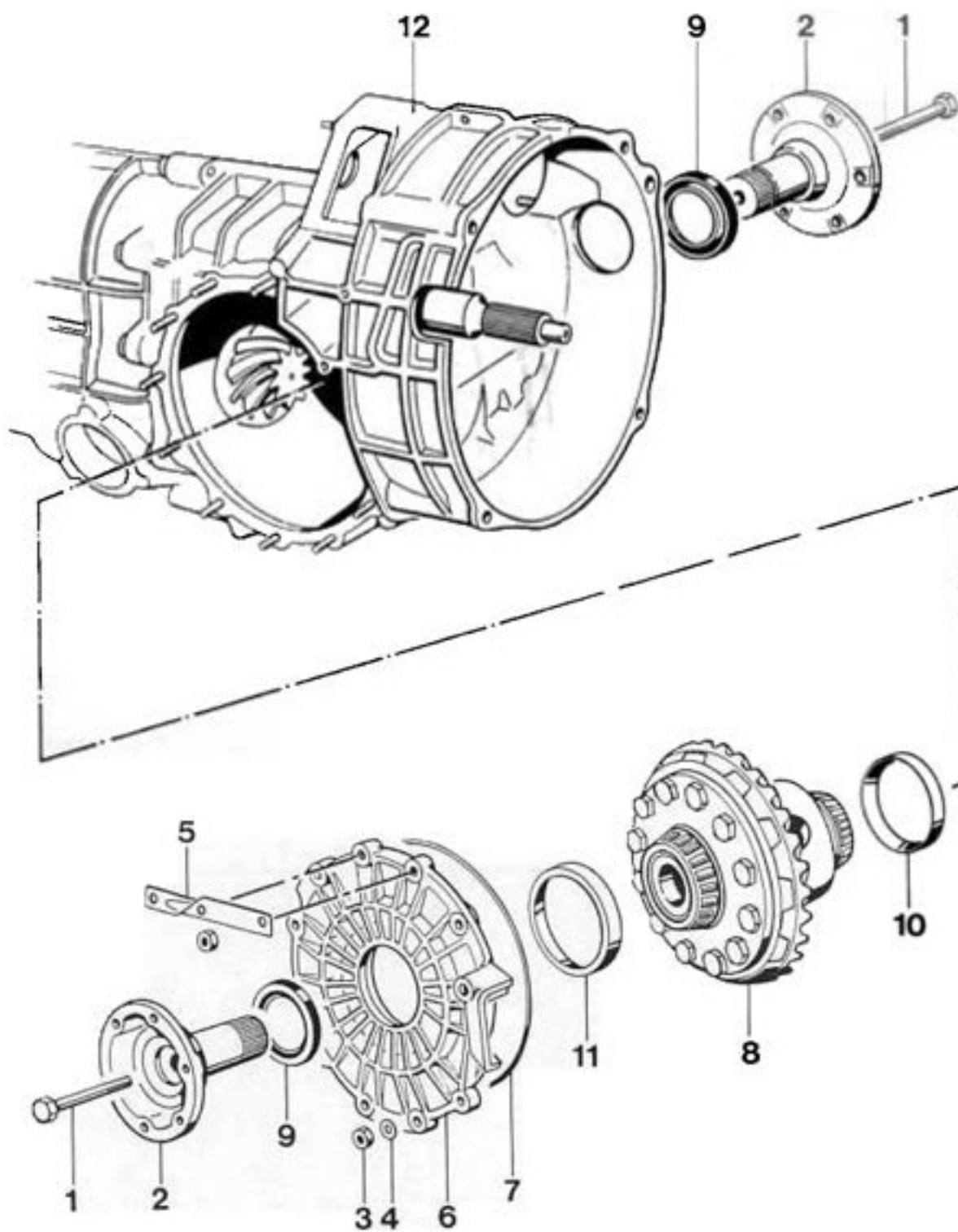
Example:

At a vehicle speed of 3 mph at the front axle, the tractive power must be between approx. 580 N and 1180 N if the viscous clutch is o.k. (cf. diagram).

39 40 19 Removing and installing differential lock**Tools**

No.	Designation	Special tool	Order number	Explanation
	Thrust piece	9252	000.721.925.20	

39 40 19 Removing and installing differential lock

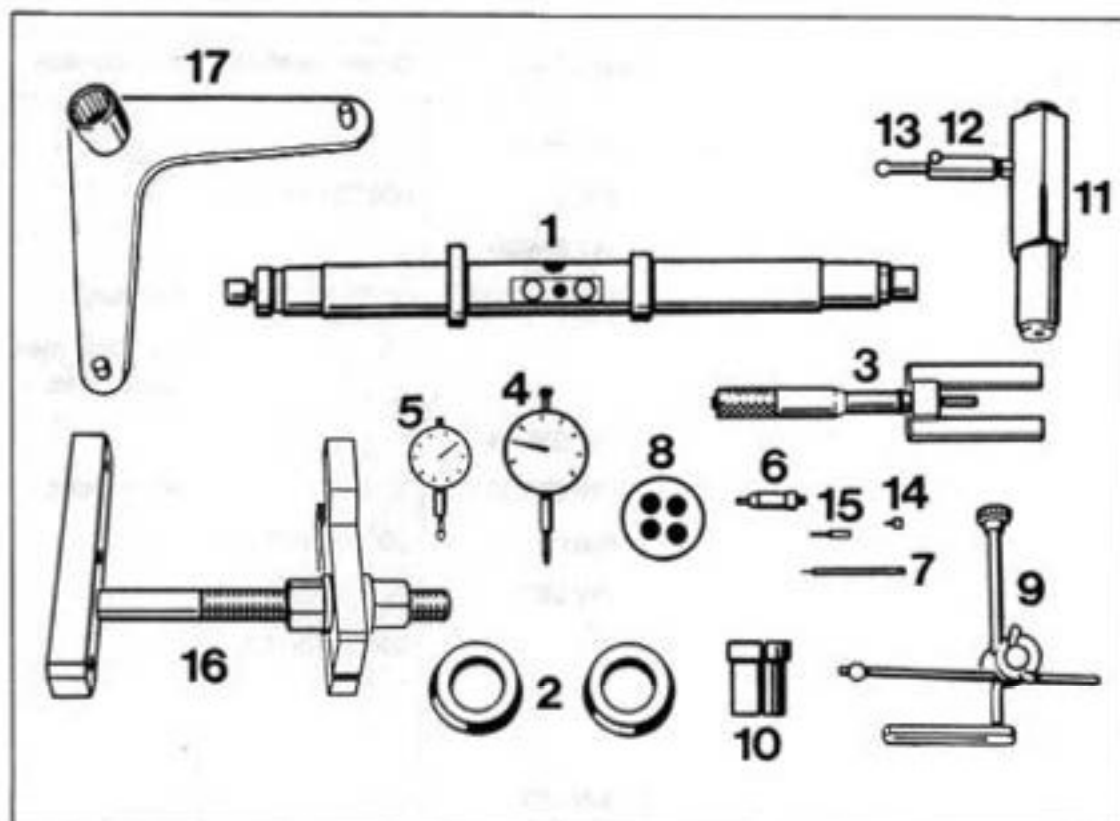


No.	Designation	Qty.	Note:	
			Removal	Installation
1	Hexagon head bolt	2		Tighten to 44 Nm (32 lbf·ft)
2	Joint flange*	2		
3	Hexagon head nut	11		Tighten to 23 Nm (17 lbf·ft)
4	Washer	11		
5	Holder	1		
6	Cover	1		
7	O-ring	1		Replace, oil lightly, do not twist
8	Differential or limited-slip differential	1		Readjust if required
9	Seal*	2		Pack space between dust and sealing lips with grease (e.g. Liqui Moly Pu 53). Use thrust piece 9252 to press in until it is seated against the stop
10	Bearing outer race	1	Pull out with suitable internal puller (e.g. Kukko 21 - 8)	Press in with suitable thrust piece
11	Bearing outer race	1	Pull out with suitable internal puller (e.g. Kukko 21 - 9)	Press in with suitable thrust piece
12	gear box	1		

* The halfshaft flange and seal ring can also be removed and installed with the transmission installed.

39 08 15 Adjusting drive set

Tools



39 08 15 Adjusting drive set**Tools**

No.	Designation	Special tool	Order number	Explanation
1	Measuring mandrel	VW 385/1		
2	Centering disks	9109	000.721.910.90	
3	Master gauge	VW 385/30		
4	Dial gauge	-		Standard
5	Dial gauge	-		Standard, measuring range 3 mm
6	Gauge plunger	VW 385/14		
7	Dial gauge extension	VW 385/56		30 mm long
8	Gauge block plate	9281	000.721.928.10	
9	Dial gauge bracket	VW 387		
10	Clamping sleeve	9145	000.721.914.50	
11	Adjusting device	VW 521/4		
12	Measuring lever	VW 388		
13	Gauge plunger	VW 388		
14	Dial gauge extension	VW 382/10		6.0 mm long
15	Dial gauge extension	VW 385/15		9.3 mm long
16	Clamping tool*	9577	000.721.957.70	
17	Holder	9253	000.721.925.30	

Practical procedure when readjusting the drive set

If it is necessary to adjust drive pinion and ring gear, follow the below sequence to ensure an efficient working procedure:

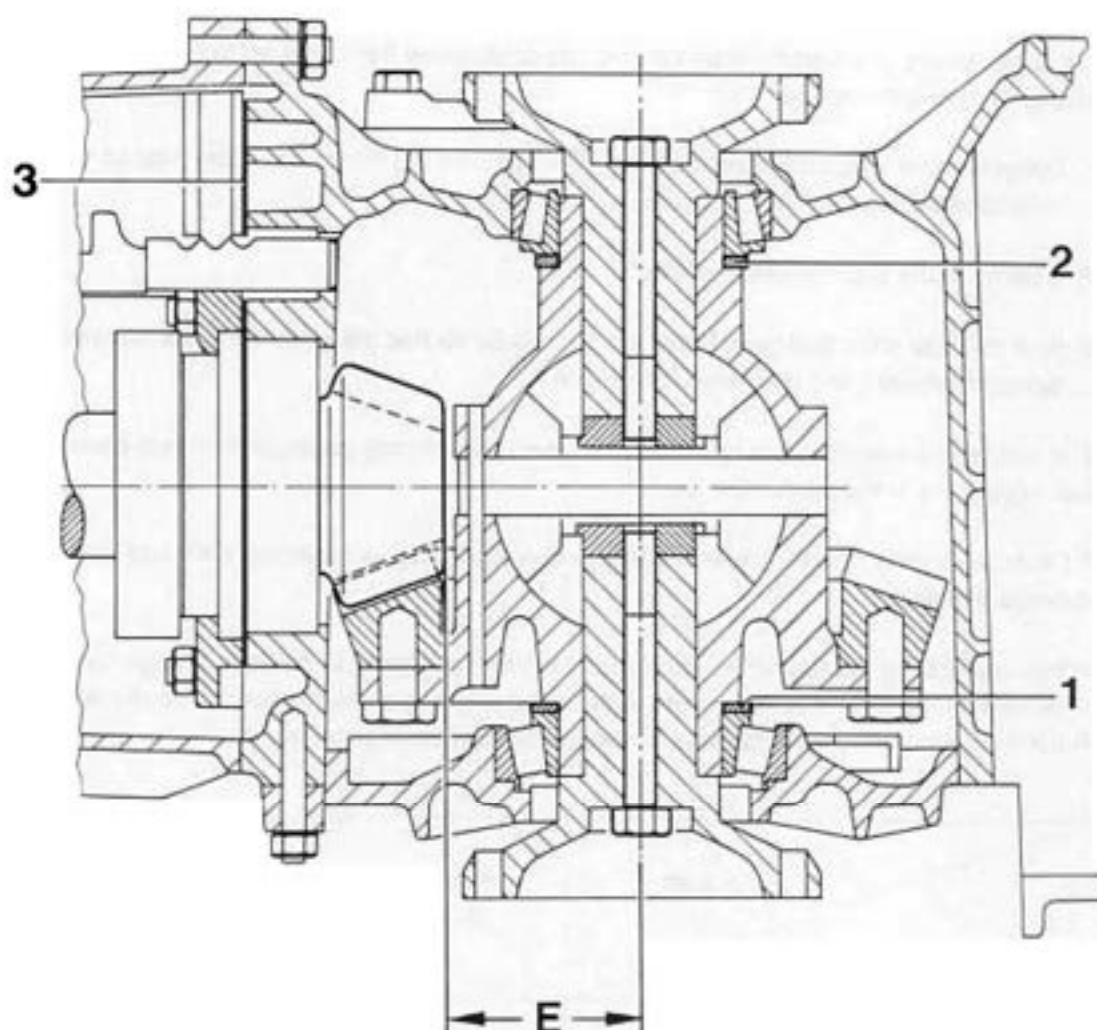
1. Determine the total thickness of shims "Stot" (S_1 plus S_2) for the specified preload on the tapered roller bearings/differential.
2. Determine the thickness of shim " S_3 ".
3. Split the total shim thickness "Stot" into S_1 and S_2 so that the specified circumferential backlash is present between ring gear and drive pinion.

The aim of this adjustment is to restore the smoothest running position which has been achieved on test equipment in the production line.

To achieve correct results, greatest possible cleanliness for all assembly work and measuring procedures is essential.

When assembling the final drive assembly, it is only necessary to readjust drive pinion and ring gear or drive set if components have been replaced which have a direct influence on the adjustment. Refer to the following table to avoid unnecessary adjustment procedures.

Adjust:	Ring gear ($S_1 + S_2$)	Drive pinion (S_3)
Replaced component		
Transmission case	x	x
Lateral transmission cover	x	
Large cylindrical roller bearing and four-point bearing for drive pinion	x	x
Drive set	x	x
Differential housing	x	
Tapered roller bearing for differential	x	

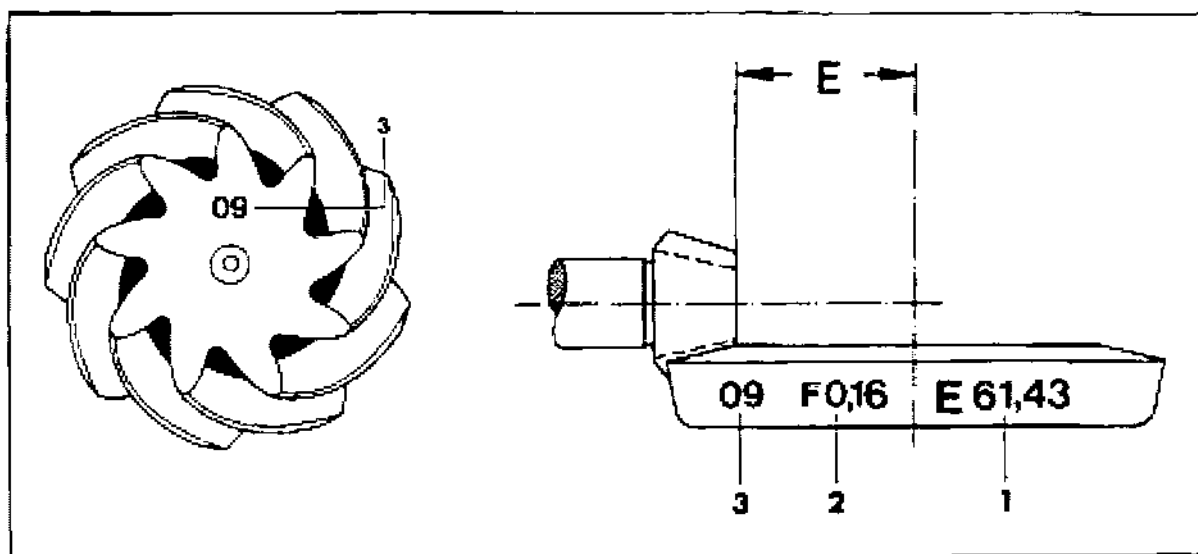


- 1 – Shim S_1
- 2 – Shim S_2
- 3 – Adjusting shim S_3
- E – Setting value

Correct results may only be achieved if assembly work and measuring procedures are carried out carefully and with maximum cleanliness.

General

The setting of drive pinion and ring gear is a determining factor for the service life and smooth running of the rear-axle drive. Drive pinions and ring gears that have been checked for good tooth contact pattern and low noise in both directions of rotation on special test equipment are therefore matched during production. The position at which smoothest running can be achieved is determined by shifting the drive pinion axially, and embossed on the ring gear as setting value "E".



1 = setting "E" (e.g. 61.43 mm)

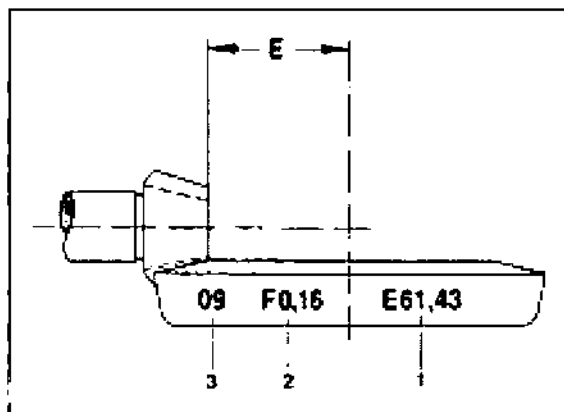
2 = circumferential backlash "F"
(e.g. 0.16 mm)

3 = matching number

39 08 15 Adjusting drive pinion

Note

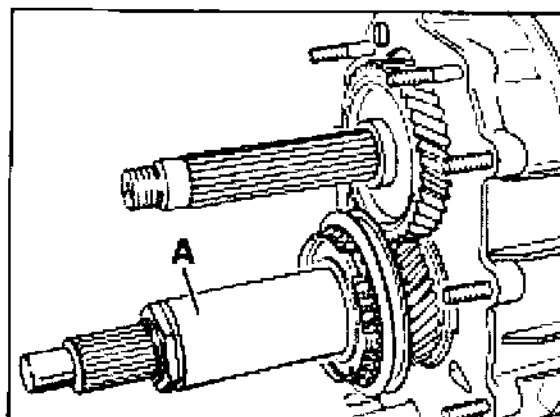
The setting value "E" is indicated on the ring gear.



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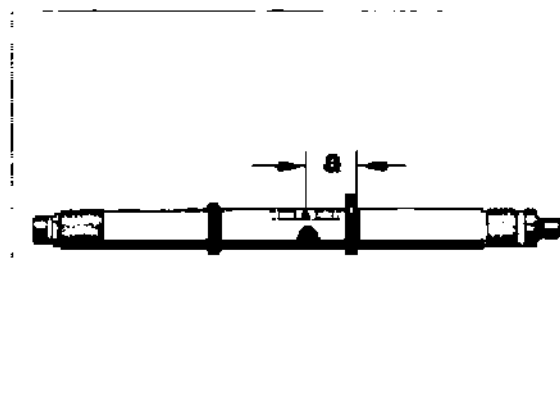
1 = Setting value "E"

1. Install complete gear set without "S3" shims and tighten all tensioning plate hexagon-head nuts to **23 Nm** (17 ftlb.).
2. Fit gear housing and locate with three nuts.
3. Install and engage sixth gear.
4. Block input shaft with Special Tool 9253 and tighten drive pinion collar nut to **300 Nm** (221 ftlb.).



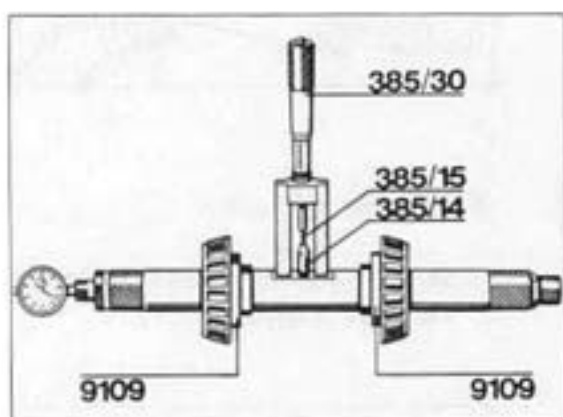
A = Suitable pipe section

5. Set adjusting ring of measuring mandrel **VW 385/1** to dimension "a".



a = approx. 65 mm

6. Assemble measuring mandrel with centering disks **9109**, tapered roller bearing, plunger **VW 385 /14** and dial gauge extension **VW 385/15**.
7. Set measuring mandrel with master gauge **385/30** to the setting value (61.43 mm in the example). Set dial gauge (3 mm measuring range) to zero with 1 mm preload.



543.39

8. Put gauge block plate **9281** on drive pinion head and insert measuring mandrel into transmission case. Dial gauge extension is located in the area of the gauge block plate.



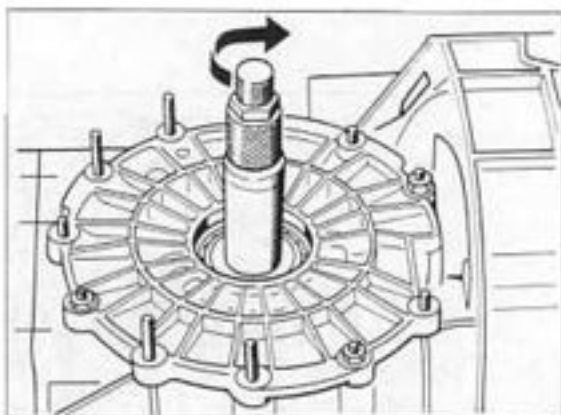
544.39

9. Fit lateral transmission cover without O-ring and tighten crosswise with 4 nuts.

Note

Do not use a hammer when fitting the lateral transmission cover (the gauge block plate held by magnets might fall off). Fit cover in installation position only by tightening the nuts uniformly.

10. Pull second centering disk with spindle towards the outside until the measuring mandrel can just be turned by hand.



545-20

Example:

If the small pointer on the dial gauge is between 1 and 2 and the large pointer indicates 0.37 mm, then 0.37 mm is the shim thickness (with 1mm gauge preload) to be inserted. Always round up or down to the nearest 0.05 mm (e.g. 0.37 mm to 0.35 mm).

12. After inserting the necessary shims, check the setting value "E" again. A deviation of ± 0.03 mm is permissible.

11. Turn measuring mandrel carefully until the dial gauge extension is vertical to the face of the drive pinion head. At this point, the pointer of the dial gauge reaches maximum deflection (reverse point) and the dial gauge must be read.

Note

The measured value always deviates from the set dimension clockwise (the smaller pointer on the dial gauge is between 1 and 2), i.e. if the dial gauge is set with a preload of 1 mm, the value deviating from 1 is taken as shim thickness "S₃".

Adjusting ring gear

Determine total shim thickness "S tot."
($S_1 + S_2$).

The ring gear must be adjusted, if the:

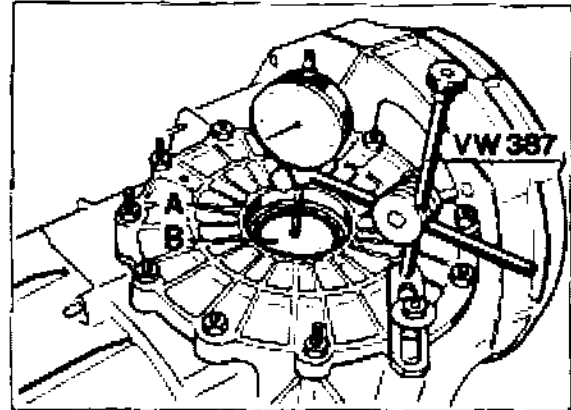
transmission case,
lateral transmission cover,
tapered roller bearing for differential,
differential housing or drive set
have been replaced.

Note

The drive pinion must be removed to determine the preload of the differential tapered roller bearings.

1. Make sure that the bearing outer races of the tapered roller bearings are well seated in the transmission case or lateral transmission cover, respectively.
2. Fit one spacer ring (2.5 mm thick) on the ring gear side and on the opposite side of the differential to be used.
3. Insert differential into transmission case and rotate several times.
4. Fit lateral transmission cover without seal and tighten all hexagon-head nuts to **23 Nm** (17 ftlb.).
5. Put gauge block plate **VW 385/17** on the collar of the differential.

6. Fasten universal dial gauge holder **VW 387** with dial gauge and extension to the case and set to 0 with 2 mm preload.



546-39

- A = Dial gauge extension (approx. 30 to 40 mm long)
B = Gauge block plate VW 385/17

7. Move differential up and down. Read off backlash on the dial gauge and note.

Note

Do not turn differential while measuring backlash as this will give an incorrect reading.

8. Calculate "S tot.".

"S tot." = Fitted shim thickness
+ Measured value
+ Pressure fit of tapered roller bearing

Example

Thickness of shims inserted	5.00 mm
Measured value	0.75 mm
Pressing (constant value)	<u>0.40 mm</u>
"S tot."	6.15 mm

9. Remove differential, pull off both tapered roller bearings and split calculated shim thickness "S tot." as follows.

Select spacer S₁ 0.70 mm thinner and S₂ 0.70 mm thicker as a starting point for subsequent adjustment of the backlash.

Example:

Calculated thicknesses

$$S_1 + S_2 = 2.375 + 3.775 = 6.15 \text{ mm}$$

Rounded thicknesses

$$S_1 + S_2 = 2.35 + 3.80 = 6.15 \text{ mm}$$

Measure shims with a micrometer in several places. Permissible deviation 0.02 mm. Also check shims for burrs and damage.

Example

Total shim thickness of spacers

$$S_1 + S_2 = 6.15 \text{ mm}$$

Thickness of spacer S₁

$$\begin{array}{rcl} \frac{6.15 \text{ mm}}{2} & = & 3.075 \text{ mm} \\ & - & 0.700 \text{ mm} \\ \hline & & 2.375 \text{ mm} \end{array}$$

Thickness of spacer S₂

$$\begin{array}{rcl} \frac{6.15 \text{ mm}}{2} & = & 3.075 \text{ mm} \\ & + & 0.700 \text{ mm} \\ \hline & & 3.775 \text{ mm} \end{array}$$

Note

Spacers are available in thicknesses of 1.6 to 3.1 mm in increments of 0.10 mm.

By using a 0.25 mm shim, the shim thicknesses may be graduated in increments of 0.05 mm.

The calculated shim thicknesses must be rounded up or down for plausible dimensions that will not alter the total thickness S₁ and S₂.

Adjusting circumferential backlash

Note

The backlash to be set is embossed on the ring gear.

1. Mount gear set using shims "S₃" determined while adjusting the drive pinion.

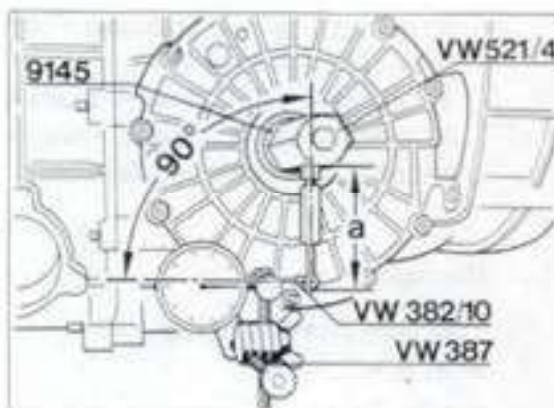
Make sure that the collar nut of the drive pinion is tightened to **300 Nm** (221 ftlb.) before measuring backlash.

2. Insert differential with tapered roller bearing and shims (S₁ + S₂) into the housing.
3. Fit lateral transmission cover and tighten all hexagon-head nuts to **23 Nm** (17 ftlb.).

Always make sure that there is a certain amount of backlash when tightening the nuts. Never allow the drive pinion to seize.

4. Assemble measuring lever **VW 388** and adjusting device **VW 521/4** and adjust lever length to 80 mm with the plunger. Refer to dimension "a" in the picture.
5. Insert adjusting device with clamping sleeve (Special Tool **9145**) into the differential and clamp firmly.
6. Rotate differential in both directions several times to settle the tapered roller bearings.

7. Fit universal dial gauge holder with flat extension in such a way as to produce a right angle between dial gauge axis and lever.



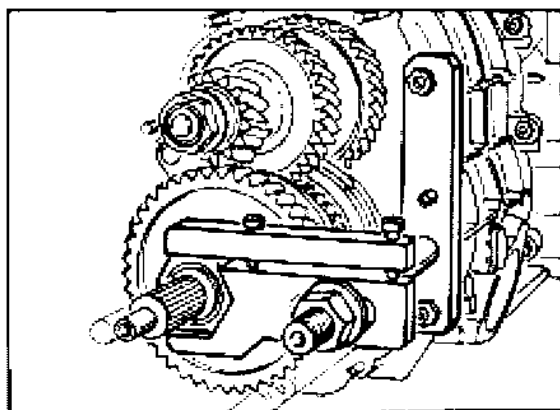
548-29

Dimension "a" = approx. 80 mm

8. Turn ring gear carefully at the clamping screw of the adjusting device as far as the stop and set the dial gauge to zero. Turn back ring gear and read off circumferential backlash. Note down the reading.

Note

When carrying out measurements, the drive pinion must be blocked with Special Tool 9562.



1710-39

9. After turning the ring gear a further 90°, repeat measuring procedures three times. The measured values must not deviate from one another by more than 0.03 mm.

Note

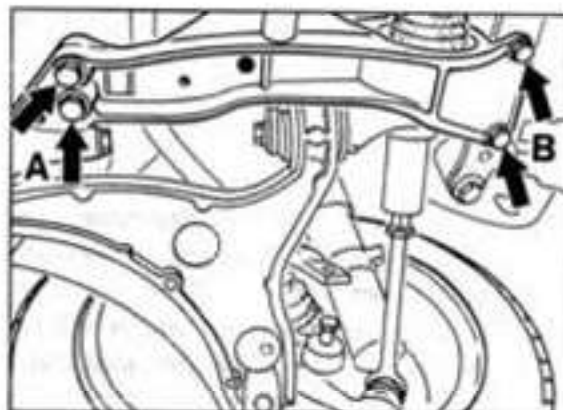
The backlash to be adjusted is embossed on the ring gear. A deviation of ± 0.03 mm is permissible

10. If the required backlash cannot be obtained, replace spacers ($S_1 + S_2$) again. The total shim thickness ("S tot.") must not be altered, however.

39 88 19 Removing and installing front-axle final drive

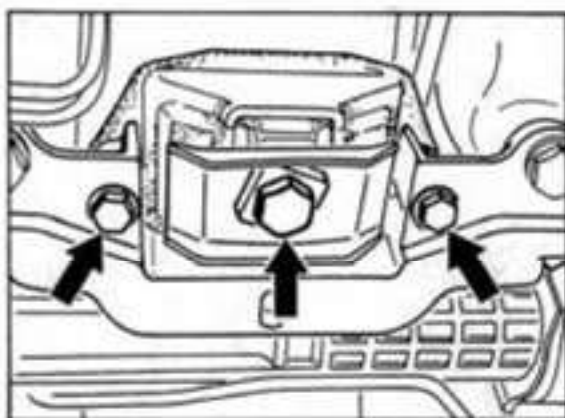
Removal

1. Remove front and center underside panel.
2. Unclip power steering lines from steering gear.
3. Separate drive shafts with Special Tool 9581 at transmission end.
4. Screw out mounting bolts of transmission mount, raise front axle final drive and take out mounts.



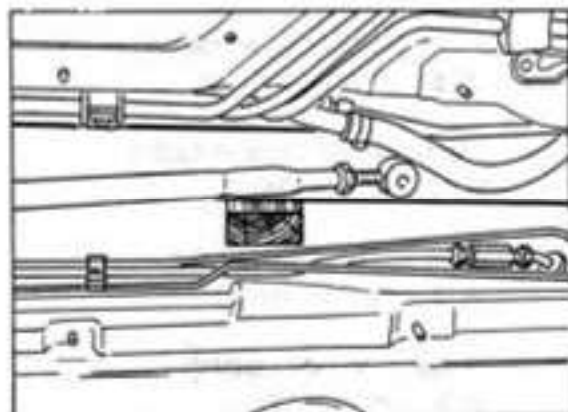
1964-39

- A - Remove bolts
B - Slacken bolts only (approx. 5 mm)



1963-39

5. Lift front-axle final drive and take out final drive mount.
6. Screw out mounting bolts "A" of side member and slacken bolts "B" only (by approx. 5 mm).



1965-39

9. Push front axle final drive out of splines and take out from below at an oblique angle. Take care not to damage the brake pipe at the spare wheel well. Place front axle out of the way so that the pinion shaft faces up.

Note

If body tolerances are unfavorable, slacken mounting bolts of front axle final drive by approx. 5 mm.
This will allow the crossmember and the central tube to move down somewhat and will make removal easier.

Installation

1. Check universal-joint shaft for true running (see page 40 - 103).
2. Check front-axle final drive oil and top up if required.
3. Measure front axle.

4. Tightening torques: (= ftlb)

Front-axle final drive mount to central tube

M 12 85 Nm (63)

Front-axle final drive mount to cross member

M 8 23 Nm (17)

Front-axle final drive to central tube

M 10 46 Nm (34)

Drive shaft to transmission

M 8 42 Nm (31)

Stabilizer to side member

M 8 23 Nm (17)

Stabilizer to stabilizer mount

M 10 46 Nm (34)

Side member to body / cross member

front / rear

M 12 90 Nm (66) / M 10 46 Nm (34)

cross member to body

outer / inner

M 12 105 Nm (77) / M 10 48 Nm (35)

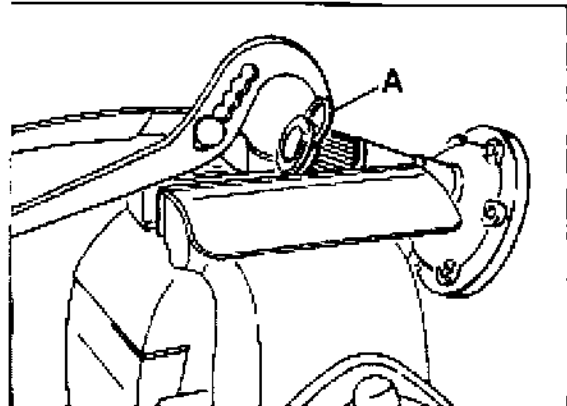
39 59 19 Removing and installing oil seal of halfshaft flange (front axle final drive)

Note

The seals may also be replaced with the front axle final drive remaining installed.

Removal

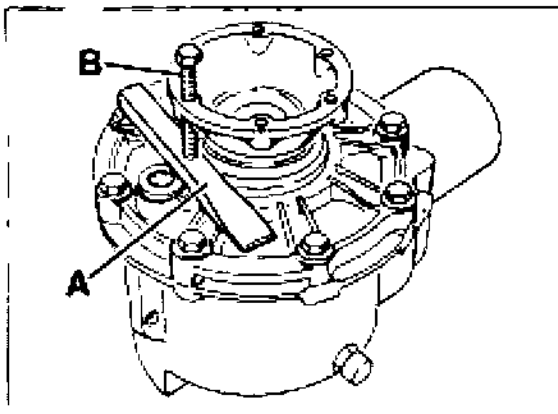
1. Remove halfshaft flange by placing a chisel or spacer „A“ below the halfshaft flange and turning bolt „B“ to pull out halfshaft flange. In case the halfshaft flange is tilted, screw another bolt into the opposite hole.



1994-39

A = New snap ring

3. Grease snap ring liberally and press in halfshaft flange with a suitable drift.



1993-39

2. Lever out seal with a suitable tool.

Installation

1. Pack space between dust and grease lips with grease (e.g. Liqui Moly Pu 53) and drive seal home to stop, using Special Tool 9537.
2. The snap ring of the halfshaft flange must always be replaced. Use protective jaws to mount flange in a vise and press out snap ring with a new snap ring „A“.

39 55 19 Removing and installing output shaft oil seal (Front axle final drive)

Removal

1. Remove front axle final drive (also refer to page 39 - 223).
2. Lever out seal with Special Tool **VW 681**.

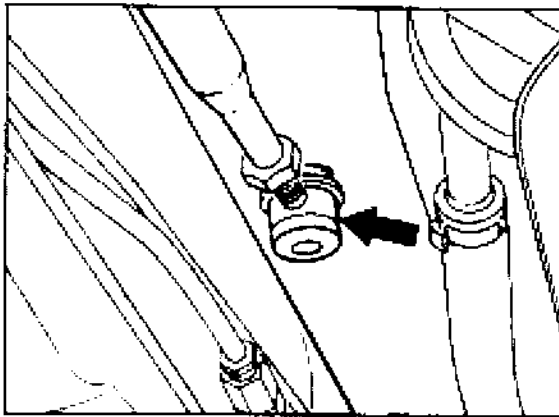
Installation

1. Using Special Tool **9545**, press in seal to stop.

39 03 19 Removing and installing central tube

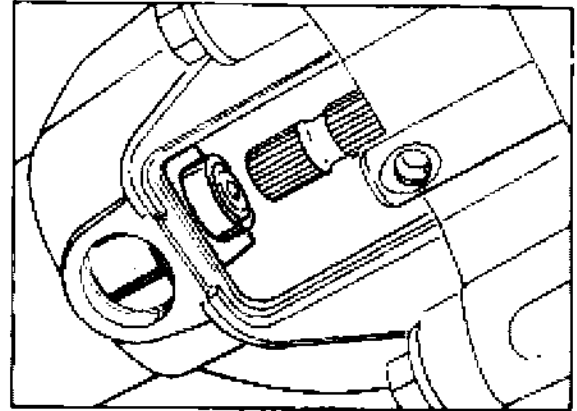
Removal

1. Remove centre console.
2. Remove underside panels.
3. Remove fit bolt for gearshift rod coupling.
4. Lift guide tube off ball joint.

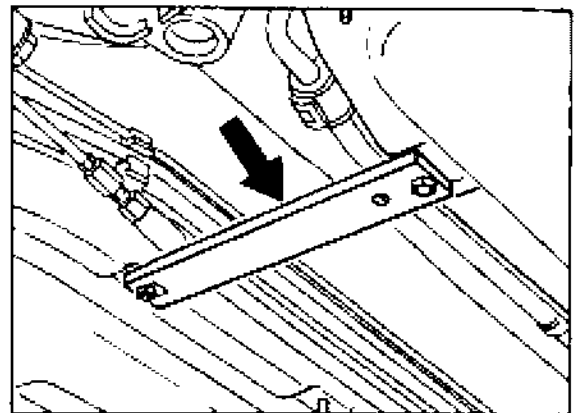


2238-39

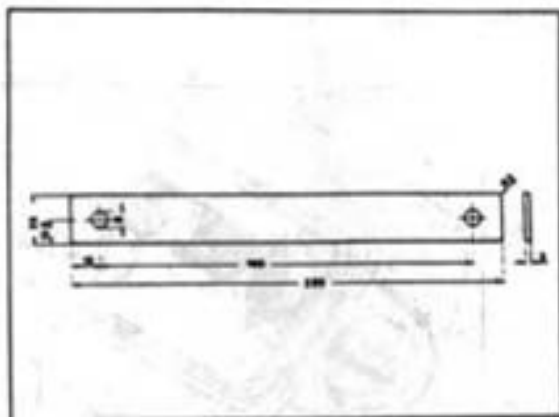
5. Remove shift lever with gearshift rod and guide tube.
6. Remove front axle final drive.
(see page 39 - 225).
7. Completely unscrew the two mounting screws of the clamping sleeve and push it forwards on the central tube.



8. Apply flat steel parts (made in workshop) to support central tube.

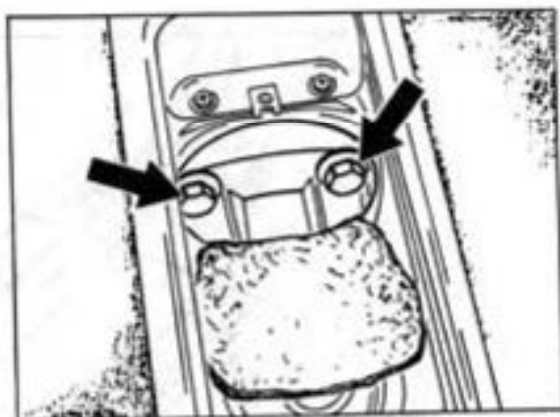


2022-10



2029-10

9. Unscrew manual transmission mounting screws (the top two screws can be reached from the passenger compartment).



2025-10

Two persons are required for the removal of the central tube.

Installation

Torque specifications

Front axle final drive and suspension
(see page 39 - 226)

Frame to tunnel:

M 6 = 10 mm

Gearshift rod to transmission - internal shift rod:

M 8 = 23 Nm (17 ftlb)

Slides to guide tube:

M 5 = 6 Nm (4.5 ftlb)

Double clamp to central tube:

M 8 = 35...40 Nm (26...29.5 ftlb)

Central tube to manual transmission

M 12 = 85 Nm (63 ftlb)

1. The parts removed must be installed in reverse order of removal.
2. Make sure that the centering pins on the transfer casing are properly positioned.



2240-39

3. Place insulation in correct position on tunnel.

4. Apply sliding compound (e.g. Contifix) to central tube in area of insulation and push central tube in carefully, making sure that the insulation is properly positioned.
5. Check adjustment of gearshift.
(see page 34 - 31).

Repair Manual

911 Carrera **(993)**

Volume III: Transmission Automatic

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	Engine – Crankshaft, pistons	13
	Engine – Cylinder head, valve drive	15
	Engine – Lubrication	17
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	Fuel supply, control	20
	Exhaust system – Turbocharging	21
	Fuel system, electronic injection	24
	Fuel system, K-Jetronic	25
	Exhaust system	26
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Volume II:	Transmission	3
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Volume III:	Transmission	3
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Preface

Structure

The "Technical Literature" for the "911 Carrera (993)" model is basically structured as before, i.e. the structure follows the familiar repair groups.

A new feature is that the structure includes the main groups 0 to 9 and the main group D.

Main groups:	0	Complete vehicle – General
	1	Engine
	2	Fuel, exhaust, engine electrical system
	3	Transmission
	4	Chassis
	5	Body
	6	Body equipment, outside
	7	Body equipment, interior
	8	Air conditioning
	9	Electrical system
	D	Diagnosis

Layout

The layout in the below items remains unchanged throughout the repair manual

1. Table of tightening torques
2. Special tools required
3. Exploded views
4. Legends for the exploded views
5. Assembly notes / use of special tools

As a new feature, however, the former item 6 (Repair group diagnosis) is no longer filed in the volume corresponding to the respective repair group. The **Diagnosis test plans / diagnosis procedures** have been combined in a **separate Diagnosis volume** broken down according to the main groups 0 to 9.

Another new feature is that the contents of the "Service Information Technik" are indicated in the Repair Manual. This brochure concentrates on a description of the design and function of components and of the new features introduced for a particular model year.

Service Number

All major repair procedures and repair descriptions are identified by a two- or four-digit **Service Number** completed by two additional digits to identify the work that corresponds to the first six digits of the working position number in the Working Times and Damage Catalog.

Example: 30 37 37 Dismantling and assembling clutch control shaft

Explanation: 30 37 37 50 (full working position number)

Repair group

here: Clutch, control

Component designation

here: Clutch control shaft

Activity

here: Dismantling and assembling

Index

here: Removed

Presentation in the various documents

30 37 37 50	Working position no. from Working Times and Damage Catalog , consisting of repair group, component designation, activity and index
30 37 37	Six-digit number in Repair Manual , consisting of repair group, component designation and activity
	Service number in Service Information , consisting of repair group and component designation

Goal

The introduction of a service number in the "technical literature" is intended to facilitate standardization and positive identification to allow direct cross-referencing among the various documents. This is of particular importance with regard to the use of electronic media.

Survey of contents of Service Information Technik '95

The Service Information gives a detailed description of the technical features of the new 911 Carrera.

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Pistons	13	1 - 10
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Operation of the ABD	45	4 - 30
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Body - interior trim	70	7 - 1
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Particle filter	85	8 - 1
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	Rep. Gr.	Page
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III Transmission Automatic Transmission

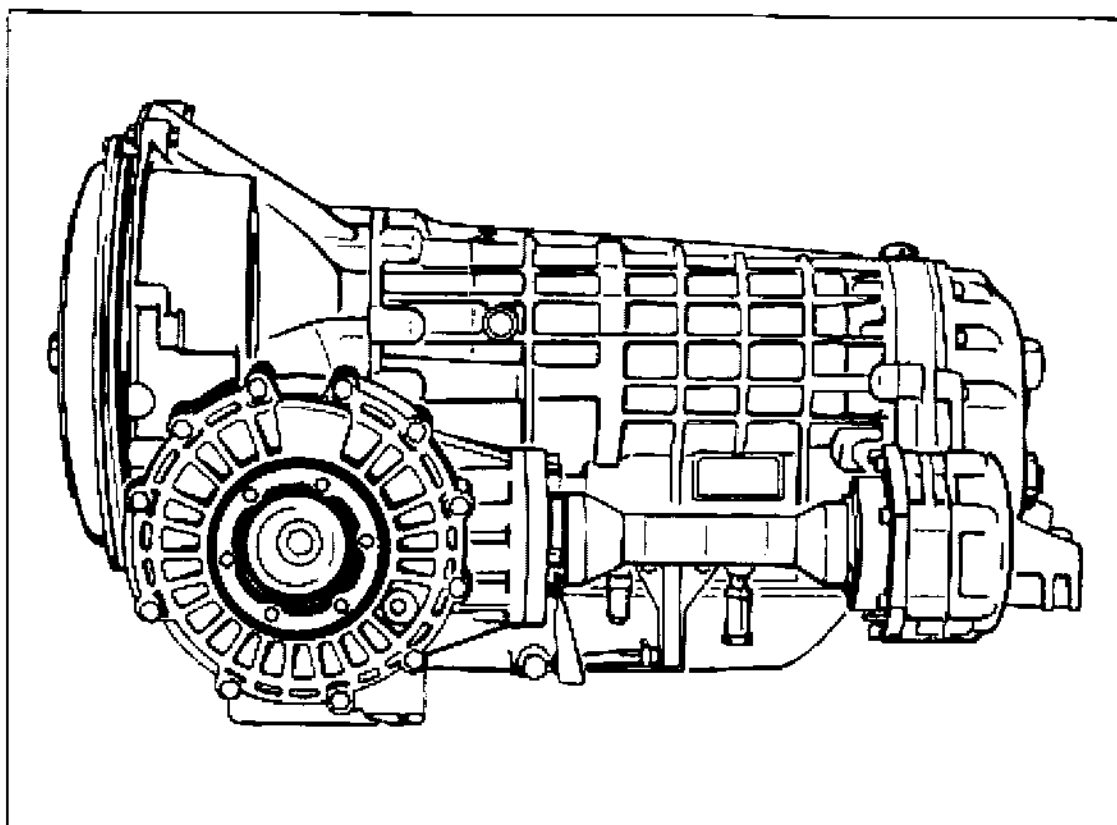
3 Transmission

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39 08 15	Adjusting drive set	39 - 127

3 Technical Data

4-speed Tiptronic transmission A50



Type	Code letter	Version	Installed in	Model year
A 50/04		4-speed	911 Carrera worldwide except USA and Taiwan	'94 / '95
A 50/05		4-speed	911 Carrera USA, Taiwan	'94 / '95

3 Technical Data

General data	A50/04	A50/05
Type	fully automatic 4-speed planetary transmission (Tiptronic)	
Gear ratios		
spur gear	1.100	1.100
1st gear	2.479	2.479
2nd gear	1.479	1.479
3rd gear	1.000	1.000
4th gear	0.728	0.728
reverse	2.086	2.086
final drive	hypoid bevel gear with 15 mm offset	
final drive ratio	9:33 = 3.667	9 : 32 = 3.556
stall speed	2300 - 400	2300 - 400
oil volume for final drive	approx. 0.9 l multi-grade transmission oil 75 W 90 API specification GL5 (MIL-L 2105 B), or SAE 90	
oil volume for automatic transmission with torque converter	total volume approx. 9.5 l oil change approx. 3.5 l ATF-Dexron II D	

3 Technical Data

Torque specifications for Tiptronic transmission

Location	Thread	Tightening Torque Nm (ftlb.)
Multifunctional switch to transmission	M 6 x 25	10 (7)
Selector lever to selector shaft	M 8 x 1	15 (11)
Long halfshaft flange to transmission housing	M 8	23 (17)
Short halfshaft flange to differential	M 10 x 60	46 (34)
Plug to rear transmission housing	M 22 x 1.5	50 (37)
Rear transmission housing to automatic transmission	M 10	46 (34)
Front transmission cover to intermediate plate	M 10 x 35	46 (34)
	M 8	23 (17)
Intermediate plate to automatic transmission	M 8	23 (17)
Drive pinion bearing assembly to front transmission cover	M 8	23 (17)
Fastening nut to helical gear	M 40 x 1.5	250 (184)
Guide part for parking lock to housing	M 6 x 20	10 (7)
Plug to ATF pan	M 14 x 1.5	40 (30)
Banjo bolt to ATF pan	M 12 x 1.5	40 (30)
Banjo bolt to housing	M 14 x 1.5	40 (30)
ATF pan to housing	M 6	6 (4)

Location	Thread	Tightening torque Nm (ftlb.)
ATF strainer to hydraulic control unit	M 6 x 65	
Plug to ATF quick-fill adapter	M 14 x 1.5	30 (22)
Hydraulic control unit to transmission	M 6	
Hexagon nut for transmission socket		12 (9)
ATF indicator tube to transmission	M 6 x 4	
Adapter of hydraulic control unit to transmission housing	M 6	
Pressure regulator and solenoid valve mount to control unit	M 6	
Solenoid valves to hydraulic control unit	M 5 x 12	
ATF pump to housing	M 6	
Oil drainage and filling plug	M 22 x 1.5	50 (37)
Side transmission cover to housing	M 8 x 35	23 (17)
Bearing cover to bearing assembly	M 6 x 15	
Drive pinion bearing assembly to housing	M 10 x 35	50 (37)
Plug for ATF ducts	M 14 x 1.5	25 (18)
Fastening nut to bearing assembly	M 36 x 1.5	250 (184)
Crown wheel to differential housing	M 10 x 1.25	85 (63), and Loctite 262

3 Technical Data

Torque specifications for transmission suspension

Location	Thread	Tightening torque Nm (ftlb.)
Transmission support to body	M 10 x 70	46 (34)
Transmission support to transmission (fastening nut)	M 12 x 1.5	85 (63)
Side member to transmission	M 12 x 1.5 x 65	85 (63)
Side member to transmission support (fastening nut)	M 10	30 (22)
Console to transmission	M 8 x 35	23 (17)

Torque specifications for ATF lines and cooler

Location	Thread	Tightening torque Nm (ftlb.)
Bracket to headlight holder	M 6	10 (7)
Console to wheel house	M 8	23 (17)
Tension strut to console	M 6	10 (7)
Bar to ATF cooler	M 6	10 (7)
ATF cooler to engine oil cooler	M 6	10 (7)
Horn holder to wheel house	M 6	10 (7)
ATF lines to transmission (union nut)	M 18	30 (22)
Joints on ATF lines (union nuts)	M 18	30 (22)

3 Technical data

Torque specifications for gear selecting system

Location	Thread	Tightening torque Nm (ftlb.)
Lock nut to clevis of selector lever cable	M 5	6 (4)
Selector lever mount to body	M 6 x 16	10 (7)
Holder for selector lever cable to transmission	M 8	23 (17)
Cable slide housing to switch plate (Keylock)	M 4	2.5 (2)
Keylock cable to ignition lock	M 10 x 1	2.5 (2)
Shiftlock to selector lever housing	M 5	6.5 (5)
Lift solenoid to Shiftlock housing	M 4	2.5 (2)

Removing and installing the torque converter

Removing

1. Remove transmission.
2. Remove converter, with transmission in horizontal position.

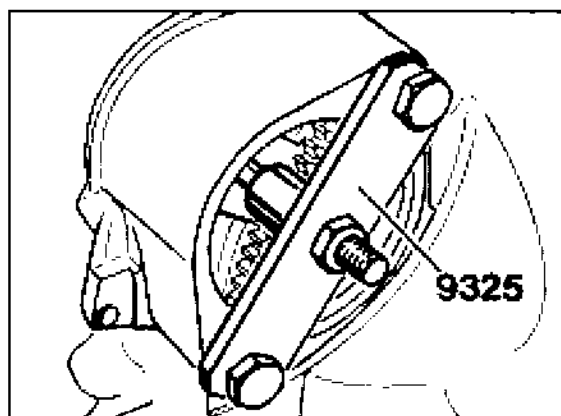
Note

Do not damage converter bearing assembly and rotary shaft seal.

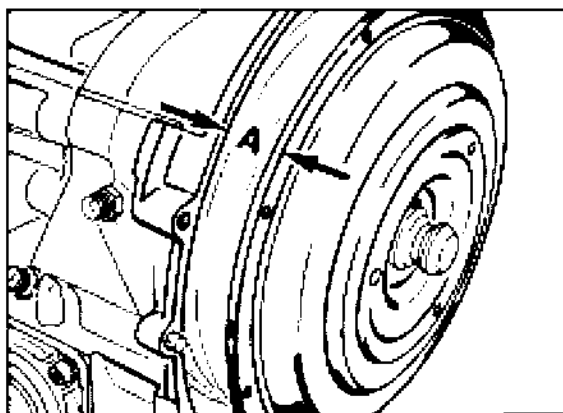
Installing

1. Carefully insert converter, with transmission in horizontal position. Turn the converter to and fro until the gear toothing engages and the installation position is reached.

2. Secure converter against falling out with special tool 9325



413-32



412-32

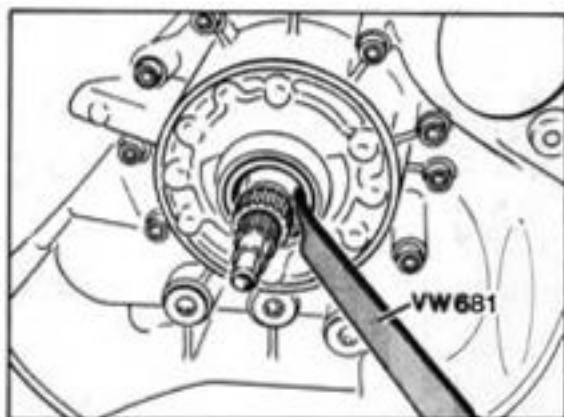
A = approx. 25 mm

Note

If the torque converter is not installed in the correct position, both the torque converter and the ATF pump may be damaged when the engine is connected to the transmission.

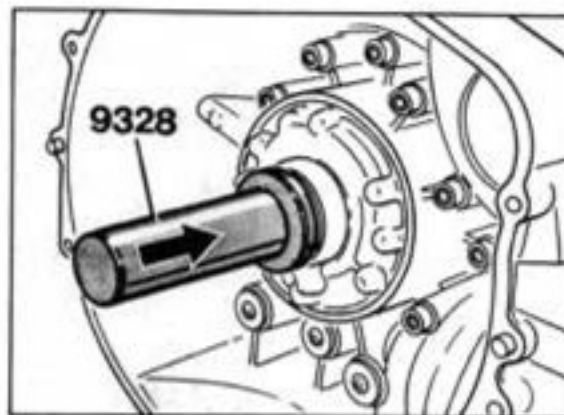
32 47 19 Removing and installing torque converter seal ring**Removing**

1. Remove transmission and converter.
2. Lever out sealing ring with VW 681



419-38

2. Press in sealing ring with special tool 9328 as far as it will go.



420-38

Installing

Installation takes place in reverse order.

1. Wet sealing lip with ATF.

Test point	DTC	Title	Fault effect	Page
31	70	Fault in torque converter clutch	Torque converter clutch always open	37 - 85

Note

Diagnosis of the torque converter clutch is active as from the 1997 model.

Fault, fault code	Possible causes, elimination, notes
Test point 31 Torque converter fault Fault code 70	Torque converter clutch always open Fault possibility: mechanical/hydraulic fault in transmission 1) Check ATF level and correct if necessary (refer to 911 Carrera (993) Workshop Manual, Page 37 - 101). 2) Erase fault memory and perform a test drive. The diagnostic test conditions are achieved if: - the torque converter clutch is electrically closed - engine speed < 3008 rpm - engine torque > 200 Nm (148 ftlb) - ATF-temperature between 40° C and 95° C 3) Read out fault memory again. The following areas have to be checked if the fault is still present: - ATF supply - torque converter - hydraulic control unit (jammed spool valves) - transmission (slipping clutches)

Checking the ATF fluid level

The prescribed fluid level is extremely important for perfect functioning of the automatic transmission.

Preconditions for checking:

Transmission underbody cladding removed

Vehicle must be horizontal

Engine operating at idling speed

Hand brake applied

Selector lever in position "P"

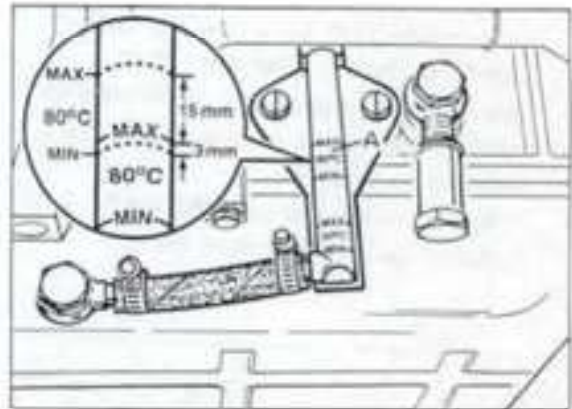
ATF temperature 80°C

Check ATF level at 80°C.

Note

The ATF capacity was increased by 0.5 l. The ATF level therefore rises and the 80°C check marks move further up on the oil level tube. Refer to imaginary and dotted lines in the close-up insert.

When checking the ATF level, make sure the fluid level is between those two lines (maximum level 15 mm above the 80°C MAX mark present, lowest level approx. 3 mm below the 80°C MAX mark present).



1958-38

A = invalid 80°C mark

The exact ATF temperature can be determined with the system tester 9288.

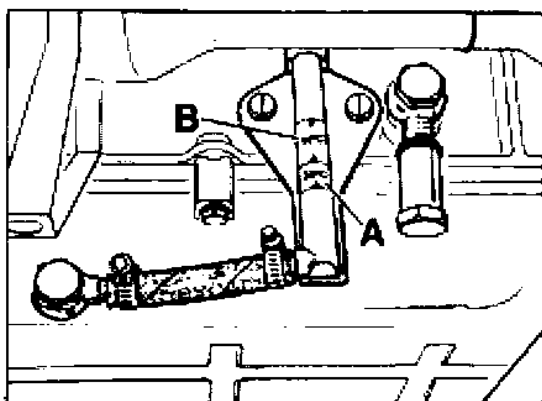
If necessary, top up missing ATF fluid via the quick-fill device (also refer to page 37-103).

Modifications from model year '95

Since model year '95, a new oil level tube with 50°C markings has been fitted.

The ATF fluid level should be checked using the same procedure as before but at an ATF temperature of 50°C.

The liquid level must be within the 50°C indication range (see B in illustration).



2036-38

A = indication range for 30°C ATF temperature

B = indication range for 50°C ATF temperature

37 02 55 Changing ATF fluid

Capacity: approx. 9.5 l

Change quantity: approx. 3.5

Oil type:

ATF-Dexron IID

The ATF fluid must be changed and the ATF strainer cleaned every 40,000 km

When changing the ATF fluid, the vehicle must be horizontal and the engine switched off.

Drain ATF fluid, remove ATF pan and ATF strainer (refer to page 38-113).

Thoroughly clean the strainer and pan.

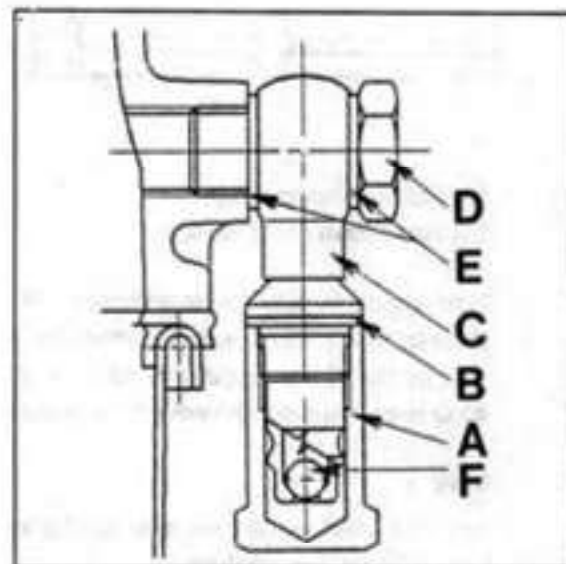
Fit the ATF strainer with a new O-ring.

Tighten the fixing screws with **8 Nm (6 ftlb)**.

Fit the ATF pan with seal. Tighten the fixing screws with **8 Nm (6 ftlb)**.

Fill with ATF fluid:

First, fill ATF fluid up to the 30°C max. mark via the quick-fill connection with the engine stationary.

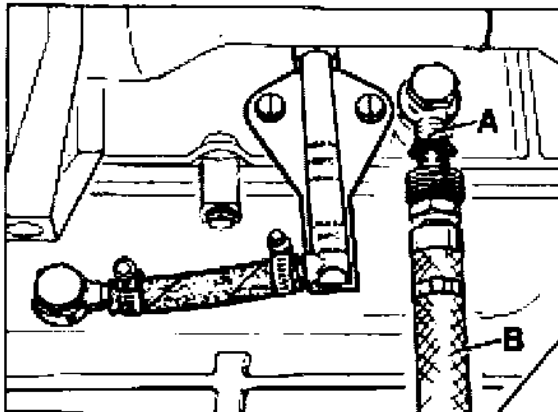


391-38

A = Hexagon cap nut (tightening torque
30 Nm = 22 ftlb)

B = Sealing ring (replace)

C = Quick-fill connection



388-38

A = Quick-fill connection

B = Hose from filling device

Start engine in selector lever position "P" and allow to run at idle speed. Observe the ATF level in the oil level pipe and top up to the 30°C max. marking immediately if required.

Note

The ATF level in the transmission changes with the fluid temperature.

Drive the transmission warm and check the ATF fluid at 80° C (from mod. '95 at 50° C).

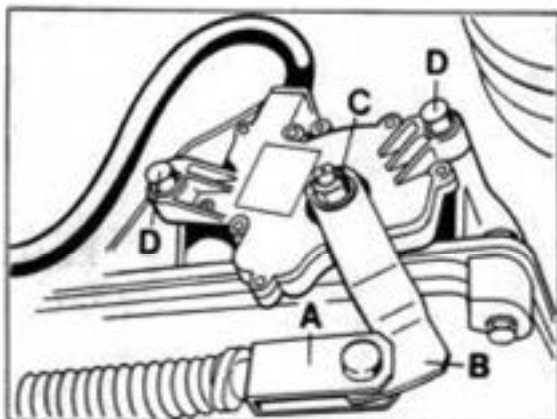
See note on page 37 - 101.

The exact ATF temperature can be determined with the system tester 9288.

37 31 19 Removing and installing multifunctional switch on transmission

Removal

1. Set selector lever to position "N".
2. Remove transmission undertray.
3. Remove rear underside panel.
4. Remove left rear hot air pipe.
5. Disconnect selector lever cable from actuating lever.
6. Remove actuating lever.

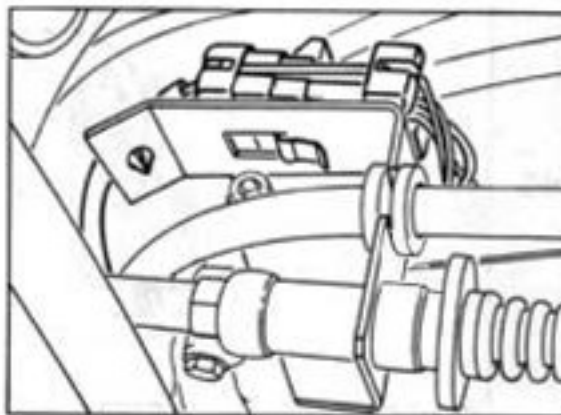


381-37a

- A = selector lever cable
 B = actuating lever
 C = hexagon nut (M 8 x 1) with washer
 D = fastening screws

7. Press retaining lugs for wire retainer "A" together and lift holder out upwards.

8. Detach, unlock and disconnect connector.



1861-37

A = wire retainer

9. Unscrew fastening screws for multifunctional switch completely and remove switch.

Installation

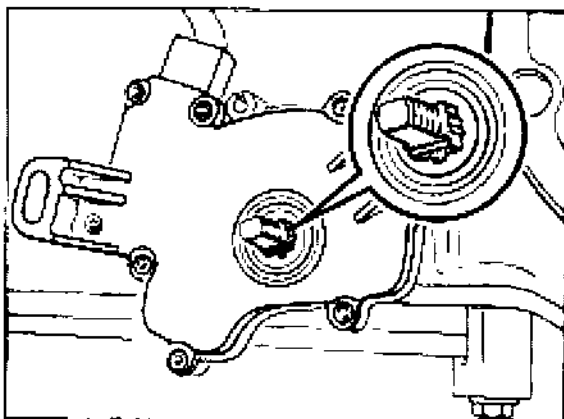
For installation, proceed in reverse order.

Tightening torques:

Multi-functional switch to transmission = 10 Nm
 Actuating lever to selector shaft = 15 Nm

1. Set selector shaft to position "N" (turn shaft anti-clockwise up to the stop and then back two clicks) and place switch in the correct position.

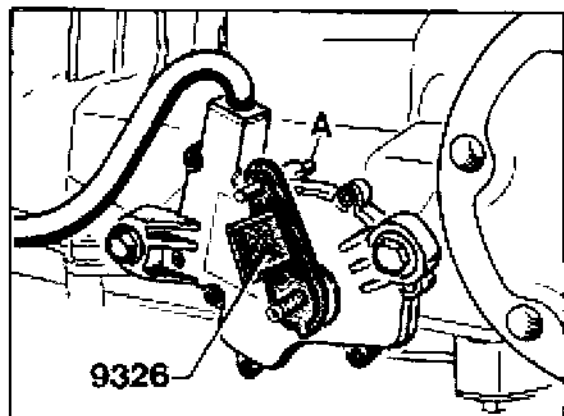
2. Check, and if necessary adjust, setting of selector lever cable.



1984-37

2. Adjusting multifunctional switch.

Push pointer of special tool 9326 onto the selector shaft and turn the switch until the locating pin can be pushed into the fixing hole of the switch. Tighten the mounting screws to 10 Nm (7 ftlb.) in this position.



A = locating pin

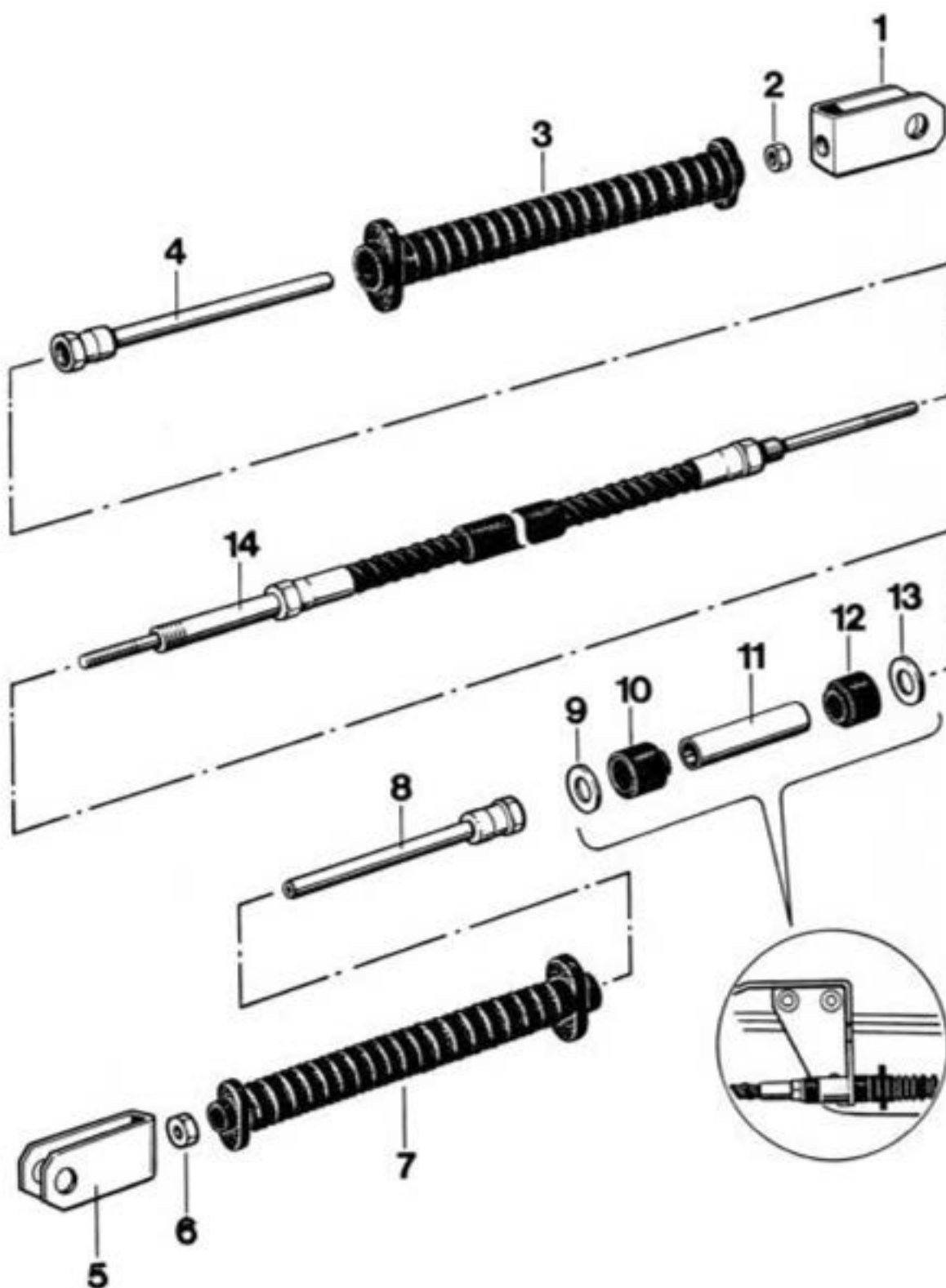
37 15 15 Adjusting cable for selector device

1. Move selector lever to position "P".
2. Set multi-function switch to position "P". To do so, press actuator lever of switch back up to stop.
3. Set cable length at clevis so that the bolt can be installed free from stress.
4. Check adjustment by shifting through all the gears and confirming that gear is displayed on speedometer. In addition change gate from "D" to "M". This must be possible with one smooth, straightline movement.
5. Mount snap ring for bolt on actuator lever.

37 15 19

Removing

installing cable for selector mechanism

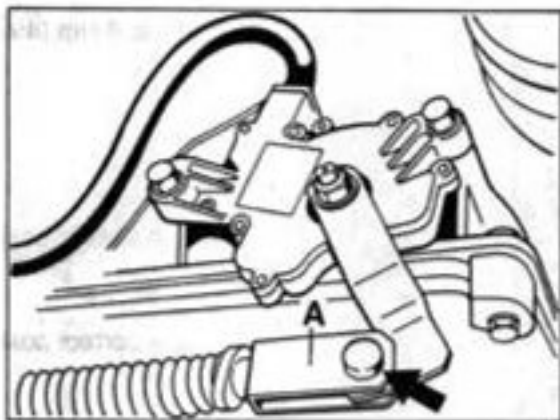


No.	Designation	Qty.	Removal	Note:	Installation
1	Fork head	1			Screw on to half length of thread on cable
2	Hexagon nut	1			Tighten to 5 Nm (4.4 ftlb)
3	Garter seal	1			
4	Guide tube	1			
5	Fork head	1			
6	Hexagon nut	1			Tighten to 6 Nm (4.4 ftlb)
7	Garter seal	1			
8	Guide tube	1			
9	Washer	1			
10	Rubber mount	1			Install in correct position
11	Spacer tube	1			
12	Rubber mount	1			Install in correct position
13	Washer	1			
14	Cable	1			Readjust

Instructions for removal and installation

Removal

1. Remove transmission undertray and underside panel.
2. Disconnect cable on operating lever and remove fork head.

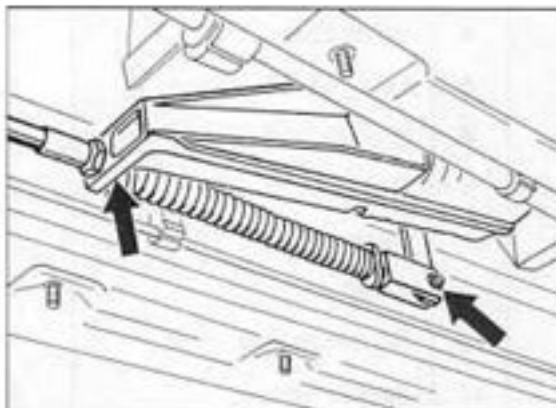


361-37

A = Fork head

3. Disconnect guide tube on bracket.

4. Disconnect cable on selector lever casing and disconnect guide tube.



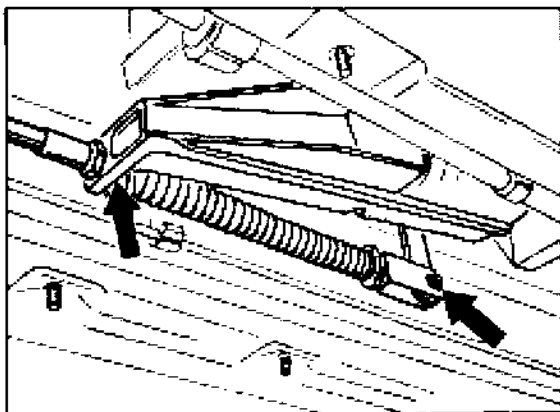
1066-37

Installation

1. Installation is carried out in reverse order.
2. Adjust cable for selector mechanism (see page 37 - 107).

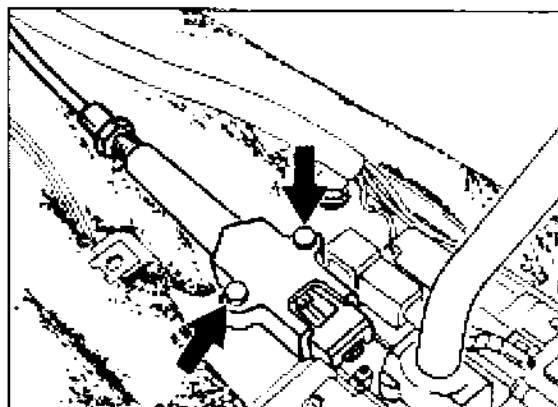
37 10 19 Removing and installing gear selecting system

1. Disconnect battery.
2. Remove center underside panel
3. Disengage selector lever cable from deflection lever and undo guide tube.



1066-37

4. Unscrew release button and pull off selector knob.
5. Remove center console.
6. Disconnect cable valve body from switch plate (only for keylock models).



1067-37

7. Disconnect connectors for switch plate.
8. Unscrew four mounting screws (M 6) and take out selector lever operator from above.

Installation

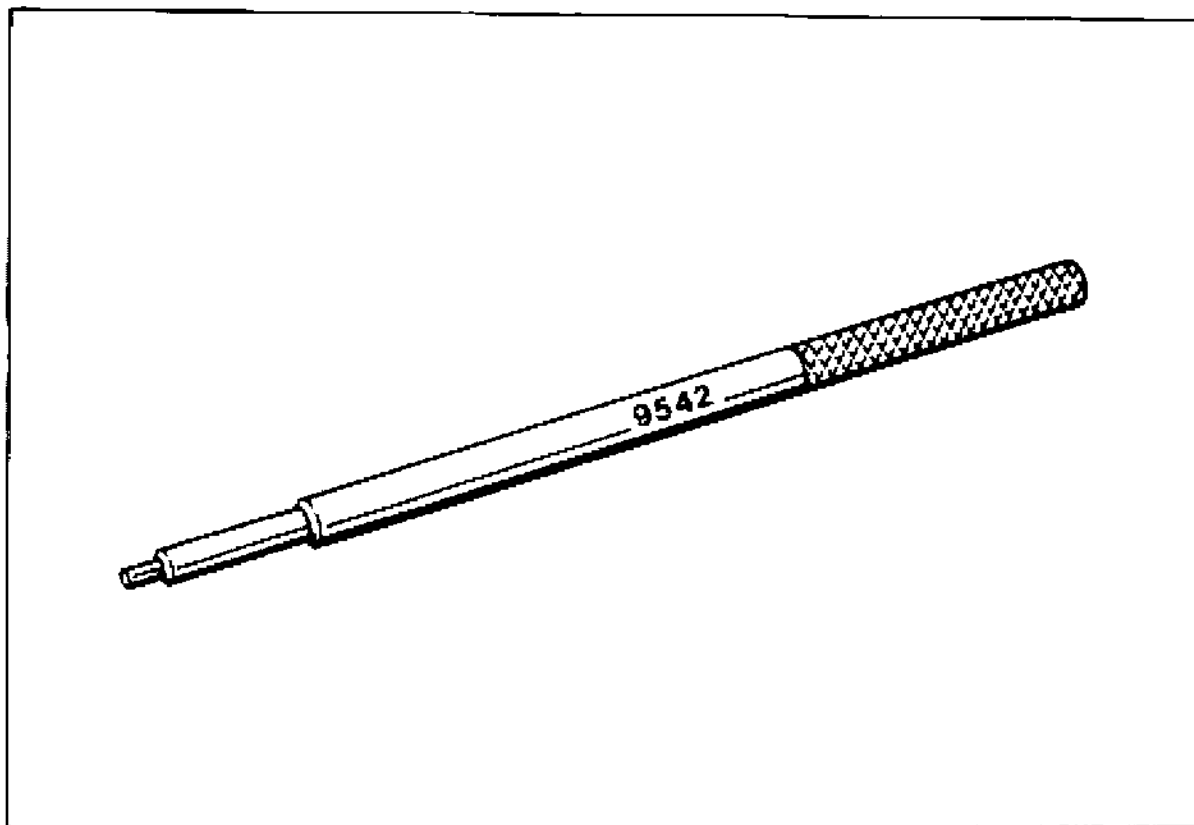
Tightening torques:

Gear selecting system to body	
	= 10 Nm (7 ftlb)
Cable valve body to switch plate	
	= 2.5 Nm (2 ftlb)

1. Install in reverse order.
2. Press selector knob manually until is up against the stop, making sure the twist lock engages in the cutout in the selector lever.
3. Check release button for smooth operation. It must return into the home position by itself.
4. Check selector lever cable adjustment and readjust if required.
5. Check operation of keylock and shiftlock

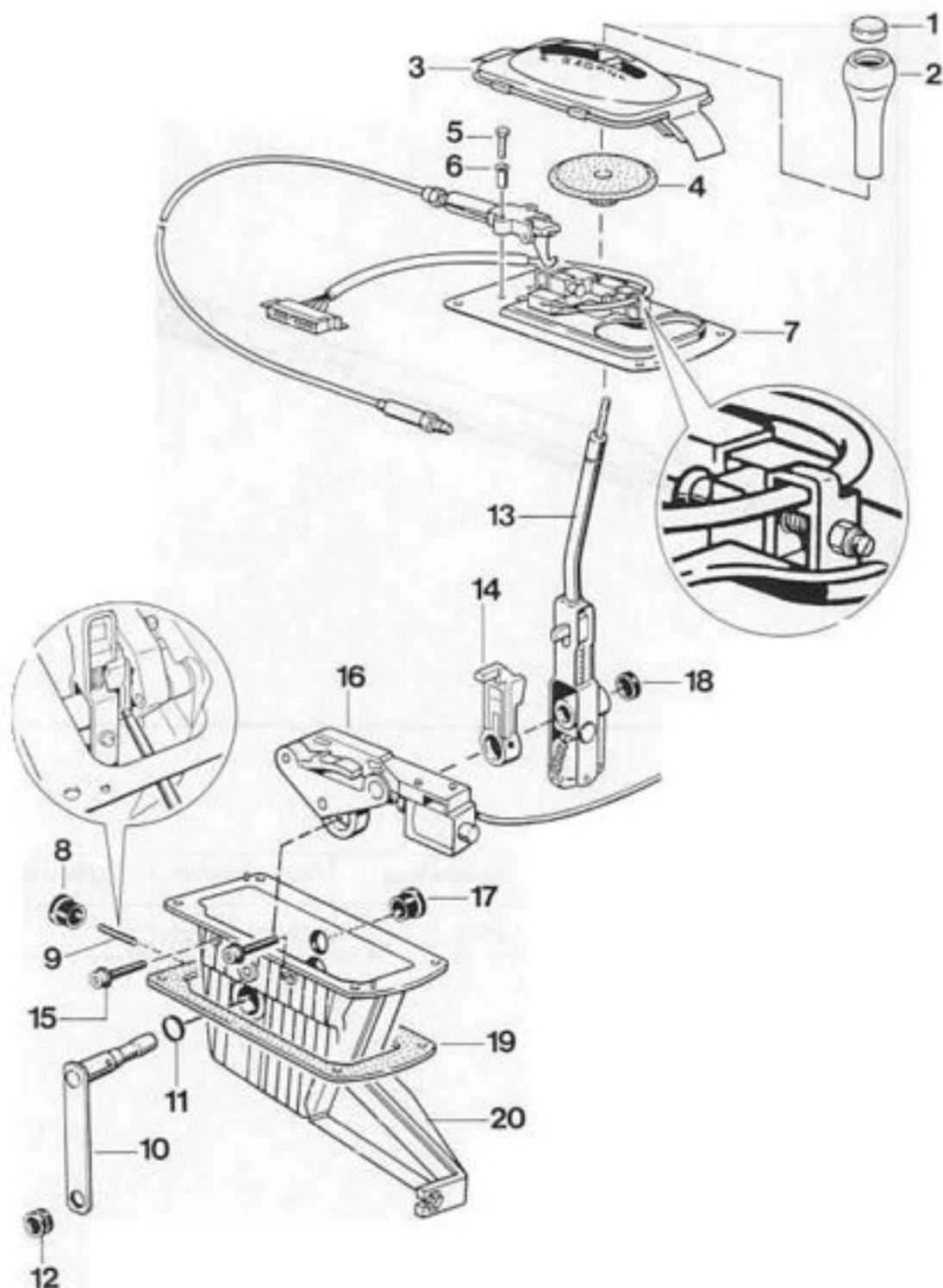
37 10 37 Dismantling and assembling gear selecting system

Tools



No.	Designation	Special tool	Order number	Explanation
	Assembly mandrel	9542	000.721.954.20	

37 10 37 Dismantling and assembling gear selecting system



No.	Designation	Qty.	Note:	
			Removal	Installation
1	Release button	1	Screw off manually	Must return to home position automatically after it has been actuated
2	Selector knob	1	Pull off manually	Push up to stop in correct position. Twist lock must engage in cutout in selector lever
3	Shroud	1	Can only be removed from below with the center console removed	
4	Cover	1		
5	Hexagon head screw	2		Tighten to 2.5 Nm (2 ftlb)
6	Sleeve	2		
7	Switch plate	1		
8	Plug	1		
9	Tensioning sleeve	1	Drive out using Special Tool 9542	Bore of driver and relay shaft must match
10	Deflection lever	1		
11	Seal	1		
12	Rubber mount	1		
13	Selector lever	1	Shift into manual speed selection gate	
14	Driver	1		
15	Hex socket head bolt	2		Tighten to 6.5 Nm
16	Shift-Lock	1		
17	Plug	1		
18	Cover	1		
19	Gasket	1		
20	Housing	1		

Note

Coat all sliding surfaces with low-temperature grease (e.g. Shell S 6508)

Notes on assembly:**Note**

Change from "D" to "M" and from "M" to "D".
Stright-line movement without catching must
be possible. In addition, there may be up to
0.4 mm play on the selector lever in the
+ and – direction with "M" selected. Adjust-
ments can be made using the selector lever
reset switch (see exploded drawing, item 7).

37 87 19 Removing and installing solenoid for shiftlock

Note

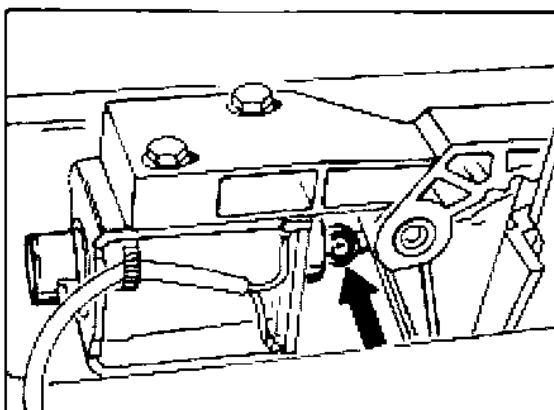
When checking the solenoid electrically, be sure to observe correct polarity

Terminal 1 = Positive

Terminal 2 = Negative

Removal

1. Remove center console and gear selecting system.
2. Remove switch plate.
3. Using a suitable tool, press connecting rod carefully off the solenoid.



4. Screw out mounting screws and take off solenoid.

Installation

1. Set selector lever to "1" position.
2. Grease ball and ball socket with low-temperature grease.
3. Clip connecting rod to solenoid, retaining the connecting rod in correct position using a suitable wire hook or marking tool and pushing the free lift solenoid carefully into the ball socket.

4. Using sleeves and hexagon head bolts, fit solenoid to gate in such a manner that it remains free to slide in an axial direction.

5. Adjusting the solenoid:

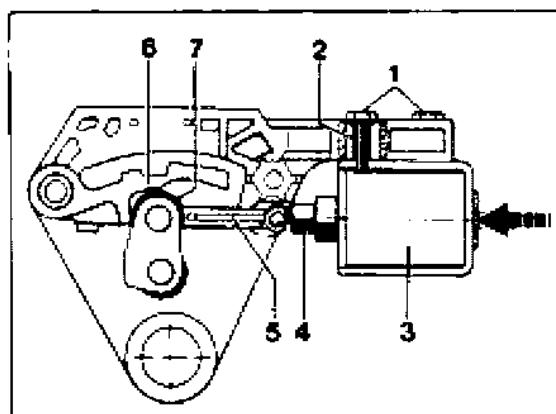
Set selector lever to position „P“.

Slide lift solenoid all the way back in the slots of the gate.

Push iron core from the solenoid towards the pawl until it contacts the stop and locate it in this position.

Slide the actuated solenoid axially until the idler contacts the stop of the pawl.

Tighten hexagon head bolt to 2.5 Nm (2 ftlb) in this position.



- | | |
|-----------------------|---------------|
| 1 - Hexagon head bolt | 2 - Sleeve |
| 3 - Solenoid | 4 - Iron core |
| 5 - Connecting rod | 6 - Pawl |
| 7 - Idler | |

6. Check operation of shiftlock.

37 13 19 Removing and installing keylock bowden cable

Removal

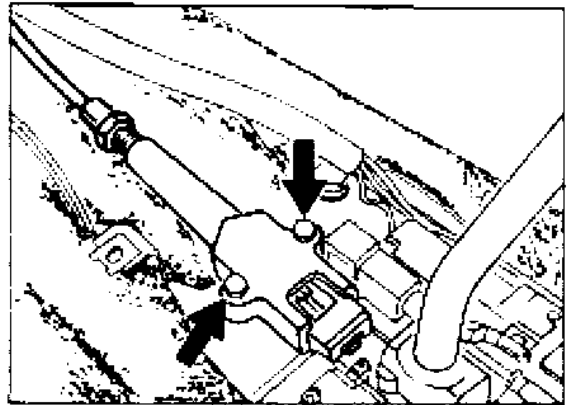
1. Disconnect battery.
2. Remove complete center console, knee guard and side nozzle.
3. Undo Central Information System and leave it suspended on the wiring.
4. Turn ignition lock to position „2“ (Ignition on).

Note

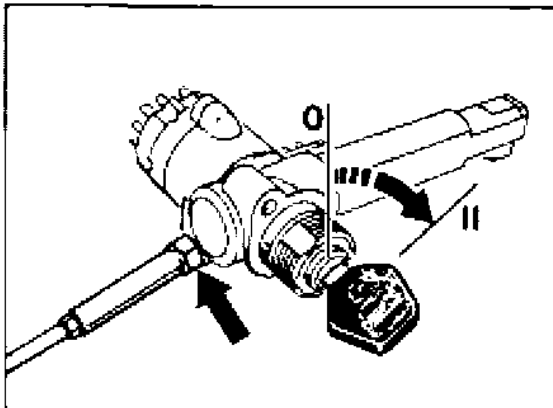
The ignition lock must be in position „2“. If it is left in any other position, both the lock and the bowden cable may be damaged.

5. Unbolt bowden cable from ignition lock.

6. Undo mounting screws of valve body.



1067-37



1068-37

7. Disengage bowden cable from pedal floorboard and from bracket for center console.

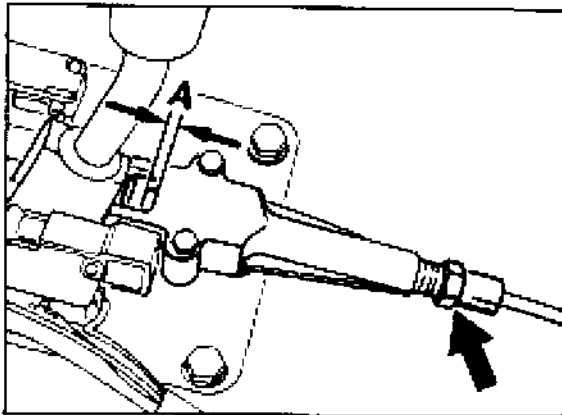
Installation

1. Assemble in reverse order, observing the following points.
2. Turn ignition lock to position „2“ (Ignition on) and fit bowden cable to ignition lock.
Tightening torque: 2.5 Nm (2 ftlb)
3. Fit bowden cable valve body using sleeves and hexagon head bolts.
Tightening torque: 2.5 Nm (2 ftlb)
4. Set selector lever to position „P“ and turn ignition lock to position „0“.

Note

If the ignition lock cannot be turned to the 0 position, the bowden cable must be re-adjusted.

5. Adjust bowden cable, turning the cable sleeve on the valve body until the lock slide reaches a setting of 2 ± 0.5 mm (refer to Fig.).



1081-37

$$A = 2 \pm 0.5 \text{ mm}$$

6. Check operation of keylock and shiftlock.

37 13 01 Checking keylock and shiftlock**Checking the keylock (ignition key lock)**

Selector lever position	Position of release button	Keylock operation	Ignition key
P	not actuated	no	rotary, may be pulled off
P	actuated	yes	cannot be turned to pull-off position
R-N-D-3-2-1	not actuated actuated	yes	cannot be turned to pull-off position

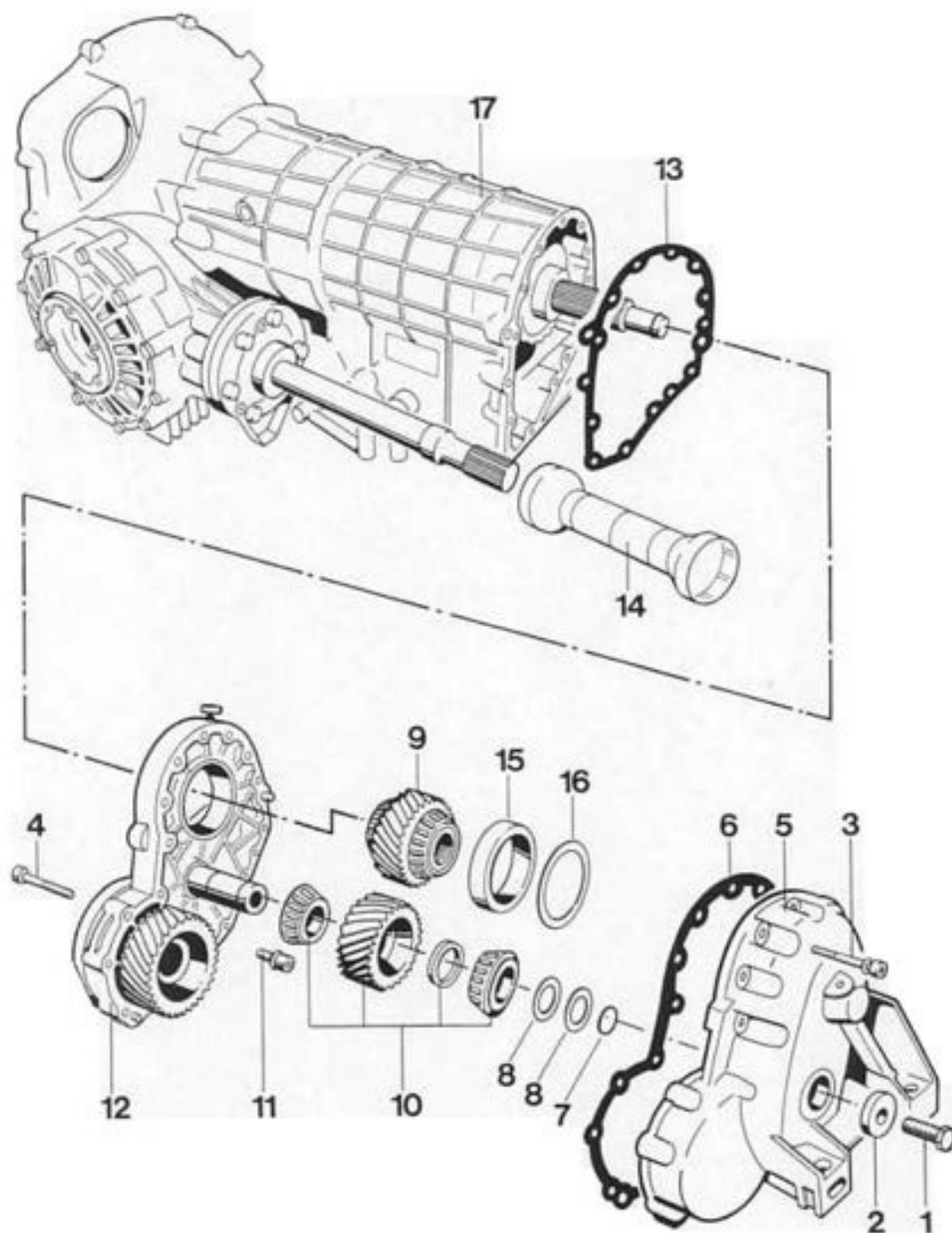
Checking the keylock (selector lever lock)

Ignition key position	Release button status	Keylock operation	Selector lever
Pulled off or pulled position	locked	yes	locked
Position 1 or 2	not locked	no	not locked

Checking the shiftlock

Selector lever position	Ignition	Shiftlock operation	Brakes	Selector lever
P-R-N-D-3-2-1	off	no	actuated not actuated	not locked
P and N	on	yes	not actuated	locked
P and N	on	yes	actuated	not locked

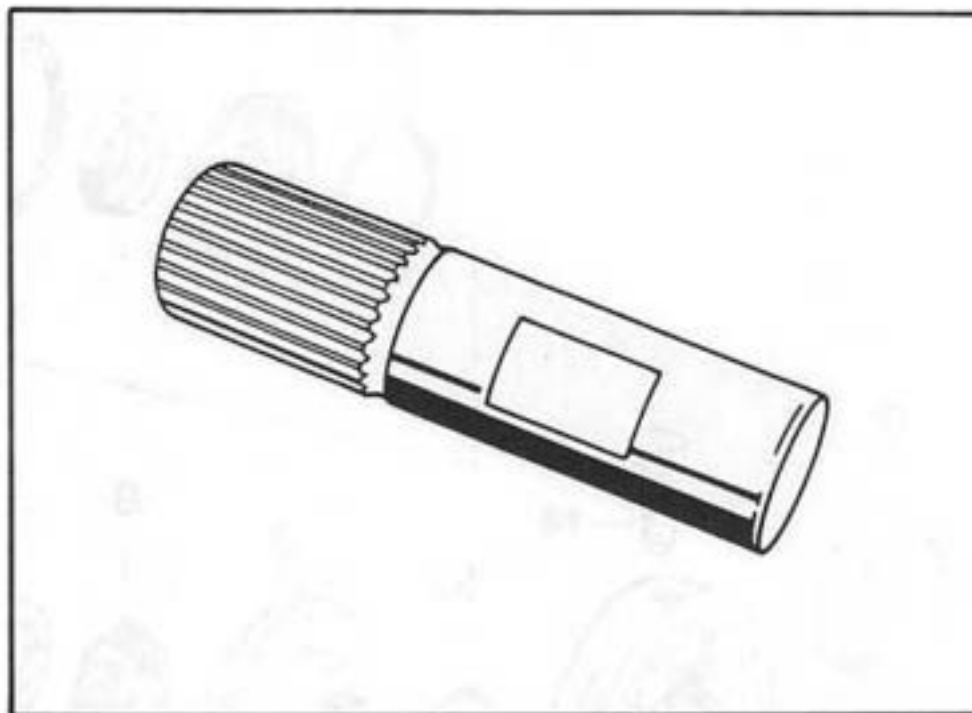
37 48 19 Removing and installing the intermediate plate



No.		Qty.	Note:	
No.	Designation		Removal	Installation
1	Hexagon screws	1		Tighten with 46 Nm
2	Washer	1		
3	Fillister head screw	11		Tighten with 23 Nm
4	Fillister head screw	6		Tighten with 23 Nm
5	Front transmission cover	1		
6	Seal	1		Replace
7	O-ring	1		Replace, oil with ATF
8	Shim	X	Note thickness for re-installation	
9	Gear wheel with tapered-roller bearings	1		
10	Gear wheel with bearing assembly	1		Set by the manufacturer.
11	Fillister head screw	2		Tighten with 23 Nm
12	Intermediate plate	1		
13	Seal	1		Replace
14	Protective tube	1		Large diameter to final drive
15	Taper roller bearing outer race	1	Remove with internal pul- ler (e.g. Schrem 60 - 70)	Heat transmission cover to approx. 120 deg. C and press in to stop
16	Adjusting shim	X	Record thickness for re-in- stallation	
17	Automatic transmission with final drive	1		

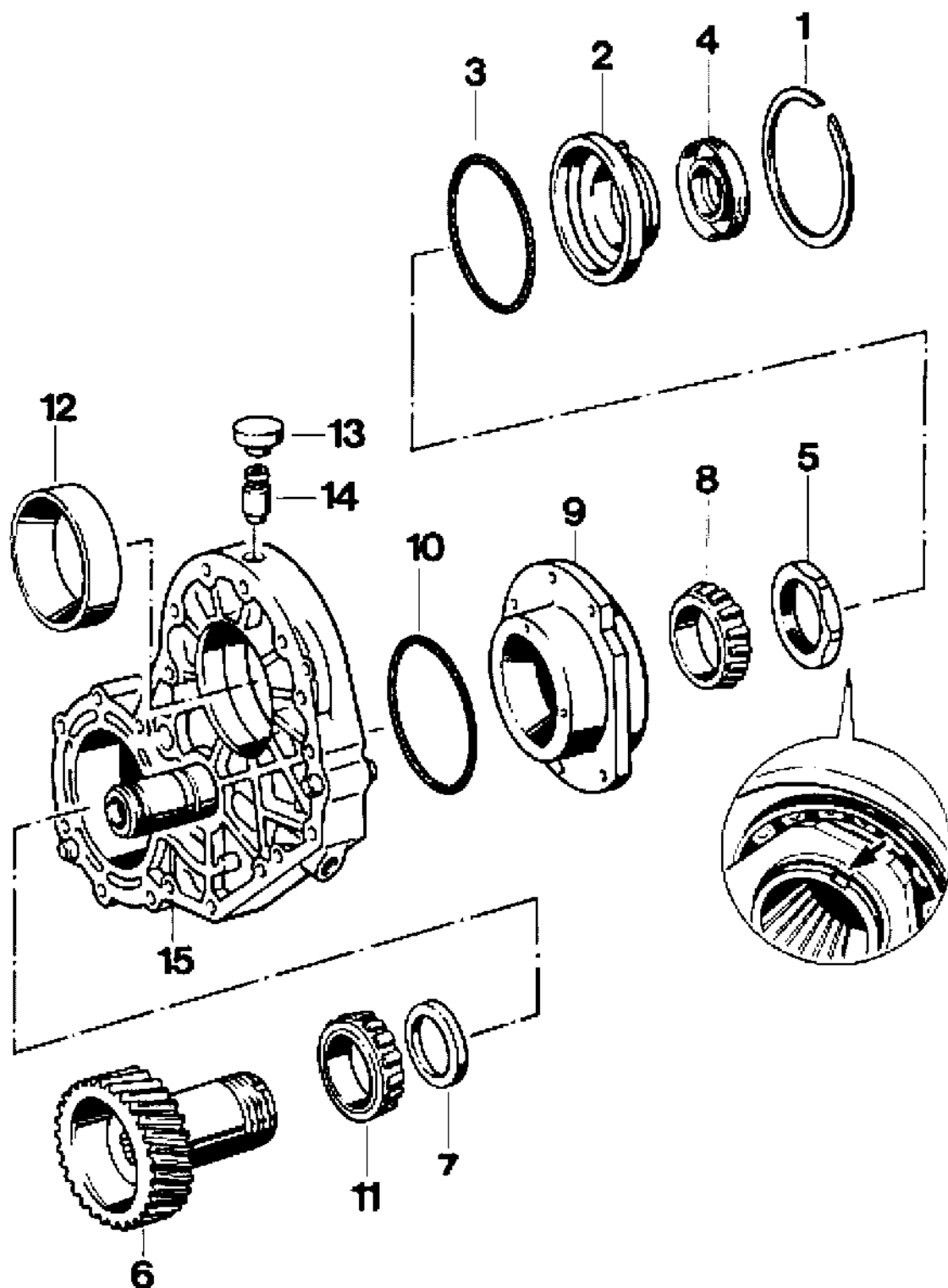
Note

The reduction gear is only available as a complete set (intermediate plate, transmission cover and adjusted spur gears). It is not necessary to adjust the tension on the tapered roller bearing.

37 48 37 Dismantling and assembling intermediate plate**Tools**

No.	Designation	Special tool	Order number	Explanation
	Retaining device	9340	000.721.934.00	

34 48 i7 D ma g and assembling intermediate plate



No.	Designation	Qty.	Note:	
			Removal	Installation
1	Snap ring	1		
2	Bearing cover	1	Grab at lugs to lift out	
3	Round seal	1		Replace, coat with ATF fluid
4	Shaft seal	1		Fitting depth 2.0 ± 0.5 mm
5	Lock nut	1	Fit Special Tool 9340 into vise. Put intermediate plate into position and undo nut.	Tighten to 250 Nm (184 ftlb). Lock by upsetting the flange (2 x 180°)
6	Helical gear	1	Press out using a hydraulic press.	Replace only as a set (along with intermediate gear and drive gear)
7	Adjuster ring*	X	Record thickness for reassembly	Thickness can only be determined by manufacturer
8	Inner race of taper roller bearing*	1	Mark for reassembly	Heat to approx. 120°C and press on
9	Bearing cover *	1		
10	Round seal	1		Replace, coat with ATF fluid
11	Inner race of taper roller bearing*	1	Remove across assembly bore of helical gear. Mark for reassembly.	Heat to approx. 120°C and press into place
12	Outer race of taper roller bearing	1		Heat intermediate plate to approx. 120°C and press into place
13	Breather cover	1	Lever off	
14	Breather tube	1	Pull out	Press in to stop
15	Intermediate plate	1		

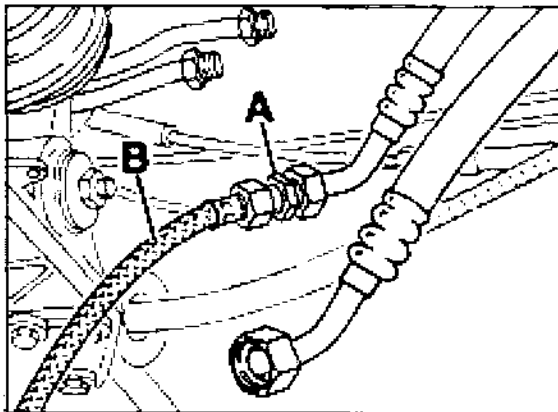
Note

The parts identified by an * have been preadjusted by the manufacturer.

38 60 29 Flushing ATF cooler and lines**Information**

If the ATF is carbonized, or if there is sludge or evidence of lining abrasion in the ATF pump, it is not sufficient merely to repair or replace the gearbox, the ATF cooler and line system must be flushed with ATF.

Attach additional hose of ATF filling device (see Workshop Manual Group 3/4) using special tool 9355/1 to ATF line with conventional twin connecting piece and flush out cooler and line system using filling device. Fluid must not be split on the ground.



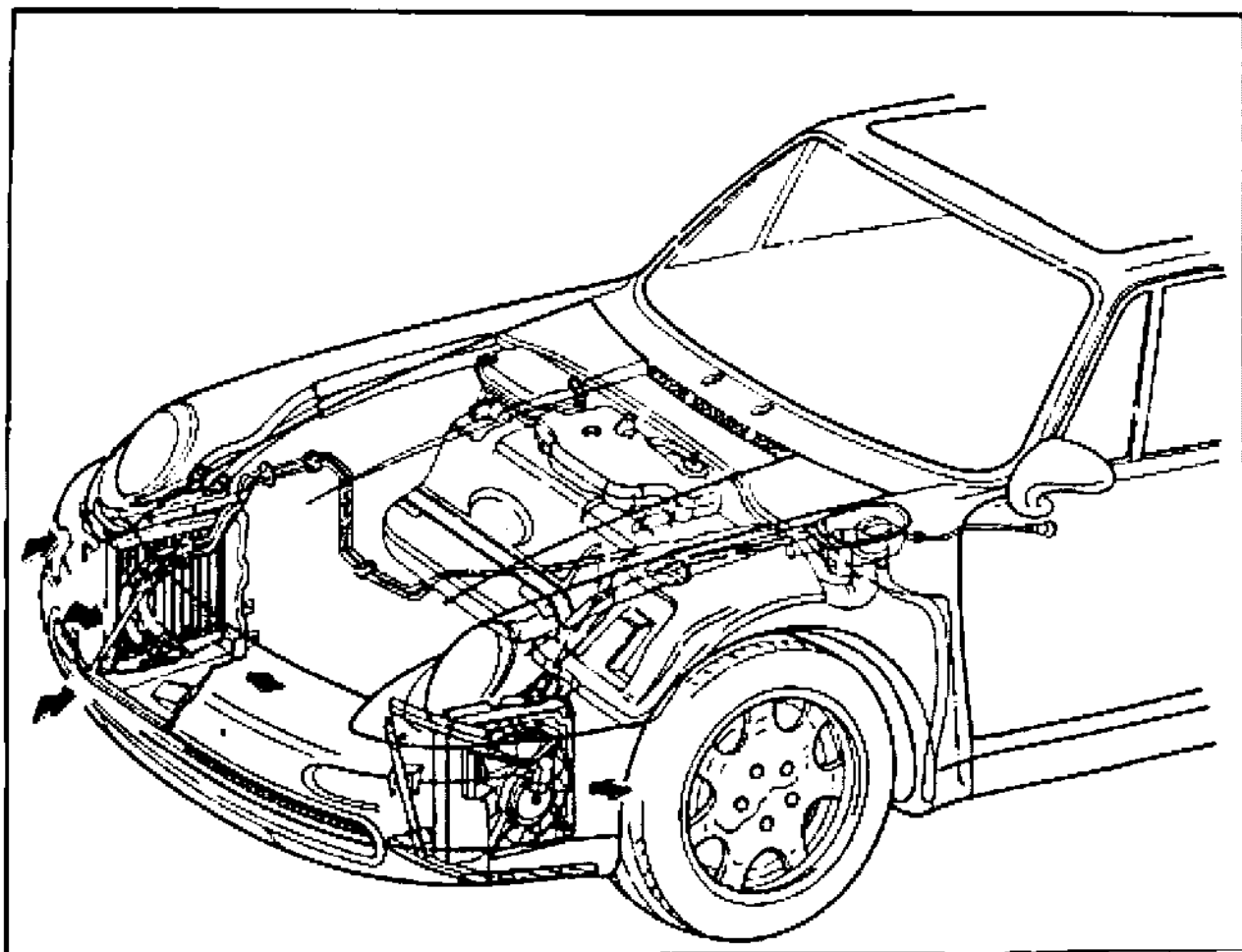
1883-37

A = Special tool 93551

B = Additional hose

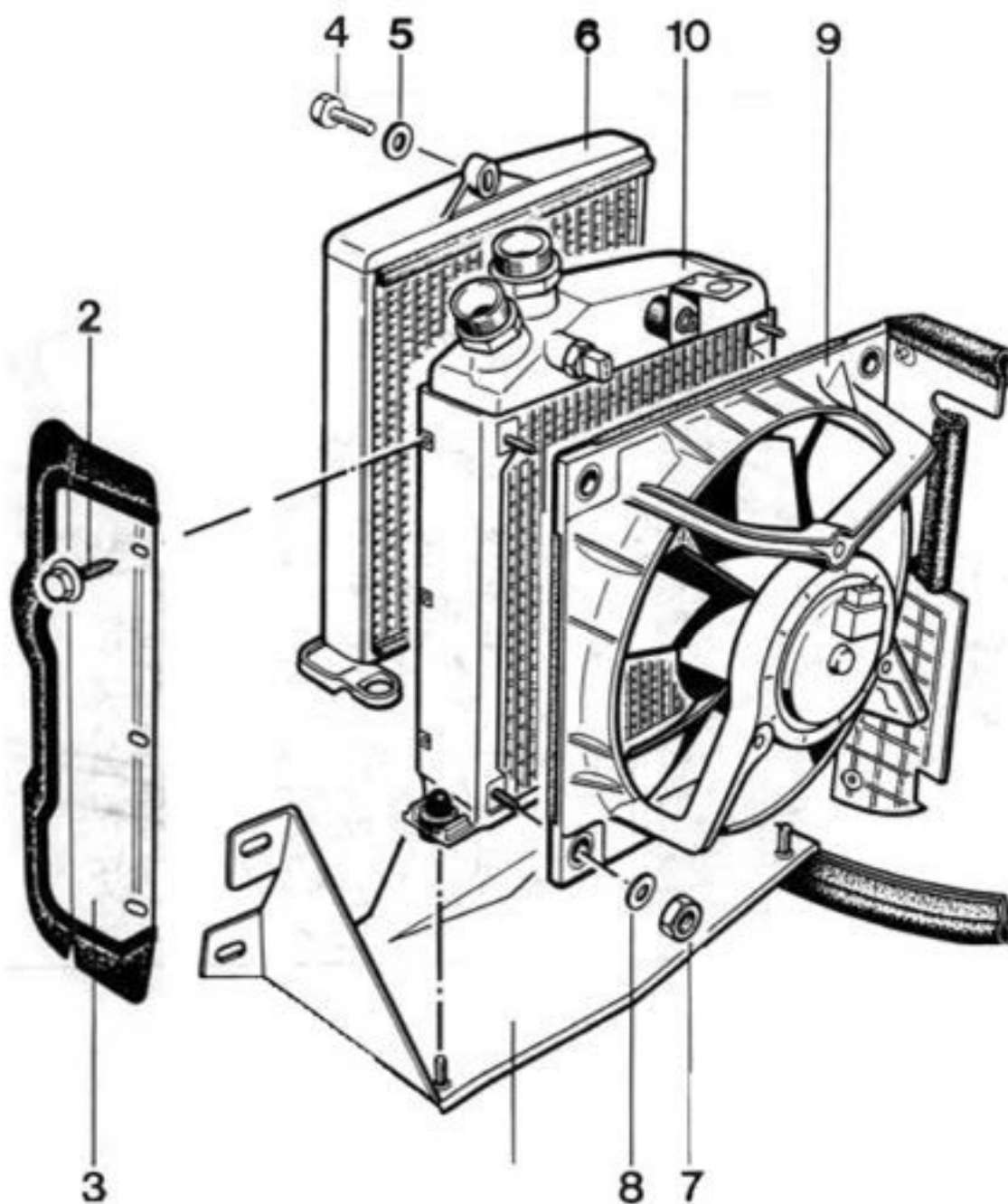
38 60

Removing and installing ATF cooler



38 60

Removing and installing ATF cooler



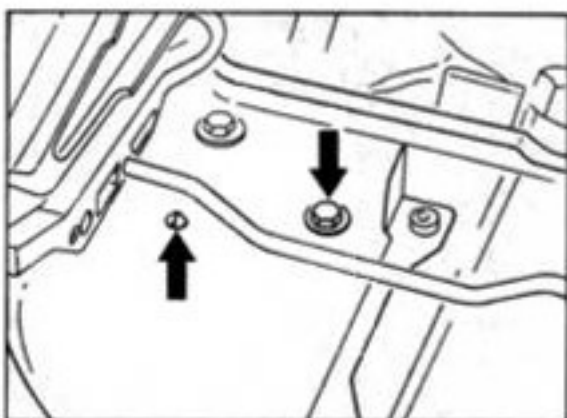
No.	Designation	Qty.	Note:	
			Removal	Installation
1	Bracket	1		
2	Washer-and-screw assembly	3		
3	Air deflector	1		
4	Hexagon screw	1		Tighten to 10 Nm (7.3 ftlb)
5	Washer	1		
6	ATF cooler	1		

38 60 19 Removing and installing ATF cooler**Notes**

The ATF cooler is installed in the wheel house in front of the right front wheel. It is mounted on the engine oil cooler and cooled by the two-stage electric fan of the engine oil cooler.

Removal

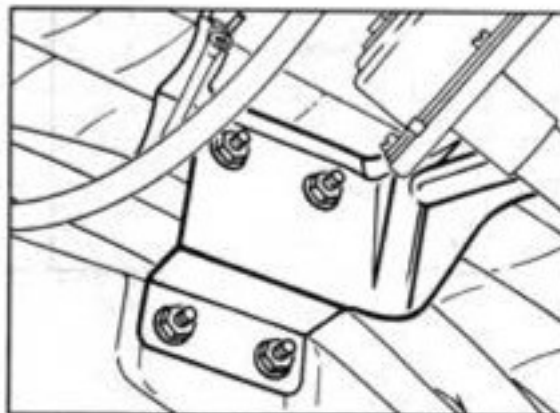
1. Remove right headlight, unclip wire retainer and unscrew hexagon head screw for cooler bracket.



1877-38

2. Remove front wheel housing liner and lower part of front spoiler.
3. Pull connector off electric fan and remove indicator mount by turning it to the left (bayonet lock).

4. Unscrew horn mount from wheel house.



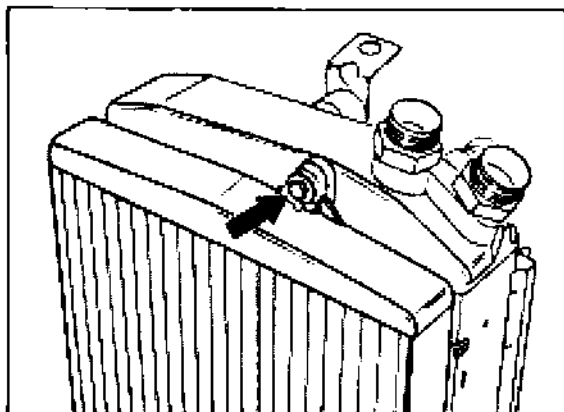
1878-38

5. Remove console. To do so, unscrew three hexagon nuts (M 8) from wheel house and one (M 6) for tension strut.
6. Remove air baffle.
7. Disconnect ATF lines from cooler.



1879-38

8. Unscrew hexagon head screw and remove ATF cooler (illustration shows cooler removed).



487-38

Note

Do not disconnect oil lines for engine oil cooler. The cooler remains in the wheel house.

Installation

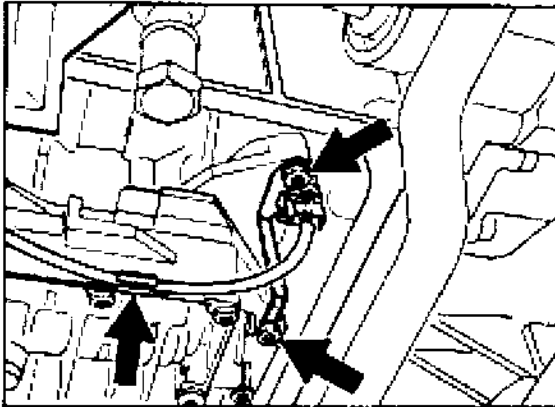
Tightening torques, Nm (ftlb).

Bracket to headlight holder	= 10 (7)
Console to wheel house	= 23 (17)
Tension strut to console	= 10 (7)
ATF cooler to engine oil cooler	= 10 (7)
Bar to ATF cooler	= 10 (7)
Horn holder to wheel house	= 10 (7)

1. Adopt the reverse procedure for installation.
2. Replace O rings for ATF lines on cooler.
3. Check that the rubber edging of the fan housing and the console is correctly positioned.
4. Check, and if necessary top up, ATF level.
(see page 37 - 101).

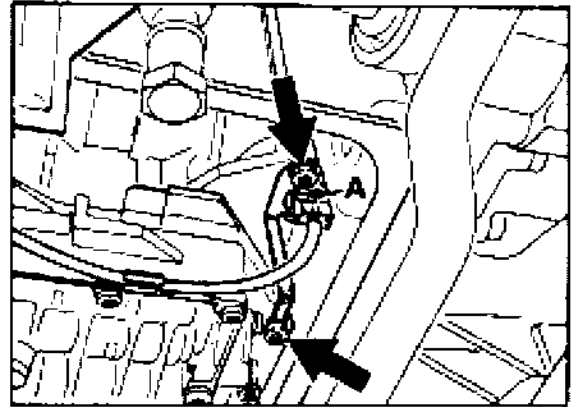
38 17 19 Removing and installing the inductive speed sensor**Removing**

1. Remove the transmission underbody cladding and ATF pan (refer to page 38 - 113).
2. Remove the holder for the inductive sensor and pull out sensor



390-38a

Insert the sensor and mount the holding plate so that the lugs engage in the groove on the connector.



386-38b

A = Connector groove

Installing

Tightening torque:

Holder to control unit = 8 Nm

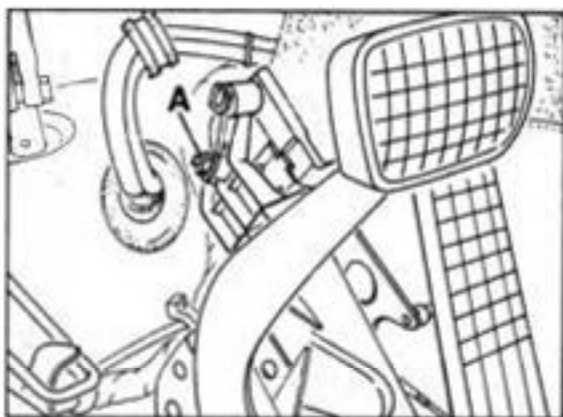
Installation takes place in reverse order.

38 90 19 Removing and installing kickdown switch

Removal

The kickdown switch is installed on a console in front of the accelerator pedal.

1. Remove floor board.
2. Remove kickdown switch with mount, unplug connector and remove switch (pressing the retaining lugs).



2244-38

A = self-locking nut

Installation

1. Install the parts removed in reverse order.
2. Use new self-locking nut for mount.
3. Place kickdown switch in installation position. With the mounting nut loosened, the mount with switch must be pushed until the actuator arm touches the console when the switch is operated.
4. Tightening torque: mount to console:
M 6 = 10 Nm (7.5 ftlb)

38 90 15 Adjusting kickdown switch

Prerequisites:

The idle position of the accelerator cable must be correctly adjusted.

Kickdown switch in installation position.
(see page 38 - point 3).

1. Connect system tester 9288.
2. Unhook pushrod from accelerator pedal.
3. Switch on system tester and select throttle valve in DME /actual values menu.
4. With the pushrod unhooked, push the accelerator pedal down to the stop (throttle valve open up to stop) and read **throttle position 1** off on system tester.

Note

With the throttle valve open (accelerator pedal fully depressed), the position is $84^\circ \pm 30$.

5. Hook pushrod back onto accelerator pedal.
6. Push gas pedal down as far as possible **without** operating kickdown switch and read off **throttle position 2** on system tester.
The position indicated must be 3° less than position 1 (tolerance ± 10).

Note

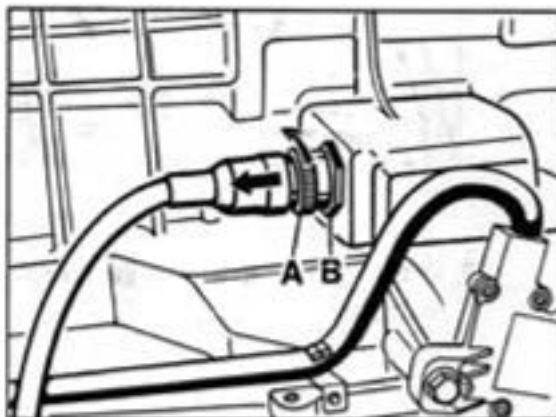
If the angle is too large, set the pushrod to a shorter position.

If the angle is too small, set the pushrod to a longer position.

After resetting the length of the pushrod, check the angle. It must not be smaller than 79° .

38 18 19 Remov. and installing the wiring harness for the transmission**Removing**

1. Remove the transmission underbody cladding.
2. Remove multi-functional switch (see page 37-105).
3. Disconnect the connector from the transmission socket. To do this, turn the bayonet lock to the left and disconnect the connector:

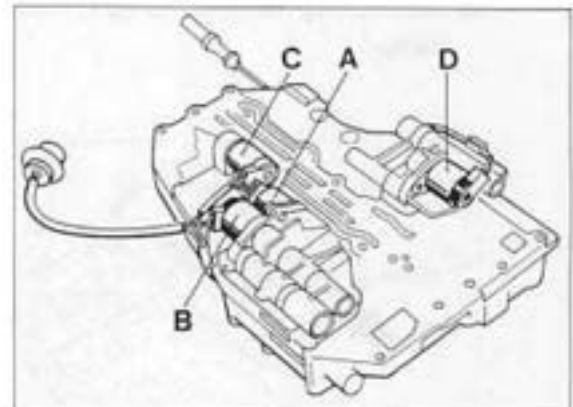


387-38a

A = Bayonet lock
B = Hexagon nut (a/f 30)

4. Unscrew the hexagon nut for the transmission socket with a suitable extension.
5. Remove hydraulic control unit with wire harness (see also page 38-113).

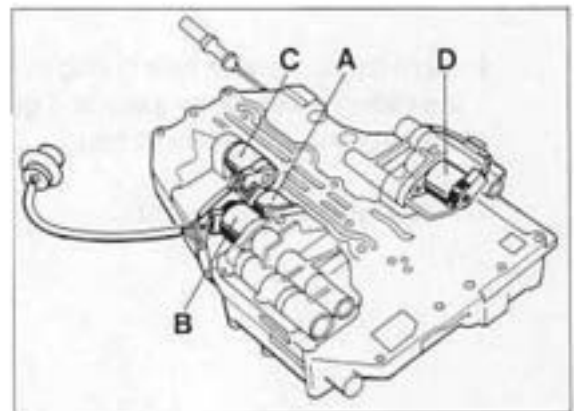
6. Mark the push-on sleeves for re-installation and pull off from the solenoid valves.



383-38

Installing

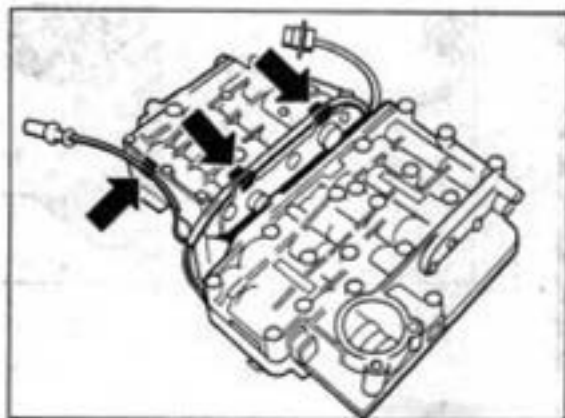
1. Push the push-on sleeves for the solenoid valves up to the stop. Pay attention to cable colors.



383-38

	Solenoid valves	Cable colors
A	Solenoid valve 1	(grey-violet)
B	Solenoid valve 2	(green-violet)
C	Solenoid valve 3	(red-violet)
D	Pressure regulator	(blue-violet)

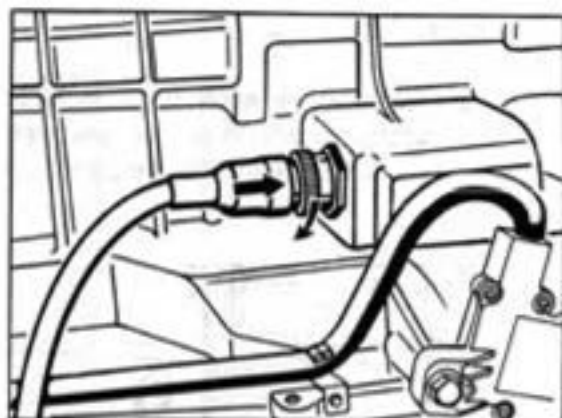
2. Route wiring harness and hang in cable clamps.



384-38

3. Place hydraulic control unit on a suitable surface (e.g. transmission jack) at installation height.
4. Insert the socket with new O-ring so that the flattened side faces upwards. Tighten hexagon nut with **12 Nm (9 ftlb)**.
5. Install hydraulic control unit.

6. Connect wiring harness with transmission socket. To do this, carefully insert the connector in the socket (fits only in one position) and lock by turning the bayonet lock to the right.

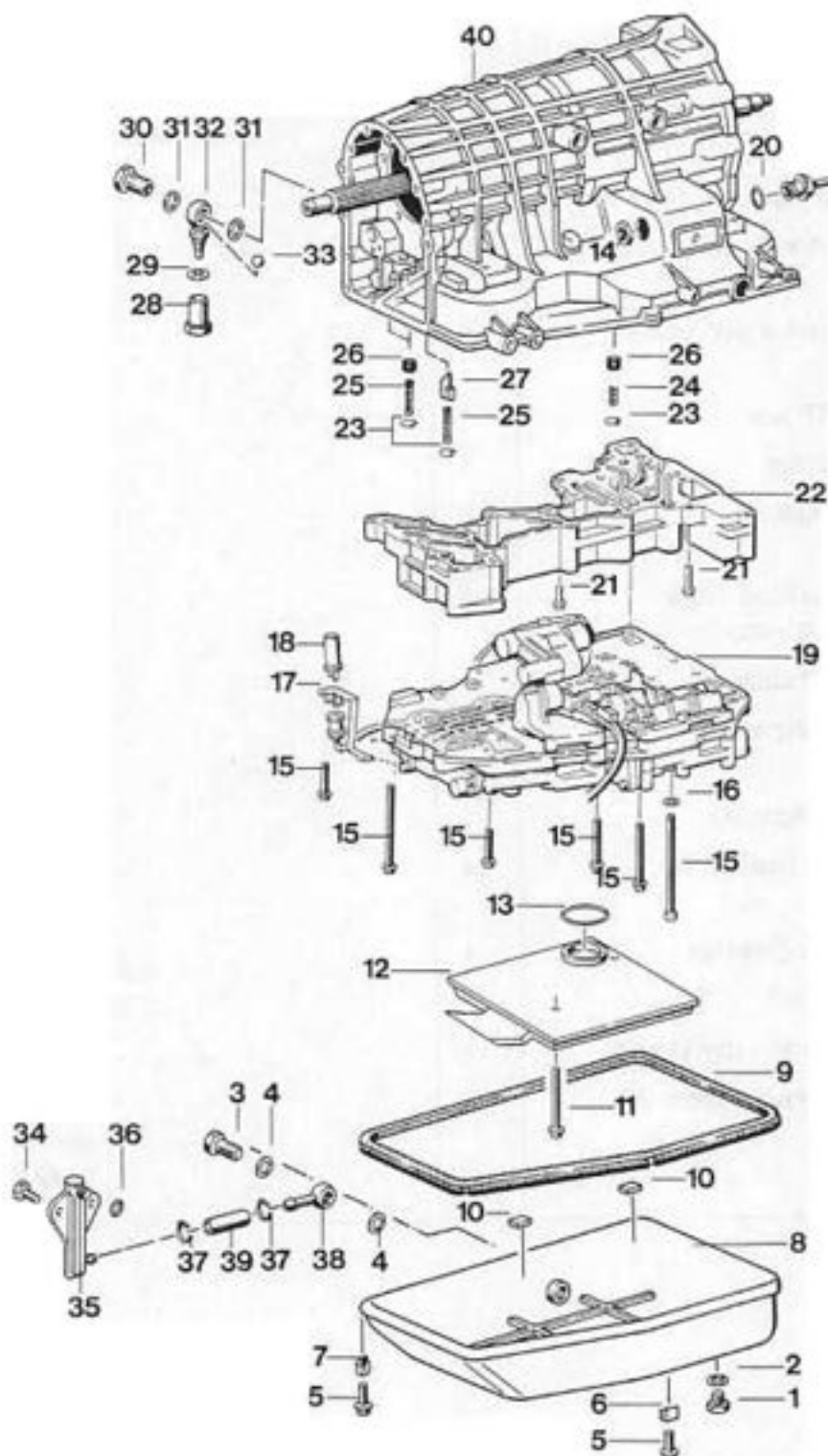


387-38b

7. Install the multi-functional switch and readjust.

38 77 19

Dismantling and assembling hydraulic control unit



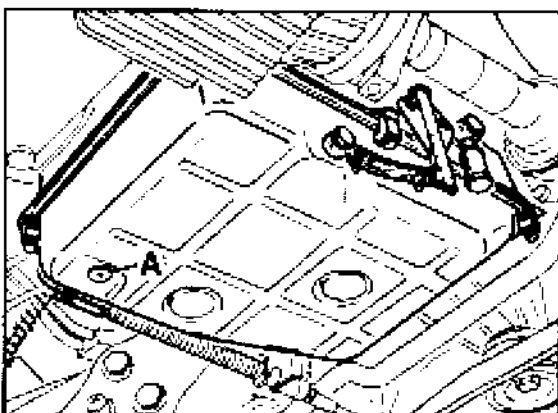
No	Designation	Qty.	Note:	
			Removal	Installation
1	Plug	1		Tighten to 40 Nm (29 ftlb)
2	Seal	1		Replace
3	Banjo bolt	1		Tighten to 40 Nm (29 ftlb)
4	Seal	2		Replace
5	Hexagon head bolt	6		Tighten to 6 Nm (4 ftlb)
6	Bracket (straight leg)	2		Short legs must force on the ATF pan
7	Bracket (curved leg)	4		Short legs must force on the ATF pan
8	ATF pan	1		
9	Gasket	1		
10	Magnet	2		Place into grooves in ATF pan
11	Pan head screw (M 6 x 65)	3		Tighten to 8 Nm (6 ftlb)
12	ATF strainer	1		
13	Round seal	1		Replace. Make sure that position is correct
14	Hexagon nut	1		Tighten to 12 Nm (9 ftlb)
15	Pan head screw	14		Observe correct length, tighten to 8 Nm (6 ftlb)
16	Spring washer	1		
17	Bracket	1		
18	Inductive rpm pickup	1		
19	Hydraulic control unit	1		The straight surface on the harness socket must point towards the outside

38 77 19 Removing and installing the hydraulic control unit

Removing

The transmission wiring harness remains on the transmission.

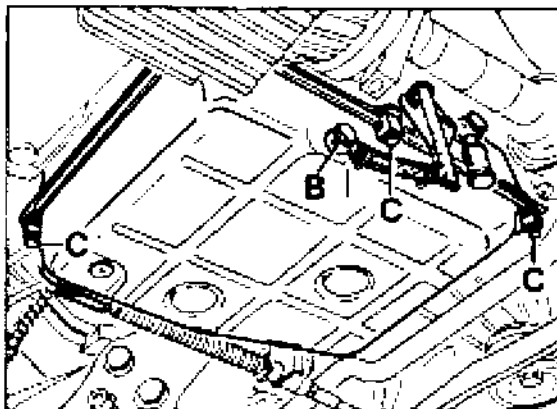
1. Remove the transmission underbody cladding.
2. Drain off ATF fluid.



386-38a

A = Drain screw with sealing ring

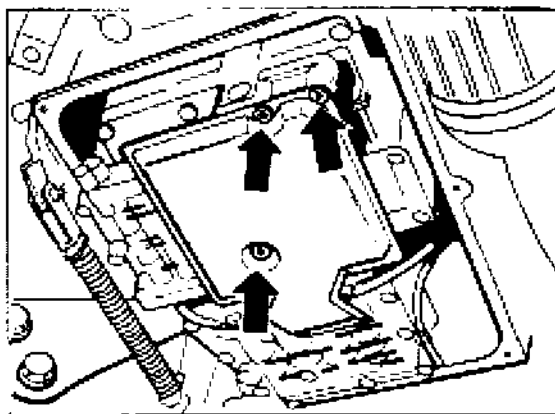
3. Remove ATF pan. To do this, unscrew hollow screw for oil level pipe and six fixing screws.



B = Hollow screw

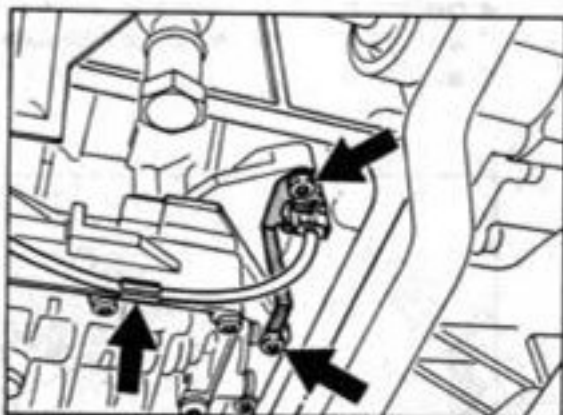
C = Fixing screws

4. Remove ATF strainer with Torx Insert T 27.



382-38

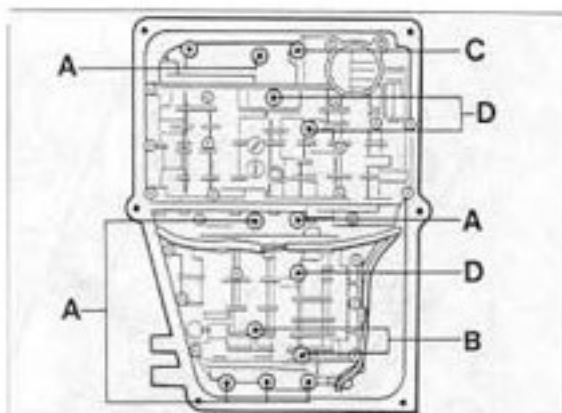
5. Remove inductive speed sensor and pull wiring harness out of holding clamps.



39G-38a

6. Unscrew 13 fixing screws (head diameter 12 mm) **with Torx Insert T 27** and lower hydraulic control unit only so far as to guarantee that the wiring harness is not subjected to any tension.

Place the hydraulic control unit on a suitable surface (e.g. transmission jack).



39G-38

7. Mark the push-on sleeves for re-installation and pull off from the solenoid valves.

8. Pull the wiring harness out of the holding clamps and remove the control unit.

Installing

Installation takes place in reverse order.

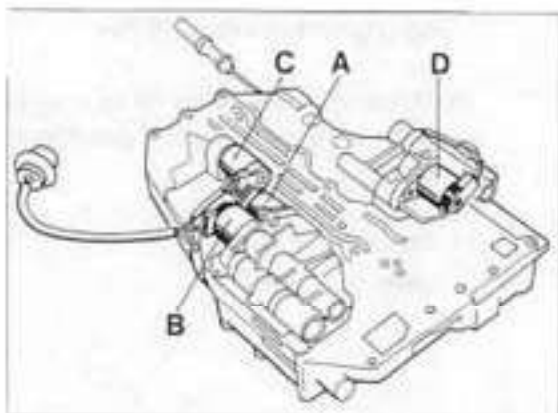
1. Tightening torques:

Control unit to transmission	= 8 Nm (6 ftlb)
ATF strainer to control unit	= 8 Nm (6 ftlb)
ATF pan to transmission	= 6 Nm (4 ftlb)
Drain screw to ATF pan	= 40 Nm (29 ftlb)
Hollow screw to ATF pa	= 40 Nm (29 ftlb)

Note

The hydraulic control unit must not be suspended from the wire harness.

2. Push on push-on sleeves for solenoid valves up to the stop. Pay attention to cable colors.

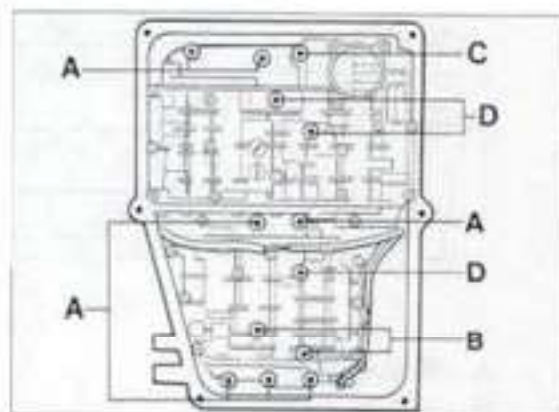


383-38

	Solenoid valves	Cable colors
A	Solenoid valve 1	(grey-violet)
B	Solenoid valve 2	(green-violet)
C	Solenoid valve 3	(red-violet)
D	Solenoid valve (pressure controller)	(blue-violet)

3. Mount the hydraulic control unit so that the pin of the notched disk projects into the recess of the selector slide.

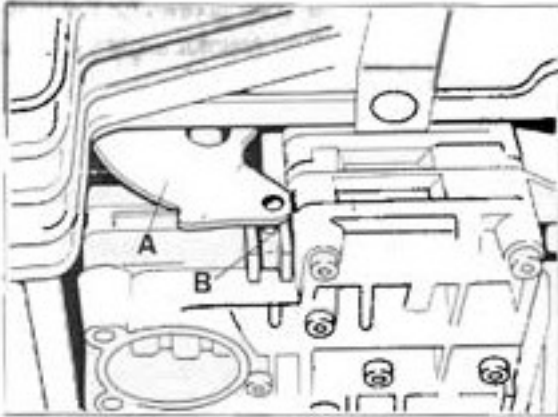
4. Screw in the fixing screws for the hydraulic control unit and counter slightly. Pay attention to screw lengths.



389-38

- A = Screw length 80 mm
 B = Screw length 65 mm
 C = Screw length 115 mm
 D = Screw length 60 mm

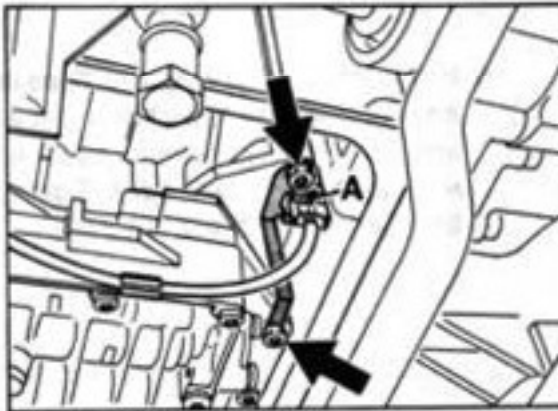
5. Position hydraulic control unit. To do this, move notched disk to position 1 (1st gear) and push hydraulic control unit back until it rests against the notched disk. Tighten fixing screws with 8 Nm in this position.



385-38

A = Notched disk
B = Selector slide

6. Insert pulse sensor and mount holding plate so that the lugs engage in the connector grooves.



390-38b

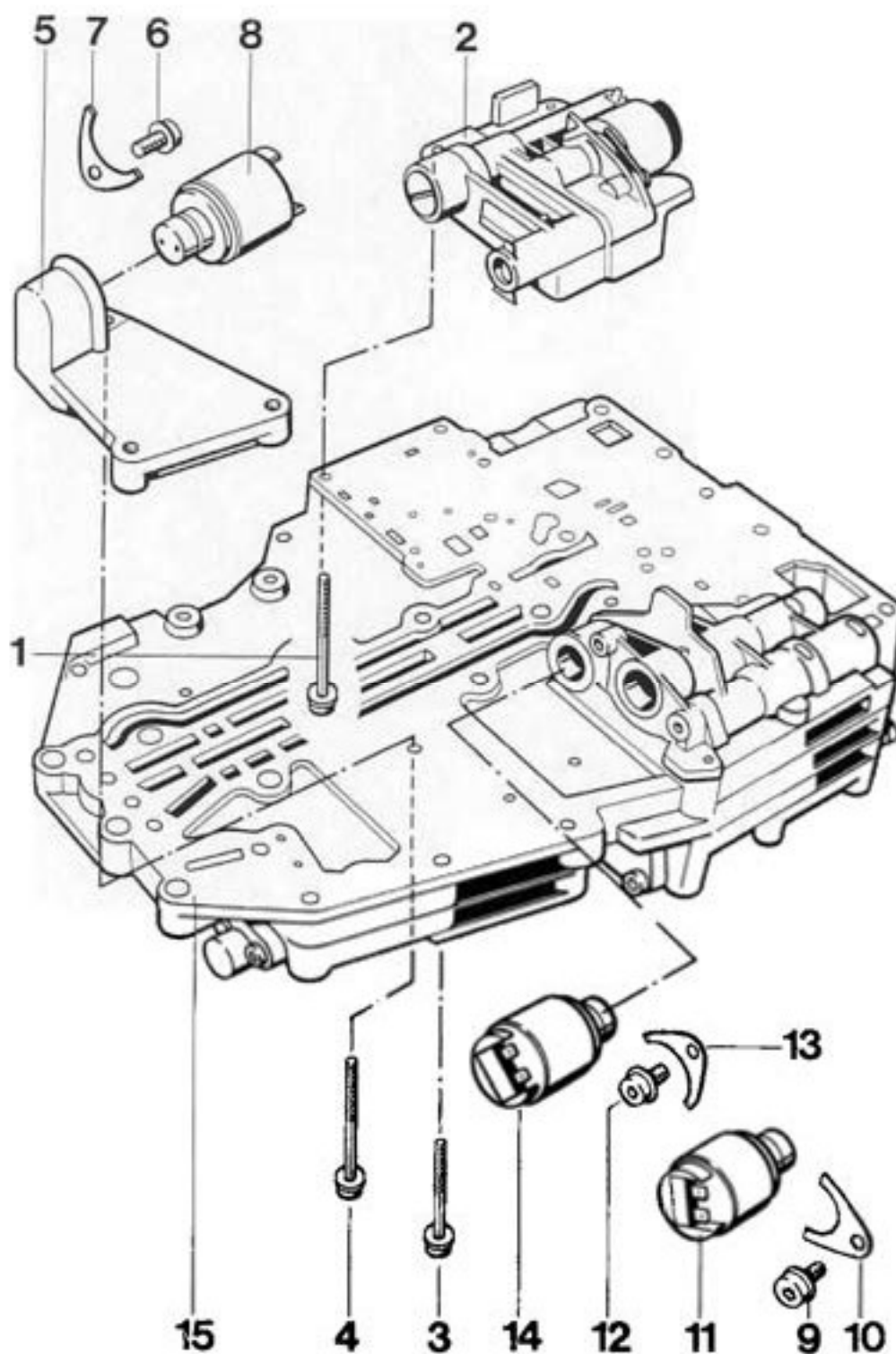
7. Fit ATF strainer with O-ring. Tightening torque **8 Nm**.
8. Place two magnets in the beads of the ATF pan, fit seal and secure the pan with the holding brackets so that the short legs press onto the ATF pan.

Note

The two holding brackets with the straight legs must be mounted at the side.

9. Screw in ATF drain screw with new sealing ring. Tightening torque **40 Nm**.
10. Screw in hollow screw for oil level pipe with new sealing rings. Tightening torque **40 Nm**.
11. Fill with ATF fluid (refer to page 38-103).

38 89 20 Removing and installing solenoid valves

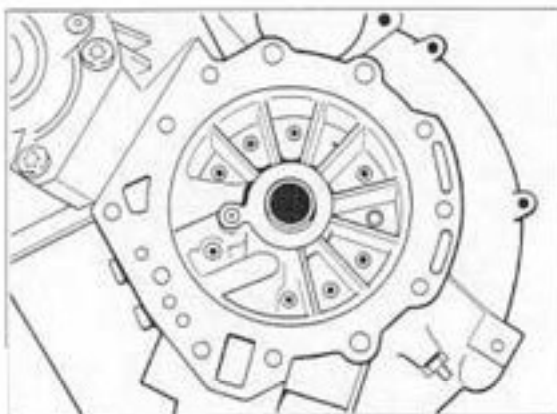


No.	Designation	Qty.	Note:	
			Removal	Installation
1	Fillister head screw	5	Remove with Torx insert T 27	Tighten with 5 Nm (4 ftlb)
2	Housing with pressure regulator	1		Set by the manufacturer, only replace complete
3	Fillister head screw	2	Remove with Torx insert T 27	Tighten with 5 Nm
4	Fillister head screw	1	Remove with Torx insert T 27	Tighten with 5 Nm
5	Holder	1		
6	Fillister head screw	1	Remove with Torx insert T 27	Tighten with 5 Nm
7	Holding plate	1		Fit in correct position, lugs point to holder No. 5
8	Solenoid valve 3	1	Mark for re-installation	Do not confuse with solenoid valves Nos. 11 and 14
9	Fillister head screw	1	Remove with Torx insert T 27	Tighten with 5 Nm
10	Holding plate	1		Fit in correct position, lugs point to housing
11	Solenoid valve 2	1	Mark for re-installation	Do not confuse with solenoid valves Nos. 14 and 8
12	Fillister head screw	1		Tighten with 5 Nm
13	Holding plate			Fit in correct position, lugs point to housing
14	Solenoid valve 1	1	Mark for re-installation	Do not confuse with solenoid valves Nos. 11 and 8
15	Hydraulic control unit	1		

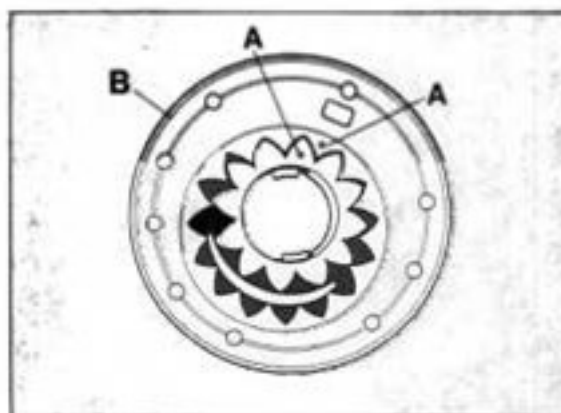
38 56 19 Removing and installing the ATF pump

Removing

1. Remove transmission.
2. Remove torque converter and final drive.
3. Unscrew fixing screws with Torx Insert T27 and remove pump. To do this, screw in two screws opposite each other and carefully drive out the pump by gentle blows on the screw heads



406-38



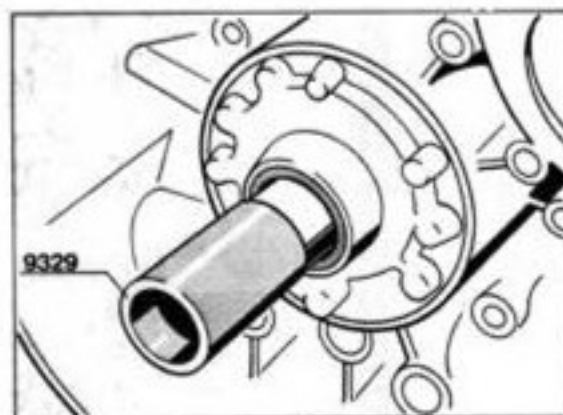
405-38

- A = Installation markings
B = O-ring

3. Install pump with new O-ring and centering sleeve. Tighten fixing screws with 10 Nm.
4. Check pump for easy operation. It must be possible to turn the pump by hand with special tool 9329 without it catching.

Installing

1. Check bearing bush for run-in scores or damage. The pump must be replaced if there are signs of damage.
2. Oil both pump wheels with ATF fluid and place in the housing so that both installation markings are visibly facing up.



421-38

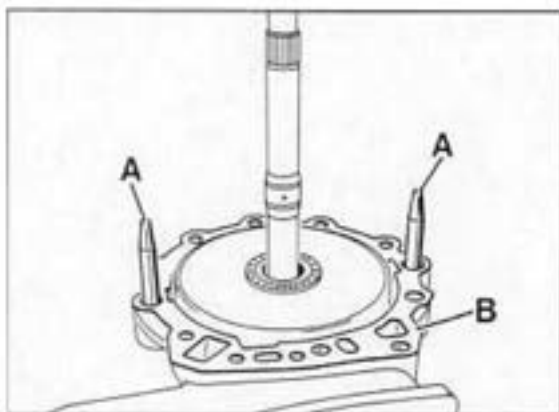
38 10 05 Determining the end clearance

Specification: 0.2...0.4 mm

Note

Check the end clearance whenever the transmission has been reassembled and readjust if required, using adjusting washer No. 12 (refer to page 39 - 111).

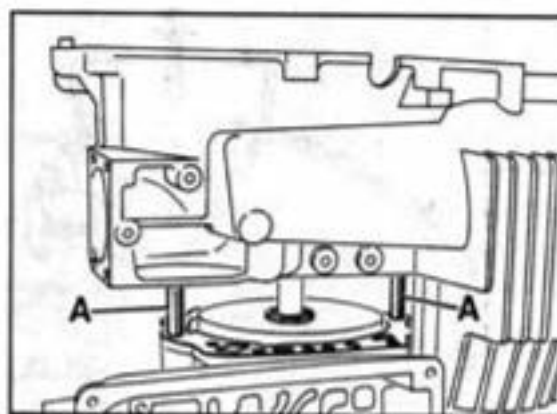
1. Assembly transmission (complete with spur gear drive and front transmission cover).
2. Screw centering pins **9321** into transmission case and apply some grease (vaseline) to stick paper gasket to sealing surface.



964-38

- A - Centering pins **9321**
B - Gasket

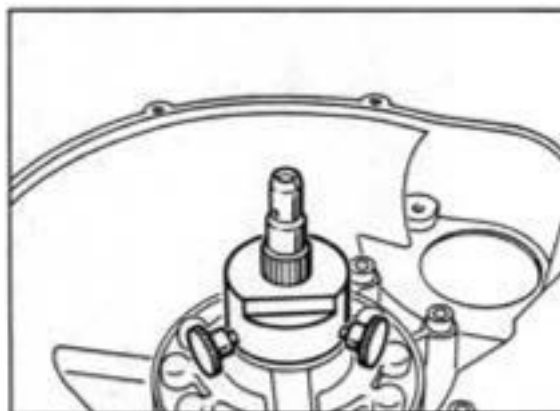
3. Using some grease (use vaseline), stick the removed adjusting washer to the final drive and put housing carefully into position.



961-38

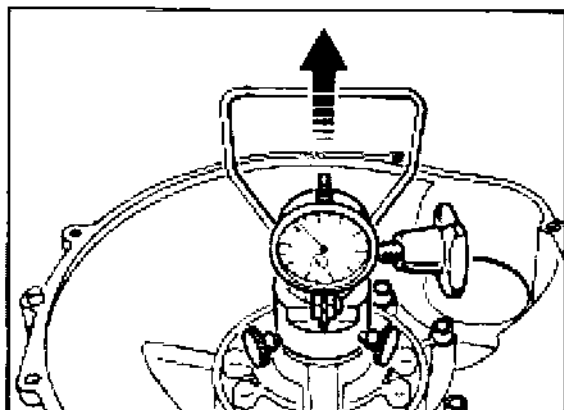
- A - Centering pins **9321**

4. Tighten all mounting screws to 46 Nm (34 ftlb).
5. Mount measuring sleeve of Special Tool **9338** with three mounting screws on the stator shaft in such a manner that no play remains.



962-38

6. Slide measuring device of Special Tool 9338 over drive shaft teeth and tighten with clamping screw so that it cannot be tilted.



963-38

7. Set measuring sleeve to zero with a certain preload.
8. Determine end clearance by pulling the handle. Repeat measurement several times.

Note

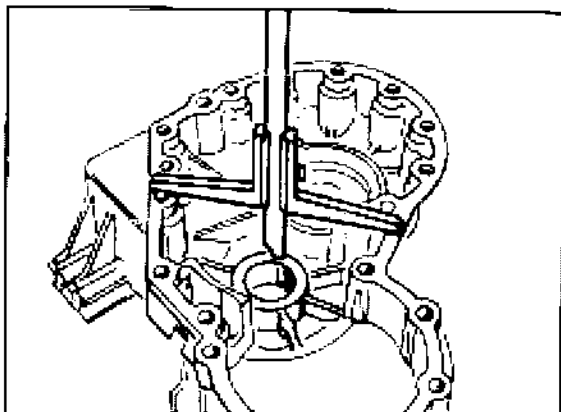
If a deviation is detected, remove final drive again and fit a thinner or thicker adjusting washer as required.

Recheck end clearance afterwards.

38 91 05 Adjusting the preload of the spur gear taper roller bearings

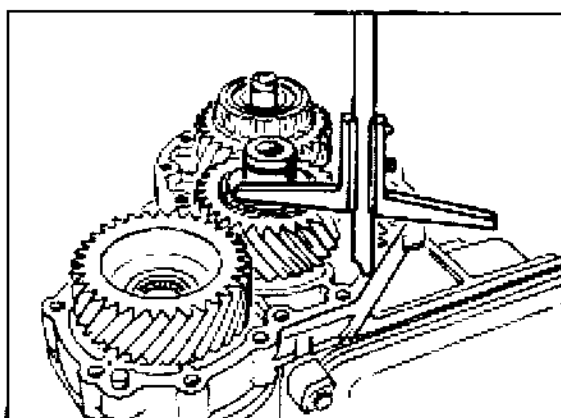
Adjusting spur gear II (intermediate gear)

1. Determine and record dimension "a" (e.g. 45.90 mm).



1030 - 39

2. Fit complete spur gear II and determine dimension "b" (e.g. 44.65 mm). Record measured value.



1031 - 39

3. Determine shim thickness by deducting dimension "b" from dimension "a" and adding a constant value of 0.22 mm (preload and thickness of compressed gasket).

Note

The constant value is 0.22 mm for all transmissions.

Example

Dimension "a" e.g.	45.90 mm
Dimension "b" e.g.	- 44.65 mm
	1.25 mm
Constant	+ 0.22 mm
Shim thickness	1.47 mm

Note

Always round shim thickness up or down to the nearest 0.05 mm figure.

Adjusting spur gear I (drive gear)

1. Remove intermediate plate.
2. Remove adjusting shim (refer to page 37 - 108, No. 16).
3. Remove spur gear II (intermediate gear) for measurement.
4. Bolt up cover with intermediate plate (but without gasket).
Tighten all screws and bolts to 23 Nm (17 ftlb).
5. Internal puller (e.g. Schrem 20 - 30) into spur gear teeth.
6. Fit dial gauge holder VW 387 with dial gauge and zero out dial gauge with a preload of 2 mm.
7. Move spur gear I up and down on internal puller and read off play on dial gauge (e.g. 1.18 mm).

Note

Do not turn or tilt the spur gear when measuring the clearance. Repeat measuring process several times.

8. Determine shim thickness. Measured value plus 0.20 mm (preload and thickness of compressed gasket) equals shim thickness

Note

The constant value is 0.20 mm for all transmissions.

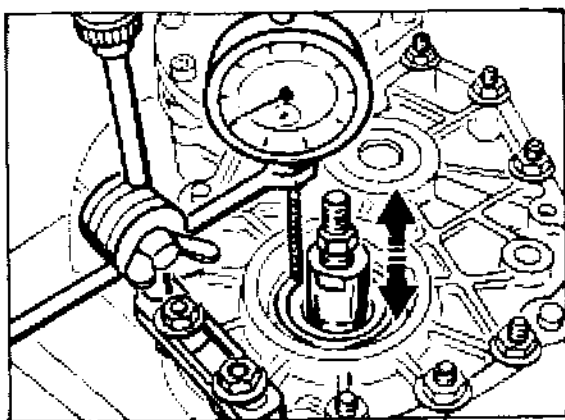
Example

Measured value	1.18 mm
Constant value	+ 0.20 mm
Shim thickness	<u>1.38 mm</u>

9. Fit a shim of the determined thickness (1.38 mm in our example).

Note

Always round shim thickness up or down to the nearest 0.05 mm figure.

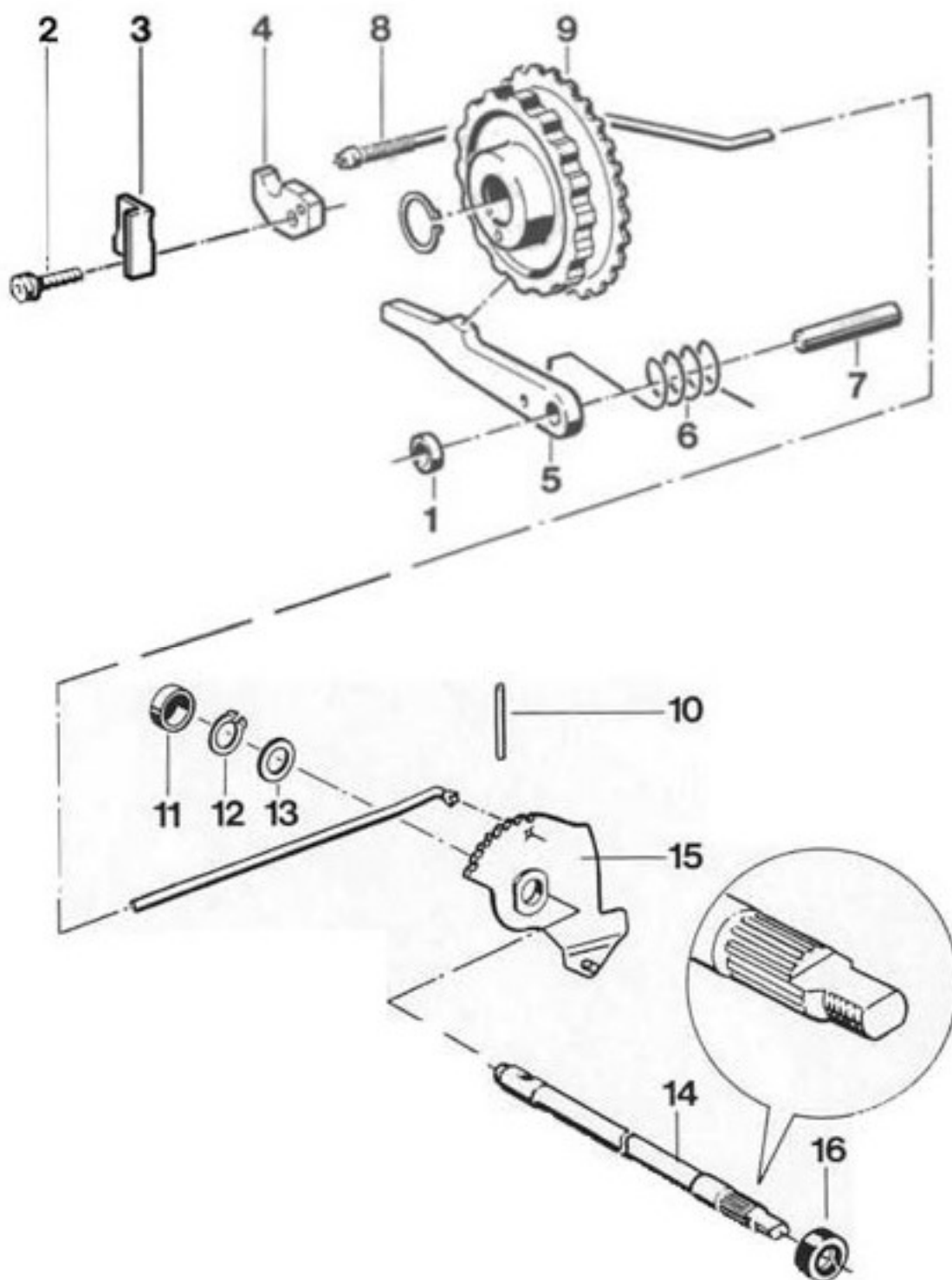


1029 - 39

A = Dial gauge extension
B = Internal puller

38 74 37

Disassembling and assembling the parking lock



No.	Designation	Qty.	Note:	
			Removal	Installation (= t/lb)
1	Bushing	1		
2	Fillister head screw	1	Remove with Torx insert T27	Tighten with 10 Nm
3	Guide plate	1		
4	Guide piece	1		
5	Catch	1		
6	Spring	1		Insert in correct position
7	Pin	1		
8	Circlip	1		
9	Parking lock wheel	1		Make sure that the square ring is correctly seated
10	Clamping pin	1		
11	Protective cap	1	Drive out to the outside by gentle blows on shaft No. 14	Replace
12	Circlip	1		
13	Shim ring	1		
14	Shaft	1	Mark installation position	Insert in correct position
15	Notched disk	1		
16	Rotary shaft seal	1	With transmission in built-in condition, remove multifunctional switch and carefully lever out with a suitable screwdriver	Wind plastic insulating tape around the shaft toothing as an assembly aid, slightly oil sealing lip and press in flush

Work instructions for disassembling and assembling

Disassembling

1. Remove transmission.

2. Insert spring for catch in correct position.

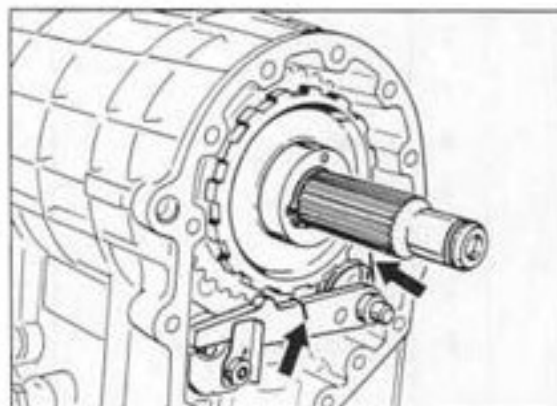
2. Remove intermediate plate.

Note

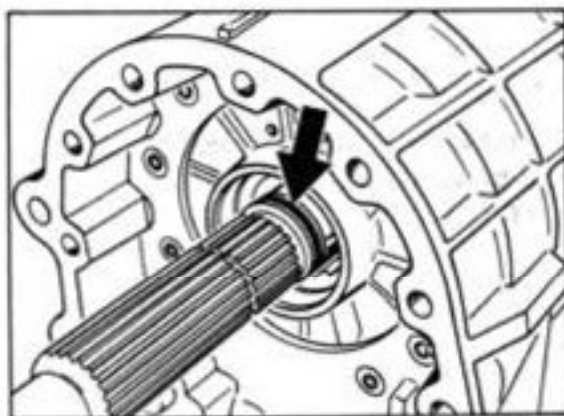
Part Nos. 11 to 15 can be removed only when the final drive and the ATF pan have been dismantled.

Assembly

1. Replace O-ring on output shaft.



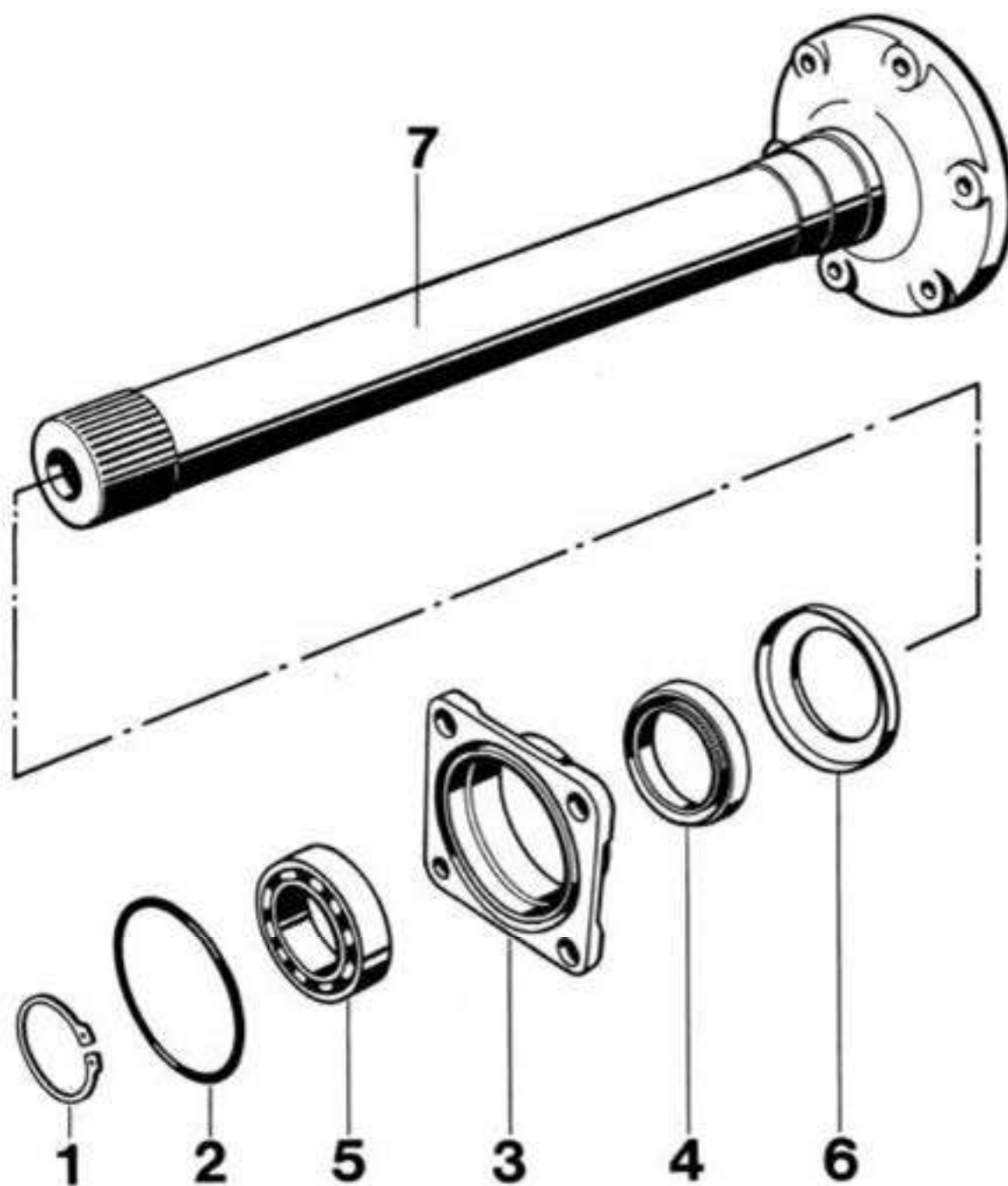
418-37



417-37

39 25 37

Disassembling and assembling the long joint flange



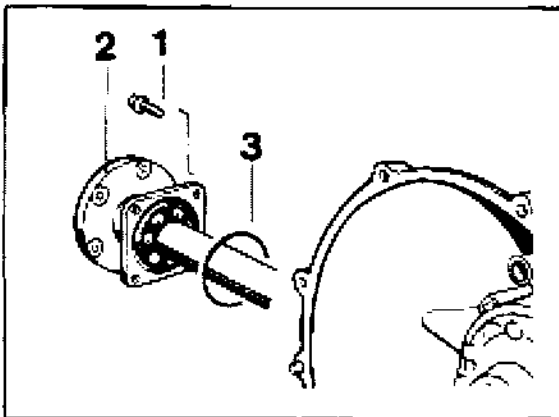
No.	Designation	Qty.	Note	
			Removal	Installation
1	Circlip	1		Replace
2	O-ring	1		
3	Bearing cap	1		
4	Rotary shaft seal	1		
5	Ball bearing	1	Press off jointly with bearing cap	Pack space between dust lip and sealing lip with grease (e.g. Shell 8420) and press in flush with suitable thrust piece Press in until it bottoms with a suitable pressure piece
6	Protective ring	1		
7	Joint flange	1		

39 25 19 Removing and installing long halfshaft flange

With transmission installed

Removal

1. Remove drive shaft (see page 42 - 17).
2. Unscrew pan head screws of halfshaft flange using special tool 9330 and pull out flange.



2184-39

- 1 = pan head screw
2 = halfshaft flange
3 = O-ring

Installation

1. Install the parts in the reverse order of dismantling
2. Replace O-ring. Slightly oil new O-ring.
3. Tighten pan head screws to **23 Nm** (17 ftlb).

39 26 19 Removing and installing short halfshaft flange

With transmission installed

Removal**Note**

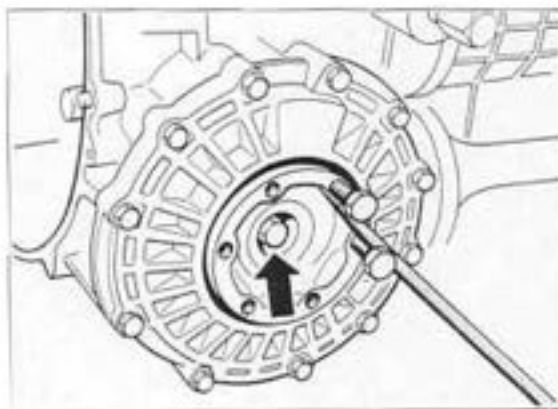
The halfshaft flange can only be removed if the suspension is partially disconnected.

1. Remove right rear wheel, engine / transmission guard and rear underside panel.
2. Remove heating pipe.
3. Remove control arm cover.
4. Disconnect drive shaft from transmission flange.
5. Partially disconnect suspension (see page 42 - 17 from point 4).

Note

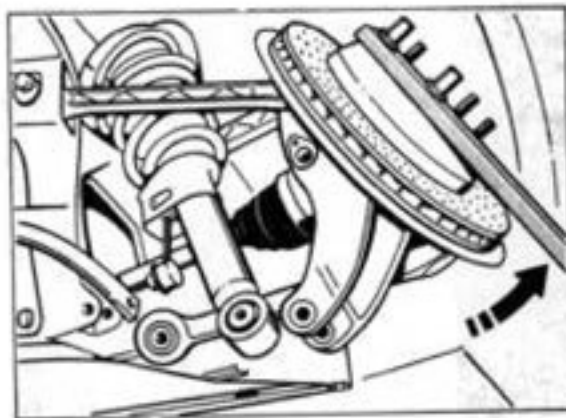
The drive shaft need not be disconnected from the wheel.

6. Unscrew fastening bolt for halfshaft flange (the illustration shows the transmission removed).



408-39

7. Lift wheel carrier using a suitable lever and take out flange (2nd person required).



2065-42

Installation

1. Install the parts in the reverse order of removal.
2. Tighten screws to specified torque figures.

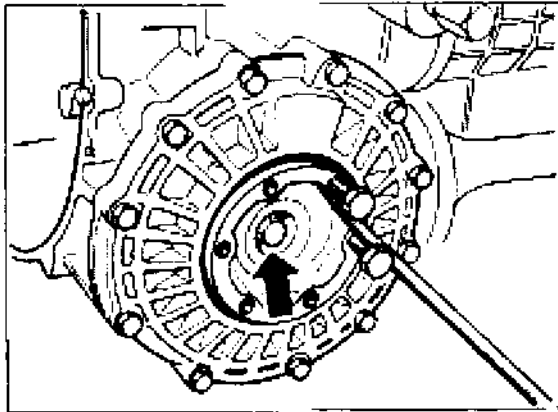
39 22 19 Removing and installing the rotary shaft seal for the short joint flange

Removing

Note

The rotary shaft seal can also be replaced with the transmission in built-in condition.

1. Unscrew the hexagon screw for the joint flange and remove flange.



408-39

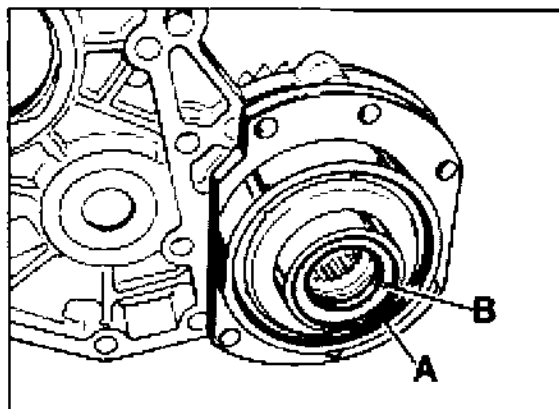
2. Lever out rotary shaft seal with VW 681

Installing

1. Press in rotary shaft seal flush with suitable pressure piece.
2. Tighten hexagon screw for joint flange with 46 Nm (34 ftlb).

39 28 19 Removing and installing drive pinion seal rings

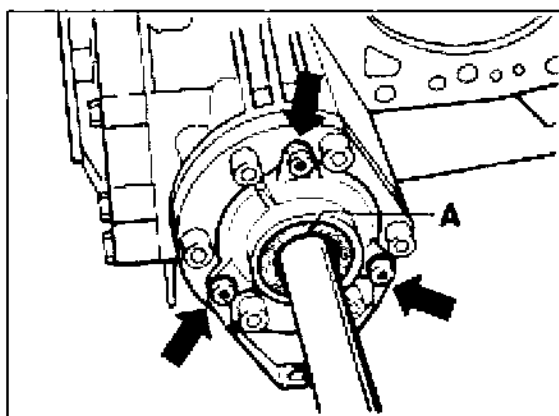
1. Drain ATF
2. Remove engine with gearbox (do not unbolt engine from gearbox).
3. Remove intermediate plate (see page 37 - 125).
4. Remove bearing cap with Torx Insert T 27 and replace rotary shaft seal. Pack space between dust lip and sealing lip with grease (e.g. Shell 8420).
Press-in depth 2.0 ± 0.5 mm.



423-39

A = Snap ring

B = Rotary shaft seal



A = Rotary shaft seal

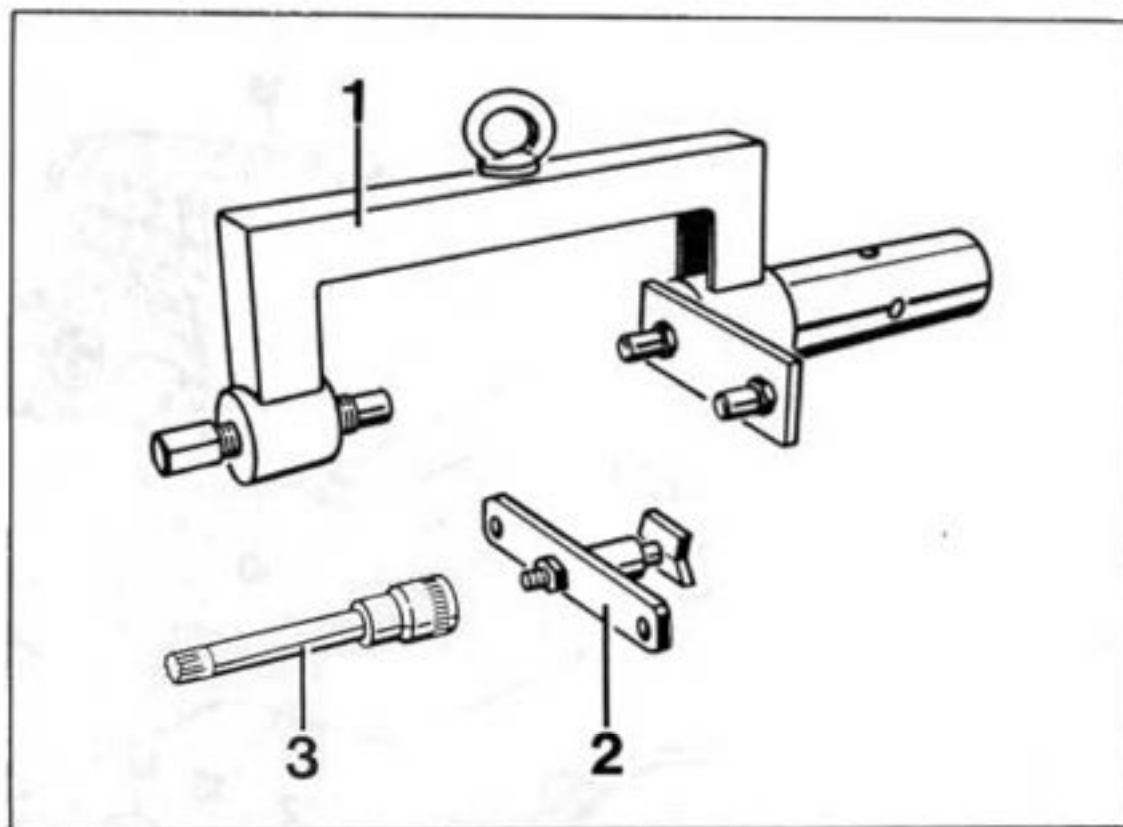
422-39

5. Fit bearing cap with new O-ring and tighten fixing screws with 10 Nm (7.5 ftlb).
6. Remove snap ring for bearing cap at intermediate plate and pull out cap.

7. Replace rotary shaft seal.
Pack space between dust lip and sealing lip with grease (e.g. Shell 8420).
Press-in depth 2.0 ± 0.5 mm
8. Fit new O-ring and wet with ATF.
9. Insert bearing cap and fit snap ring.
10. Fill with ATF.

39 01 19 Removing and installing rear transmission case

Tools



No.	Designation	Special tool	Order number	Explanation
1	Transmission holder	9324	000.721.932.40	
2	Holding device	9325	000.721.932.50	
3	socket key insert	9330	000.721.933.00	

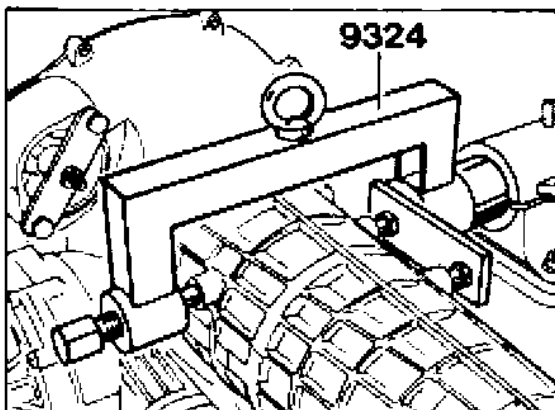
No.	Designation	Qty.	Note:	
			Removal	Installation (= ftlb)
1	Fillister head screw	4	Use Special Tool 9330 to remove	Tighten with 23 Nm (17)
2	Joint flange	1		
3	O-ring	1		Replace, oil slightly
4	Screw plug	3	Remove with Torx insert T55	Tighten with 50 Nm (36)
5	Sealing ring	3		Replace
6	Fillister head screw	3	Remove with Torx insert T50	Tighten with 46 Nm (34)
7	Fillister head screw	3	Remove with Torx insert T50	Tighten with 46 Nm (34)
8	Fillister head screw	6	Remove with Torx insert T50	Tighten with 46 Nm (34)
9	Rear transmission case	1		
10	Protective tube	1		Large diameter to final drive
11	Seal	1		Replace, glue to case with a small quantity of grease (vaseline)
12	Shim*	X	Note thickness for re-installation	Redetermine thickness if required (refer to page 38 - 171)
13	Needle cage	1		
14	Angle ring	1		Fit in correct position
15	Automatic transmission	1		Pay attention to installation position of clutch A. Make sure that the square drive shaft rings are correctly fitted. Coat the drive shaft with a small quantity of grease in the area of the square rings.

* Must be checked and, if required, corrected after each transmission installation.

Assembly instructions for removal and installation

Removal

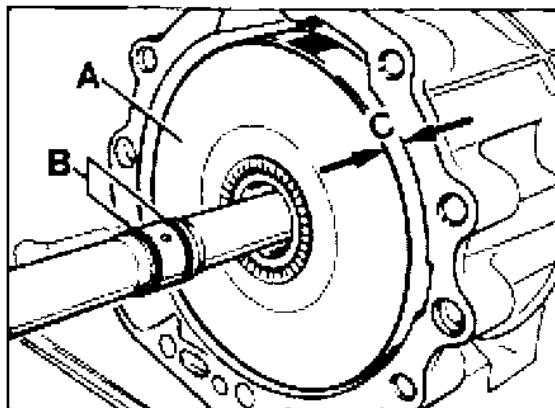
1. Remove the transmission and torque converter.
2. Secure transmission on assembly block with holding device 9324.



416-37

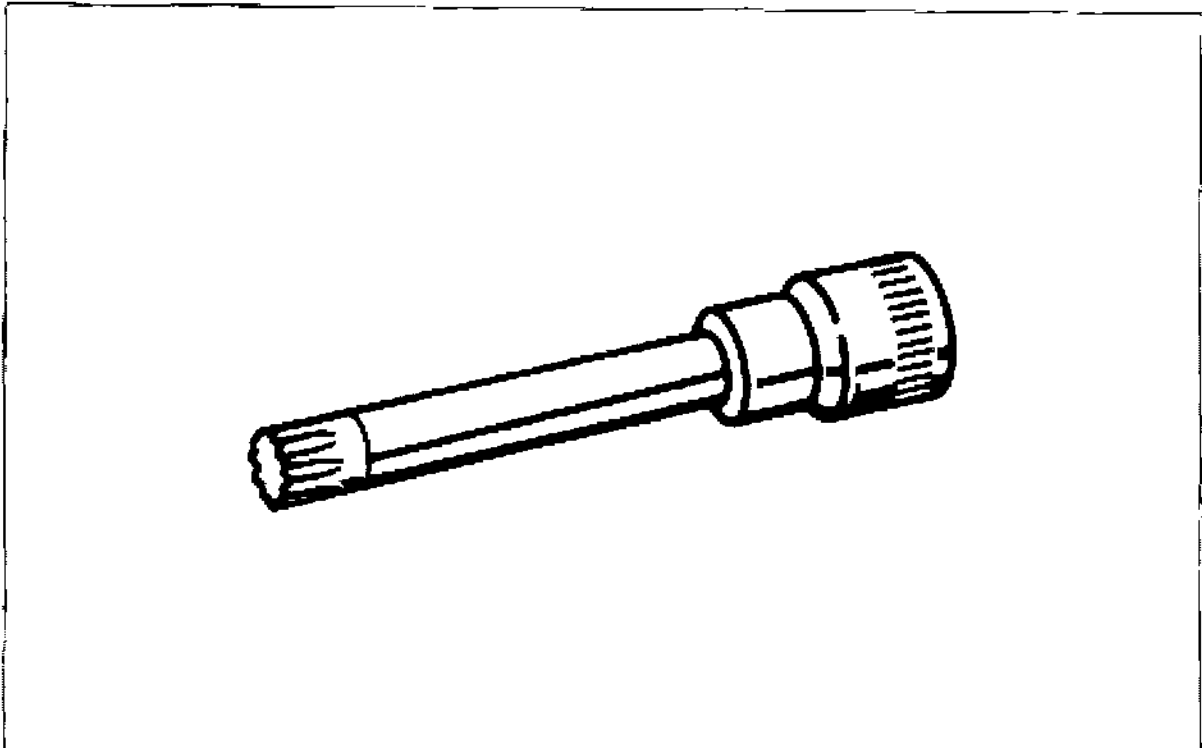
Installation

1. Check the installation position of clutch A.
The installation depth has been reached when the distance "C" is approx. 8.5 mm.



416-37

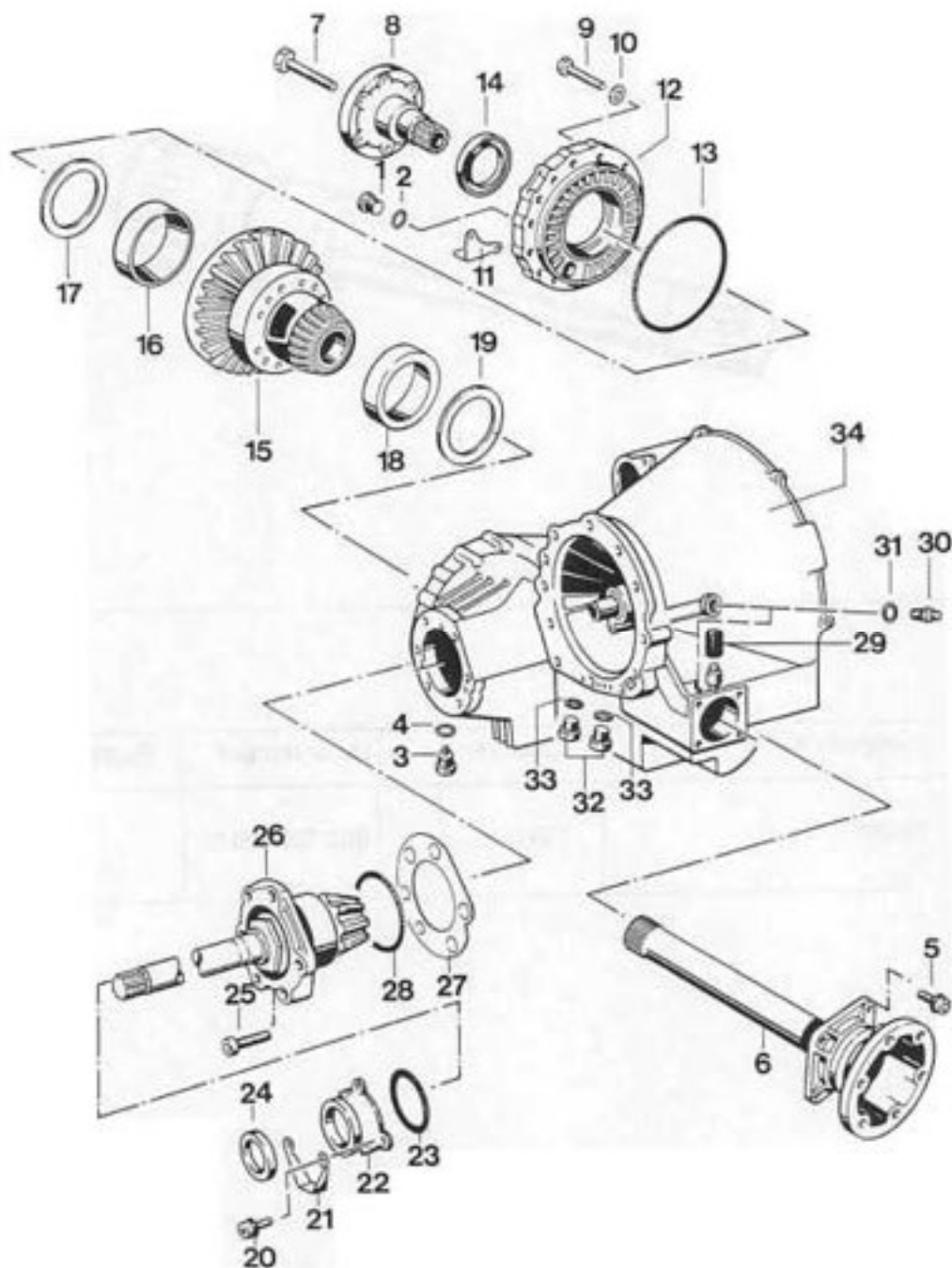
- A = Clutch A
B = Square rings
C = Installation depths

39 09 19 Removing and installing differential and drive pinion

No.	Designation	Special tool	Order number	Explanation
	Socket	9330	000.721.933.00	

39 09 19

Removing and installing differential and drive pinion



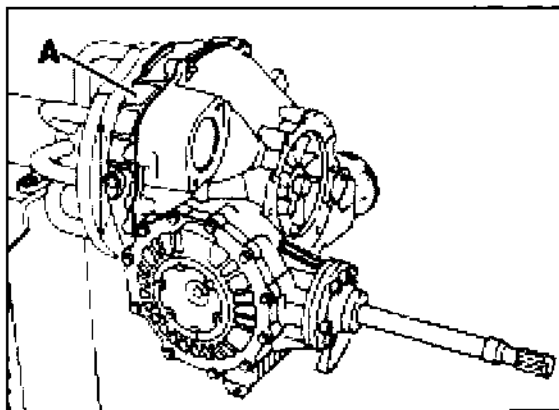
946-39

No.	Designation	Qty.	Note:	
			Removal	Installation
1	Plug	1		Tighten to 50 Nm (37 ftlb)
2	Seal	1		Replace
3	Plug with solenoid	1		Tighten to 50 Nm (37 ftlb)
4	Seal	1		Replace
5	Pan head screw	4	Remove with Special Tool 9330	Tighten to 23 Nm (17 ftlb)
6	Halfshaft flange	1		For dismantling, refer to page 39 - 102
7	Hexagon head bolt	1		Tighten to 46 Nm (34 ftlb)
8	Halfshaft flange	1		
9	Hexagon head bolt	11		Tighten to 23 Nm (17 ftlb)
10	Spring washer	11		
	Bracket	1		
12	Transmission side cover	1		
13	Round seal	1		Replace, oil lightly
14	Shaft seal	1	Replace	Pack space between dust lip and sealing lip with grease (e.g. Shell 8420) and press in flush
15	Differential	1		
16	Taper roller bearing outer race	1		
7	Adjusting shim "S ₁ "	X	Record thickness for refitting	Redetermine thickness if required
18	Taper roller bearing outer race	1		
19	Adjusting shim "S ₂ "	X	Record thickness for refitting	Redetermine thickness if required
20	Pan head screw	3		Tighten to 10 Nm (7 ftlb)

No.	Designation	Qty.	Note:	
			Removal	Installation
21	Bracket	1		
22	Bearing cover	1		
23	Round seal	1		Replace
24	Shaft seal	1		Replace. Pack space between dust lip and sealing lip with grease (e.g. Shell 8420). Press in to depth of 2.0 ± 0.5 mm
25	Pan head screw	6		Tighten to 50 Nm (37 ftlb)
26	Drive pinion with bearing assembly	1	To drive out, use a plastic hammer to apply light blows on the drive pinion head	Observe matching number, readjust if required
27	Adjusting shim "S3"	X	Record thickness for reassembly	Redetermine thickness if required
28	Round seal	1		Replace, oil lightly
29	Breather	1		
30	Screw-in flange	2		Tighten to 35 Nm (26 ftlb)
31	Seal	2		Replace
32	Plug	2		Tighten to 25 Nm (18 ftlb)
33	Seal	2		Replace
34	Final drive	1		

Removal note

Mount rear transmission case with converter housing on assembly support.

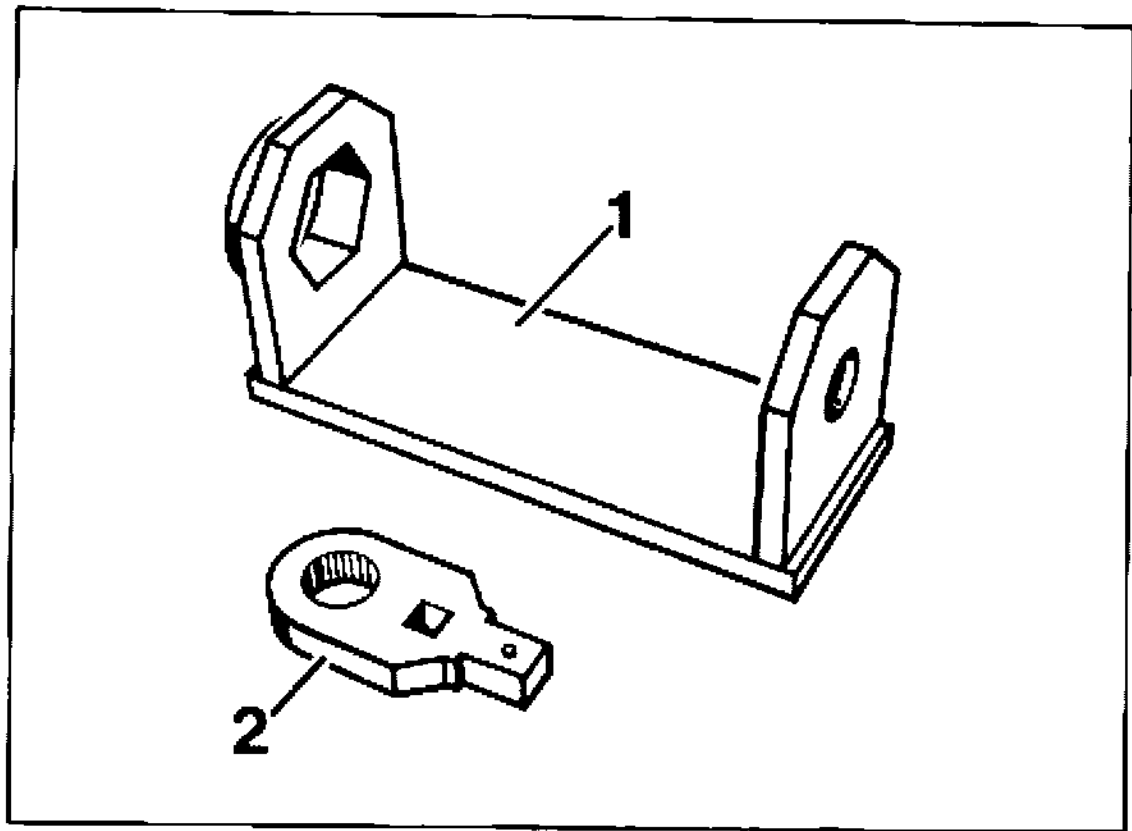


965-39

A - Converter housing (spare part)

No.	Designation	Qty.	Note:	
			Removal	Installation
1	Hexagon nut	12		replace. Thread must be dry and free from grease. Secure with Loctite 262. Tightening torque: 85 Nm (62.7 ftlb)
2	Lock screw	12		replace. Thread must be dry and free from grease. Install in correct position.
3	Ring gear	1	Separate from housing by applying light blows with a plastic hammer	Observe matching number. Readjust if required
4	Thrust washer	1		
5	Shaft bevel gear	1		
6	Nut	1		
7	Roll pin	3		Press in place in correct position
8	Stud (long)	1		Lock with roll pin
9	Stud (short)	2		Lock with roll pin
10	Cross fitting	1		
11	Bevel gear	4		
12	Thrust washer	4		
13	Shaft bevel gear	1		
14	Thrust washer	1		
15	Taper roller bearing inner race	1	Pull off with suitable puller, or remove ring gear and take out from inside through bores	Heat to approx. 120° C and press on
16	Taper roller bearing inner race	1	Pull off with P 263	Heat to approx. 120° C and press on
17	Housing	1		Fit centering pin with Loctite 262

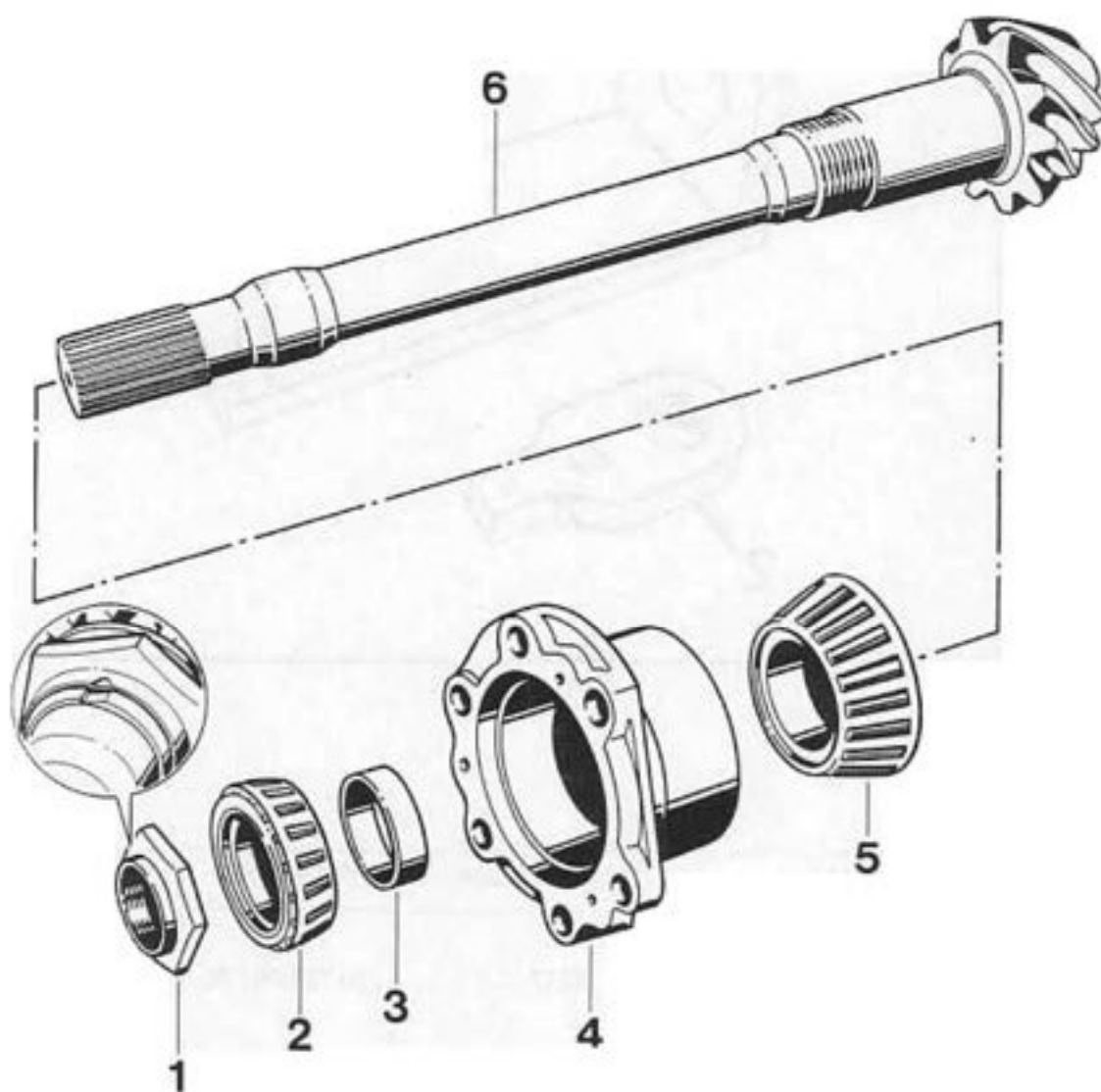
39 30 19 Dismantling and assembling drive pinion



967-30

No.	Designation	Special tool	Order number	Explanation
1	Retainer bracket	9337	000.721.933.70	
2	Insert	9282	000.721.928.20	

39 30 19 Dismantling and assembling drive pinion



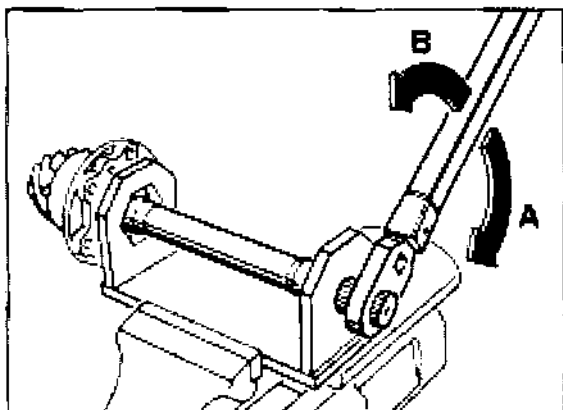
944-30

No.	Designation	Qty.	Note:	
			Removal	Installation
1	Lock nut	1	Undo with Special Tool 9337 and 9282	Tighten to 250 Nm (184 ftlb) and secure by upsetting the flange in two places
2	Taper roller bearing inner race	1	Press off with bearing flange	Heat to approx. 120° C and press on
3	Adjusting ring	X		
4	Bearing flange	1		
5	Taper roller bearing inner race	1	Press off with separating device	Heat to approx. 120 °C and press on
6	Drive pinion	1		

Notes on removal

Dismantling

Undo lock nut with Special Tools 9337 and 9282.



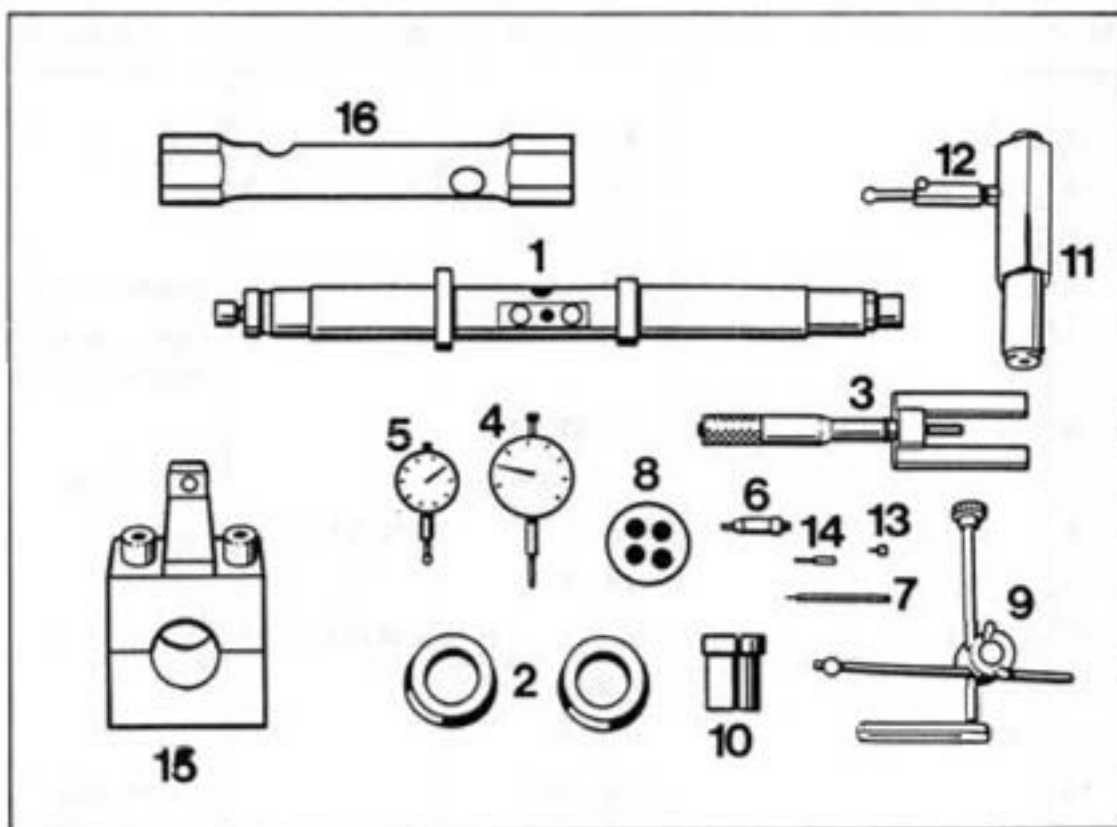
968-39

A - Undoing

B - Tightening

39 08 15 Adjusting drive set

Tools



Adjusting drive set

Tools

No.	Designation	Special tool	Order number	Explanation
1	Measuring mandrel	VW 385/1	000.721.932.70	commercially available
2	Centering discs	9327		
3	Master gauge	VW 385/30		
4	Dial gauge	-		
5	Dial gauge	-		
6	Gauge plunger	VW 385/14	000.721.928.10	commercially available, measuring range 3mm
7	Dial gauge extension	VW 385/56		
8	Gauge block plate	9281		
9	Dial gauge bracket	VW 387		
10	Clamping sleeve	9145		
11	Adjusting device	VW 521	000.721.914.50	30 mm long
12	Measuring lever	VW 388		
13	Dial gauge extension	VW 382/10		
14	Dial gauge extension	VW 385/53		
15	Clamping device	9339		
16	Socket wrench (24 mm A/F)	-	000.721.933.90	6.0 mm long
				14 mm long
				commercially available (e.g. Stahlwille No. 10 750)

Practical procedure when readjusting the drive set

If it becomes necessary to adjust drive pinion and ring gear, follow the below sequence to ensure an efficient working procedure:

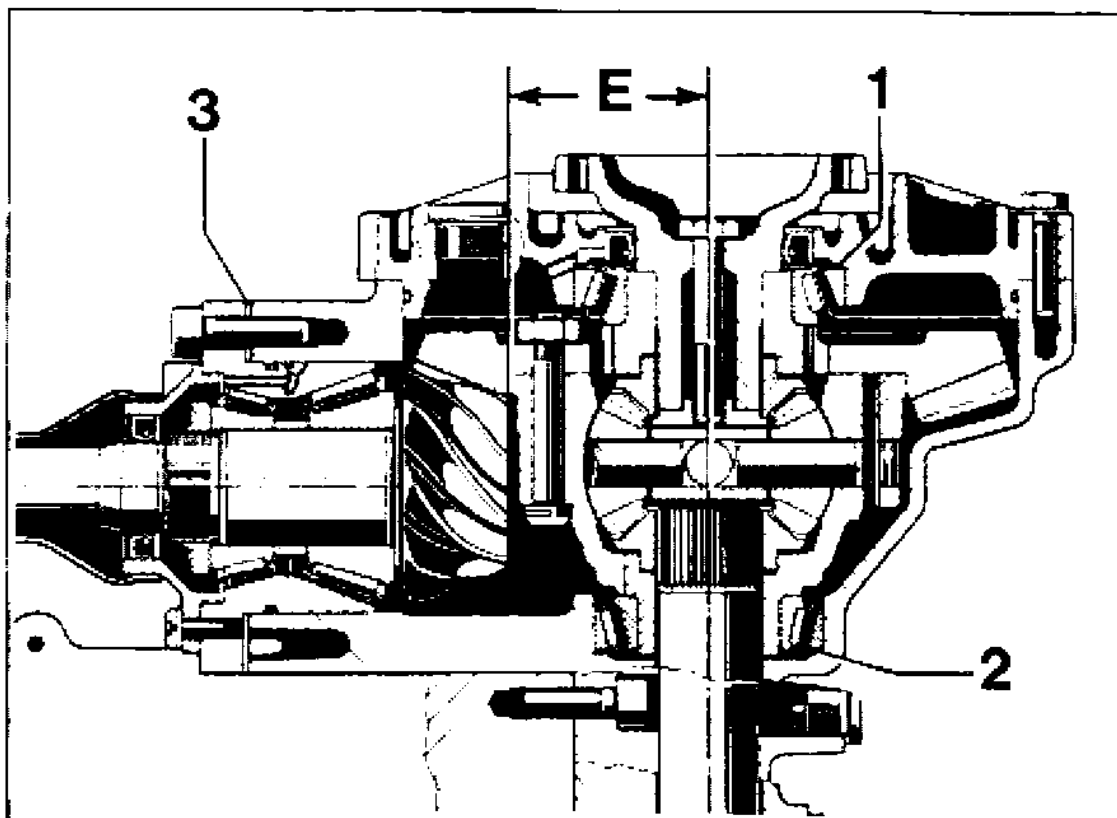
1. Determine the total shim thickness "Stot" (S_1 plus S_2) for the specified preload on the taper roller bearings/differential.
2. Determine the thickness of shim " S_3 ".
3. Split the total shim thickness "Stot" into S_1 and S_2 so that the specified circumferential backlash is present between ring gear and drive pinion.

The aim of this adjustment is to restore the smoothest running position which has been achieved on test equipment in the production line.

To achieve correct results, greatest possible cleanliness for all assembly work and measuring procedures is essential.

When assembling the final drive assembly, it is only necessary to readjust drive pinion and ring gear or drive set if components have been replaced which have a direct influence on the adjustment. Refer to the following table to avoid unnecessary adjustment procedures!

Adjust: Replaced component	Ring gear ($S_1 + S_2$)	Drive pinion (S_3)
Rear transmission case	x	x
Transmission side cover	x	
Bearing assembly for drive pinion	x	x
Drive set	x	x
Differential housing	x	
Taper roller bearing for differential	x	



- 1 – Spacer S₁
- 2 – Spacer S₂
- 3 – Adjusting shim S₃
- E – Adjustment dimension

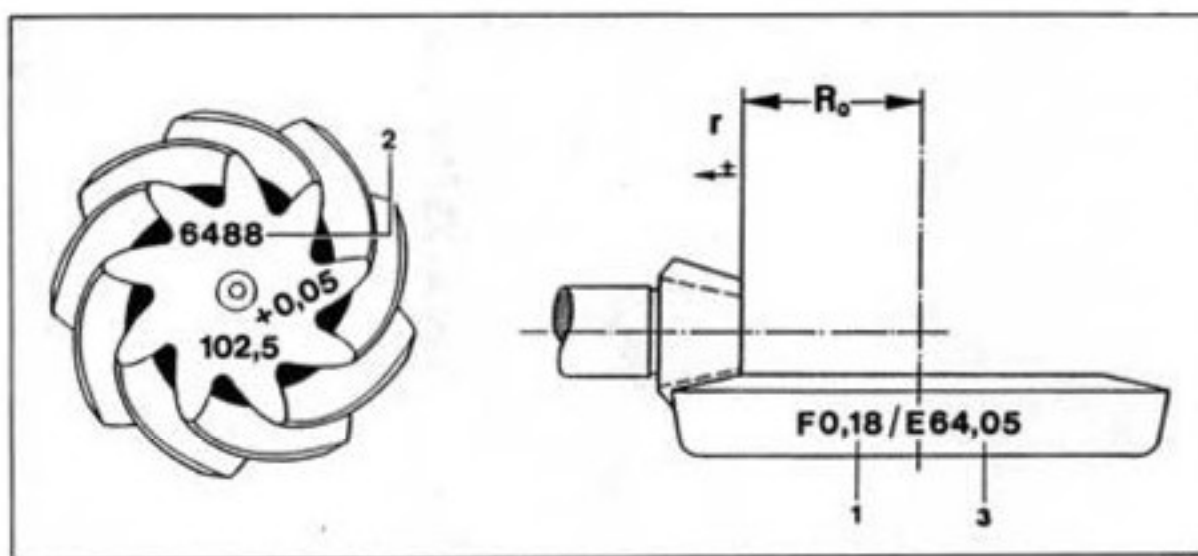
945-36

Correct results can only be achieved if assembly work and measuring procedures are carried out carefully and with maximum cleanliness.

Adjusting drive set

General

The setting of drive pinion and ring gear is a determining factor for the service life and smooth running of the rear-axle drive. Drive pinions and ring gear that have been checked for good tooth contact pattern and low noise in both directions of rotation on special test equipment are therefore matched during production. The position at which smoothest running can be achieved is determined by shifting the drive pinion axially, with the ring gear being kept within the tolerance of the prescribed tooth backlash. The deviation "r" from the specified design dimension "Ro" is measured, added to the design dimension "Ro" and engraved on the ring gear as setting value "E".



996-30

R_o - design dimension (64.00 mm)

r - deviation r

1 - Backlash "F" (e.g. 0.18 mm)

2 - Matching number

3 - Setting value "E" ($R_o + r$)

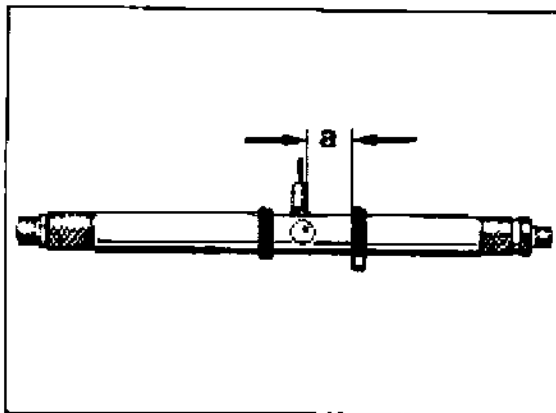
All other characters are used for adjustment in production.

Adjusting drive pinion

Note

The setting dimension "E" is indicated on the ring gear.

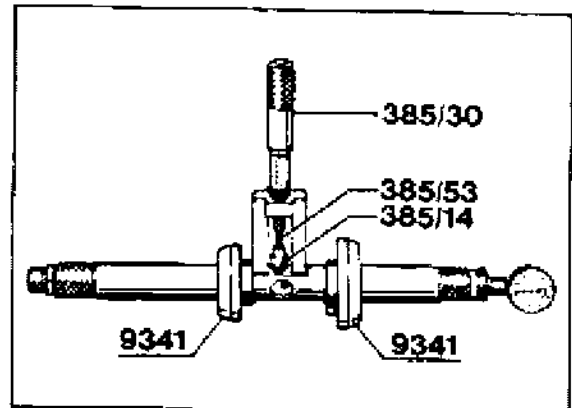
1. Install drive pinion **without** shims "S₃" and tighten all pan head screws of bearing assembly to 50 Nm (37 ftlb).
2. Rotate adjustable stop ring along with spindle towards measuring plunger as far as it will go and set second setting ring to dimension "a".



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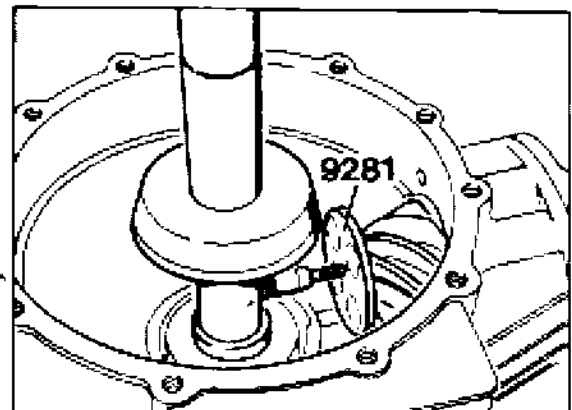
a = 30 mm

3. Assemble measuring mandrel and set with master gauge VW 385/30 to setting dimension "E" (64.05 mm in the example). Set dial gauge (3 mm measuring range) to zero with 1 mm preload.



911-39

4. Put gauge block plate 9281 on drive pinion head and insert measuring mandrel with dial gauge towards transmission side cover into transmission case. Dial gauge extension points towards center of drive pinion.



5. Fit transmission side cover without shaft seal and sealing ring and tighten crosswise with 4 hexagon head bolts.

Note

Do not use a hammer when fitting the transmission side cover (the gauge block plate held by magnets might fall off). Locate cover in correct installation position only by tightening the hexagon head bolts uniformly.

6. Using a socket wrench (24 mm A/F, e.g. Stahlwille No. 10 750), pull adjustable centering disc over spindle towards the outside until the measuring mandrel can just about be turned.
7. Turn measuring mandrel carefully until the dial gauge extension is vertical to the face of the drive pinion head. At this point, the pointer of the dial gauge is at its maximum deflection (reversing point) and the dial gauge must now be read.

Note

The measured value always deviates from the set dimension in clockwise direction (small pointer on the dial gauge is between 1 and 3), i.e. if the dial gauge is set with a preload of 1 mm, the value deviating from 1 is taken as the shim thickness "S₃" to be inserted.

Always round up or down to the nearest 0.05 mm (e.g. 1.63 to 1.60 mm).

8. After inserting the required shims, check the setting value "E" once again. A deviation of ± 0.03 mm is permissible.

Adjusting ring gear

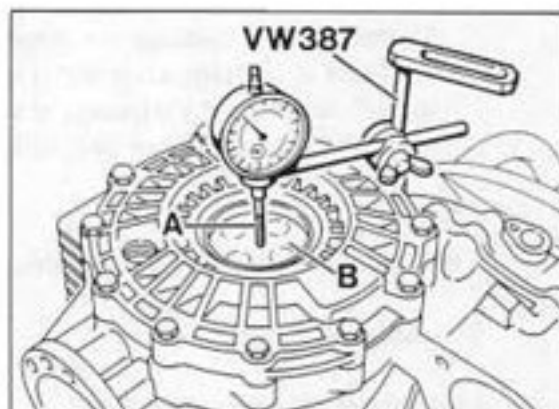
Determining total shim thickness "Stot." ($S_1 + S_2$)

The ring gear must be readjusted if the transmission case, transmission side cover, taper roller bearing for differential, differential housing or drive set have been replaced.

Note

The drive pinion must be removed in order to determine the preload of the differential tapered roller bearings.

1. Remove adjusting shim "S₁" (in transmission side cover).
2. Adjusting shim "S₂" remains in the transmission case.
3. Make sure that the bearing outer races of the tapered roller bearings are well seated in the transmission case or in the transmission side cover.
4. Insert differential into transmission case and rotate several times.
5. Fit transmission side cover without seals and tighten all hexagon head bolts to **23 Nm (17 ftlb)**.
6. Place gauge block plate 9281 on the collar of the differential.
7. Fasten universal dial gauge holder **VW 387** with dial gauge and extension to the case and set to 0 with 2 mm preload.



973-30

- A - Dial gauge extension (approx. 30 mm long)
B - Gauge block plate 9281

8. Using a suitable tool, move differential up and down.
Read off backlash on the dial gauge and note.

Note

Do not turn differential while measuring backlash as this will give an incorrect reading.

9. Remove adjusting shim "S₂" and determine thickness using a micrometer.
10. Calculate "Stot".
"Stot." = Thickness of adjusting shim "S₂"
+ measured value
+ pressure fit of taper roller bearings

Example

Thickness of adj. shim "S ₂ "	1.70 mm
Measured value	0.91 mm
Press fit (constant value)	0.24 mm
"Stot"	2.85 mm

11. Spread calculated shim thickness "Stot" as follows.

To start with the backlash adjustment, the thickness of adjusting shim "S₁" is reduced by 0.40 mm while the thickness of adjusting shim "S₂" is increased by 0.40 mm.

Example

Total shim thickness of adjusting shims

$$S_1 + S_2 = 2.85 \text{ mm}$$

Thickness of adjusting shim "S₁"

$$\begin{array}{r} 2.85 \text{ mm} \\ \hline 2 \\ \hline \end{array} \qquad \begin{array}{r} 1.425 \text{ mm} \\ - 0.40 \text{ mm} \\ \hline 1.025 \text{ mm} \end{array}$$

Thickness of adjusting shim "S₂"

$$\begin{array}{r} 2.85 \text{ mm} \\ \hline 2 \\ \hline \end{array} = \begin{array}{r} 1.425 \text{ mm} \\ + 0.40 \text{ mm} \\ \hline 1.825 \text{ mm} \end{array}$$

Note

The adjusting shims are avail. in thicknesses of 1.0...2.0 mm in increments of 0.05 mm.

The shim thicknesses calculated must be rounded up or down for plausible dimensions that will not alter the total shim thickness S₁ and S₂.

Example

Calculated shim thickness

$$S_1 + S_2 = 1.025 + 1.825 = 2.85 \text{ mm}$$

Rounded down shim thickness

$$S_1 + S_2 = 1.00 + 1.85 = 2.85 \text{ mm}$$

Adjusting circumferential backlash

Note

The backlash to be set is embossed on the ring gear.

1. Drive pinion set using the shims "S3" determined while adjusting the drive pinion and tighten all mounting screws to 50 Nm (37 ftlb).

Note

Make sure the collar nut of the drive pinion is tightened to 250 Nm (184 ftlb).

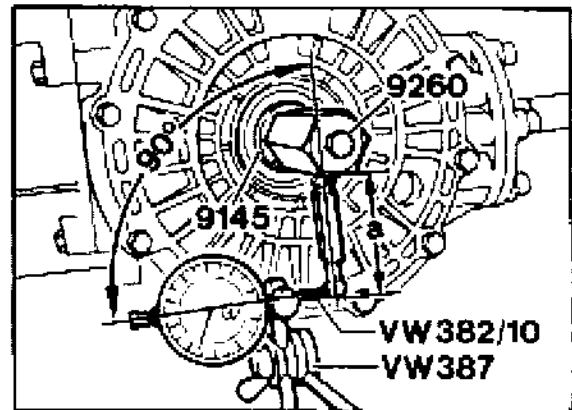
2. Fit the adjusting shims ($S_1 + S_2$) determined into the transmission case and the transmission side cover, respectively.
3. Fit differential and transmission side cover and tighten all hexagon head bolts of the cover to 23 Nm (17 ftlb).

Note

Always make sure that there is a certain amount of backlash when tightening the hex bolts. Never allow the drive pinion to bind.

4. Assemble measuring lever VW 388 and adjusting device VW 521/4 and adjust lever length to 80 mm with the plunger. Refer to dimension "a" in the picture.
5. Insert adjusting device with clamping sleeve (Special Tool 9145) into the differential and clamp firmly.
6. Rotate differential in both directions several times to settle the tapered roller bearings.

7. Fit universal dial gauge holder with flat extension in such a way as to produce a right angle between dial gauge axis and lever.



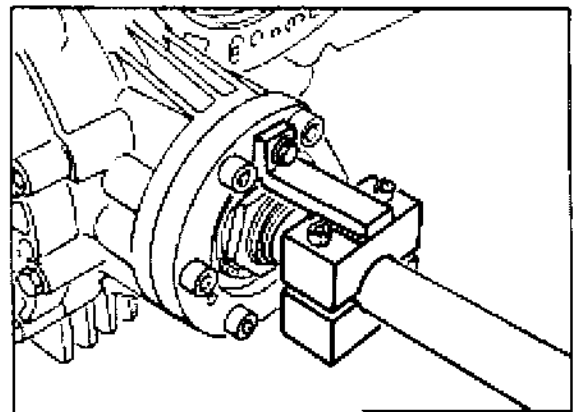
972-39

Dimension "a" = approx. 80 mm

8. Turn ring gear carefully at the clamping screw of the adjusting device up to the stop and set the dial gauge to zero. Turn back ring gear and read off circumferential backlash. Note the reading.

Note

When carrying out measurements, the drive pinion must be blocked with Special Tool 9339.



A129-39

9. After turning the ring gear a further 90° each, repeat measuring procedures three times. The measured values must not deviate from one another by more than 0.05 mm.

Note

The backlash to be adjusted is embossed on the ring gear. The actual value may be less than the specified value by - 0.05 mm. Under no circumstances must the backlash be greater than the specified value.

10. If the required backlash cannot be obtained, replace spacers (S1 + S2) again. The total shim thickness (S tot.) must not be altered, however.

Note

Changing the shim thickness of "S1" or "S2" by 0.05 mm shim results in a change of backlash by approx. 0.1 mm.